**Radiation Oncology Education Collaborative Study Group (ROECSG)**

**Graduate Medical Education (GME)**

# Physics Study Guide

**(August 2026)**

The following ROECSG physics study guide is an educational supplement for radiation oncology trainees. This document is provided as an educational tool and does not intend to mimic nor anticipate questions on board examinations.

A needs assessment of radiation oncology residents revealed that the preferred form for this study guide is a question and multiple choice answer format (with detailed answers). This study guide is the result of a collaborative effort among radiation medical physicists, radiation oncologists and trainees (medical physics and radiation oncology residents).

The ROECSG physics study guide has been formatted in accordance to the American Association of Physicists in Medicine (AAPM)/ American Society for Radiation Oncology (ASTRO) Core Physics Curriculum for Radiation Oncology Residents[1].

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[1] Studenski MT, Cetnar A, Derosiers CM, Dooley S, Gagneur JD, Galavis PE, Kainz KK, Lamichhane N, Sandwall PA, Shen J, Tien CJ, Wang D, Wang IZ, Warkentin HK, McAvoy S. The AAPM/ASTRO 2023 Core Physics Curriculum for Radiation Oncology Residents. Int J Radiat Oncol Biol Phys. 2024 Feb 1;118(2):325-329. doi: 10.1016/j.ijrobp.2023.08.063. Epub 2023 Sep 9. PMID: 37689369.

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# Chapter 1: Fundamental physics

Yunjie Yang , Peter Ferjancic,

Ria Mulherkar Kraft, May Tsao

**1.1 SI Units and Prefixes**

**Question 1.1:**

What is the International System (SI) unit for **electric current**?

A. Sievert (Sv)
B. Becquerel (Bq)
C. Coulombs per kilogram (C/kg)
D. Ampere (A)
E. Gray (Gy)

**Answer Question 1.1: D**

Electric Current:

Ampere (A) – equivalent to 1 coulomb per second; coulomb (C) is the unit of electric charge

**Question 1.2:**

What is the International System (SI) unit for **absorbed radiation dose**?

A. Sievert (Sv)
B. Becquerel (Bq)
C. Coulombs per kilogram (C/kg)
D. Ampere (A)
E. Gray (Gy)

**Answer Question 1.2: E**

Absorbed radiation dose:

Gray (Gy) - energy deposited in a kilogram of a substance by radiation or joules/kilogram (J/kg)

**Question 1.3:**

What is the International System (SI) unit for **radiation exposure**?

A. Sievert (Sv)
B. Becquerel (Bq)
C. Coulombs per kilogram (C/kg)

D. Ampere (A)
E. Gray (Gy)

**Answer Question 1.3: C**

Radiation exposure:

Coulombs per kilogram (C/kg) in air - this is currently the ICRU SI unit for radiation exposure which has replaced Roentgen (R). The old unit Roentgen = 0.000258 C/kg

**Question 1.4:**

What is the International System (SI) unit for **radiation activity**?

A. Sievert (Sv)
B. Becquerel (Bq)
C. Coulombs per kilogram (C/kg)
D. Ampere (A)
E. Gray (Gy)

**Answer Question 1.4: B**

Radiation Activity:

Becquerel (Bq) – measures how many particles or photons are emitted per second by a source measured with a Geiger counter. 1 Bq = 1/s

**Question 1.5:**

What is the International System (SI) unit for **equivalent radiation dose?**

A. Sievert (Sv)
B. Becquerel (Bq)
C. Coulombs per kilogram (C/kg)
D. Ampere (A)
E. Gray (Gy)

**Answer Question 1.5: A**

Equivalent radiation dose:

Sievert (Sv)- the absorbed dose in Gy multiplied by a radiation weighting factor ($W_R$); for x rays, gamma rays and electrons in human tissue $W_R$ =1; For alpha particles $W_{R=20}$; $W_R$ is dependent on the type and energy of the radiation

Dose Equivalent:

$$H_T = \sum_R w_R \cdot D_{T,R}$$

$H_T$ is the dose equivalent
$D_{T,R}$ is the absorbed dose
$W_R$ is *the radiation weighting factor for radiation R, and T is the tissue*

•calculated for individual organs

•accounts for the particular sensitivity of certain organs and body parts to radiation

•dose equivalent limits are in place to avoid deterministic effects (where severity of deterministic effect increases with radiation dose above a threshold, below which tissue reactions are not detectable nor observed; eg. radiation skin reactions)

**Question 1.6:**

What is the International System (SI) unit for **effective radiation dose**?

A. Sievert (Sv)
B. Becquerel (Bq)
C. Coulombs per kilogram (C/kg)
D. Ampere (A)
E. Gray (Gy)

**Answer Question 1.6: A**

Effective radiation dose:

Sievert (Sv)- Effective radiation dose is the equivalent dose multiplies by the tissue weighting factor $W_T$ (eg. the tissue weighting factors for the following organs, colon, skin and whole body are 0.12, 0.01 and 1 respectively). For example the effective dose to skin with the dose equivalent of 100 mSV is 100 mSV X 0.01 =1 mSv

Effective Dose (*E*):

$$E = \sum_T w_T \cdot H_T$$

$w_T$ is the tissue-weighting factor

- Calculated for the whole body
- Accounts for sensitivity of the organ to radiation
- Effective dose limits are in place to avoid stochastic effects (where severity does not increase with dose (eg. radiation induced cancer)

References:

Bjärngard BE. Radiation therapy and SI units. Int J Radiat Oncol Biol Phys. 1981 Feb;7(2):283-5. doi: 10.1016/0360-3016(81)90450-8. PMID: 7216865.

ICRU report 51, quantities and units in radiation protection dosimetry

## 1.2 Classical Mechanics

**Waves (transverse and longitudinal):**

**Question 2:**

Which statement regarding waves is true?

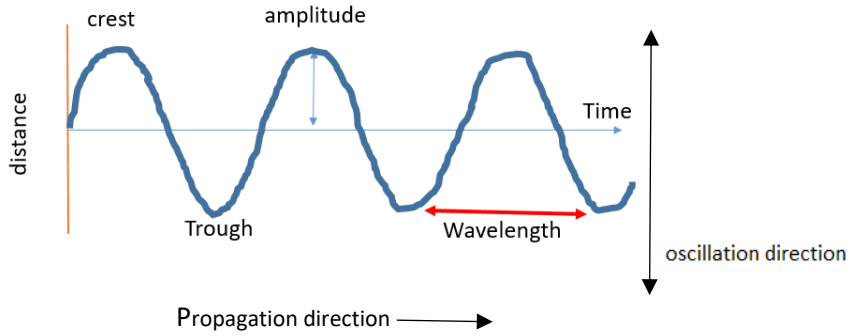
A. Light waves are longitudinal
B. Electromagnetic waves are transverse
C. Sound waves are transverse
D. For transverse waves, the particles move parallel to the direction of the wave

**Answer Question 2: B**

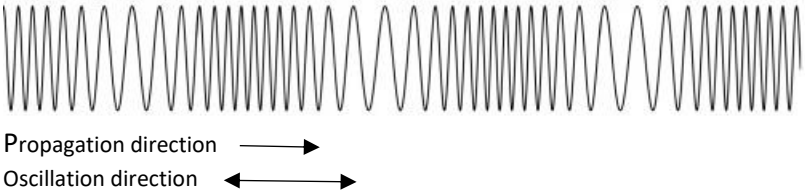
Electromagnetic waves, including light waves are transverse.

Sound waves are longitudinal

For transverse waves, the particles move perpendicular to the direction of the wave:




For longitudinal waves, the particles move parallel to the direction of the wave:




Reference:

https://openstax.org/books/physics/pages/13-1-types-of-waves

**Relationship between wavelength and frequency:**

**Question 3:**

Using the quantum model for electromagnetic radiation such as the photoelectric effect or Compton scattering, which of the following statement is true:

A) As wavelength becomes longer, frequency becomes lower and energy of the photon becomes greater
B) As the wavelength becomes longer, frequency becomes higher and the energy of the photon becomes greater
C) As the wavelength becomes shorter, frequency becomes higher, and the energy of the photon becomes greater
D) As the wavelength becomes shorter, frequency becomes lower and the energy of the photon becomes greater

**Answer Question 3: C**

As the wavelength becomes shorter, frequency becomes higher, and the energy of the photon becomes greater

Using the quantum model, the amount of energy (E) carried by a packet of energy or photon is:

E = h $v$

h = Planck's constant = 6.62 x $10^{-34}$ J-sec
$v$ = frequency or cycles per second

The relationship between wavelength (λ) in metres, frequency ($v$) in cycles per second or hertz and velocity of propagation (c) in metres per second is:

$$c = v\,\lambda$$

Therefore:

$$E = \frac{\mathrm{h\,c}}{\lambda}$$

Wavelength is inversely proportional to energy (E) and frequency ($v$).

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**1.3 Electricity and Magnetism**

**Charge and Charge Conservation:**

**Question 4:**

Which statement is **false** for electric charge?

A. An electric charge is a physical property of matter that causes it to experience a force when applied in an electromagnetic field
B. Electric charge is a conserved physical property of matter.
C. Negative charge is carried by electrons, positive charge is carried by protons.
D. The smallest charge is approximately 1.602 x $10^{-19}$ Coulombs

E. A voltmeter measures the flow of electric charge through a circuit

**Answer Question 4: E**

All the statements are true except E.

A voltmeter measures voltage or potential difference across 2 points in a circuit, measured with the SI unit volt (V).

An ammeter measures current or the flow of electric charge through a circuit, measured in amperes (A), the SI unit for electric current.

A galvanometer also measures current but this method involves breaking the electrical circuit.

Reference:

Purcell, E. M., & Morin, D. J. (2013). Electricity and Magnetism (3rd ed.). Cambridge: Cambridge University Press

**Electric Fields, Coulomb Force**

**Question 5:**

The SI unit for electric field is:

A. Coulomb per meter (C/m)
B. Volt per meter (V/m)
C. Newtons per meter (N/m)
D. Amperes per meter (A/m)

**Answer Question 5: B**

The electric field is defined as a vector field that relates a point in space to force per unit charge. The SI unit is volt per meter (V/m). This is equivalent to newtons per coulomb (N/C).

Reference:

Newell, D. and Tiesinga, E. (2019), The International System of Units (SI), 2019 Edition, Special Publication (NIST SP), National Institute of Standards and Technology, Gaithersburg, MD, [online], https://doi.org/10.6028/NIST.SP.330-2019

**Question 6:**

According to Coulomb's law, which of the following is **false** regarding the electrostatic force between two point charges:

A. The force is a vector product, proportional to the product of the magnitudes of the charges and proportional to the inverse square of the distance between them
B. Will repel if the charges have the same electric charge or sign
C. Will attract if the charges have different electric charge or sign
D. The force is along a straight line joining the two charges

**Answer Question 6: A**

All of the above are true except for A.

The force is a vector product, proportional to the product of the magnitudes of the charges and ***inversely*** proportional to the inverse square of the distance between them.

Thus, A is false.

Reference:
Fulcher LP. Improved result for the accuracy of Coulomb's law: A review of the Williams, Faller, and Hill experiment. Phys Rev A Gen Phys. 1986 Jan;33(1):759-761.

**Current and Potential Difference**

**Question 7:**

Which of the following is **false** regarding an electric current?

A. Current flows from higher potential to lower potential
B. It is the number of coulombs of charge that move past a point per unit of time
C. It is measured in amperes (A) or coulombs per second
D. Electric current that flows only in one direction is an alternating current
E. Most clinical linear accelerators use an alternating electric field for accelerating of electrons

**Answer Question 7: D**

All the above are true except for D.

Electric current that flows in one direction is a direct current.

For alternating current, the flow of electrons changes direction back and forth at regular intervals or cycles.

Reference:

Semat H, Katz R. Physics, Chapter 26: The electric current. 1958. Robert Katz Publications. 156.

**Question 8:**

How much work is done when 2 coulomb charge moves against a potential difference of 1 Volt?

A. 1 Joule
B. 2 Joules
C. 3 Joules
D. 4 Joules

**Answer Question 8: B**

The work (W in joules) done to move a charge (q in coulombs) through a potential difference ( $\triangle$ V in volts) is directly proportional to the charge and the potential difference, where:

W= q * $\triangle$ V

Reference:

https://digestiblenotes.com/physics/current_electricity/charge.php

**The Electron Volt**

**Question 9:**

What is the definition of an electron volt (eV)?

A. It is the SI unit of energy
B. It is the amount of kinetic energy gained by a single electron accelerating through an electric potential difference of 1 volt in a vacuum
C. It is a measure of voltage or electric potential difference, dependent on current and resistance
D. It is a measure of electric charge, a physical property of matter that causes it to experience a force when placed in an electromagnetic field

**Answer Question 9: B**

It is the amount of kinetic energy gained by a single electron accelerating through an electric potential difference of 1 volt in a vacuum.

The SI unit of energy is Joule (J). 1 eV= 1.602 x $10^{-19}$ J.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Magnetic Fields, Motion of a Charged Particle in a Magnetic Field**

**Question 10:**

Which of the following is **false** regarding magnetic fields:

A: Regions where magnetic field lines are closer together represent stronger magnetic field
B: Magnetic field lines cannot intersect
C: Magnetic fields have both magnitude (strength) and direction
D: the SI unit is gauss
E: Magnetic field lines of a moving charge form closed loops

**Answer Question 10: D**

The SI unit for magnetic field is tesla (T) and not gauss. 1 T = 10 000 G= 1 newton/ampere-meter

All the other options are true.

Reference:

Purcell, E. M., & Morin, D. J. (2013). Electricity and Magnetism (3rd ed.). Cambridge: Cambridge University Press.

**Question 11:**

When a charged particle enters into a uniform magnetic field, the charge particle will:

A. Start turning in a circular path
B. Bend 90 degrees or perpendicular to its direction of motion
C. Lose kinetic energy
D. Move in a straight line

**Answer Question 11: A**

Will start to travel in a circular path. No kinetic energy is lost.

Reference:

Purcell, E.M. (1965) Electricity and Magnetism. Berkeley Physics Course, Vol. 2, McGraw-Hill Book Company, New York.

**Electromagnetic Spectrum: Energy and Frequency for Photons (Planck Relationship)**

**Question 12:**

Order the following electromagnetic radiation types in order of **frequency** (number of electromagnetic wave oscillations per second or Hertz-Hz) **from lowest to highest**.

Choose the correct answer **A, B, C or D** below:

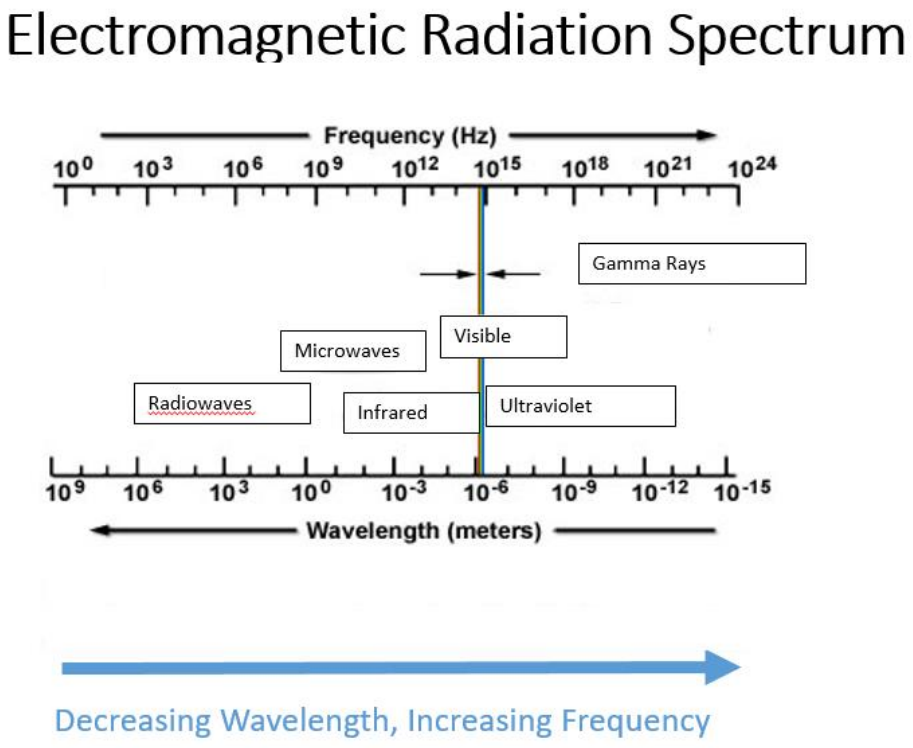
1. Gamma rays
2. Microwave
3. Infrared
4. Visible
5. Ultraviolet
6. Radiowaves

**A. 6, 2, 3, 4, 5, 1**
**B. 1,2, 3, 4, 5, 6**
**C. 3, 2, 1, 6, 5, 4**
**D. 6, 2, 3, 5, 4, 1**

**Answer Question 12: A**

**6, 2, 3, 4, 5, 1**

Radiowaves, Microwave, Infrared, Visible, Ultraviolet, Gamma Rays




References:

https://micro.magnet.fsu.edu/primer/lightandcolor/electromagintro.html

https://en.wikipedia.org/wiki/Electromagnetic_spectrum

**Question 13:**

The energy of a single photon **in a vacuum** is proportional to which of the following quantity of the photon?

A. Speed
B. Frequency
C. Wavelength
D. Intrinsic spin

**Answer Question 13: B**

In quantum mechanics, the energy of a photon is given by $E = h\nu$, where $E$ is the photon energy, $h$ is the Planck constant ($6.626 \times 10^{-34}$ Joule-second), and $\nu$ is the frequency of the photon. Hence, the correct answer is B. Note that the frequency of the photon can be written as $\nu = \frac{c}{\lambda}$, where $c$ is the speed, and $\lambda$ is the wavelength. As a result, the energy of the photon is *inversely* proportional to the

wavelength, so C is incorrect. The speed of a photon (i.e., speed of light) in vacuum is a constant (about $3 \times 10^8\ m/s$), so A is incorrect. The intrinsic spin is an intrinsic quantum mechanical property of a given particle, and it is equal to 1 for *any* photon, regardless of its energy, so D is incorrect.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

## 1.4 Basic Special Relativity

### Universal Speed limit and $E=mc^2$

**Question 14:**

What is the universal speed limit?

A. The speed of light in a vacuum
B. The speed of sound in a vacuum
C. 399 792 458 meters per second
D. The speed of electrons in a vacuum

**Answer Question 14: A**

The speed of light is the universal speed limit. This relates to Einstein's theory of relativity $E= mc^2$. The speed of sound in a vacuum is zero. The speed of electrons in a vacuum approaches but does not exceed the speed of light.

The speed of light is: $3 \times 10^8$ m/s or 2.99792458 m/s.

Einstein's Theory of relativity relating to energy and mass indicates that the speed of light in a vacuum is the same for all observers regardless of their relative motion or of the motion of the light source. The speed of light is the universal speed limit. Energy and mass are interchangeable and related to the speed of light. Pure energy is electromagnetic radiation, which travels at the speed of light.

$E = mc^2$

E = energy in joules
m= mass in kg
c = speed of light in m/s

Reference:

Feigenbaum MJ, Mermin ND. E=mc$^2$. Am J Phys 56: 18-21, 1988.

**Relativistic Mass Increase**

**Question 15:**

What is the relativistic mass increase in particle accelerators?

A. The increase in mass or inertia as a particle's velocity approaches the speed of light
B. The increase in mass of the particle when the particles collide
C. The relative increase in mass for $^1$H protons, $^4$He helium, $^{12}$C carbon used in proton radiotherapy
D. The relative increase in weight of the particle due to the effect of gravity as compared to a vacuum

**Answer Question 15: A**
The increase in mass or inertia as the particle's velocity approaches the speed of light.

Reference:

https://www.energy.gov/science/doe-explainsrelativity

**Rest Mass**

**Question 16:**

What is the relativistic rest mass of a photon, with an energy of 5 joules?

A. $\frac{5}{(3 \text{ x } 10^8)^2}$ kg

B. $5 \times (3 \times 10^8)$ kg

C. $\frac{5}{3 \times 10^8}$ kg

E. $5 \times (3 \times 10^8)^2$ kg

**Answer Question 16: A**

$E = mc^2$

$m = E/c^2$

$$= \frac{5\ \text{J}}{(3 \text{ x } 10^8\ \text{m/s})^2}$$

$$= \frac{5}{(3 \text{ x } 10^8)^2}\ \text{kg}$$

m=mass in kg
E= energy in J
c=speed of light

The relativistic rest mass is also the invariant mass and it is the mass of an object measured when it is at rest relative to an observer and remains constant regardless of the object's velocity.

Reference:

Feigenbaum MJ, Mermin ND. $E=mc^2$. Am J Phys 56: 18-21, 1988.

**Total Energy**

**Question 17:**

Which of the following statement regarding elastic and inelastic collisions is true?

A. An elastic collision conserves both kinetic energy and momentum
B. An inelastic collision conserves both kinetic energy and momentum
C. An elastic collision converses kinetic energy and not momentum
D. An inelastic collision conserves kinetic energy but not momentum

**Answer Question 17: A**

An elastic collision conserves both kinetic energy and momentum.

An inelastic collision conserves momentum but not kinetic energy

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 18:**

The amount of energy transferred from ionizing radiation to tissue per unit length of the radiation track is:

A. Relative biological effectiveness
B. Linear energy transfer
C. Mass stopping power
D. Mass energy transfer coefficient
E. Kerma

**Answer Question 18: B**

Linear energy transfer: is the amount of energy transferred from ionizing radiation ***to tissue*** (ie. the rate of energy loss per unit path length in collisions in which energy ***is locally absorbed***, rather than carried away by secondary electrons. Other commonly used units for linear energy transfer is kiloelectronvolts per micrometer (keV/μm) or megaelecronvolts per centimeter (MeV/cm).

Whereas:

Relative biological effectiveness is the ratio of biological effectiveness of one type of ionizing radiation relative to another, given the same amount of absorbed energy. For example the relative biological effectiveness of protons in clinical radiotherapy has been quoted to be 1.1 relative to photon beams.

Mass stopping power is the rate of kinetic energy ***loss*** per unit path length of a particle divided by the density of the medium, usually expressed as MeV $cm^2$ /g. This could be considered as linear energy transfer divided by the density of the medium.

Mass energy transfer coefficient is the quantity of energy ***transferred*** from photons ***to charged particles*** per unit mass, expressed as $cm^2/g$.

Kerma (kinetic energy released in the medium) is the total kinetic energy of charged particles liberated by ionizing radiation (eg. photons or neutrons) per unit mass of a material and is measured in Gy or J/kg. Kerma is approximately the same as absorbed dose for low energies, but kerma is much higher than absorbed dose for higher energies because some of the energy escapes from the absorbing volume in the form of bremsstrahlung (x-rays) or fast moving electrons, not accounted as absorbed dose.

References:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

The 2007 Recommendations of the International Commission on Radiological Protection". *Annals of the ICRP*. ICRP publication 103. **37** (2–4). 2007.

Paganetti H, Blakely E, Carabe-Fernandez A, Carlson DJ, Das IJ, Dong L, Grosshans D, Held KD, Mohan R, Moiseenko V, Niemierko A, Stewart RD, Willers H. Report of the AAPM TG-256 on the relative biological effectiveness of proton beams in radiation therapy. Med Phys. 2019 Mar;46(3):e53-e78.

Muhammad W, Maqbool M, Hussain A. Basic concepts in radiation dosimetry. In: An introduction to medical physics. 2017, 9-41.

**Question 19:**

A photon interacts with an atom and undergoes Compton scattering. The electron rest mass energy is 0.511 MeV or 511 keV. If the photon is backscattered 180 degrees, what is the residual energy of the scattered electron?

A. 0.256 MeV
B. 0.511 MeV
C. 1.022 MeV
D. 2.044 MeV

**Answer Question 19: A**

When a photon collides with an electron in a Compton interaction, the photon and electron may be scattered in various directions depending on the energy of the incident photon, and the amount of energy transferred to an electron. An electron has a rest mass energy of 0.511 MeV (511 keV). One possible scenario is that the photon is scattered 0 degrees (retains its forward motion).

For a backscattered photon (180 degrees), half the energy of the electron is imparted to the photon. Due to conservation of energy, the residual energy of the electron would be 0.511/2= 0.256 MeV.

Reference:

Bichsel H, Schindler H. The interaction of radiation with matter. In: *Particle Physics Reference Library*. Springer International Publishing; 2020:5-44.

# Chapter 2: Atomic and nuclear structure

Peter Ferjancic, Victoria (Tori) Doss, May Tsao

## 2.1 The atom

### Protons, Neutrons, Electrons

**Question 20:**

For an atom described as $^{60}_{27}Co$, which of the following is correct:

A. Cobalt is the name of the atom, 60 is the number of protons and 27 is the number of neutrons

B. Cobalt is the name of the atom, 60 is the number of neutrons and 27 is the number of electrons

C. Cobalt is the chemical symbol of the element, 60 is the mass number and 27 is the atomic number

D. Cobalt is the chemical symbol of the element, 60 is the mass number and 27 is the number of neutrons

**Answer Question 20: C**

For $^{A}_{Z}X$:

X is the chemical symbol for the element (ie.cobalt), A is the mass number (number of nucleons or number of neutrons and protons in the nucleus. The mass number in this example is 60. Z is the atomic number (or number or protons in the nucleus or number of electrons outside the nucleus). In this case, it is 27.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

### Atomic number and atomic mass

**Questions 21:**

For the following categories of atoms, indicate which is correct (A, B, C or D):

A**. Isotopes** (same number of protons but different number of neutrons); **Isotones** (same number of neutrons but different number of protons); **Isobars** (same number of nucleons but different number of protons); **Isomers** (Identical atoms but differs in their nuclear energy states)

B**. Isotopes** (same number of neutrons but different number of protons); **Isotones** (same number of nucleons but different number of protons); **Isobars** (Identical atoms but differs in their nuclear energy states); **Isomers** (same number of protons but different number of neutrons)

C**. Isotopes** (Identical atoms but differs in their nuclear energy states); **Isotones** (same number of protons but different number of neutrons); **Isobars** (same number of neutrons but different number of protons); **Isomers** (same number of nucleons but different number of protons)

D**. Isotopes** (same number of nucleons but different number of protons); **Isotones** (Identical atoms but differs in their nuclear energy states), **Isobars** (same number of protons but different number of neutrons), **Isomers** (same number of neutrons but different number of protons)

**Answer Question 21: A**

Isotopes- Atoms having nuclei with the same number of protons but different number of neutrons (eg. hydrogen $^{1}_{1}H$, deuterium $^{2}_{1}H$, tritium $^{3}_{1}H$)

Isotones- The same number of neutrons but different number of protons (eg. oxygen $^{16}_{8}O$, nitrogen $^{15}_{7}N$, carbon $^{14}_{6}C$)

Isobars – The same number of nucleons but different number of protons (eg. Argon $^{40}_{18}Ar$, Potassium $^{40}_{19}K$, $^{40}_{20}Ca$ Calcium)

Isomers –Identical atoms but differs in their nuclear energy states (eg. $^{131m}_{54}XE$ is an isomer of $^{131}_{54}XE$, where $^{m}$ represents the metastable state)

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Strong and Weak Nuclear Forces:**

**Question 22:**

Sort the following forces in physics from **weakest to strongest**.

Choose the correct answer **A, B, C or D**.

1. Strong nuclear force
2. Gravity on particles with mass
3. Electromagnetic forces on electrically charged particles
4. Weak nuclear force

**A. 4, 3, 2, 1**
**B. 2, 4, 3, 1**
**C. 2, 4, 1, 3**
**D. 2, 3, 4, 1**

**Answer Question 22: B**

Gravity on particles, Weak nuclear force, Electromagnetic forces on electrically charged particles, strong nuclear

Gravity (2) is the weakest of the four interactions at the atomic scale with a relative strength of $10^{-38}$.

Gravity is followed by the weak nuclear force (4), which is responsible for causing radioactive decay of atoms (relative strength $10^{-13}$).

Next, electromagnetic forces (3) has a relative strength of $10^{-2}$.

Lastly, strong nuclear force (1), responsible for holding quarks and nucleons together is the strongest at relative strength of 1.

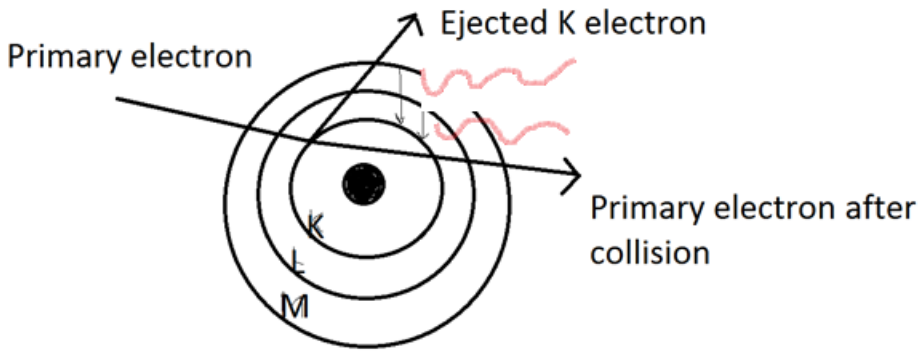
References:
Urone PP, Hinrichs R. OpenStax. Physics; March 26, 2020. Houston, Tx.
https://openstax.org/books/physics/pages/23-1-the-four-fundamental-forces

Podgorsak EB. Basic Radiation Physics. Chapter 1.
https://www.irsn.fr/sites/default/files/documents/professionnels_sante/documentation/syllabus_chapitre_1.pdf

**Orbital Electron Shells (binding energy, transitions, characteristic x rays, Auger electrons)**

**Question 23:**

The following diagram depicts:



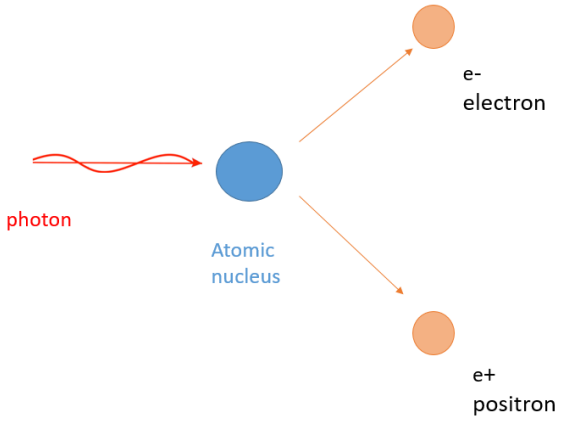

A. Characteristic radiation
B. Pair production

C. Rayleigh scattering
D. Electromagnetic effect

**Answer Question 23: A**

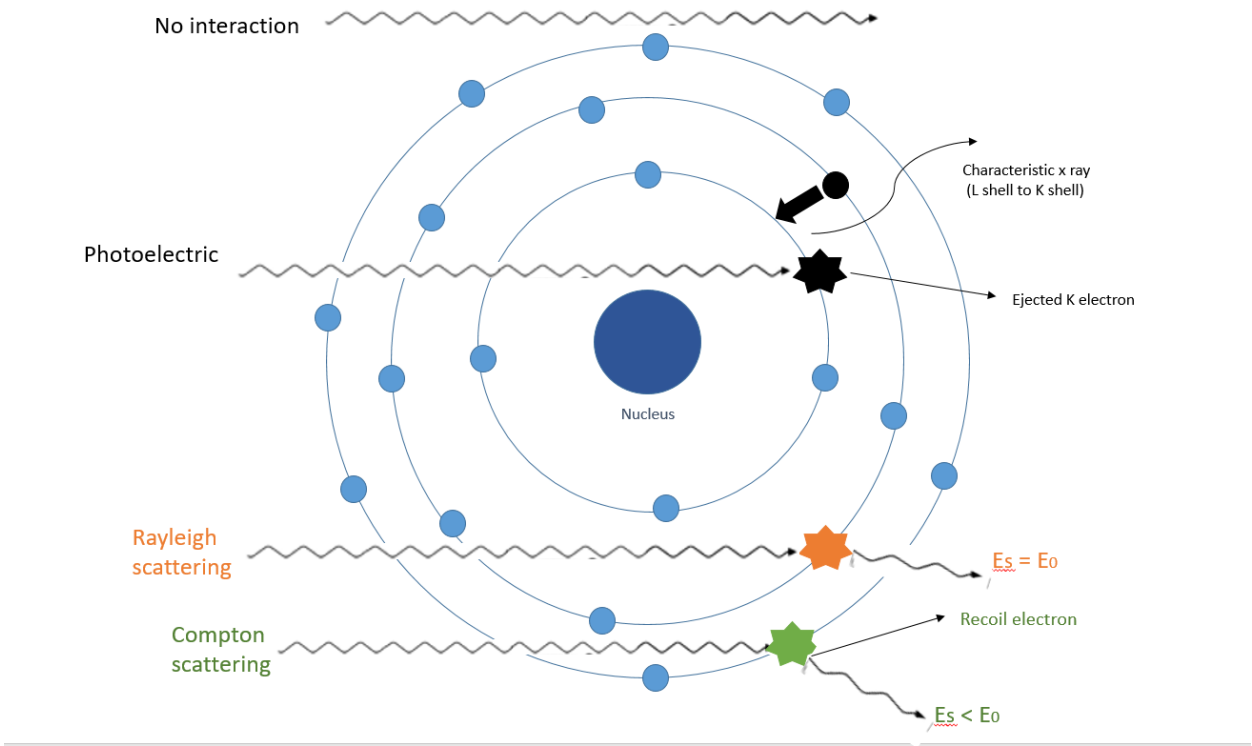
The diagram depicts characteristic radiation or the photoelectric effect. Characteristic radiation occurs when an electron interacts with atoms by ejecting an orbital electron (eg. K, L or M orbital electron), leaving the atom ionized. The incoming electron energy needs to be greater than the binding energy of the electron for that orbiting electron to be ejected. The vacancy left by the ejected electron will result in outer orbital electrons to fall down to fill that vacancy. As a result, energy is radiated as electromagnetic radiation, specifically characteristic radiation.

Pair production occurs when a high energy photon (energy above 1.022 MeV) converts into a electron-positron pair when passing through the Coulomb field of a nucleus. Representative diagram below:




Rayleigh scattering is the deflection of electromagnetic radiation by particles much smaller than the wavelength of the radiation. Rayleigh scattering of light off oxygen and nitrogen molecules in the Earth's atmosphere causes the blue color of the sky and the yellowish and reddish sky in the setting sun.

The diagram below summarizes a summary of **x ray and gamma ray** interactions.




In Rayleigh scattering, the scattered photon energy Es is equal to the initial photon energy $E_0$.

In Compton scattering, the scattered photon energy Es is smaller than the initial photon energy $E_0$. The initial photon energy in Compton scattering is equal to the sum of the energy of the recoil electron and the scattered photon Es.

Electromagnetic effect is the interaction between particles with electric charge through electromagnetic fields.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 24:**

Tungsten has the following binding energies:

K shell =69.5 keV

L shell = 11.5 keV

M shell = 2 keV

A 70 keV electron strikes a tungsten target.

Which of the following can be produced?

Choose the correct answer **A, B, C or D** below.

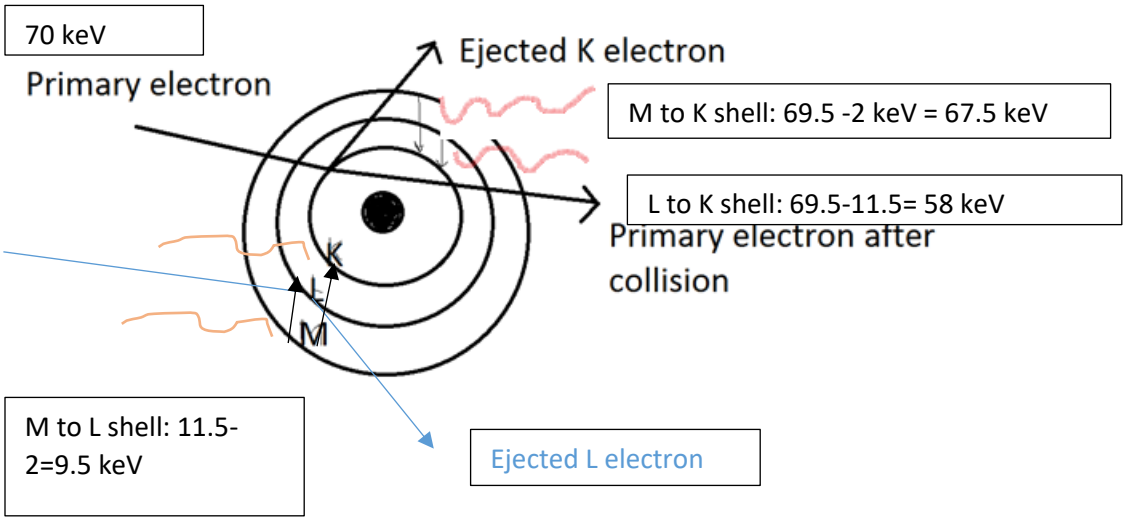
1) 58 keV characteristic radiation
2) 2 keV characteristic radiation
3) 9.5 keV characteristic radiation
4) 67.5 keV characteristic radiation

**A. 1, 2, 3, 4**
**B. 1, 2, 3**
**C. 1, 3, 4**
**D. 1, 2**

**Answer Question 24: C**

**1, 3, 4**

58 keV characteristic radiation,
9.5 keV characteristic radiation
67.5 keV characteristic radiation




Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 25:**

In characteristic radiation due to the photoelectric effect, when an inner-shell is filled with an electron and energy released causes another electron from a different shell to be ejected, that ejected electron is **best described** as a:

A. Auger electron
B. Core electron
C. Valence electron
D. Lepton

**Answer Question 25: A**

Auger electron

A core electron is an electron that is in the inner shell closer to the nucleus.

A valence electron is an electron in the outermost shell of an atom that can form chemical bonds.

A lepton is an elementary particle that either carry one unit of electric charge (eg. electron) or can be neutral.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**2.1 Basic Radioactivity and Decay**

**Topic: Decay Processes (isotopes for imaging/therapy), alpha, beta, electromagnetic (gamma, internal conversion, Auger emission)**

**Question 26:**

Which of the following is **not** a decay process?

A. Alpha decay
B. Beta decay
C. Delta decay
D. Gamma decay
E. Internal conversion

**Answer Question 26: C**

Radioactive decay is the process by which an unstable atomic nucleus loses energy by emitting radiation.

Alpha decay involves the emission of an alpha particle (helium nucleus consisting of 2 protons and 2 neutrons). The atomic number of the nucleus is reduced by 2 and the mass number reduces by 4.

Beta-minus decay occurs when a neutron in the nucleus is converted into a proton, emitting an electron (beta particle) and an antineutrino. The atomic number of the nucleus increases by 1, while the mass number remains the same.

Beta-plus decay occurs when a proton transforms to a neutron, releasing a positron and a neutrino. The atomic number decreases by 1, while the mass number stays the same.

Gamma decay often follows alpha or beta decay. The excited nucleus releases energy in the form of high-energy photons (gamma rays). Atomic number and mass number do not change.

Internal conversion is a radioactive decay process that occurs when an excited nucleus transfers its excess energy directly to an inner-shell electron causing ejection of the electron from the atom.

Reference:

**https://www.arpansa.gov.au/understanding-radiation/what-is-radiation/ionising-radiation/radiation-decay**

**Question 27:**

Which of the following statements regarding $^{60}_{27}$Co or cobalt-60 is **true**?

A. The half-life of cobalt-60 is 74 days. Co-60 decays to photons (0.2 MeV and 0.6 MeV).

B. The half-life of cobalt-60 is 7.2 years. Co-60 decays to 2 photons (1.2 MeV and 1.3 MeV).

C. The dose rate from cobalt decreases with time and results in increased monitor units for the same dose prescription.

D. Cobalt-60 is the most commonly used isotope for intracavitary high dose rate (HDR) brachytherapy worldwide.

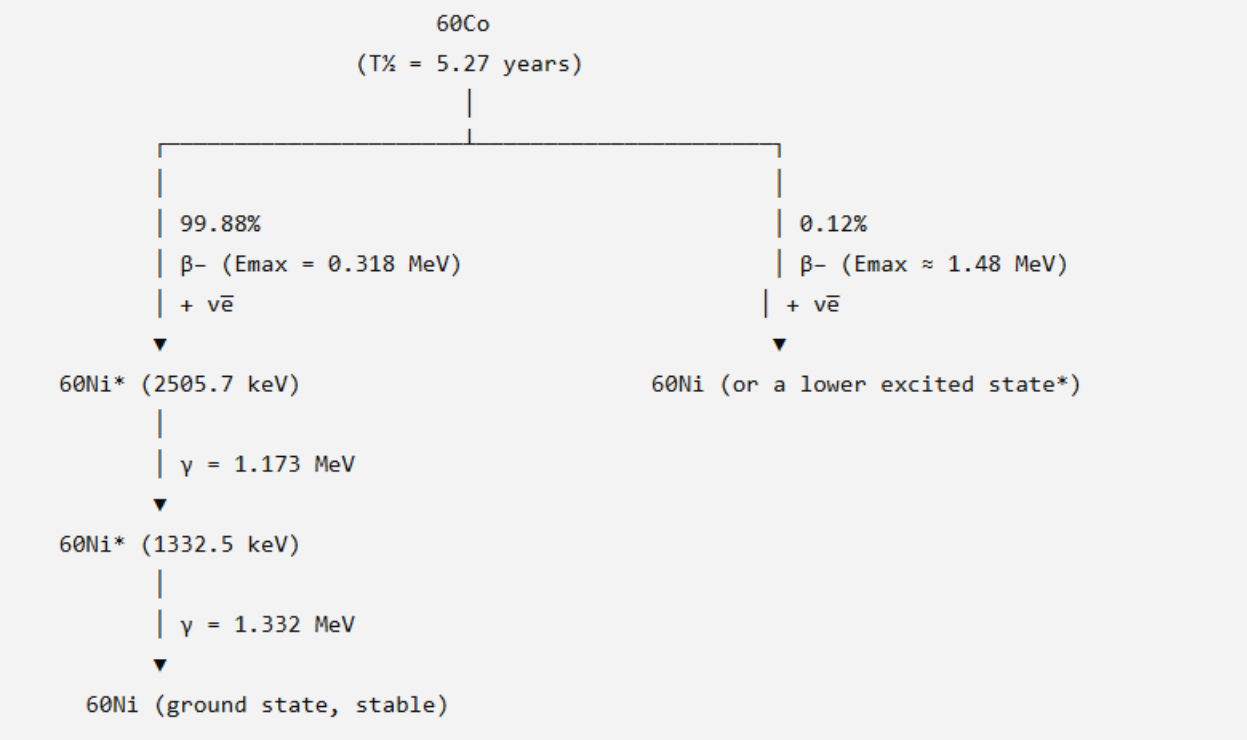
**Answer Question 27: C**

The dose rate from cobalt decreases with time due to decay. As a result, more treatment time is needed for the same radiation prescription. So, C is true.

Answers A and B are incorrect:

The half-life of cobalt-60 is 5.27 years (often quoted as 5.26 years). Cobalt-60 decays predominantly (99.88%) by $\beta^-$ emission with a maximum β energy of 0.318 MeV and an antineutrino. The resulting excited state of $^{60}$Ni emits two γ-rays of 1.173 MeV and 1.332 MeV.

A rare branch (≈0.12%) emits a $\beta^-$ particle with a maximum energy of 1.48 MeV.




Answer D is incorrect. Ir 192 is the most commonly used isotope for intracavitary HDR brachytherapy worldwide.

References:

Strohmaier S, Zwierzchowski G. Comparison of (60)Co and (192)Ir sources in HDR brachytherapy. J Contemp Brachytherapy. 2011 Dec;3(4):199-208. doi: 10.5114/jcb.2011.26471. Epub 2011 Dec 30. PMID: 23346129; PMCID: PMC3551360.

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Activity, Half Life, Mean Life**

**Question 28:**

What is the half-life of Ir-192?

A. 28 days

B. 35 days
C. 74 days
D. 80 days

**Answer Question 28: C**

74 days

Reference: Podgorsak MB, DeWerd LA, Paliwal BR. The half-life of high dose rate Ir-192 sources. Medical Physics 20(4). 1993, 1257-1259.

**Question 29:**

Radium-223 has a **half-life** of 11.4 days. What is the probability of a radioactive Radium-22 atom decaying within 11.4 days?

a. 50%
b. More than 50%
c. Less than 50%
d. -50%

**Answer Question 29: A**

Half-life is defined as the time required for one-half of the atomic nuclei to decay. So, in one half life, the probability of a radioactive atom decaying is 50%. Probabilities cannot be negative numbers.

The half-life of a specific radioactive substance is constant throughout time.

The half-life of a specific radioactive substance does **not** depend on concentration, volume, temperature nor pressure.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 30:**

What is the approximate half-life of a radioactive substance, if its mean life is 30 seconds?

A. 43.35 seconds

B. 20.76 seconds

C. 0.02 seconds

D. 29.31 seconds

**Answer Question 30: B**

Mean life is the average lifetime of all nuclei of a specific radioactive substance. Mathematically, it is the sum of lifetimes of all the particular unstable nuclei in a given sample divided by the total number of present unstable nuclei.

Mean life is related to half-life in *this approximation:*

$t_{1/2} \approx \ln 2\ \tau$

≈ 0.692 x 30 seconds

≈ 20.76 seconds

$t_{1/2}$ = half life

ln2 =0.693

τ = mean life

Reference:

http://hyperphysics.phy-astr.gsu.edu/hbase/Nuclear/halfli2.html

**Probability and Decay Constant**

**Question 31:**

How is the decay constant related to half-life of a radioactive substance?

A. The decay constant is approximately 0.693 multiplied by the half life
B. The decay constant is the activity of the radioactive substance divided by its initial activity at 1 second
C. The decay constant is approximately 0.693 divided by the half life
D. The decay constant is natural logarithm of the number of radioactive nuclei at a certain time divided by the initial number radionuclei

**Answer Question 31: C**

Radioactive decay is a random process and is described as the probability of decay within a given time interval.

The decay constant of a radioactive material is:

$\lambda = \ln(2) / T_{1/2}$

$\lambda$ = decay constant
ln (2) = 0.693
$T_{1/2}$ = half life

Radioactive decay is exponential and is also expressed mathematically as:

$$N_t = N_0 * e^{-\lambda t}$$

Or

$$A_t = A_0 * e^{-\lambda t}$$

Where:
$N_t$ = the number of nuclei remaining at time t
$N_0$ = the initial number of nuclei
e = natural logarithm

$A_t$ = the activity at time t
$A_0$ = the initial activity

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Mathematical Calculations of Radioactive Decay**

**Question 32:**

A radioactive substance has a half-life of 10 seconds. The sample contains $10 \times 10^{20}$ radioactive atoms. How many radioactive atoms will be left after 30 seconds?

A. $10 \times 10^{20} * e^{-6.93 \times 30}$

B. $10 \times 10^{20} * e^{-0.0693 \times 30}$

C. $10 \times 10^{20} / e^{-0.693 \times 30}$

D. $10 \times 10^{20} + e^{-0.693 \times 30}$

**Answer Question 32: B**

$\lambda = \ln(2) / T_{1/2}$
$=0.693 / 10 \text{ s}$
$=0.0693 /s$

$N_t = N_0 * e^{-\lambda t}$
$= 10 \times 10^{20} * e^{-0.0693 \times 30}$

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 33:**

For 100 g of a radioactive isotope, how much of the isotope will remain after 3 half-lives?

A. 12.5 g
B. 50 g
C. 25 g
D. 6.25 g

**Answer Question 33: A**

At 1 half-life, there will be 50 g of the radioactive isotope.
After another half-life (2 half-lives), there will be 25 g of the radioactive isotope.
At 3 half-lives, there will be 12.5 g of the radioactive isotope.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 34:**

Calculate the activity of 1 g of $^{226}Ra$ (half-life 1622 years or 1622 x 3.15 x10 $^{7}$ sec).

A. 0.975 Ci
B. 0.9 Ci
C. 2 Ci
D. 1.356 Ci

**Answer Question 34: A**

0.975 Ci/g

Activity = λN

$$= \frac{0.693}{1622 \times 3.15 \times 10^{7}\ sec} \times \frac{6.02 \times 10^{23}}{226}$$

=3.61 x $10^{10}$ disintegrations per second per gram

=0.975 Ci/g

1 Ci = 3.7 x $10^{10}$ disintegrations per second

N = number of atoms per gram = Avogardo's number/mass number

Avogardo's number = 6.02 x $10^{23}$

λ = 0.693/ $T_{1/2}$

$T_{1/2}$ = half life

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Nuclear Binding Energy/Nuclear Stability/ Chart of Nuclides**

**Question 35:**

Which isotope below has the **highest** nuclear binding energy per nucleon?

A. $^{2}H$
B. $^{6}Li$
C. $^{12}C$
D. $^{19}F$
E. $^{56}Fe$

**Answer Question 35: E**

Nuclear binding energy is the minimum energy needed to disassemble the nucleus of an atom into protons and neutrons (nucleons).

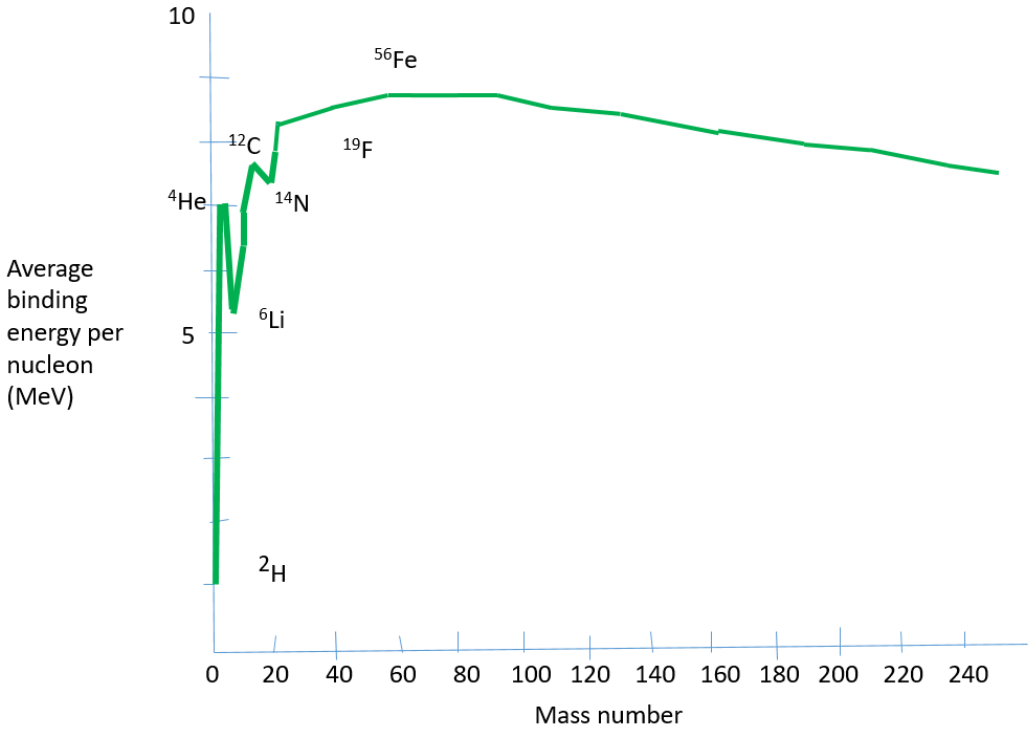
$E=mc^2$

E= nuclear binding energy

c=speed of light

m=difference in mass

With the exception of $^4$He in relation to $^6$Li and $^{12}$C with $^{14}$N, nuclear Binding energy per nucleon generally increases with mass number up to $^{56}$Fe and then gradually decreases as shown in the graph below. In other words, most elements will release energy as they decay towards iron-lighter elements through fusion, heavier through fission.




Reference:

https://en.wikipedia.org/wiki/Nuclear_binding_energy

**Question 36:**

Which statement is **false** regarding nuclear **stability**?

A. Stable nuclei tend to have approximately equivalent numbers of protons and neutrons
B. Neutron to proton ratio tends to be more than 1 in the majority of stable nuclei
C. Atoms with certain numbers of nucleons, known as magic numbers are especially unstable

D. The nuclear stability curve shows unstable isotopes to the right and left of the curve.

**Answer Question 36: C**

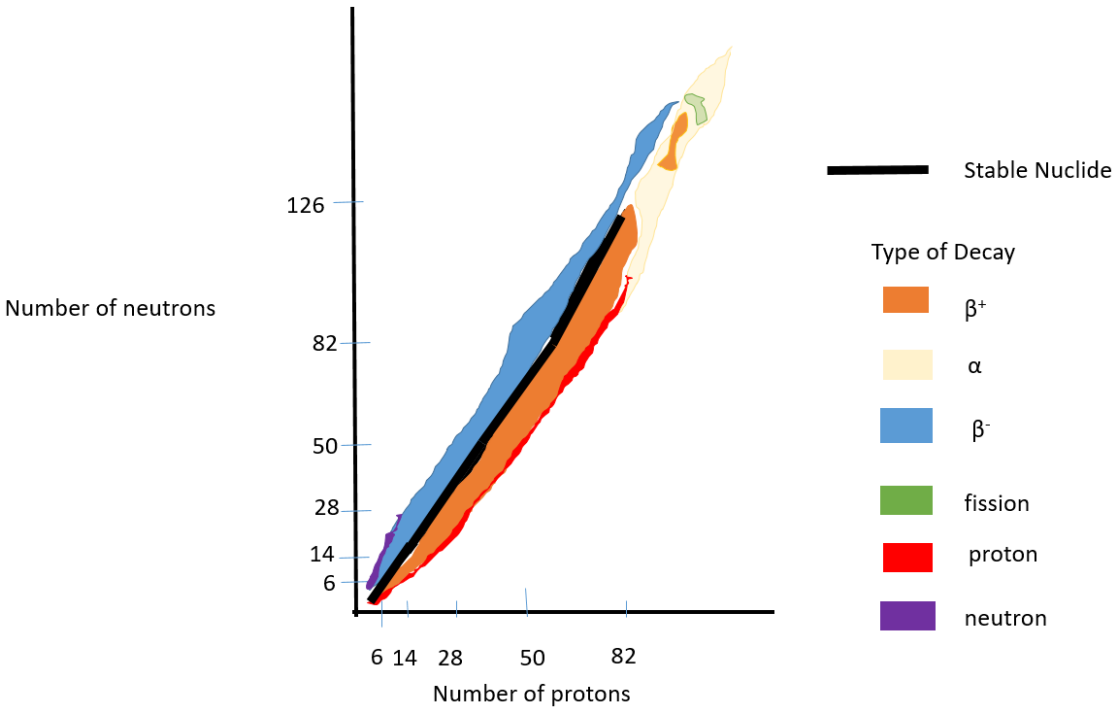
Nuclear stability is dependent on nuclear forces and the balance of attractive strong nuclear force and the repulsive electrostatic force, influenced by the neutron to proton ratio.

Nuclei with magic number of protons and neutrons (2, 8, 20, 28, 50, 82, 126) are particularly **stable**. These numbers correspond to filled energy shells within the nucleus.

Thus, C is false.

The nuclear stability curve shows the line of stability in which the number of neutrons is equivalent to the number of protons up to approximately Z=20, the atomic number or the number of protons. However above Z=20, stable nuclei have increasingly more neutrons (N) than protons (P), the stable nuclide curve gets steeper and N/P >1. For example, the stable nuclide $^{207}$Pb has 125 neutrons and 82 protons for an N/P ratio of 1.52. Larger nuclei have more proton-proton repulsions and require a larger number of neutrons to overcome the electrostatic repulsions and hold the nucleus together.

As such, as stable nucleus may have more neutrons than protons.




Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Radioactive Series/ Parent-Daughter Equilibrium**

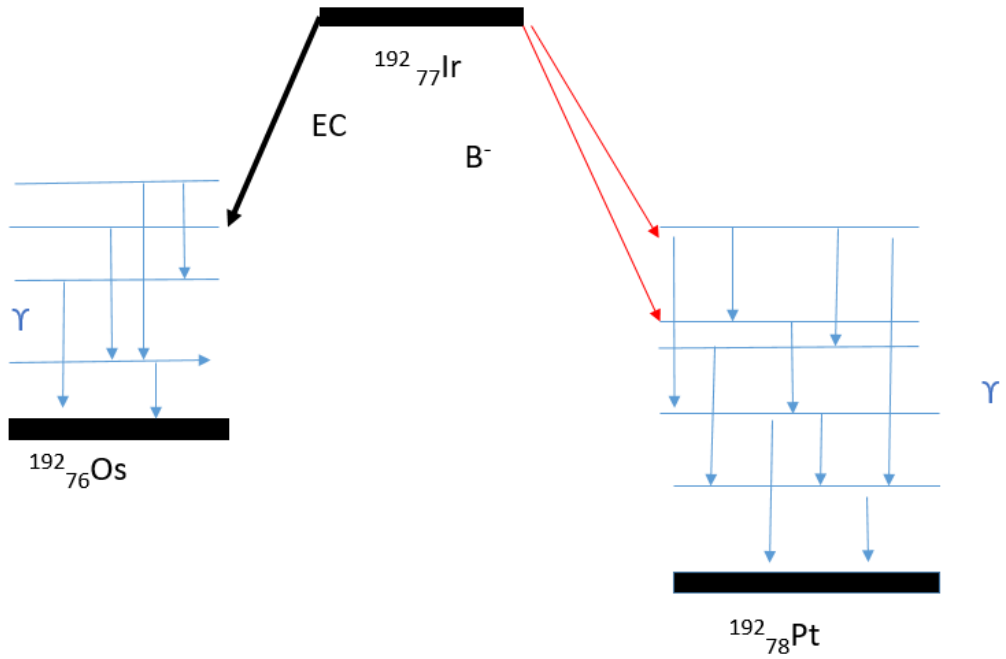
**Question 37:**

$^{192}_{77}$Ir decays to $^{192}_{78}$Pt by the emission of what particle?

A. $\beta^+$
B. $\beta^-$
C. $\alpha$
D. neutron
E. proton

**Answer Question 37: B**

Iridium 192 decays through $\beta^-$ decay (95.1%) to form platinum 192. Iridium 192 also decays through electron capture (4.9%) to excited osmium $^{192}$Os.




Reference:

Nath R, Anderson L L, Luxton G, Weaver K A, Williamson J F, Meigooni A S. *Dosimetry of Interstitial Brachytherapy Sources: Recommendations of the AAPM Radiation Therapy Committee Task Group No. 43*. Medical Physics; 1995:209-234.

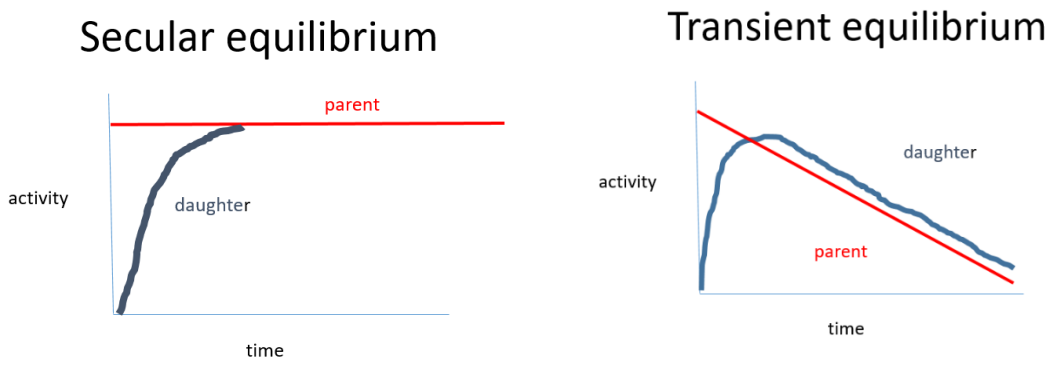
**Question 38:**

What is the definition of parent-daughter equilibrium?

A. A state in which the decay rates of the parent and daughter isotope are equal resulting in a constant ratio of their activities
B. A state in which the daughter's half-life is equal to that of the parent
C. A state in which the decay constant is equivalent between the parent and daughter
D. A state in which the parent and daughter have decayed one half-life

**Answer question 38: A**

In radioactive decay, parent-daughter equilibrium occurs when the decay rate of the parent and daughter isotope are equal. This equilibrium can either be secular (parent half-life much longer than the daughter) or transient (parent half-life significantly longer).




Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Nuclear Reactions**

**Question 39:**

Which of the following is **not** a nuclear reaction?

A. Fission
B. Fusion
C. Nuclear decay
D. Transmutation
E. Electron transport

**Answer Question 39: E**

Electron transport involves proteins or other molecules, which transfer electrons via reduction and oxidation reactions across a cellular membrane. Electron transport is not a nuclear reaction.

Fission occurs when a neutron bombards a larger atom, splitting the nucleus of the atom into two or more smaller nuclei releasing a large amount of energy.

Fusion occurs when 2 or more atomic nuclei combine to form larger nuclei, nuclei/neutron by-products, associated with release or absorption of energy.

Radioactive decay is a nuclear reaction whereby an unstable atomic nucleus loses energy in the form of ionizing radiation.

Transmutation is a nuclear reaction in which an isotope transforms into another by altering the number of protons or neutrons in the nucleus (eg. electron capture radioactive decay where a nucleus absorbs an inner shell electron converting a proton into a neutron and emitting a neutrino).

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

# Chapter 3: Production of kilovoltage xray beams

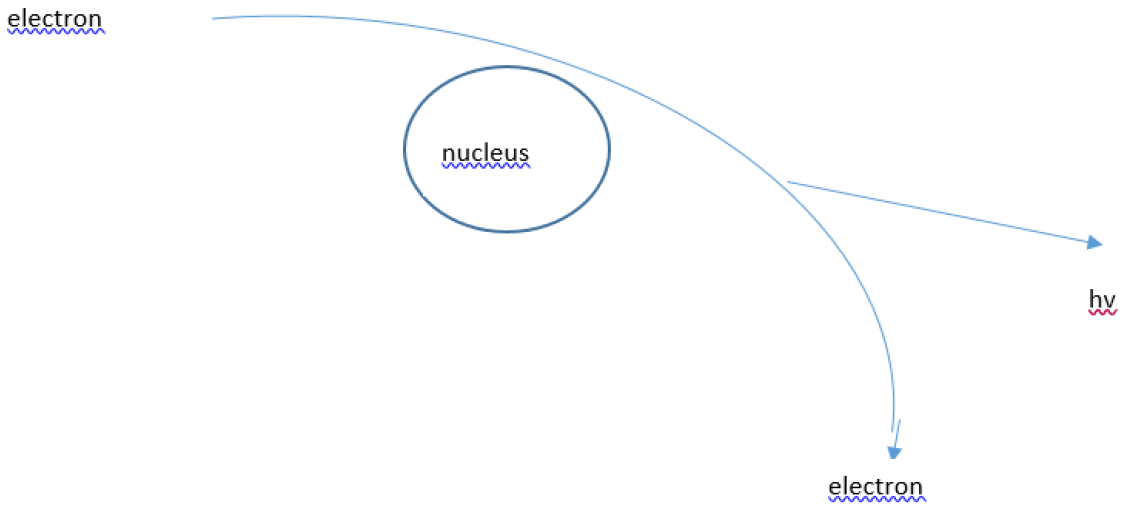
Erika Kollitz, Peter Ferjancic,
Michael Weisman, May Tsao

### 3.1 Basic Physics of X-ray Beam Production

**Bremsstrahlung production of x rays, directionality as a function of energy**

**Question 40:**

The following diagram represents:




A. Characteristic radiation
B. Pair production
C. Raleigh scattering
D. Bremsstrahlung radiation

**Answer Question 40: D**

Bremsstrahlung (braking) radiation occurs when **an electron** passes near a nucleus and is deflected from its path as a result of Coulomb forces of attraction. A part or all of the electron's energy is dissociated from it and propagates in space as electromagnetic radiation or Bremsstrahlung radiation. This phenomenon is predicted by Maxwell's general theory of electromagnetic radiation.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**X-Ray Energy Spectrum, Theoretical (Unfiltered) Spectrum, Effective Photon Energy**

**Question 41:**

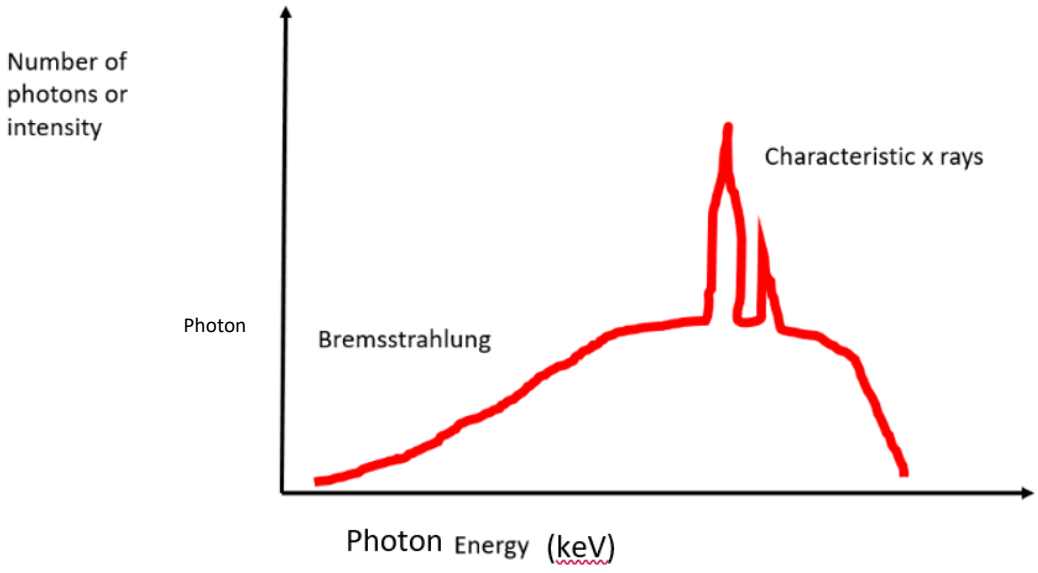
Which is **false** regarding the x ray energy spectrum?

A. It is a plot of the number of photons on the y axis and energy on the x axis
B. It shows both a continuous spectrum (Bremsstrahlung) and characteristic peaks, arising from atomic process within an x ray tube
C. The shortest wavelength limit or maximum energy of the continuous x-ray spectrum is proportional to the potential difference
D. The x ray energy spectrum's shape and characteristics change with x-ray tube current, voltage settings and filters

**Answer Question 41: C**

The shortest wavelength (or maximum energy) of a continuous x-ray spectrum is inversely proportional to the square of the potential difference. Thus, C is incorrect.

An x-ray spectrum is obtained when energetic electrons strike material (anode). The smooth (continuous) part of the spectrum is due to Bremsstrahlung interactions and the peaks are due to characteristic x-rays, specific for the anode material.




Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 42:**

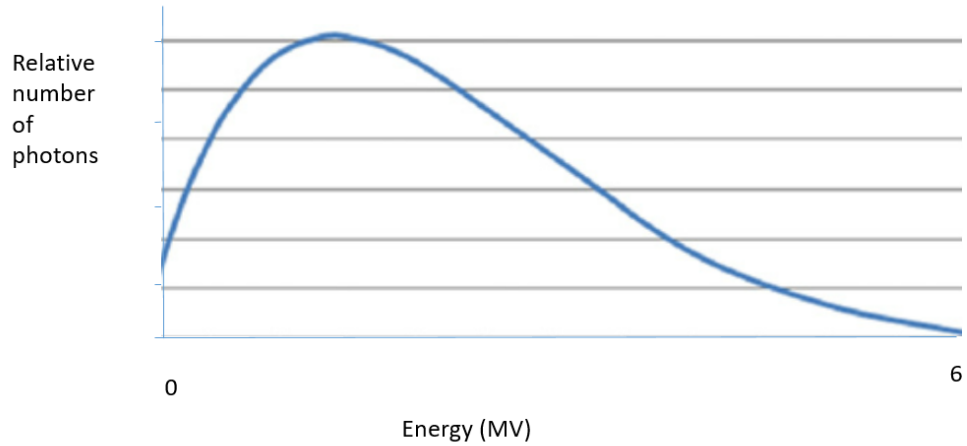
Which statement is **true** regarding a 6 MV photon beam generated from a linear accelerator used for radiation therapy?

A. The 6 MV beam is a monoenergetic beam
B. The nominal peak energy is 6 MeV
C. The effective energy is 6 MeV
D. The half value layer is 3.3 cm lead

**Answer Question 42: B**

The half value layer for 6 MV photons is 4.1 ***mm*** lead.

A 6 MV beam is poly energetic with 6 MeV peak energy. For a poly energetic beam, the effective energy is the energy of the poly energetic beam that has the same half-value layer as a monoenergetic beam. For poly energetic 6 MV, the effective energy is approximately 5.3 +/- 0.3 MV on the central axis.




Reference:

Energy spectrum of 6 MV photon beam from the Pinnacle 3$^{TM}$ treatment planning system v9.0 from a Varian iX linear accelerator (Varian$^{R}$ Medical System, Palo Alto, CA)

**Question 43:**

Which statement is **false** regarding a filtered photon beam versus unfiltered beam from a linear accelerator?

A. For a filtered photon beam, the average energy of the photon beam increases compared to an unfiltered beam
B. The dose rate for a filtered photon beam is decreased compared to an unfiltered beam
C. Overall treatment time is increased for a filtered beam compared to an unfiltered beam for the same dose delivery
D. There is less scatter at the head of the linear accelerator with a filtered beam compared to unfiltered.
E. The filtered beam is more penetrating compared to an unfiltered beam

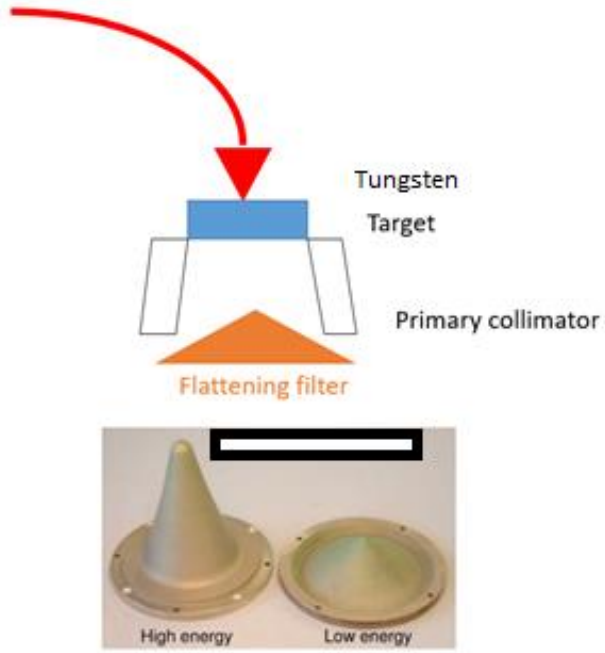
**Answer Question 43: D**

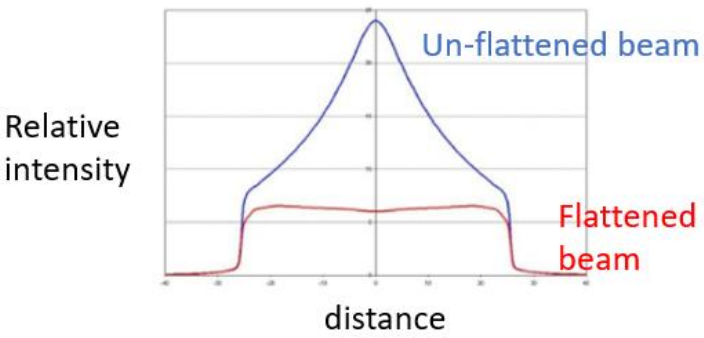
An unfiltered beam contains a wide range of photon energies, including low energy photons. In a linear accelerator, the flattening filter is located between the primary collimator and the monitor chamber. The flattening filter is made of high Z materials and is usually conical in shape to flatten the forward peaked bremsstrahlung spectrum of megavoltage photons. The flat dose profiles with a homogeneous dose variation across the beam provide ease in patient dose calculation during treatment planning.

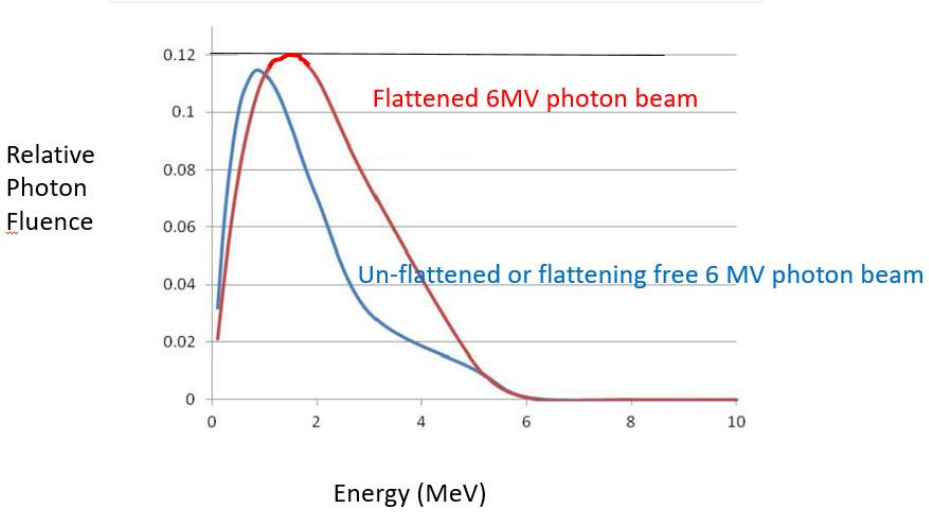
Also, by using a filter, lower energy photons are absorbed by the filter leaving higher average energy photons in the primary beam. As a result, a filtered beam is more penetrating.

The dose rate for a filtered beam is also decreased and treatment time is increased.

The flattening filter is a major source of linac head scattered photons. Thus, D is incorrect.


Tungsten
Target
Primary collimator
Flattening filter
High energy
Low energy


Relative intensity
Un-flattened beam
Flattened beam
distance




References:

Foster RD, Speiser M, Solberg TD. Commissioning and verification of the collapsed cone convolution superposition algorithm for SBRT delivery using flattening filter-free beams. J Applied Clinical Medical Physics 2014; 15(4): 4631.

Saroj DK, Yadav S, Paliwal N, Haldar S, Jagtap A, Kumar A. Assessment of Treatment Plan Quality between Flattening Filter and Flattening Filter Free Photon Beam for Carcinoma of the Esophagus with IMRT Technique. J Biomed Phys Eng. 2023 Jun 1;13(3):227-238.

Sharma SD. Unflattened photon beams from the standard flattening filter free accelerators for radiotherapy: Advantages, limitations and challenges. J Med Phys 2011 36(3): 123-125.

**Characteristic radiation**

**Questions 44:**

Match the following classifications of ionizing photon radiation to its corresponding effect:

**Question 44.1:**

Which of the following is correct regarding **characteristic radiation?**

A. Results from positron-electron collision
B. Results from electron transitions between atomic shells
C. Results from electron-nucleus Coulomb interactions
D. Results from nuclear transitions

**Answer Question 44.1: B**

Characteristic x-rays results from electron transitions between atomic shells

**Question 44.2:**

Which of the following is correct regarding **Bremsstrahlung radiation?**

A. Results from positron-electron collision
B. Results from electron transitions between atomic shells
C. Results from electron-nucleus Coulomb interactions
D. Results from nuclear transitions

**Answer Question 44.2: C**

Bremsstrahlung results from electron-nucleus Coulomb interactions

**Question 44.3:**

Which of the following is correct regarding **Gamma rays?**

A. Results from positron-electron collision
B. Results from electron transitions between atomic shells
C. Results from electron-nucleus Coulomb interactions
D. Results from nuclear transitions

**Answer Question 44.3: D**

Gamma rays results from nuclear transitions

**Question 44.4:**

Which of the following is correct regarding **annihilation quanta?**

A. Results from positron-electron collision
B. Results from electron transitions between atomic shells
C. Results from electron-nucleus Coulomb interactions
D. Results from nuclear transitions

**Answer Question 44.4: A**

Annihilation quanta results from positron-electron collision

Reference:

Podgorsak EB. Basic Radiation Physics. Chapter 1.
https://www.irsn.fr/sites/default/files/documents/professionnels_sante/documentation/syllabus_chapitre_1.pdf

**Efficiency and directionality as a function of energy (keV vs MeV)**

**Question 45:**

Regarding the physics of x-ray production, when electrons hit a tungsten target, what mathematical relationship is there with efficiency (the ratio of output energy emitted as x-rays to the input energy deposited by electrons)?

A. Efficiency is inversely proportional to Z of the target and inversely proportional to the tube voltage
B. Efficiency is proportional to Z of the target and inversely proportional to the tube voltage
C. Efficiency is proportional to Z of the target and is proportional to the tube voltage
D. Efficiency is independent of the tube voltage

**Answer Question 45: C**

Efficiency is proportional to Z of the target and efficiency is proportional to the tube voltage. All other statements are incorrect.

Efficiency = $9 \times 10^{-10}$ ZV

Z= 74 for Tungten
V = tube voltage

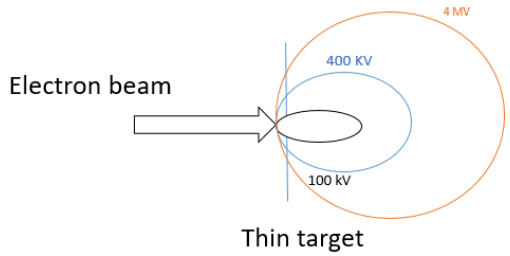
Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

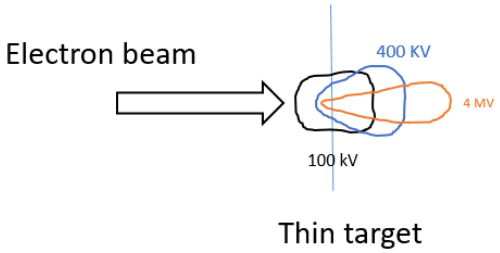
**Question 46:**

Which diagram below best represents the spatial distribution of x-rays around a thin target?

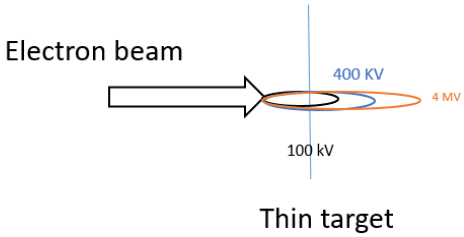
A.



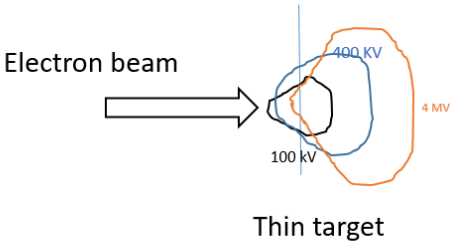

B.




C.




D.




**Answer Question 46: B**

The spatial distribution of x rays around a thin target describes how electrons may have one or more bremsstrahlung interactions with a thin metal target. Electron interaction with a thin target causes partial or complete loss of electron energy resulting in bremsstrahlung photons. The bremsstrahlung photons

may have any energy up to the energy of the incident electron. The direction of the bremsstrahlung photon depends on the incident electron energy. For electron energies below 100 keV, x-rays are emitted approximately equally in all directions. With increased electron energy, the direction of the x-ray emission becomes increasingly forward.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**3.2 Generation of Beams**

**X-ray tube**

**Question 47:**

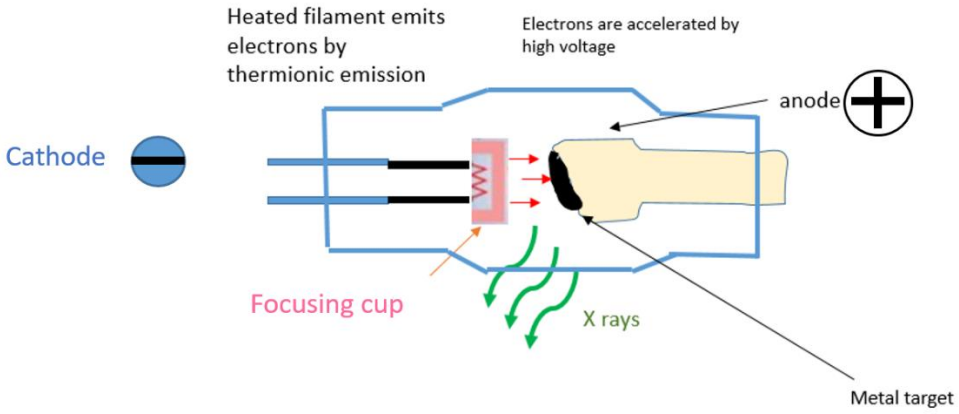
Which one of the following is **not** a component of an x-ray tube?

A. Cathode
B. Anode
C. Metal target
D. Focusing cup
E. Ion chamber
F.

**Answer Question 47: E**

Ion chamber is not part of an x-ray tube.

X-ray tube:



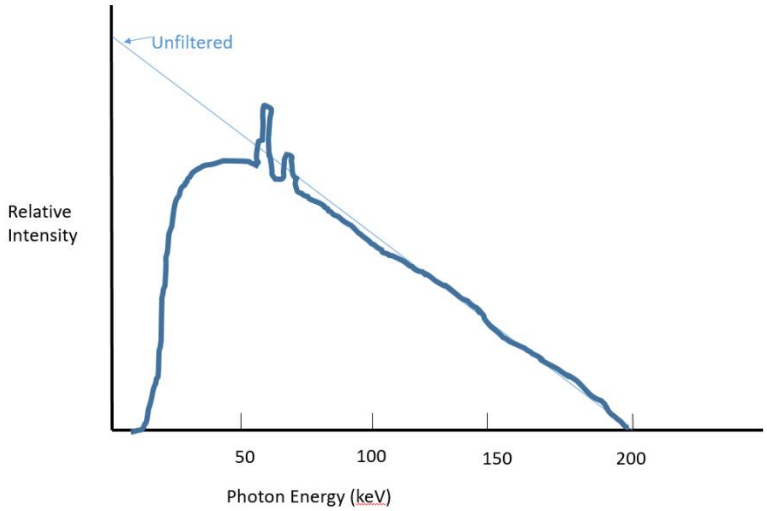

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Anode, Cathode kVp, mA, Focal Spot, Line Focus Principle, Heel effect, Rotating Anode, thermionic emissions**

**Question 48:**

What option below is the best approximation for **average** x-ray energy with respect to the following spectral distribution of x rays, ignoring filtration effects?




A. 200 keV
B. 50 keV
C. 67 keV
D. 167 keV

**Answer Question 48: C**

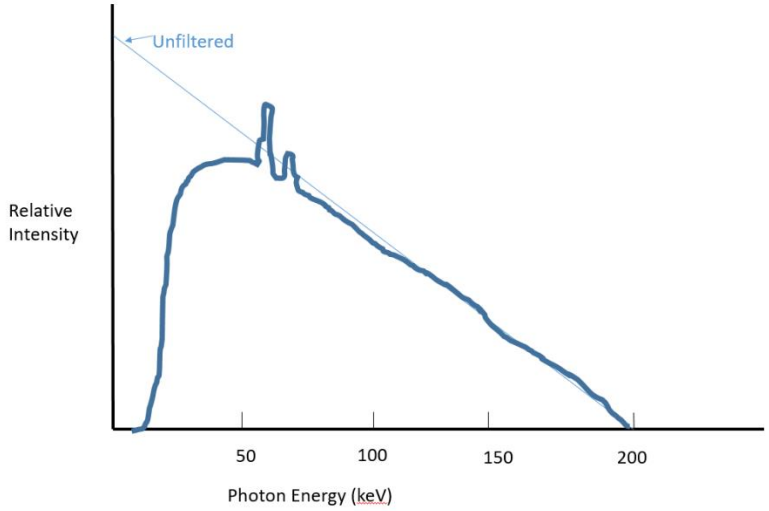
The general rule of thumb is that the average x-ray energy is approximately 1/3 the maximum energy or the applied kilovolts across the x ray tube (kilovolts peak or kVp). In this case, the best approximation for average x-ray energy is: 200 keV/3= 67 keV.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 49:**

What is the energy of the applied kilovoltage across the x ray tube for the following x-ray spectrum




A.200 keV

B.50 keV

C. 67 keV

D. 167 keV

**Answer Question 49: A**

The maximum energy of a kiloelectron volt is equivalent to the applied kilovolts across the x-ray tube. In this case, it is 200 keV.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 50:**

Which of the following is **false** regarding a filtered photon beam?

A. Filters absorb high-energy photons
B. The average energy of a filtered beam is higher compared to unfiltered beams
C. The filtered beam is hardened
D. Filtered beams reduce overall photon fluence

**Answer Question 50: A**

All are true EXCEPT for A. Filters absorb low-energy photons (not high energy).

References:

Meshram MN, Pramanik S, Ranjith CP et al. Dosimetric characteristics of equivalent quality flattening filter free (FFF) and flatted photon beams. Int J Radiat Oncol Biol Phys 93 (3) Suppl;E548-549, 2015.

https://www.radiologycafe.com/frcr-physics-notes/x-ray-imaging/production-of-x-rays/

**Question 51:**

What is the line focus principle?

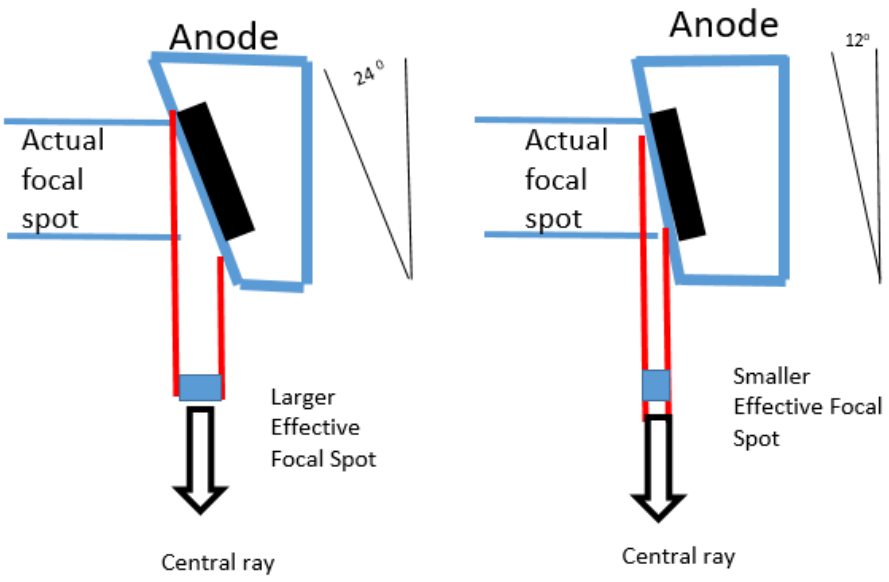
It is an x-ray tube principle...

A. where the focusing cup must be in line with the anode and cathode
B. which describes the relationship between the actual focal spot and the effective focal spot
C. where the angle of the anode target results in a larger effective focal spot with a steeper angle
D. which describes the requirement for the actual focal spot and the effective focal spot to be equivalent
E. which describes that a larger effective focal spot leads to a thinner x-ray beam

**Answer Question 51: B**

The line focus principle describes the relationship between the actual focal spot (where the electrons hit the anode) and the effective focal spot (area projected toward the patient). A smaller effective focal spot leads to a thinner x-ray beam (and better image sharpness or better spatial resolution in x-ray imaging).

A steeper anode anode (ie. smaller angle degree) results in a smaller effective focal spot.




Actual focal spot equal

a= A SinΘ

a=effective focal spot length
A = actual focal spot length
Θ= target angle

Reference:

McKay L, Line focus principle (illustration). Case study, Radiopaedia.org https://doi.org/10.53347/rID-185896

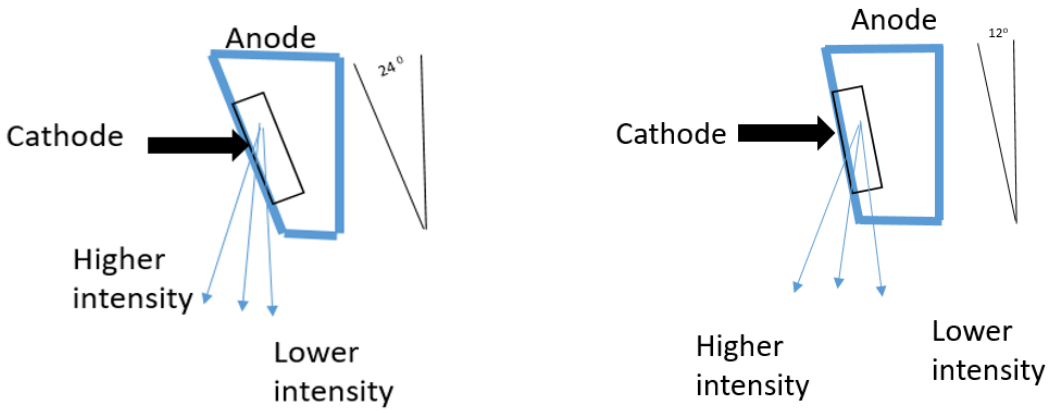
**Question 52:**

Which of the following is **true** regarding the heel effect?

A. Lower x-ray intensity occurs closer to the anode
B. Is independent of the anode angle
C. Lower x-ray intensity occurs closer to the cathode
D. Is independent of field size
E. Electron beam conversion to x-rays occur only on the surface of the metal target

**Answer Question 52: A**

The electron beam is converted to x-rays within the metal target. As such, the x-rays that are produced in the metal target must traverse through the target before emerging. The longer the x-rays must traverse through the target, the more resorption of the x-rays occurring in the target, resulting in a lower intensity of x-rays. X-rays that are emitted closer to the anode traverse a longer path in the target before emerging as compared to x-rays closer to the cathode. The path length that the x-ray beam must pass before emerging the metal target is affected by the anode angle.




Reference:

https://radiopaedia.org/articles/anode-heel-effect

**Question 53:**

What is the main purpose of a rotating anode?

A. Disperse heat over a larger area
B. Disperse x-rays
C. Reduce the size of the effective focal spot
D. To reduce pitting on the surface of the anode exposed to the electron beam

**Answer Question 53: A**

When electrons from the cathode strike the anode, a significant amount of heat is generated which may damage the anode. A rotating anode dissipates heat.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 54:**

Where does thermionic emission occur?

A. Cathode
B. Anode
C. Focusing cup
D. Voltage source

**Answer Question 54: A**

The cathode (tungsten filament) is heated to emit electrons. This process is called thermionic emission

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Effects of Electron Energy (kVp), Electron Tube Current (mA), Target Material, Filtration (inherent and added)**

**Question 55:**

Which one of the following is **not** a source of **inherent** filtration in an x-ray tube?

A. X-ray window
B. X-ray housing
C. Anode
D. Insulating oil
E. Aluminum sheet

**Answer Question 55: E**

Inherent filtration includes the x-ray tube components which absorbs lower energy photons, increasing the average energy and penetrating ability of the x-ray beam.

Aluminum sheet is an example of added filtration (not inherent).

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Topic: Differences between Diagnostic Tubes and Therapy Tubes (hooded anode)**

**Question 56:**

Which one of the following is **false** regarding a diagnostic/imaging x-ray tube as compared to a radiation therapy tube?

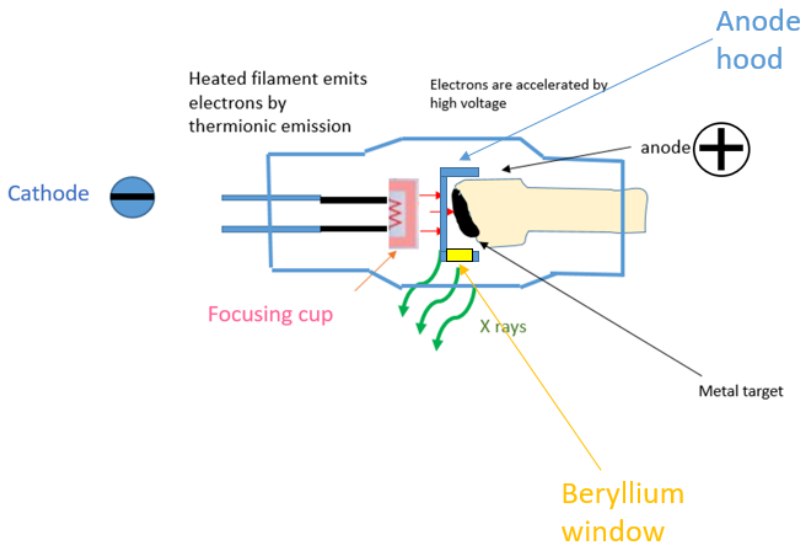
A. Imaging x-rays are lower in energy compared to radiation therapy x-rays
B. Hooded anodes are used in imaging x-rays, and also with radiation therapy linear accelerators
C. The photoelectric effect dominates with imaging x-rays
D. A much higher voltage difference is used in a linear accelerator to accelerate electrons for radiation therapy compared to a much smaller voltage potential needed for imaging.

**Answer Question 56: B**

All are true except for B.

Hooded anodes are used in imaging x rays but not in radiation therapy linear accelerator.

Hooded anodes are anodes surrounded by copper and tungsten (hood) to minimize stray secondary electrons and unwanted secondary electrons to improve image quality and increase the heat capacity of the target. Hooded anodes are typicallyused in imaging. Hooded anodes maybe used in some orthovoltage machines but are not used in linear accelerators.




References:

https://oncohemakey.com/principles-of-radiologic-physics-and-dosimetry/

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

# Chapter 4: Production of megavoltage beams

Erika Kollitz, Peter Ferjancic,
Michael Weisman, May Tsao

**Basic Layout, Types of Mechanical Motion Possible, Isocenter, Source-to-Axis Distance (SAD) as a Fundamental Mechanical Characteristic of a Linac**

**Question 57:**
The order of a linear accelerator's components, with the farthest component from the patient (or from the power source) starts with:

Pulsed modulator, Electron gun, Accelerating waveguide...

Select **A, B, C or D** below, which indicates the correct order of the remaining components.

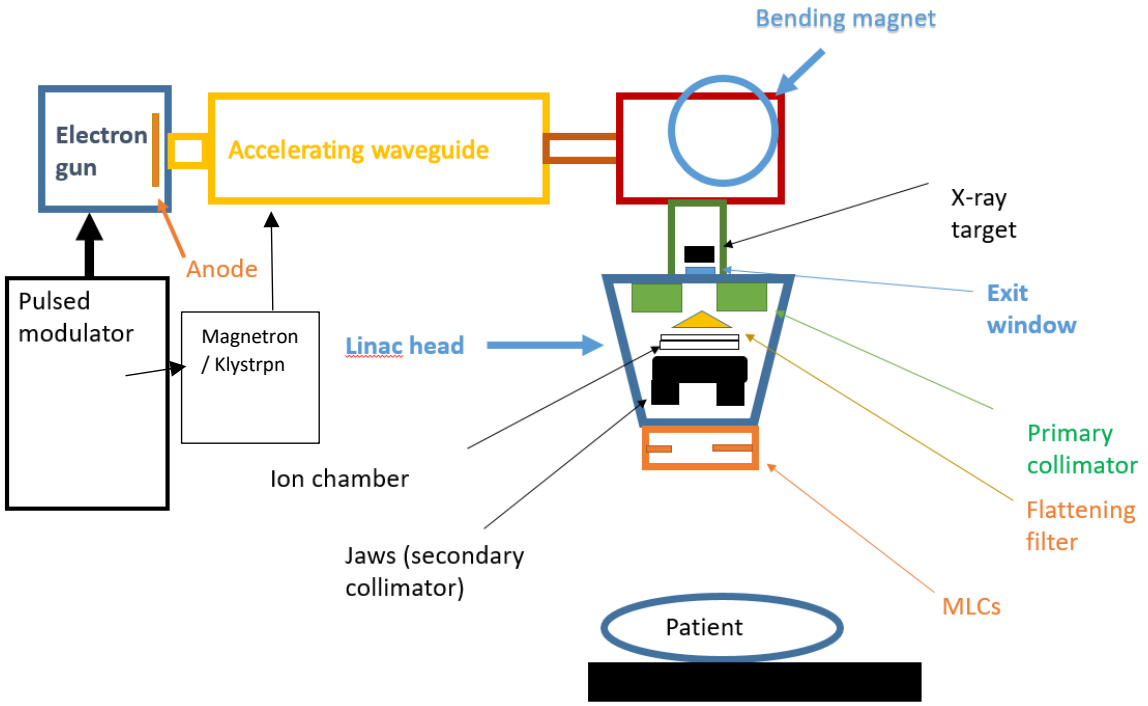
1. Primary collimator
2. Flattening filter
3. Dual ionization chamber
4. Multi-leaf collimator and Jaws (secondary collimator)
5. Exit window
6. X ray target
7. Bending magnet

**A. 6, 7, 5, 1, 2, 3, 4**
**B. 7, 6, 5, 2, 3, 1, 4**
**C. 1, 2, 3, 4, 5, 6, 7**
**D. 7, 6, 5, 1, 2, 3, 4**

**Answer Question 57: D**

Pulsed modulator, Electron gun, Anode, Accelerating waveguide,

Then:
7 bending magnet, 6 x ray target, 5 exit window, 1 primary collimator, 2 flattening filter, 3 dual ion chamber, 4 multi-leaf collimator and jaws




Reference:

Van Dyk, J. The modern technology of radiation oncology Madison, WI, USA: Medical Physics Publishing; 1999. p1073

**Question 58:**

What is the maximum number of degrees of freedom (axes) that can be used to position a patient on a linear accelerator couch?

A. 3
B. 4
C. 5
D. 6
E. 7

**Answer Question 58: D**

3 linear axes or translational motions (X, Y, Z) and 3 rotary axes or rotational motions (ΘX, ΘY, ΘZ)

Reference:

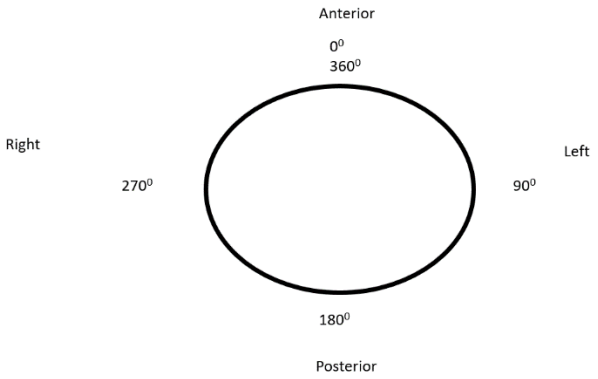
Rajaguru P, Caudell J, Kanakamedala M et al. Clinical use of hexapod couch in combination with Elekta XVI-CBCT for head and neck IGRT treatments. Int J Radiat Oncol Biol Phys 2010 78(3), S440.

**Question 59:**

Where are the 0, 90, 180, 270 and 360 degree gantry angles in relation to a head-first supine patient, respectively?

A. Directly anterior, left lateral, directly posterior, right lateral, directly anterior
B. Directly anterior, right lateral, directly posterior, left lateral, directly anterior
C. Directly posterior, right lateral, directly anterior, left lateral, directly posterior
D. Directly posterior, right lateral, directly anterior, left lateral, directly anterior

**Answer question 59: A**




Reference:

Haneefa A, Cyriac S, Musthafa M et al. Pretreatment patient specific quality assurance and gamma index variation study in gantry dependent EPID Positions for IMRT prostate treatments. Journal of Radiotherapy 2014. 1-6.

**Question 60:**

Which of the following are **true** statements regarding the isocenter?

Select **A, B, C, or D**.

1. The radiation isocenter is defined as the point in space where the patient receives 100% of the prescription dose
2. The mechanical isocenter is the centroid of a sphere about which the gantry, collimator and couch rotate
3. The radiation isocenter is the point in space where the central axes of the radiation beams intersect

4. The mechanical isocenter is the centroid of a sphere where the central axes of the radiation beams intersect
5. The mechanical and radiation isocenters are distinct and should ideally coincide

**A. 1, 2, 5**
**B. 2, 3, 4**
**C. 2,3,5**
**D. 1,2,3,5**

**Answer Question 60: C**

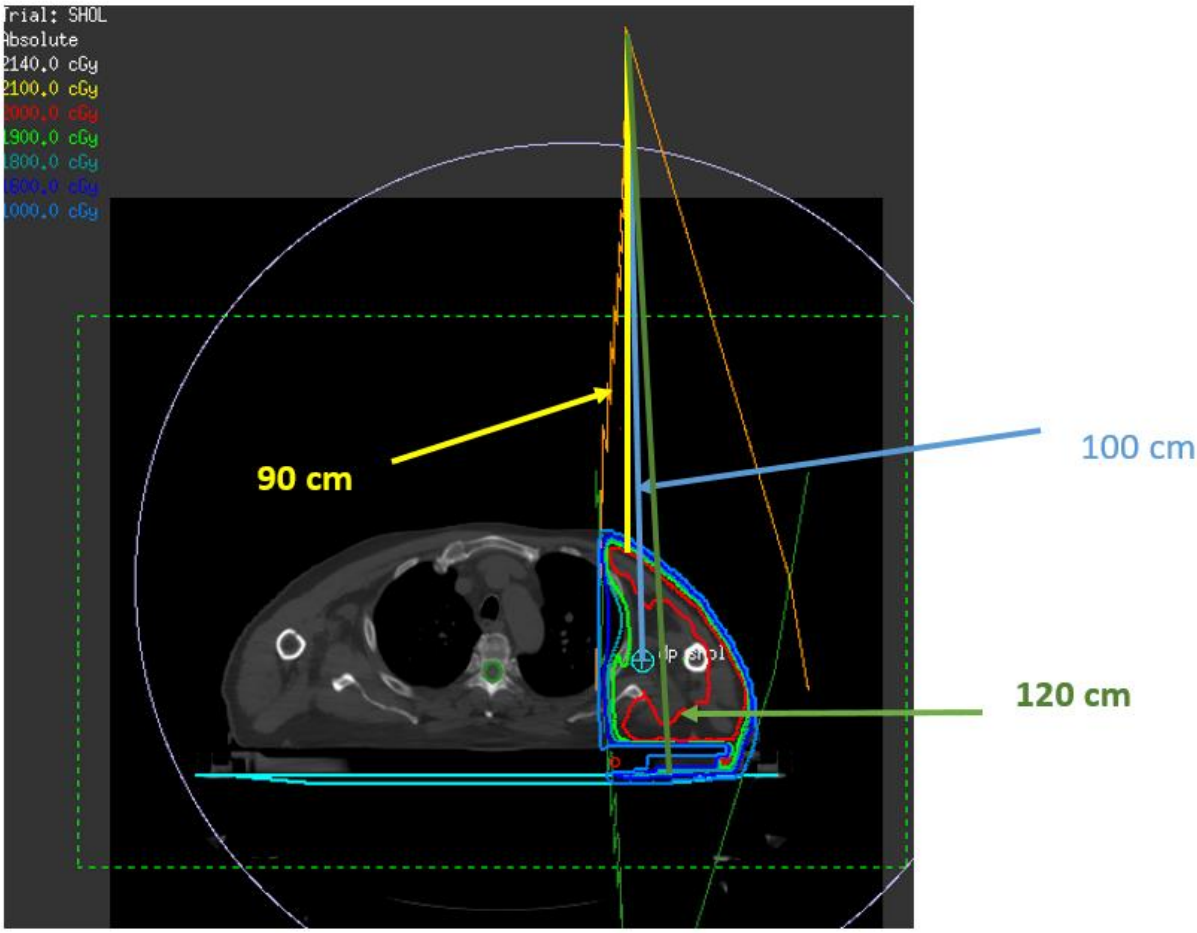
The radiation isocenter is only defined on the machine and is independent of patient prescription and positioning. However, it is common to place the treatment isocenter in the center of the treated volume.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 61:**

A patient is planned for palliative radiation to treat a bone metastasis to the left shoulder. What is the source axis distance (SAD), in this set-up?




A. 90 cm
B. 100 cm
C. 120 cm
D. 20 cm

**Answer Question 61: B 100 cm**

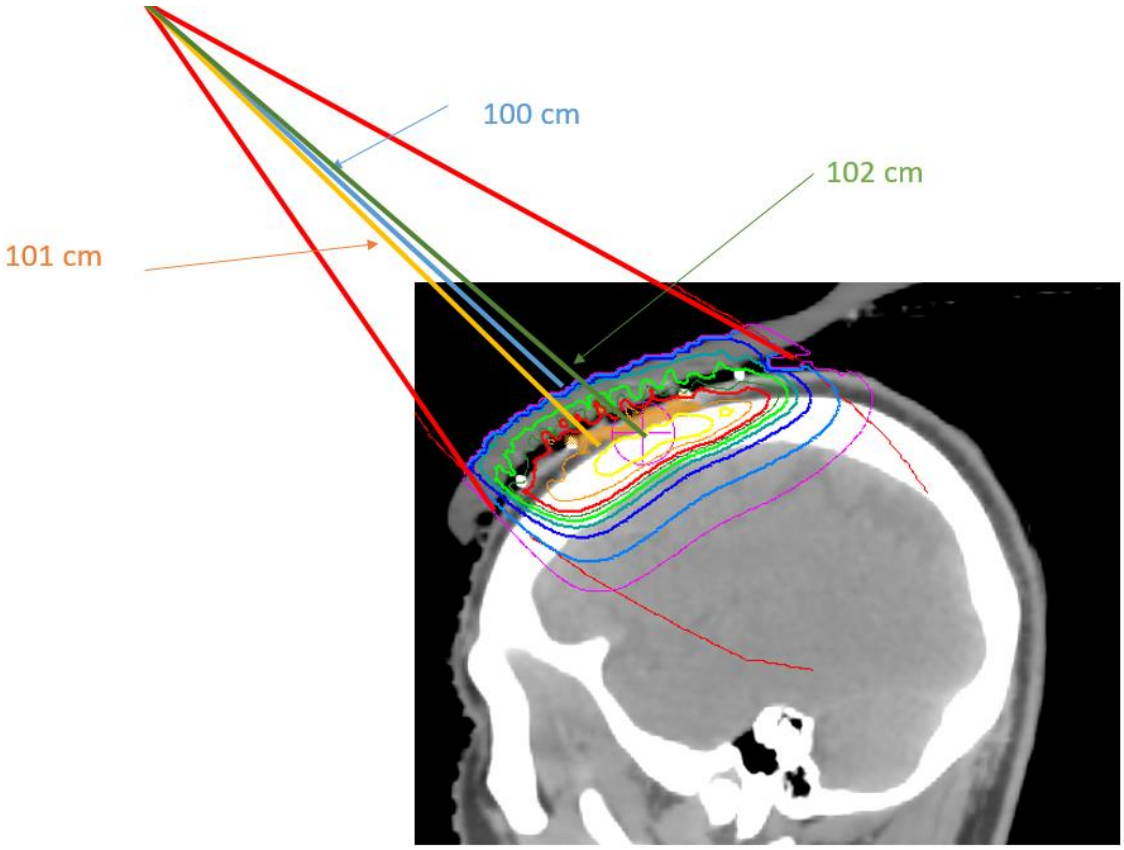
The SAD or source-axis distance is defined based on machine parameters to be the distance from the radiation source to the linac's isocenter. For standard C-arm linacs, this is 100 cm.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 62:**

This patient was treated for an indolent scalp cutaneous lymphoma using a direct electron beam with bolus. What was the source surface distance (SSD) used?




A. 100 cm
B. 101 cm
C. 102 cm
D. 101.5 cm

**Answer Question 62: A**

The direct electron was set up so that 100 cm SSD is from the source to the surface of the bolus.

Reference:

Gunhan B, Kemikler G, Koca A. Determination of surface dose and the effect of bolus to surface dose in electron beams. Medical Dosimetry 2003; 28(3): 193-8

**Photon and Electron Beam Energy Spectrum**

**Question 63:**

Which type of radiation is the **least** penetrating?

A. Alpha particles
B. Beta particles
C. Megavoltage photons
D. Orthovoltage photons
E. Neutrons

**Answer Question 63: A**

From least to most penetrating:

Alpha particles, beta particles, neutrons, orthovoltage photons, megavoltage photons

Reference:

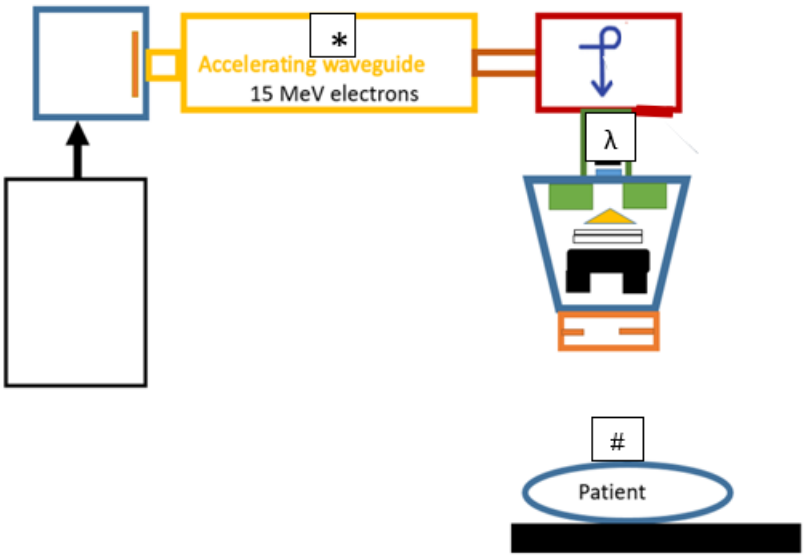
https://sciencedemonstrations.fas.harvard.edu/presentations/%CE%B1-%CE%B2-%CE%B3-penetration-and-shielding

**Question 64:**

Electron energy spectrums can vary depending on location in the linac. Match the energy spectrum to the corresponding location in this treatment setup. The locations are:

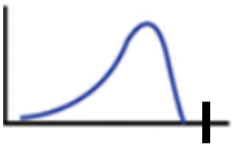
Inside the linear accelerator waveguide (*)
After the scattering foil (λ)
At the patient's surface (#)

Choose the correct answer below: **A, B, C or D**


*
Accelerating waveguide
15 MeV electrons
λ
#
Patient

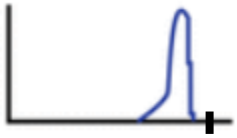
The following electron energy spectra show normalized planar fluence/energy bin (on the x-axis) and energy in MeV (on the y-axis).

**1.**



15 MeV

**2.**



15 MeV

**3.**



15 MeV

| | |
|---|---|
| **A.** Inside the linear accelerator waveguide: | 3 |
| After the bending magnet: | 2 |
| At the patient's surface | 1 |
| **B.** Inside the linear accelerator waveguide | 1 |
| After the bending magnet | 2 |
| At the patient's surface | 3 |
| **C.** Inside the linear accelerator waveguide | 2 |
| After the bending magnet | 3 |
| At the patient's surface | 1 |
| **D.** Inside the linear accelerator waveguide | 1 |
| After the bending magnet | 3 |
| At the patient's surface | 2 |

**Answer Question 64: A**

**3** (inside the linear accelerator waveguide), **2** (after the bending magnet)**, 1** (at the surface of the patient)

Electrons in the linear accelerator waveguide have a single well-defined and largely monoenergetic energy. These 15 MeV electrons have 15 MeV in the waveguide as shown in figure **3**.

The electrons lose energy when they bend resulting in a poly-energetic spectrum, called energy "straggling" (figure **2**).

As the electrons pass through the scattering foil, monitor chamber and through air, a more poly-energetic spectrum beam emerges at the surface of the patient (figure **1**).

Reference:
https://radiologykey.com/dosimetry-of-electron-beams/

**Waveguides, Traveling Wave, Standing Wave (Side Cavity Coupling), Electron Gun, Waveguide Orientation**

**Question 65:**

What type of wave can be used in linear accelerators to accelerate electrons? Choose the correct answer **A, B, C or D** below.

1. Travelling wave
2. Standing wave
3. Longitudinal wave
4. Mechanical wave

**A. 1 and 2**
**B. 2 and 3**
**C. 1 and 3**
**D. 3 and 4**

**Answer Question 65: A**

Travelling waves are carry energy and momentum through a medium and thus allow for efficient and continuous acceleration of electrons through the linear accelerator. Travelling waveguides are used in Elekta linacs.

Standing waves are two waves of equal amplitude and frequency, which travel in opposite directions, creating stationary nodes, by timing the electron bunches with the wave frequency. Standing waves can accelerate electrons as they pass through cavities while the wave is amplified. Standing waveguides are used in Varian linacs and are typically much shorter than travelling waveguides.

Longitudinal waves, like sound, are waves where the propagation of the wave is in the same direction as the vibration of the medium through which it travels.

Mechanical waves are waves that need a medium for propagation (sound waves, water waves, seismic waves). Non-mechanical waves do not need any medium for propagation (eg. electromagnetic waves)

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Sources of Microwave Power: Magnetrons, Klystrons, Relationship Between Microwave Frequency and Accelerator Length**

**Question 66:**

In a linear accelerator, select which component(s) feed high frequency microwaves to the accelerating waveguide.

Choose the correct answer **A, B, C or D** below.

1. Modulator
2. Magnetron
3. Klystron

**A. 1 and 3**
**B. 2 and 3**
**C. 2 only**
**D. 1,2, and 3**

**Answer Question 66: B**

Both a Magnetron and Klystron provide high frequency microwaves to the accelerating waveguide in a linear accelerator.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 67:**

For a given energy of accelerated electrons, if the microwave frequency increase, what happens to the length of the linear accelerator?

A. Increases
B. Decreases
C. Increases to a point then stays the same
D. Stays the same

**Answer Question 67: B**

Higher microwave frequency is associated with shorter wavelengths and thereby shorter accelerator lengths.

The length of a linear accelerator is determined by the number of accelerations (or pushes) needed to reach the desired electron energy.

A higher microwave frequency (and therefore shorter wavelength) allows for more acceleration pushes within the same length, leading to shorter accelerators, to achieve the same electron energy.

Reference:
Zhu, T.C., Wang, K.KH. (2013). Linear Accelerators (LINAC). In: Brady, L.W., Yaeger, T.E. (eds) Encyclopedia of Radiation Oncology. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-540-85516-3_37

**Block Diagram and Overall Operation: Modulator, Pulse Forming Network, Thyratron**

**Question 68:**

Which of the following is **not** a function of a thyratron in a linear accelerator?

A. High voltage switching
B. Pulse generation

C. Controlled rectifier
D. Provides radiofrequency power

**Answer Question 68: D**

A thyratron switches high voltages and currents in a linear accelerator. Thyratrons control the release of energy stored in the pulse forming network, creating high power pulses that are then used to power the Klystron. Thyatrons also act as rectifiers [converts alternating current (AC) to direct current (DC)], allowing the current to flow in only one direction.

The magnetron (not the thyatron) generates the radio frequency waves.

Reference:

Balakrishnan IS, Jayaraman R, Bhaskar BK, Lakshmanan AV. Thyratron performance—dependence on mounting in a medical linear accelerator, *British Journal of Radiology*, Volume 68, Issue 815, 1 November 1995, Pages 1242–1243

**Treatment Head: Bending Magnet, X-Ray Target, Scattering Foils, Flattening Filter, Flattening Filter Free, Monitor Ion Chambers, Collimation System, Light Localizing System, Optical Distance Indicator**

**Question 69:**

Which of the following is **not** true regarding flattening filter-free (FFF) beams?

A. FFF beams deliver a higher dose rate
B. FFF beams allow for faster treatment delivery
C. FFF beams have reduced peripheral dose
D. FFF beams have a sharper penumbra
E. FFF beams have more linear accelerator head scatter

**Answer Question 69: E**

A-D are correct.

Removal of the flattening filter reduces linear accelerator head scatter.

Reference:

Zhang R, Xie Y, DiTusa C, Ohler R, Heins D, Bourgeois D, Guo B. Flattening Filter-Free Volumetric-Modulated Arc Radiotherapy for Left-Sided Whole-Breast, Partial-Breast, and Postmastectomy Irradiations. J Med Phys. 2022 Apr-Jun;47(2):166-172.

**Electron vs X-ray Mode: Beam Current, Electron Applicators**

**Question 70:**

Which of the following is **retracted** from the path of the electron beam when a linear accelerator is in **electron mode**?

A. Ion chamber
B. Scattering foil
C. Heavy metal target (eg. tungsten or copper)
D. Waveguide

**Answer Question 70: C**

Heavy metal target (eg. tungsten or copper)

In a linear accelerator, electrons are accelerated in a wave guide. The scattering foil is used as well as ionization chambers in electron mode. The scattering foil (eg. made of tantalum or gold) is used to broaden the narrow electron beam into a wider more uniform clinical electron beam. Ionization chambers (or monitor chambers) monitor radiation beam output and radiation dose.

However, the heavy metal target (eg. tungsten or copper) is retracted, so that photons do not get produced.

Reference:

Van Dyk J: The modern technology of radiation oncology. Madison, WI, USA. Medical Physics Publishing, 1999, 1073.

**Linac Subsystems: Cooling, Chiller, Vacuum, $SF_6$**

**Question 71:**

What is the purpose of sulfur hexafluoride ($SF_6$) in a linear accelerator?

A. It is used as an insulating gas
B. It is used to absorb stray scattered photons
C. It is used to absorb stray scattered electrons
D. It is used to reduce greenhouse emissions from linear accelerators

**Answer Question 71: A**

It is used as an insulating gas to prevent electrical breakdown and to maintain stable electric fields in the acceleration process in linear accelerator waveguides.

$SF_6$ is a potent greenhouse gas with very high global warming potential.

Reference:

Lichter KE, Bloom JR, Sheu R-D et al. Tracking and reducing $SF_6$ usage in radiation oncology: A step toward net-zero health care emissions. Prac Radiat Oncol 13 (6): e471-3474, 2023.

**Treatment couch**

**Question 72:**

What is the usual maximum number of degrees a linear accelerator couch may rotate in the plane parallel to the floor?

A. 90 degrees
B. 180 degrees
C. 270 degrees
D. 360 degrees

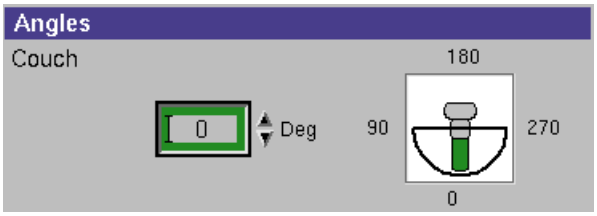
**Answer Question 72: B**

zero degrees to 90 degrees = 90 degrees

270 degrees to 360 degrees = 90 degrees

Total: 180 degrees

Top/down view below:




**Field Shaping: Block, Collimators, MLC**

**Question 73:**

Which of the following is **not** used for field shaping in linear accelerator photon or electron treatments?

A. Primary collimator
B. Secondary collimator
C. Multi-leaf collimation
D. Flattening filter
E. Lead shielding on the patient's skin surface

**Answer Question 73: D**

The primary collimator, secondary collimator, multi-leaf collimator are used to shape linear accelerator beams. Lead shielding on a patient's surface may also be used to shield electron beams.

A flattening filter does not shape the beam, but instead creates a more uniform dose distribution across the beam.

Reference:

Verhaegen F, Buffa FM, Deehan C. Quantifying effects of lead shielding in electron beams: a Monte Carlo study. Phys Med Biol. 2001 Mar;46(3):757-69.

**Safety Systems: Collision Avoidance, Interlocks**

**Question 74:**

Which of the following parameters will not trigger a safety interlock on a linear accelerator?

A. Gantry, collimator, couch angle
B. Jaws and MLCs
C. Temperature, Pressure, Water
D. Beam output
E. Door interlock
F. Dose to personnel thermoluminescent dosimeters (TLD)

**Answer Question 74: F**

All of the above are examples of safety interlocks on a linac except for personnel dose, which is monitored with personnel dosimetry badges (OSLDs, TLDs etc).

Reference:
Liu JC, Bull JS, Drozdoff J, May R, Vylet V. Radiation safety systems for accelerator facilities. Radiat Prot Dosimetry. 2001;96(4):429-39.

# Chapter 5: Radiation interactions

Ria Mulherkar Kraft, Yunjie Yang,
Peter Ferjancic, May Tsao

**5.1 Directly and Indirectly ionizing radiation**

**Question 75. 1:**

Electrons are:

A. Directly ionizing
B. Indirectly ionizing

**Answer Question 75.1: A**

Electrons: directly ionizing

**Question 75.2:**

Neutrons are:

A. Directly ionizing
B. Indirectly ionizing

**Answer Question 75.2: B**

Neutrons: indirectly ionizing

**Question 75.3:**

Protons are:

A. Directly ionizing
B. Indirectly ionizing

**Answer Question 75.3: A**

Protons: directly ionizing

**Question 75. 4:**

Alpha particles are:

A. Directly ionizing
B. Indirectly ionizing

**Answer Question 75.4: A**

Alpha particles: directly ionizing

**Question 75. 5:**

Photons are:

A. Directly ionizing
B. Indirectly ionizing

**Answer Question 75.5: B**

Photons: indirectly ionizing

Charged particles such as electrons, protons and alpha particles are directly ionizing, as long as these charged particles have enough kinetic energy to produce ionization by collision as they interact with matter. Uncharged particles such as neutrons and photons are indirectly ionizing as they produce directly ionizing particles when they interact with matter.

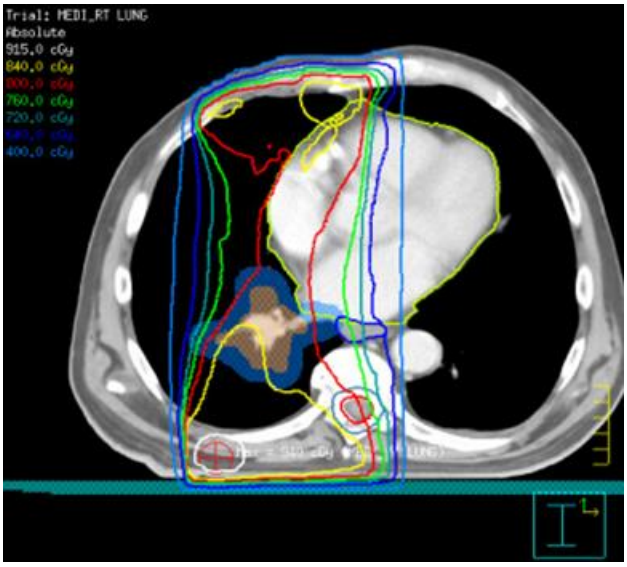
Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

## 5.2 Interactions of x and gamma rays with matter (scatter vs absorption of radiation, photoelectric effect, Compton effect, pair production, relative dependence on atomic number, energy and density, photonuclear disintegration)

**Question 76:**

This patient with stage IV non-small cell lung cancer was planned and treated with a parallel opposed beam technique to help palliate hemoptysis.




Which of the following is true regarding Compton scattering in lung?

A. Compton scattering is proportional to the atomic number (Z) cubed
B. Compton scattering is dependent on electron density
C. Compton scattering results in the creation of a characteristic X-ray
D. Compton scattering results from the interaction of the incident photon with the nucleus of an atom

**Answer Question 76: B**

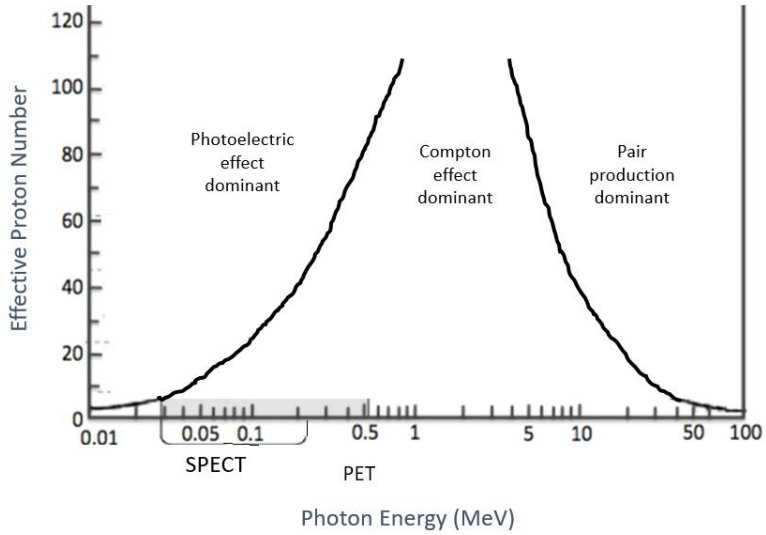
Compton scattering describes photon colliding with an atom, releasing loosely bound electrons from the outer shell of the atom, resulting in a scattered photon and a scattered electron.

Compton scattering is **independent** of the atomic number (Z or the number of protons in the nucleus of an atom). However, Compton scattering is **dependent** on the electron density.

Because Compton scattering involves the interaction of a photon with a single electron, the electron density or number of electrons per unit mass is the most important tissue property for Compton scattering.

The electron density of lung ($2.29 \times 10^{23}/cm^3$) is much lower than soft tissue ($3.51 \times 10^{23}/cm^3$). The probability of Compton scattering is greater in higher density.

Compton scattering effect is the most probable photon interaction in the range of energies used in therapeutic radiation.




The photoelectric effect, on the other hand is proportional to the atomic number (Z or the number of protons in the nucleus of an atom) cubed. The energy of the photon absorbed is used to overcome the electron's force that binds the electron to the nucleus of the atom. Photoelectric absorption more frequently occurs with absorbing material of high atomic number and low energies of gamma rays.

The creation of characteristic X-rays also results from the photoelectric effect (not from the Compton effect). This is because in the photoelectric effect, an incident photon causes an inner shell electron to leave its orbit, thus ionizing the atom. When there is a vacancy in the inner shell, an outer shell electron will release some energy and take the place of the inner shell electron. The energy that is released as the outer shell electron moves to the inner shell is a quantized amount, and it represents the energy difference between the two electron shells. Because this energy difference will always be a set quantized amount, it is known as a "characteristic" X-ray.

Photoelectric effect and Compton scattering are both photon-electron interactions. Pair production occurs from interaction of the incident photon with the nucleus of an atom.

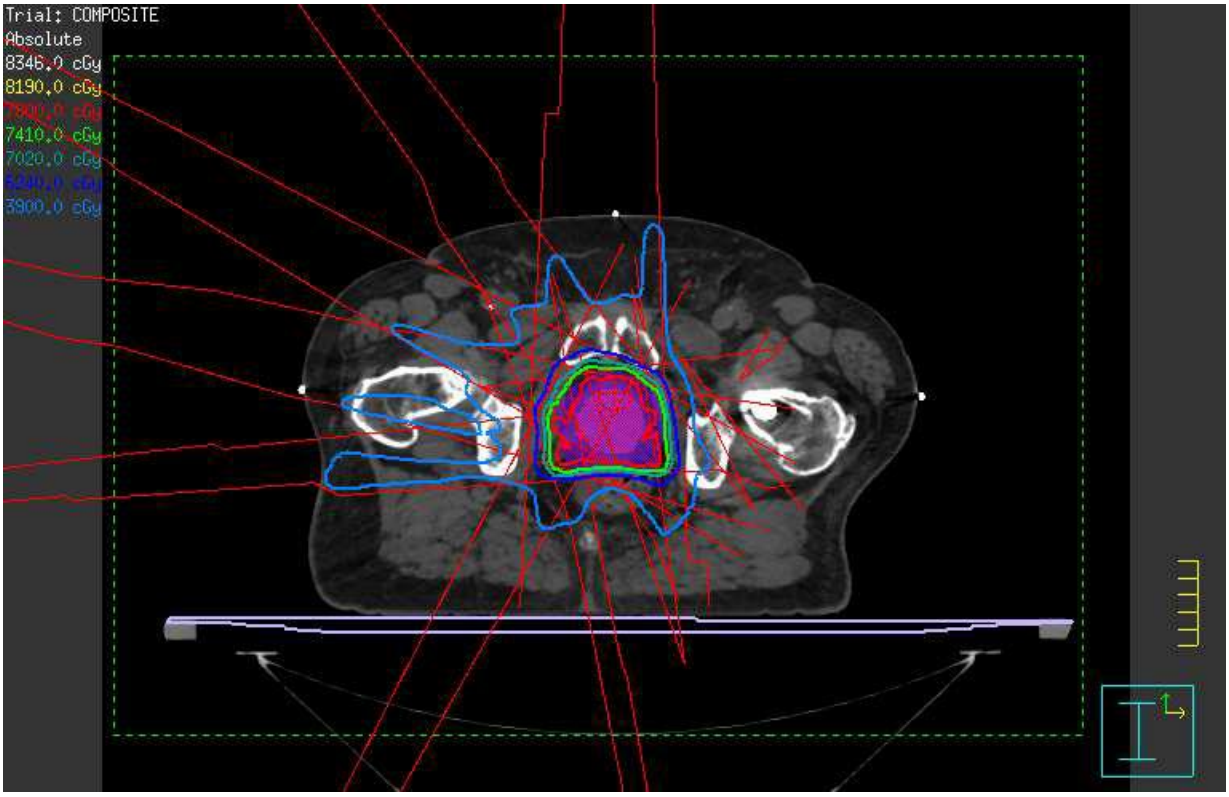
References:

Bichsel H, Schindler H. The interaction of radiation with matter. In: *Particle Physics Reference Library*. Springer International Publishing; 2020:5-44.

Shrimpton PC. Electron density values of various human tissues: in vitro Compton scatter measurements and calculated ranges. Phys Med Biol 26(5):907-913, 1981.

**Question 77:**

This patient with a hip prosthetic rod was planned for intensity modulated radiation therapy (IMRT) to treat his prostate cancer.




The IMRT beams were arranged to avoid the left hip metallic rod. As well 6 MV photon beams were used.

High energy photon beams >=10 MV were intentionally avoided in this IMRT plan to avoid secondary neutron contamination, which would occur through which photon interaction with the metallic rod?

A. Photonuclear disintegration
B. Pair production
C. Compton
D. Photoelectric effect

**Answer Question 77: A**

The photonuclear effect occurs when high energy photons interact with the metallic rod producing secondary neutrons. High energy photons can activate prostheses via the photonuclear effect (causing the emission of neutrons from within the patient).

Reference:

Keehan S, Smith RL, Millar J, Esser M, Taylor ML, Lonski P, Kron T, Franich RD. Activation of hip prostheses in high energy radiotherapy and resultant dose to nearby tissue. J Appl Clin Med Phys. 2017 Mar;18(2):100-105.

## 5.3 Interactions with Charged particles and Neutrons with Matter (interactions with orbital electrons and nucleus- collisional versus radiative energy loss, linear energy transfer, stopping power, range, interactions of electrons and positrons, interactions of heavy charged particles, interactions of neutrons)

**Question 78:**

Linear energy transfer quantifies:

A. The energy transferred by a photon in tissue along its path in target material (measured in Gray/cm)
B. The energy transferred by a neutron along its path in target material (measured in Newtons/cm)
C. The linear energy path of a charged particle in target material (measured in cm/Joules)
D. The energy transferred by an ionizing particle to the target material along the particle's path (measured in keV/µm)

**Answer Question 78: D**

The energy transferred by an ionizing particle to the target material along the particle's path (measured in keV/µm)

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 79:**

Order the following types of radiation from lowest linear energy transfer to highest linear energy transfer.

Choose the correct answer **A, B, C or D** below.

1) 250 kV X rays
2) Cobalt 60 gamma rays
3) 10 MeV protons
4) 2.5 MeV alpha particles
5) 14 MeV neutrons

**A. 2, 1, 3, 5,4**

**B. 2, 1, 5, 3, 4**
**C. 1, 2, 3, 5, 4**
**D. 2, 1, 3, 4, 5**

**Answer Question 79: A**

Cobalt 60 gamma rays, 250 KV x rays, 10 MeV protons, 14 MeV neutrons, 2.5 MeV alpha particles

| | Approximate LET (keV/µm) |
|---|---|
| $^{60}$Co | 0.2 |
| 250 KeV | 2 |
| 10 MeV protons | 4.7 |
| 14 MeV neutrons | 75 |
| 2.5 MeV alpha | 166 |

Reference:

https://www.sciencedirect.com/topics/medicine-and-dentistry/linear-energy-transfer

**Question 80:**

What is the continuous slowing down approximation (CSDA) range?

A. the average fraction of the initial kinetic energy of an electron that is converted to bremsstrahlung energy as a particle slows down to rest.

B. The average value of depth to which a charged particle will penetrate in the course of slowing down to rest, measured along the initial direction of the particle.

C. A close estimate of the average path traveled by a charged particle as it slows down, obtained by integrating the reciprocal of the total stopping power with respect to energy.

D. The average rate of energy loss per unit path length due to Coulomb collisions that result in the ionization and excitation of atoms

**Answer Question 80: C**

The *CSDA range* is a close estimate of the average **path** traveled by a charged particle as it slows down to rest. In this estimate, the rate of energy loss at every point along the track is assumed to be equal to the total stopping power, where energy-loss fluctuations are neglected. The CSDA range is calculated by integrating the reciprocal of the total stopping power with respect to energy. With knowledge of energy loss, the range (path length) a heavy charged particle travels before stopping can be calculated;

CSDA charged particle range:

$R = \int_{E}^{E_0} dE / S(E)$

R= CSDA range

$E_0$ =initial kinetic energy of the particle

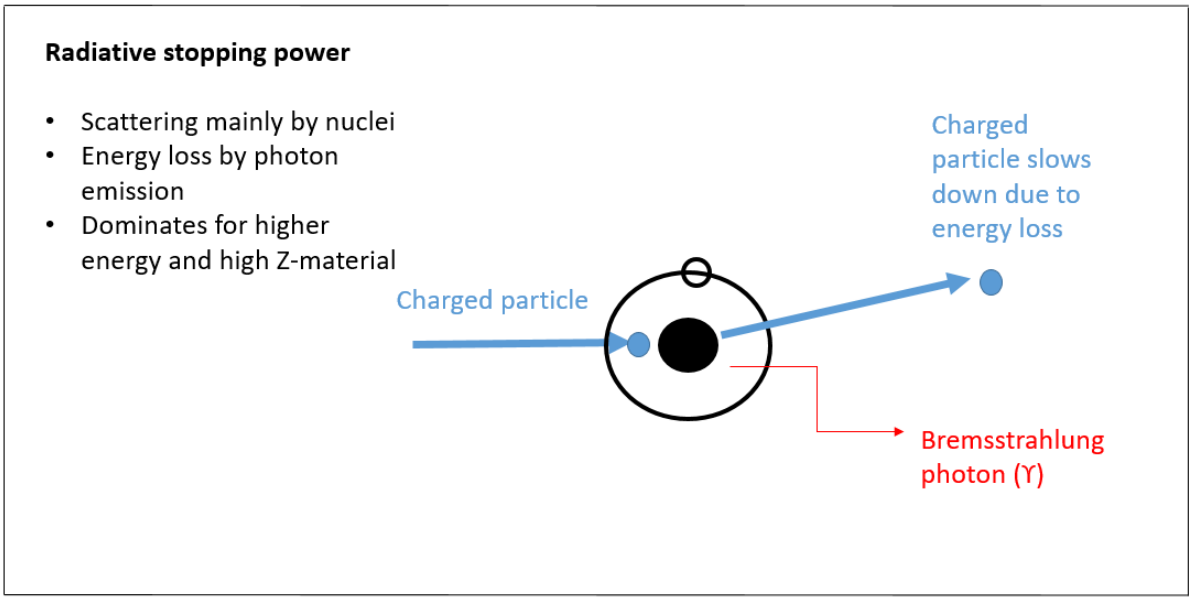
E = final kinetic energy of the particle (typically 0)

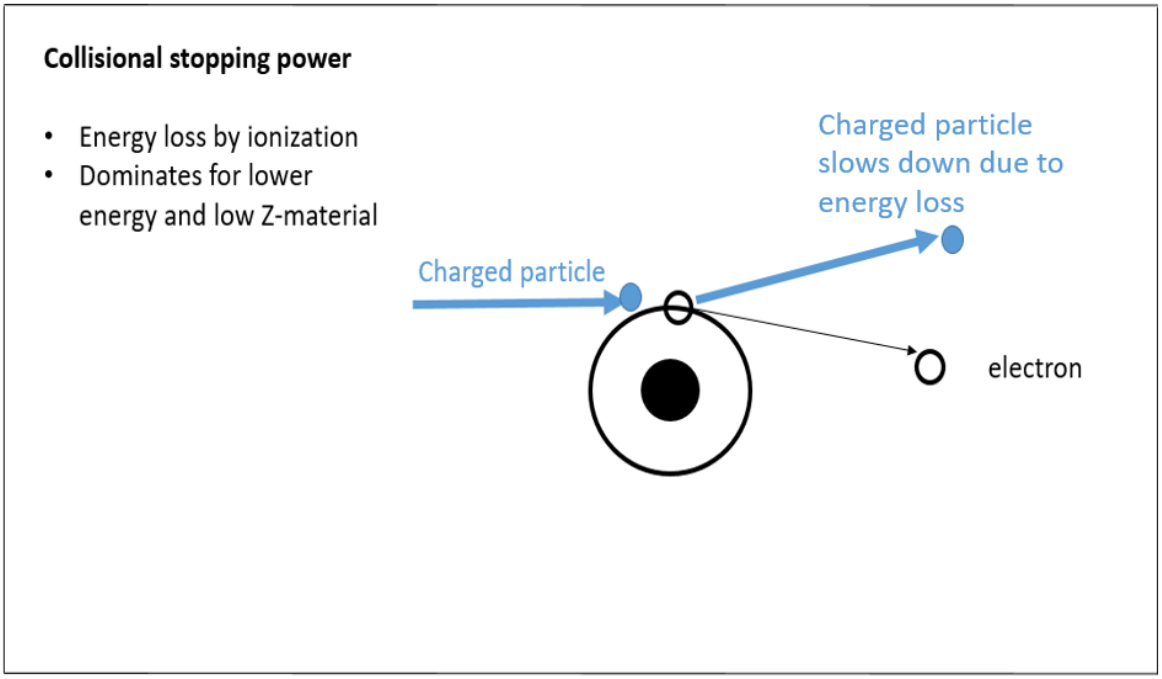
S(E)= the total stopping poser of the medium at energy E (includes both collisional and radiative stopping power

*Projected range* is the average value of depth to which a charged particle will penetrate to slow down, measured along the initial direction of the particle. Of note, particle paths are not straight due to scattering. As such, the particle's **path** length is the distance travelled by the particle, but the **range** distance is smaller as range distance measures the depth in the initial direction of motion before the particle comes to a rest. Definition of option b.

*Radiation yield or radiative stopping power* is the average fraction of the initial kinetic energy of a charged particle that is converted to bremsstrahlung energy as a particle slows down to rest. Definition of option a.

*Collision stopping power* is the average rate of energy loss per unit path length due to Coulomb collisions that result in the ionization and excitation of atoms. For heavy charged particles, collision stopping power is also *electronic stopping power*. Definition of option d.






References:

National Institute of Standards and Technology. Physical Measurement Laboratory. https://physics.nist.gov/PhysRefData/Star/Text/appendix.html#:~:text=The%20CSDA%20range%20is%20obtained,initial%20direction%20of%20the%20particle.

Usta M. Continuous slowing-down approximation ranges of biological materials for 0.05-10 MeV alpha particles by using different approach methods. Appl Radiat Isot. 2021 Dec;178:109951. doi: 10.1016/j.apradiso.2021.109951. Epub 2021 Sep 15. PMID: 34537564.

Tufan MC, Gumus H. A study on the calculation of stopping power and CSDA range for incident positrons. Journal Nuclear Materials 412(3): 308-314, 2011.

**Question 81:**

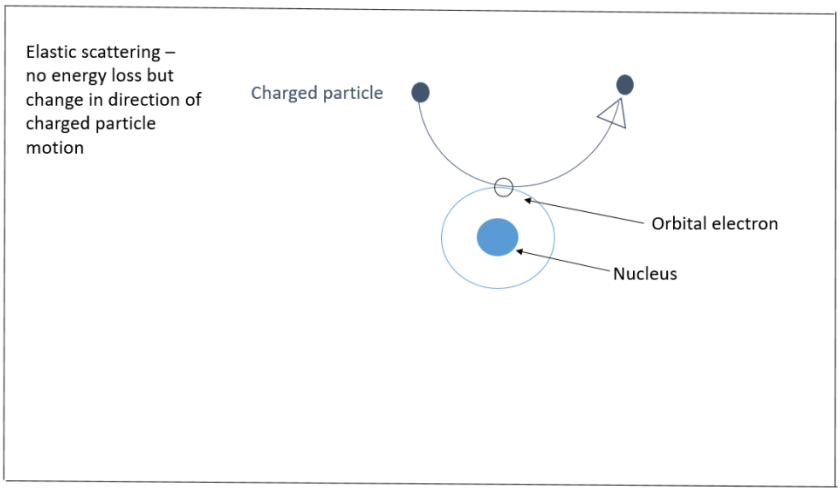
When **neutrons** or **protons** interact with atoms, all of the following are possible **interactions with orbital electrons** of the absorber atom, **EXCEPT:**

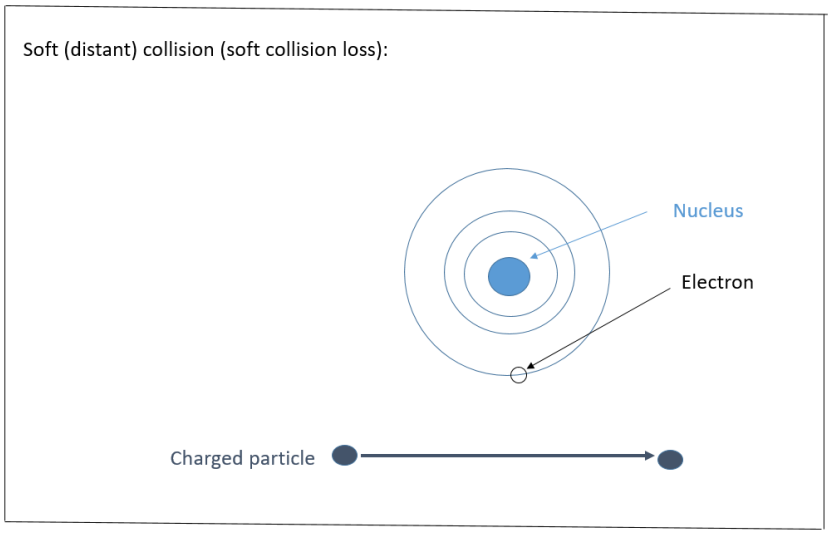
A. Elastic scattering (no energy loss but change in direction of motion
B. Soft (distant collision)
C. Hard (direct) collision
D. In flight annihilation (radiation loss with emission of annihilation photons)

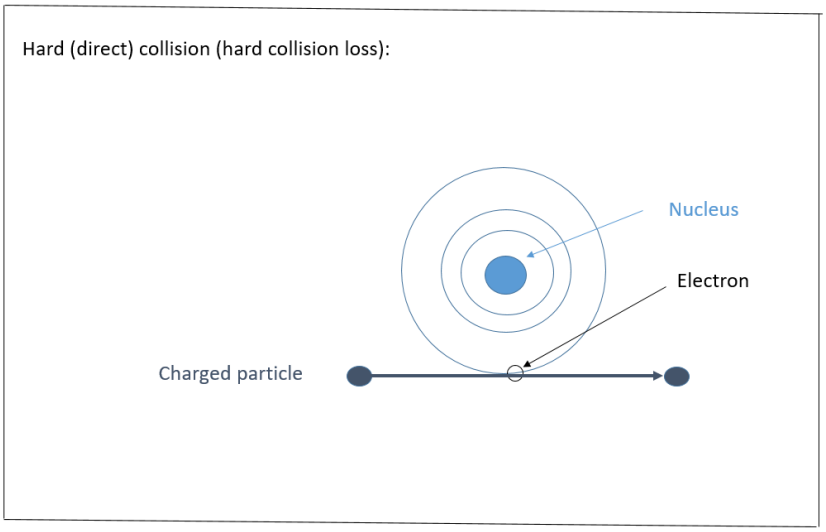
**Answer Question 81: D**

In-flight annihilation (radiation loss with emission of annihilation photons) only occur with positrons (not neutrons or protons). A, B, and C are correct.

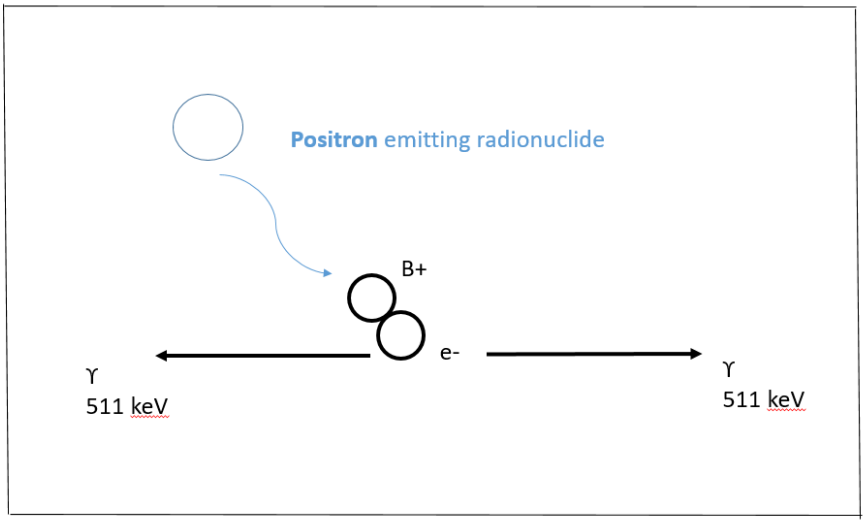
Neutron and proton interaction with orbital electrons:






Annihilation (radiation loss with emission of annihilation photons) occur with positrons:




Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 82:**

When charged particles interact with atoms, **which of following** are possible interactions with the **nucleus** of the absorber atom.

Choose the correct answer **A, B, C or D** below.

1. Elastic scattering (no energy loss but change in the direction of motion
2. Inelastic scattering (radiation loss with emission of Bremsstrahlung or gamma rays)
3. Penetration of nucleus (nuclear reaction)
4. In flight annihilation (radiation loss with emission of annihilation photons)

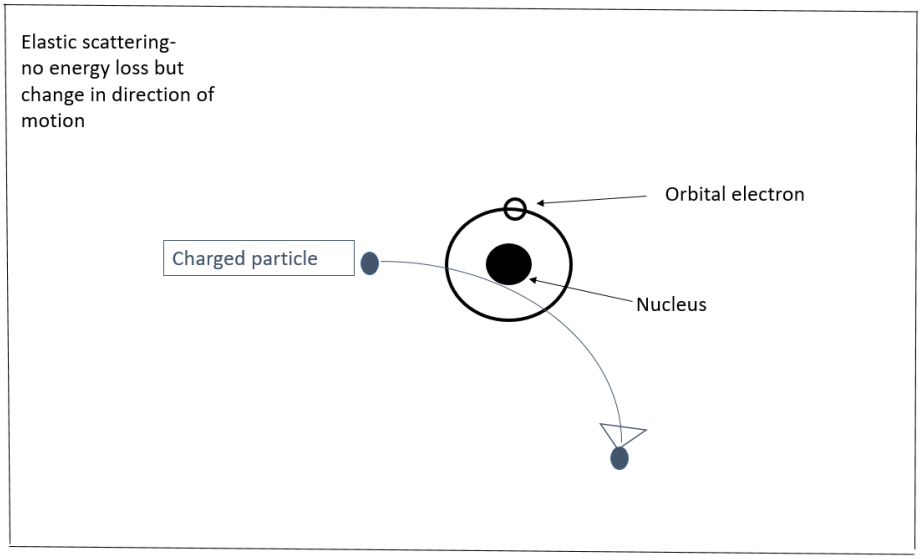
**A. 1, 2, 3, 4**
**B. 1, 2, 3**
**C. None**
**D. 4 only**

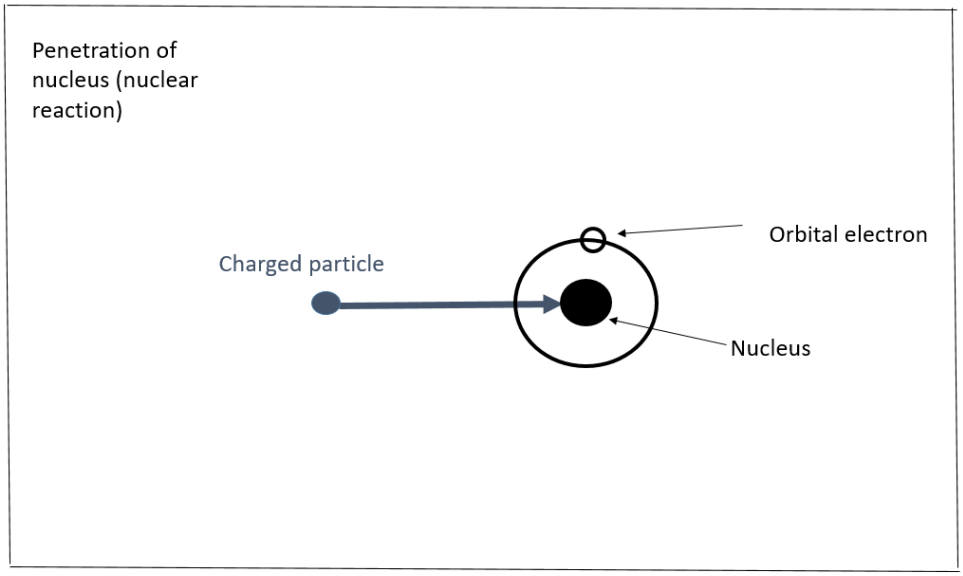
**Answer Question 82: B**

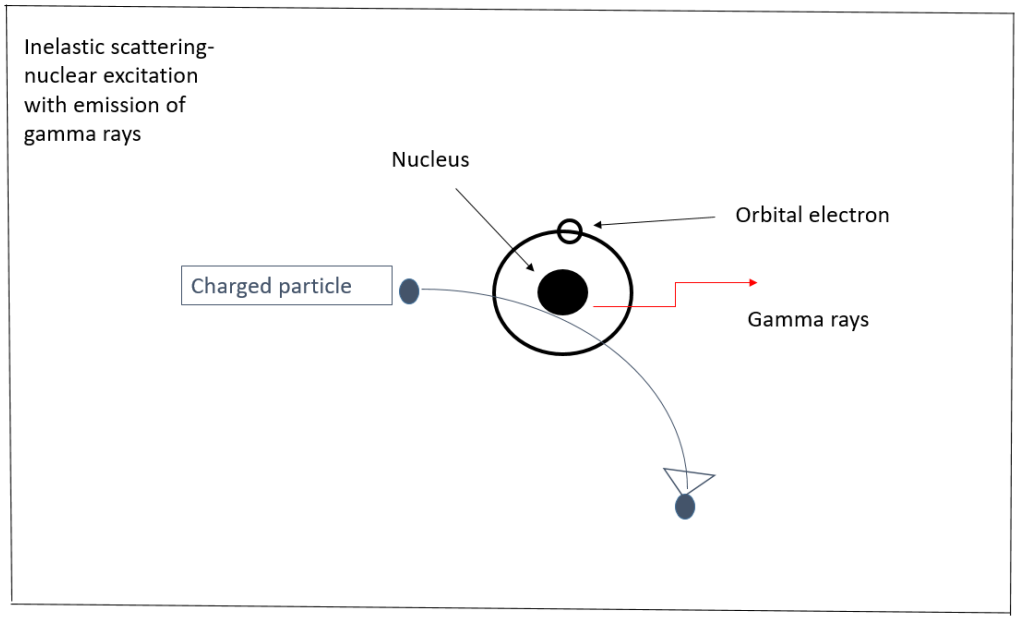
1, 2, 3 are all possible interactions of charged particles with the nucleus of the absorber atom.

In-flight annihilation (radiation loss with emission of annihilation photons) only occur with positrons interacting with the **electrons** of the absorber atom (not the nucleus). Option D is incorrect.

Charged particle interactions with nucleus:




Penetration of nucleus (nuclear reaction)
Orbital electron
Charged particle
Nucleus


Inelastic scattering-nuclear excitation with emission of gamma rays
Nucleus
Orbital electron
Charged particle
Gamma rays

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**5.4 X-ray attenuation (exponential attenuation equation, attenuation coefficients, half-value layer – monoenergetic vs polyenergetic beams, heterogeneity coefficient, beam geometry –narrow beam vs broad beam, inverse square law)**

**Question 83:**

Which of the following is the exponential attenuation equation (which quantifies intensity reduction of an x-ray beam as it traverses matter)?

I=intensity (number of photons that traverse a given area per second or photons $cm^{-2}$ $s^{-1}$)

$I_0$ = initial intensity

e = exponential

$\mu$ = linear attenuation coefficient in $cm^{-1}$

x = distance travelled in cm

A. $I = I_0 \ x / e^{-\mu}$

B. $I = I_0 \ e^{\mu x}$

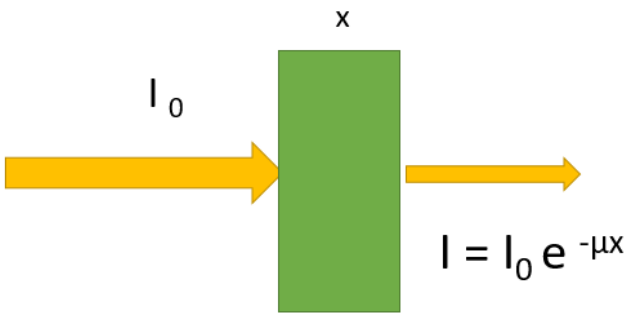
C. $I = I_0 e^{-\mu x}$

D. $I = I_0 e^{-x} / \mu$

**Answer Question 83: C**

$I = I_0 e^{-\mu x}$

This exponential relationship between incident primary photons and transmitted photons for a mono-energetic beam may be used to calculate the attenuation of any thickness of material, x.




Reference:

McKetty MH. The AAPM/RSNA physics tutorial for residents. X-ray attenuation. Radiographics. 1998 Jan-Feb;18(1):151-63; quiz 149. doi: 10.1148/radiographics.18.1.9460114. PMID: 9460114.

**Question 84:**

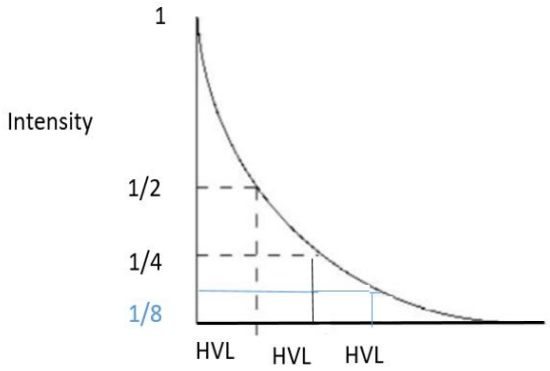
For a monoenergetic beam, by what proportion does the intensity of the beam decrease after 3 half value layers of material, (in relation to the initial intensity)?

A. 1/3
B. 1/6
C. 1/2
D. 1/8

**Answer Question 84: D**

1/8

For a monoenergetic beam, the intensity of the beam decreases by half for each successive half-value layer of material.




Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

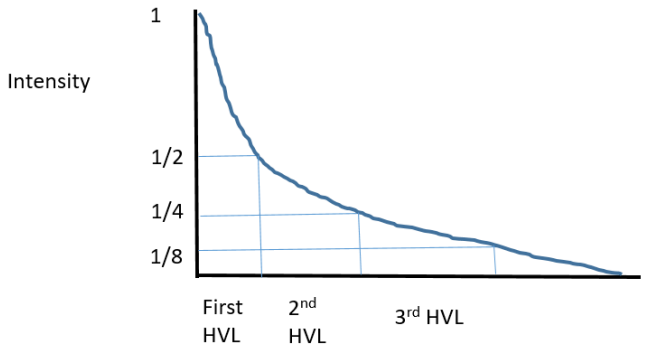
**Question 85:**

For a narrow polyenergetic beam, the successive half value layers distances generally:

A. Remain the same
B. Decrease
C. Increase
D. Are not measurable

**Answer Question 85: C**

A clinical photon beam consists of a spectrum of photon energies or is a polyenergetic beam.
In the first half value layer, lower-energy photons are preferentially absorbed leaving the radiation beam "harder", with a higher proportion of higher energy photons. An increase in half value layer thickness is needed with successive half value layer distances.




Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 86:**

Which statement below is **false** regarding tissue heterogeneity coefficients?

A. Improve the accuracy of dose calculations especially in lung and bone
B. Are derived from CT scans
C. Depend on electron density of different tissues
D. Affects the determination of clinical target volume

**Answer Question 86: D**

All the above options are true except D.

Reference:

Morill SM, Langer ML, Lane RG, Rosen II. Tissue heterogeneity effects in treatment plan optimization. Int J Radiat Oncol Biol Phys 30(3): 699-706, 1994.

**Question 87:**

Which of the following is **NOT** affected by the inverse square law?

A. Increasing the source surface distance for a direct orthovoltage beam, increases the percentage depth dose
B. The widest vaginal cylinder that can be comfortably inserted is chosen for vaginal vault brachytherapy.

C. Dose across a non-flat surface for a direct orthovoltage beam.
D. Increasing photon field size, increases the percentage depth dose
E. Uneven air gaps with electron beam therapy

**Answer Question 87: D**

The inverse square law states the intensity of radiation is inversely proportional to the square of the distance from the ionizing radiation source.

$$\frac{\text{Intensity 1}}{\text{Intensity 2}} = \frac{\text{Distance } 2^2}{\text{Distance } 1^2}$$

All the examples except D, involve varying source surface distance which will affect dose by the inverse square law.

Increasing photon field size increases percentage depth dose due to increased scatter to the central axis, not due to the inverse square law.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 88:**

What is the key difference between narrow-beam geometry and broad-beam geometry?

A. A narrow beam is the primary beam without significant scatter. A broad beam includes the primary beam and scattered radiation produced in the material being irradiated.

B. A narrow beam is a photon beam, which is less than 4 cm wide. A broad beam is a photon beam more than 4 cm wide. A photon beam field which is 4 cm or smaller has significant lateral scatter.

C. A narrow beam is a pencil beam, only a few millimeters wide and is used, for example, with pencil beam proton therapy. Broad beams are photon beams.

D. Narrow beam geometry are narrow beams with small penumbra. Broad beams are beams with large penumbra.

**Answer Question 88: A**

A narrow beam is the primary beam without significant scatter. Narrow beam geometry is used for measuring half value layer and beam attenuation.

A broad beam includes the primary beam as well as scattered radiation produced in the material irradiated. Broad beam geometry is used for shielding requirements.

Reference:

Chang, D.S., Lasley, F.D., Das, I.J., Mendonca, M.S., Dynlacht, J.R. (2014). Characteristics of Photon Beams. In: Basic Radiotherapy Physics and Biology. Springer, Cham. https://doi.org/10.1007/978-3-319-06841-1_7

# Chapter 6: Radiation quantities and units

Victoria (Tori) Doss, Seyi Oderinde, May Tsao

## 6.1 Fluence and Energy Fluence, Exposure, Kerma, Mean Energy to Create an Ion Pair, Absorbed Dose (Dose in Free Space, Medium)

**Question 89:**

What is the definition for **photon fluence**?

A. The number of photons incident per unit area (photons/$m^2$)
B. The number of photons incident per unit volume (photons/ $m^3$)
C. The number of photons incident per unit area per unit time (photons/$m^2$/s)
D. The number of photons incident per unit volume per unit time (photons/$m^3$/s)

**Answer Question 89: A**

Photon fluence is the number of photons incident per unit area.

$$\Phi = \frac{dN}{da}$$

Where:
Ø = photon fluence
*d*N = number of photons
*d*a = cross-sectional area

Photon fluence **rate** is defined as the photon fluence per unit time.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 90:**

What is the definition for **energy fluence**?

A. The sum of the energies of all photons incident per unit area (J/$m^2$)
B. The sum of the energies of all photons incident per unit volume (J/$m^3$)
C. The sum of the energies of all photons incident per unit area per unit time (J/$m^2$/s)
D. The sum of the energies of all photons incident per unit volume per unit time (J/$m^3$/s)

**Answer Question 90: A**

Energy fluence is the sum of the energies of all photons incident per unit area (J/$m^2$).

$$\Psi = \frac{dE}{da}$$

Ψ = energy fluence
*d*E = sum of the energies of all photons
*d*a = cross sectional area

Energy fluence **rate** is defined as the energy fluence per unit time.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 91:**

The metric unit for exposure is:

A. Ci/g

B. J/kg

C. Gy/kg

D. C/kg

E.A/kg

**Answer Question 91: D**

The Roentgen R (coulomb/kg) is the metric unit for exposure. This is the measure of ionizations produced in **air** by photons. 1R=2.58 x $10^{-4}$ C/kg air.

Exposure is defined for photons with energy less than 3 MeV.

**Exposure** measures the ionization (or electric charge) produced per kg of air, by incident photons.

On the other hand, **air kerma** measures the kinetic energy of ions per unit mass of air and is measured in Gy or J/kg.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 92:**

Which of the following statements regarding KERMA is **false**?

A. Kerma is the sum of the initial kinetic energies of all charged particles liberated by uncharged ionizing radiation (ie. indirectly ionizing radiation such as photons and neutrons) measured in Gy or J/kg
B. Kerma is approximately equal to absorbed dose at **high** energies but kerma is much higher than absorbed dose at **lower** energies
C. Total kerma is the sum of collision kerma and radiative kerma
D. Kerma is used in the calculation of absorbed dose in tissue

**Answer Question 92: B**

B is false. The reverse is true:

Kerma is the Kinetic Energy Released in Medium per unit Mass. It has the unit of J/kg, which is the same as absorbed dose.

Kerma is approximately equal to absorbed dose at **low** energies but kerma is much higher than absorbed dose at **higher** energies in the buildup region.

Photon energy is transferred to matter in a two-step process. As a first step, photon energy is transferred to charged particles through various interactions such as photoelectric effect, Compton scattering, pair production. These secondary charged particles then transfer their energy to matter through atomic excitations and ionizations.

At low energies, kerma is approximately equal to absorbed dose. However, at higher energy, some energy escapes in the form of bremsstrahlung x rays or fast-moving electrons not accounted for as absorbed dose.

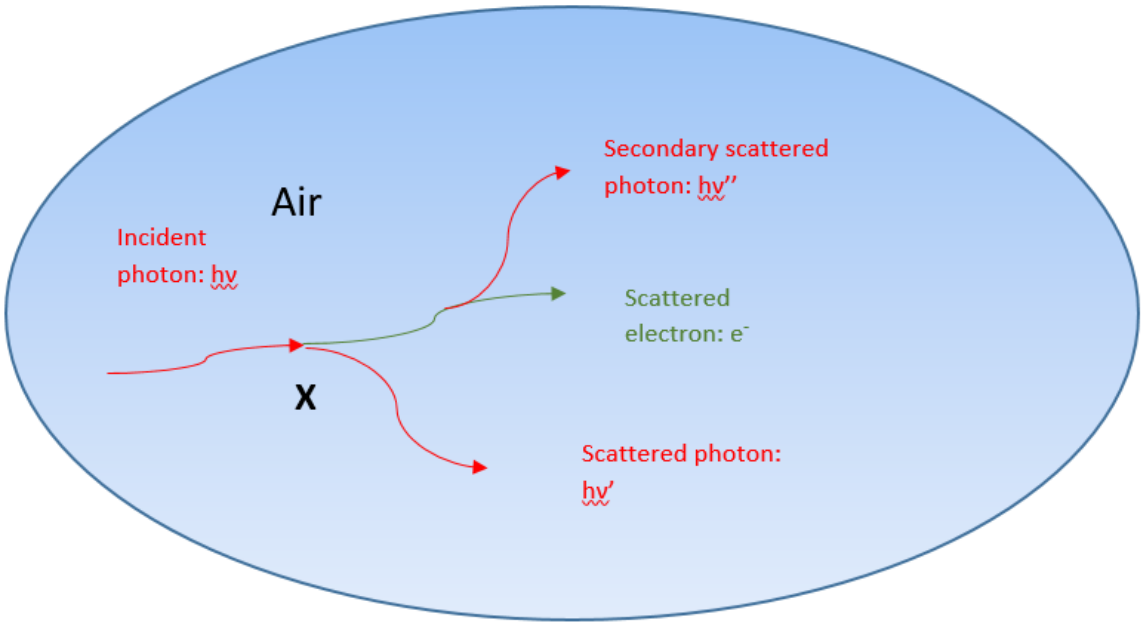
Collision kerma occurs when uncharged particles (eg. photons or neutrons) interact or collide with atoms producing ionization and excitation.

Radiative kerma occurs with the production of radiative photons when the charged particle interacts with atomic nuclei producing Bremsstrahlung radiation or photons from annihilation of positrons.

Total kerma is the sum of collision kerma and radiative kerma.

The following 2 figures illustrate:

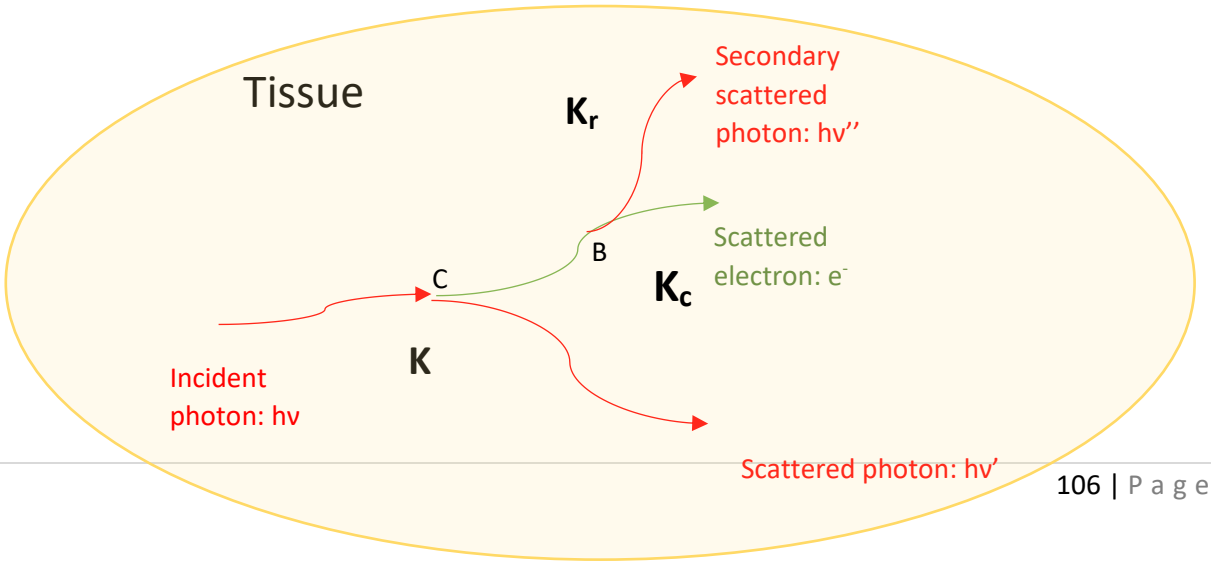
**Exposure in air:**




Exposure is measured at **X.** Exposure measures the ionizations produced in **air** by photons.

$$X = (Kc)_{air} \times \left(\frac{e}{\bar{W}}\right)$$

**Kerma and absorbed dose:**



Kerma is measured at **K**.

Collision kerma (**Kc**) occurs when the incident photon collides and produces a scattered electron [eg. through ionization and excitation]. Collision kerma is proportional to photon fluence and energy fluence.

Radiative kerma (**Kr**) occurs when the scattered **electron** produces a scattered photon [eg. through Bremstrahlung (B) radiation]

At low photon energies (e.g., diagnostic x-rays up to ~150 keV), **Kc** dominates due to minimal bremsstrahlung production, while at high photon energies (e.g., megavoltage photon beams used in radiotherapy), **Kr** becomes increasingly significant.

Compton interaction occurs at C, where the incident **photon** (hv) interacts with matter producing a scattered photon (hv') and scattered electron ($e^-$).

Reference:

Ma C-M Charlie. AAPM. Continuing Education Therapy Physics Review Course. Measurement of Radiation. Pages 5-8, 2005.

**Question 93:**

Which of the following statements is **false** regarding the mean energy to produce an ion pair?

A. The mean energy to produce an ion pair in air is defined as the average energy needed for an incident particle (eg. electron, alpha particles, beta particles etc.) or incident photon to separate an electron from an atom, resulting in a positively charged ion and negatively charged electron
B. The mean energy to produce an ion pair in dry air (air that lacks humidity) is almost constant for all electron energies
C. The mean energy for an x-ray to produce an ion pair in dry air is 33.97eV/ion pair or 33.97 J/C at standard temperature and pressure
D. The mean energy to produce an ion pair is independent of the type of incident particle and independent on the medium being ionized.

**Answer Question 93: D**

All are true except for D. The mean energy to produce an ion pair is **dependent** on the type of charged particle and medium being ionized.

References:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

Bronic IK, Srdoc D, Obelic B. The mean energy required to form an ion pair for low-energy photons and electrons in polyatomic gases.

**Question 94:**

Which of the following are **true** regarding dose in free space?

Choose the correct answer **A, B, C or D** below.

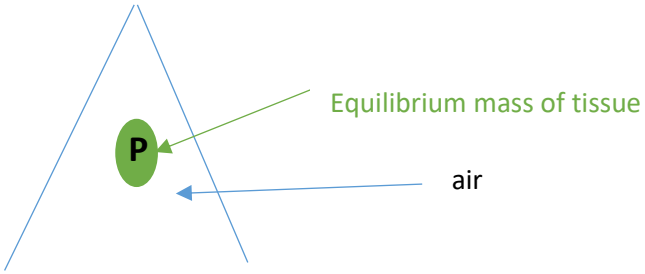
1. Dose in free space is based on measurement of kerma in air
2. Dose in free space is practically defined only for photons with energies below 3 MeV
3. Dose in free space is measured in an ionization chamber
4. Dose in free space relates to dose in tissue medium by means of tissue-air ratios [TAR]

**A. 1, 2, 3, 4**
**B. None**
**C. Only 1 and 2**
**D. Only 3 and 4**

**Answer Question 94: A**

All are true.

Dose in free space is measured when a small amount of soft tissue at point P is just large enough to provide electronic equilibrium at its centre. This small amount of soft tissue is surrounded by air. The dose at the centre of this equilibrium mass of tissue surrounded by air is the "dose in free space" or "dose to small mass of medium, in air".




Tissue-Air Ratio (TAR) $= \frac{Dose\ @\ depth\ in\ water}{Dose\ in\ air\ with\ buildup\ cap}$

Since dose in free space is not well-defined at photon energies above 3 MeV, the Tissue-Air Ratio (TAR) method is not used for point dose calculations in patients treated with high-energy photon beams on a linear accelerator (LINAC). Instead, TMR or TPR methods are employed.

References:
Cunningham JR, Johns HE. The calculation of absorbed dose from exposure measurements: practical problems in dosimetry. Phys Med Biol. 1970 Jan;15(1):71-7. doi: 10.1088/0031-9155/15/1/003. PMID: 5418323.

Measurement of Absorbed dose, 2016. https://radiologykey.com/measurement-of-absorbed-dose/

https://international.anl.gov/training/materials/IAEA%20Publications/Radiation%20Oncology%20Physics%20Handbook/Radiation%20Oncology%20Physics%20-%20Slides%20-%20pdf/Chapter_06_Photon_beams.pdf

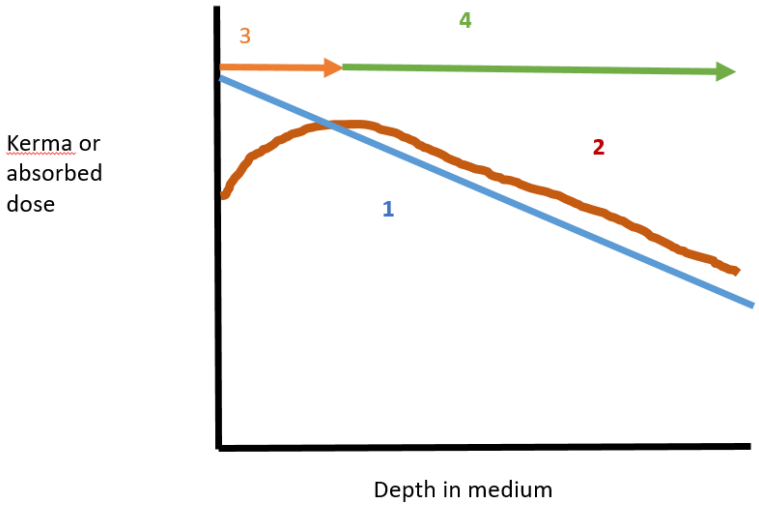
**6.2 Relationship of Kerma, Dose, Exposure (Charged Particle Equilibrium and Transient Charged Particle Equilibrium, Calculation of Absorbed Dose from Kerma and Exposure)**

**Question 95:**

The diagram below describes the relationship between absorbed dose and kerma (for photons and neutrons) with depth.

What do the labels 1, 2, 3, 4 represent?

Choose the correct answer **A, B ,C or D** below.



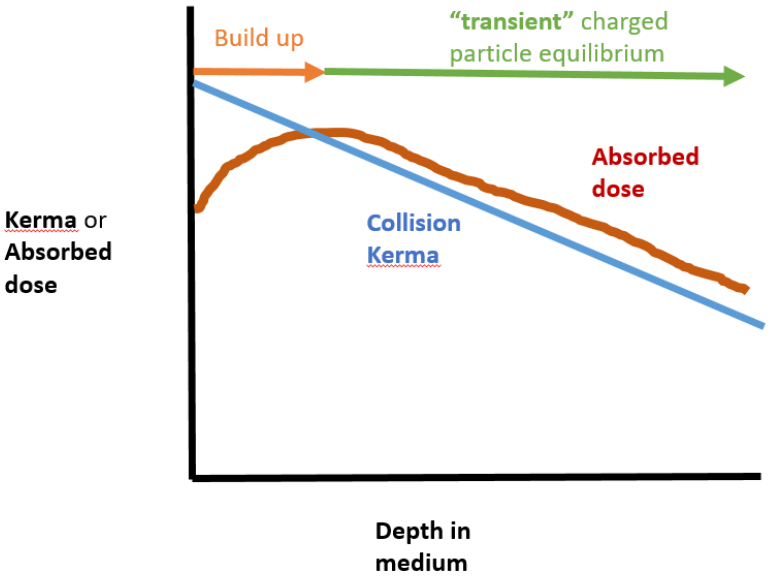

A. 1= collision kerma, 2= absorbed dose, 3= build-up region, 4=transient charged particle equilibrium region

B. 1= bulid-up region, 2= collision kerma, 3=Transient charged particle equilibrium region, 4=absorbed dose

C. 1 = build-up region, 2 = transient charged particle equilibrium region, 3= collision kerma, 4 = absorbed dose

D. 1 = absorbed dose, 2= build-up region, 3= collision kerma, 4=transient charged particle equilibrium region

**Answer Question 95: A**




This diagram illustrates that in many situations, the absorbed dose is approximately the same as kerma in the region of charged particle equilibrium.

For gamma rays in air, 1 Roentgen of exposure = 0.88 cGy absorbed dose = 0.88 cGy kerma.

For neutrons, the absorbed dose is equal to kerma because the range of charged particles is short and there is no radiative loss (Bremsstrahlung).

Reference:

Batho HF. Relationships between exposure, kerma and absorbed dose in a medium exposed to megavoltage photons from an external source. Phys Med Biol. 1968 Jul;13(3):335-46. doi: 10.1088/0031-9155/13/3/301. PMID: 5700173.

**Question 96:**

What is the difference between charged particle equilibrium (CPE) and transient charged particle (TCPE) equilibrium?

A. CPE is a state where each charged particle of a given type and energy leaving a volume is replaced by an identical particle entering the volume. TCPE is a state where the number of charged particles entering and leaving a volume is not fully equal but the imbalance of charged particles entering and leaving a volume is small.

B. CPE is a state where the number of charged particles in a volume have a net zero charge. TCPE is a state where the charged particles are in a transient net zero charge in a volume.

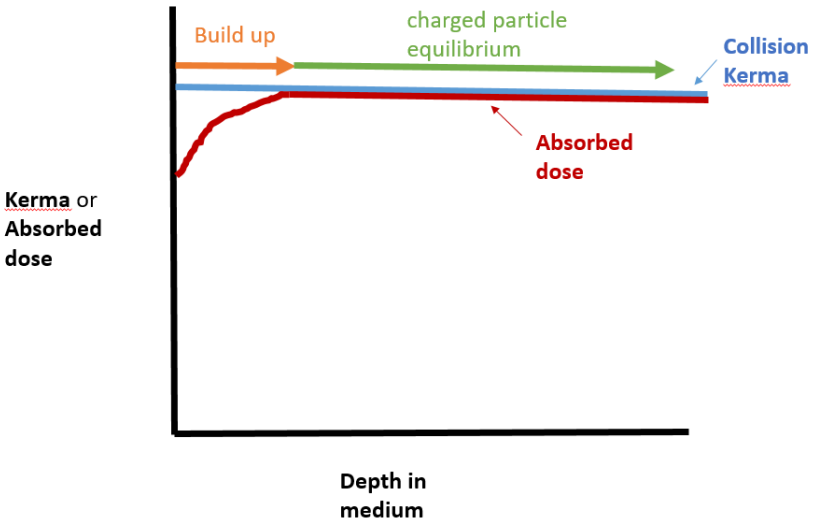
C. CPE is an equilibrium state where the sum of the energies of the charged particles in a volume is equal to the incident photon energy. TCPE occurs when the sum of energies of the charged particles in a volume is close but not equal to the incident photon energy.

D. CPE occurs when the absorbed dose in the volume is equal to the radiative kerma. TCPE occurs when the absorbed dose in the volume is in the buildup region.

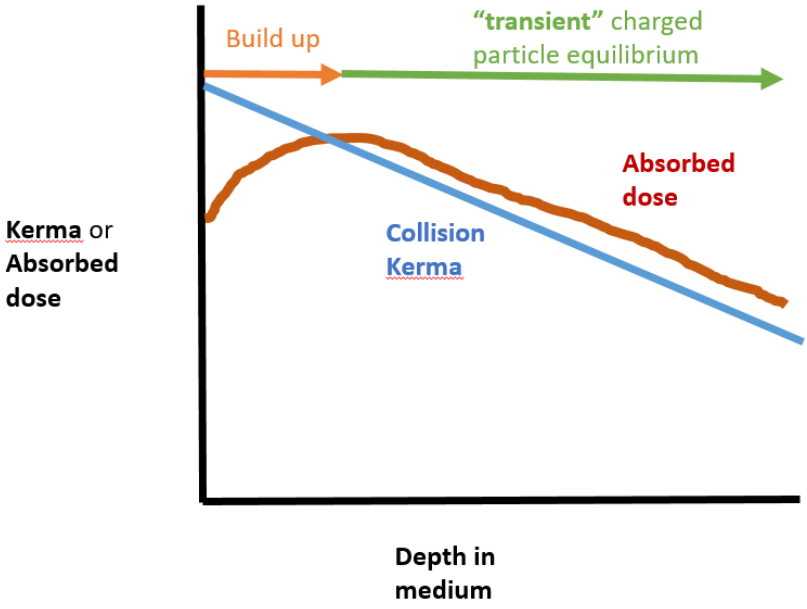
**Answer Question 96: A**

CPE is a state where each charged particle of a given type and energy leaving a volume is replaced by an identical particle entering the volume. TCPE is a state where the number of charged particles entering and leaving a volume is not fully equal but the imbalance of charged particles entering and leaving a volume is small.

CPE exists in a medium when the absorbed dose in the volume is equal to the collision kerma.




Absorbed dose (D) = β × collision kerma (Kc)
β is the quotient of absorbed dose at a point, a ratio of absorbed dose to collision kerma
β=1 in the situation of charged particle equilibrium




In the build up region, β <1.

In the transient charged particle equilibrium region, β>1.

References:

Ma C-M Charlie. AAPM. Continuing Education Therapy Physics Review Course. Measurement of Radiation. Pages 14- 17, 2005.

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 97:**

Which of the following statements is **false** regarding calculation of absorbed dose from exposure?

A. The absorbed dose in tissue (medium) is proportional to exposure
B. The absorbed dose is related to the mass energy absorption coefficients
C. The absorbed dose is dependent on the f factor or roentgen to rad conversion factor
D. The absorbed dose calculation as it relates to exposure is dependent on field size

**Answer Question 97: D**

Dis false. The calculation is not dependent on field size.

The absorbed dose to human tissue (medium) is calculated by the following formula:

$$D_{\text{med}} = \left[0.876 \frac{(\overline{\mu}_{en}/\rho)_{\text{med}}}{(\overline{\mu}_{en}/\rho)_{\text{air}}}\right] \cdot X \cdot A$$

$D_{\text{med}}$ = dose to tissue (medium) (in cGy)

X = exposure in Roentgen

A = transmission factor = energy fluence in medium divided by the energy fluence in air

$\frac{(\overline{\mu}_{en}/\rho)_{\text{med}}}{(\overline{\mu}_{en}/\rho)_{\text{air}}}$ = mass energy coefficient in tissue divided by the mass energy coefficient in air

$$f_{\text{med}} = 0.876 \frac{(\overline{\mu}_{en}/\rho)_{\text{med}}}{(\overline{\mu}_{en}/\rho)_{\text{air}}}$$

$f_{\text{med}}$ is also known as the f factor which varies for water, bone and muscle depending on photon energy.

**f-factor examples for orthovoltage energies:**

| **Energy (kV)** | **f factor (bone/water)** |
|---|---|
| 100 | 4.2 |
| 180 | 2.9 |
| 250 | 1.6 |
| 300 | 1.3 |

The equation for $\mathbf{D_{med}}$ was historically used to calibrate low-energy photon beams (up to 1.25 MeV) in air or in a medium such as water. However, current best practice recommends using standardized protocols such as AAPM TG-51 or IAEA TRS-398 for clinical linear accelerator calibration.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

# Chapter 7: Radiation measurement and calibration

Yunjie Yang, Jessica Fagerstrom, May Tsao

### 7.1 Phantoms (composition, electron density, effective Z value, anthropomorphic and geometric)

**Question 98:**

A water phantom could be used for basic dose distribution data because:

A. It is more convenient to set up the water tank than solid water (epoxy resin based mixture)
B. Its mass density is 2 g/cm$^3$, similar to soft tissue
C. It approximates the radiation absorption and scattering properties of muscle and soft tissue
D. It has the same effective atomic number, number of electrons per gram and mass density as the human body

**Answer Question 98: C**

All statements are false except for C. C is a correct statement.

A water phantom is approximately equivalent to a solid water phantom for dose distribution accuracy. Although there may be a 1% different attenuation difference between solid water and a water phantom.

Solid dry phantoms such as solid water allow for measurements using radiation detectors that are not waterproof. So, it is more convenient to set up solid water (epoxy resin based mixture), rather than a water phantom. Option A is false.

The mass density of water is 1 g/cm$^3$. Option B is incorrect.

The human body has various tissue densities (muscle, soft tissue, lung, bone) and as such water does not have the same effective atomic number, nor number of electrons per gram, nor mass density as all the tissues in the human body. Option D is incorrect.

Anthropomorphic phantoms are phantoms which mimic the various tissue types in the human body for both physical properties (composition, electron density, effective Z value) and for geometry.

References:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

Yadav N, Singh M,Mishra SP. Tissue-equivalent materials used to develop phantoms in radiation dosimetry. Materials Today: Proceedings 47(19): 7170-7173, 2021

## 7.2 Ionization Chambers

**Types of ionization chambers (cylindrical, parallel-plate)**

**Question 99:**

Ionization chambers measure:

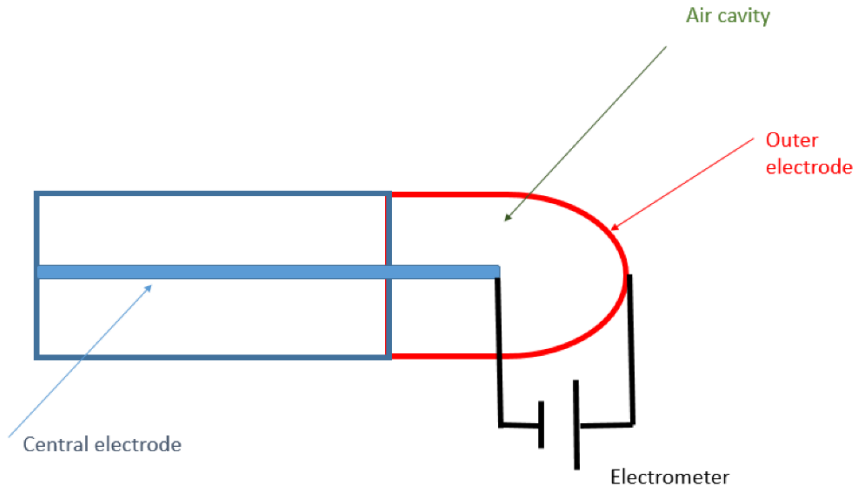
A. Dose deposited in soft tissue
B. Energy fluence within a gas caused by incident radiation
C. Charge from the number of ion pairs created within a gas caused by incident radiation
D. Kerma in air caused by incident radiation

**Answer Question 99: C**
Ionization chambers measure charge from the number of ion pairs created in a gas caused by incident radiation.

The two types of ionization chambers are cylindrical (eg. thimble) or parallel-plate.

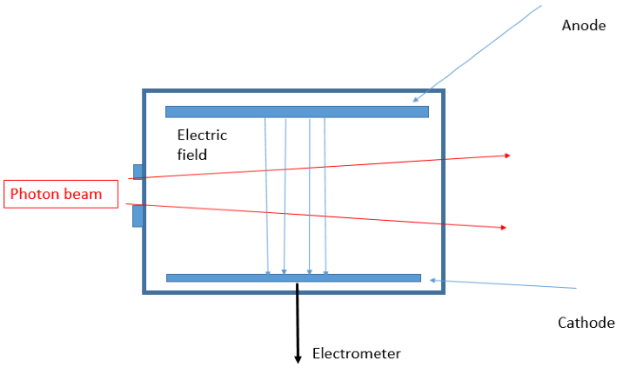
Cylindrical (thimble) ionization chamber:






Parallel plate ionization chamber:






References:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 100:**
Ionization chamber correction factors take into account the following EXCEPT:

A. beam quality
B. magnetic field
C. low density medium
D. geometry
E. absence of charged particle equilibrium

**Answer question 100: E**

All the factors listed are correct except for E.

Ion chambers are designed to operate under conditions of electronic equilibrium.

Reference:

Pisaturo O, Pachoud M, Bochud FO, Moeckli R. Calculation of correction factors for ionization chamber measurements with small fields in low-density media. Phys Med Biol. 2012 Jul 21;57(14):4589-98.

### 7.3 Other Dosimetry Instrumentation

**Absolute versus Relative Dosimetry**

**Question 101:**

What is the difference between absolute versus relative dosimetry?

A. Absolute dosimetry measures absorbed dose directly. Relative dosimetry uses dosimeters which are calibrated for a known reference dose
B. Absolute dosimetry uses Monte Carlo methods to calculate dose. Relative dosimetry uses convolution superposition
C. Absolute dosimetry uses an anthropomorphic phantom. Relative dosimetry uses a water phantom
D. Absolute dosimetry uses a thimble chamber. Relative dosimetry uses a parallel plate ionization chamber

**Answer Question 101: A**

Absolute dosimetry measures absorbed dose directly. An example is a calorimeter which measures the heat generated by absorbed dose. Another example is an ionization chamber which measures the charge produced by ionizing radiation.

Relative dosimetry uses dosimeters, which are calibrated for a known reference dose. An example is film dosimetry where the opacity of the film is related to the dose absorbed. The dose is measured by a reference dosimeter, which is calibrated against an absolute dosimeter.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Calorimetry, Chemical Dosimetry (Fricke, alanine)**

**Question 102:**

Fricke calorimetry relates what chemical ions to absorbed dose?

A. Cerium (ceric) sulphate ions
B. Ferric ions
C. Alanine ions
D. Glutamine ions

**Answer Question 102: B**

Fricke calorimetry (also known as ferrous sulfate dosimetry) uses a chemical method to measure absorbed dose to water by spectrophotometrically measuring oxidation of ferrous ions ($Fe^{2+}$ ) to ferric ions ($Fe^{3+}$).
An FBX calorimeter consists of ferrous sulphate, benzoic acid andxylenol orange to improve sensitivity, accuracy and precision of absorbed dose measurements. The ferric ions formed produce a colour change (colour complex with xylenol orange). The colour change is measured spectrophotometrically to estimate the number of ferric ions formed.

An alanine calorimeter measures the formation of free radicals in alanine when exposed to radiation.

Reference:

Ross CK, Klassen NV, Shortt KR, Smith GD. A direct comparison of water calorimetry and Fricke dosimetry. Phys Med Biol. 1989 Jan;34(1):23-42.

**Solid State Detectors (Thermoluminescent dosimeters – TLD, Optically Stimulated Luminescence dosimeters – OSLD, Diode detectors, Diamond detectors, Scintillators, Metal Oxide Field Effect Transitor (MOSFET) dosimeters**

**Question 103.1:**

Thermoluminescent dosimeter (TLD)

A. Measures radiation dose by releasing visible light (**when heated**). The intensity of emitted light is proportional to the radiation dose received.

B. Measures radiation dose by releasing visible light of a specific wavelength (**when stimulated with light**). The light of specific wavelength emitted is proportional to the radiation dose.

C. Measures radiation dose by generating a charge when photons or electrons interact with the detector. The charge is then measured using an electrometer.

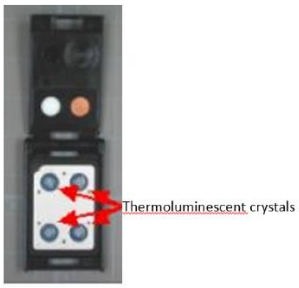
D. Measures radiation dose by the generation of electron-hole pairs, which are charge carriers. These charge carries create current which is measured and correlated to absorbed dose.

E. Measures radiation dose when radiation interacts with a silicone chip creating a shift in threshold voltage. The shift in threshold voltage is proportional to absorbed dose.

**Answer Question 103.1: A**

TLDs are commonly used for radiation worker dose monitoring. TLDs are also used for example to measure the skin dose to patients and for audits of radiotherapy external beam calibration.

A TLD measures radiation exposure by using a material that emits light when heated after being exposed to ionizing radiation.




**Question 103. 2:**

Optically stimulated luminescence dosimeter (OSLD)

A. Measures radiation dose by releasing visible light (**when heated**). The intensity of emitted light is proportional to the radiation dose received.

B. Measures radiation dose by releasing visible light of a specific wavelength (**when stimulated with light**). The light of specific wavelength emitted is proportional to the radiation dose.

C. Measures radiation dose by generating a charge when photons or electrons interact with the detector. The charge is then measured using an electrometer.

D. Measures radiation dose by the generation of electron-hole pairs, which are charge carriers. These charge carries create current which is measured and correlated to absorbed dose.

E. Measures radiation dose when radiation interacts with a silicone chip creating a shift in threshold voltage. The shift in threshold voltage is proportional to absorbed dose.

**Answer Question 103.2: B**

OSLD does not require heat (unlike TLDs).



**Question 103. 3:**

Diode detectors

A. Measures radiation dose by releasing visible light (**when heated**). The intensity of emitted light is proportional to the radiation dose received.

B. Measures radiation dose by releasing visible light of a specific wavelength (**when stimulated with light**). The light of specific wavelength emitted is proportional to the radiation dose.

C. Measures radiation dose by generating a charge when photons or electrons interact with the detector. The charge is then measured using an electrometer.

D. Measures radiation dose by amplifying the input signal.

E. Measures radiation dose when radiation interacts with a silicone chip creating a shift in threshold voltage. The shift in threshold voltage is proportional to absorbed dose.

**Answer Question 103.3: C**

Diode detectors are usually silicone based devices that convert radiation into electrical signal, used to measure dose of radiation delivered to a patient. Diode detectors also measure radiation dose by the generation of *electron-hole pairs,* which are charge carriers. These charge carriers create current which is measured and correlated to absorbed dose.



**Question 103. 4:**

Diamond detectors

A. Measures radiation dose by releasing visible light (**when heated**). The intensity of emitted light is proportional to the radiation dose received.

B. Measures radiation dose by releasing visible light of a specific wavelength (**when stimulated with light**). The light of specific wavelength emitted is proportional to the radiation dose.

C. Measures radiation dose by generating a charge when photons or electrons interact with the detector. The charge is then measured using an electrometer.

D. Measures radiation dose by the generation of electron-hole pairs, which are charge carriers. These charge carries create current which is measured and correlated to absorbed dose.

E. Measures radiation dose when radiation interacts with a silicone chip creating a shift in threshold voltage. The shift in threshold voltage is proportional to absorbed dose.

**Answer Question 103.4: D**

When a diamond detector is exposed to radiation, charge carriers /electric current gets produced which is measured and correlated to absorbed dose. Diamond detectors have several advantages to traditional detectors due to its high sensitivity, good spatial resolution, and applicability to high and low dose rates (eg. FLASH radiotherapy). Diamond detectors also allow for real-time measurements.



**Question 103. 5**

Metal oxide field effect transistor (MOSFET) dosimeter

A. Measures radiation dose by releasing visible light (**when heated**). The intensity of emitted light is proportional to the radiation dose received.

B. Measures radiation dose by releasing visible light of a specific wavelength (**when stimulated with light**). The light of specific wavelength emitted is proportional to the radiation dose.

C. Measures radiation dose by generating a charge when photons or electrons interact with the detector. The charge is then measured using an electrometer.

D. Measures radiation dose by the generation of electron-hole pairs, which are charge carriers. These charge carries create current which is measured and correlated to absorbed dose.

E. Measures radiation dose when radiation interacts with a silicone chip creating a shift in threshold voltage. The shift in threshold voltage is proportional to absorbed dose.

**Answer Question 103.5: E**

In a metal oxide field effect transistor (MOSFET) dosimeter, radiation interacts with a silicone chip creating a shift in threshold voltage. The shift in threshold voltage is proportional to absorbed dose.




Reference:

Angelou C, Patallo IS, Doherty D et al. A review of diamond dosimeters in advanced radiotherapy techniques. Medical Physics 51(12): 9230-9249, 2024.

**Point versus Spatial Dosimetry, Volume averaging, 2D (Film, detector array), 3D (gels)**

**Question 104:**

Thermoluminescent dosimeters (TLD) and optically simulated luminescent dosimeters (OSLD) are examples of:

A. 3D gel dosimetry
B. Spatial dosimetry
C. Point dosimetry
D. 2D film dosimetry

**Answer Question 104: C**

Both TLDs and OSLD's are examples of point dosimetry. Both measure dose at a particular point.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 105:**

Which of the following is **NOT** impacted by volume averaging in dosimetry?

A. Dose in the centre of a field
B. Penumbra size
C. Small radiation field size
D. Clinical target volume

**Answer Question 105: D**

Volume averaging in dosimetry occurs when a detector measures an average dose over its volume rather than an accurate dose at a single point. This occurs especially in small fields (eg. stereotactic radiosurgery). As a result, the dose in the center of the field could be underestimated and the size of the penumbra could be measured wider than the actual penumbra.

Reference:

Parwaie W, Refahi S, Ardekani MA, Farhood B. Different Dosimeters/Detectors Used in Small-Field Dosimetry: Pros and Cons. J Med Signals Sens. 2018 Jul-Sep;8(3):195-203.

**Gas Filled Detectors (Geiger-Muller counters, Proportional counters)**

**Question 106:**

Which of the following is **false** regarding a Geiger-Muller counter?

A. It is a proportional counter
B. It is a gas filled detector
C. It measures current when irradiated
D. It displays radiation levels as audible clicks or counts per minute (ion pairs detected per minute)

**Answer Question 106: A**
All the above are true, except Geiger-Muller counters are not proportional counters.

Gieger-Muller counter:



Proportional counters produce an electrical pulse that is proportional to the energy of the detected radiation. Not only can proportional counters measure absorbed dose, but it can also distinguish

between different types of radiation (eg. alpha and beta particles based on the size of the produced pulse).

Example of proportional counters:



Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Neutron Detectors**

**Question 107:**

Which of the following is **not** measured by neutron detectors?

A. Neutron contamination produced by high energy photon beams
B. Monitoring dose delivered during neutron therapy
C. Neutron contamination produced by proton therapy
D. Energy of incoming neutrons

**Answer Question 107: D**

Neutron detectors do not measure the energy of incoming neutrons directly.

References:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

https://en.wikipedia.org/wiki/Neutron_detection

**7.4 Calibration of Megavoltage Beams**

**TG-51 calibration protocol-measurement of absorbed dose in water using calibrated ion chambers**

**Question 108:**

Which of the following methods is primarily used for the calibration of megavoltage photon beams in a linear accelerator?

A) TG-51 protocol with an in-air calibration using an ionization chamber
B) TG-51 protocol using a water phantom and ionization chamber
C) Film dosimetry with radiochromic film
D) Semiconductor diode measurement in a solid phantom

**Answer Question 108: B**

TG-51 protocol using a water phantom and ionization chamber.

The TG-51 protocol, established by the AAPM, is the standard method for the calibration of megavoltage photon beams. It involves using a water phantom and a calibrated ionization chamber, such as a Farmer-type chamber, to measure dose in water at a reference depth. This process ensures accurate dose delivery by providing a consistent and traceable way to calibrate the linear accelerator output. Option A describes an in-air calibration, which is not the standard method for clinical calibration. Option C (film dosimetry) and option D (semiconductor diode measurement) are used for relative dose measurements or specific QA tasks but not for primary beam calibration.

Reference:

Muir B, Culberson W, Davis S, Kim GG, Lee SW, Lowenstein J, Renaud J, Sarfehnia A, Siebers J, Tantôt L, Tolani N. AAPM WGTG51 Report 374: Guidance for TG-51 reference dosimetry. Med Phys. 2022 Nov;49(11):6739-6764.

**Question 109:**

According to AAPM TG-51 protocol, which of the following is an appropriate detector choice for performing Reference Dosimetry?

A. Farmer-type ionization chamber
B. Stereotactic diode detector
C. Radiochromic film
D. Geiger-Muller counter

**Answer Question 109: A**

The AAPM TG-51 protocol restricts the choices of detectors used for Reference Dosimetry to be Farmer-type ionization chambers. Other detectors, while useful for certain clinical tasks in the clinics, are not suitable for this purpose.

Reference:

Muir B, Culberson W, Davis S, Kim GG, Lee SW, Lowenstein J, Renaud J, Sarfehnia A, Siebers J, Tantôt L, Tolani N. AAPM WGTG51 Report 374: Guidance for TG-51 reference dosimetry. Med Phys. 2022 Nov;49(11):6739-6764.

**Basic Cavity Theory- Bragg-Gray and Spencer-Attix**

**Question 110:**

What is the basic cavity theory?

A. The basic cavity theory relies on the Monte Carlo method of dosimetry to evaluate dose in air equivalent cavities.

B. The basic cavity theory relates dose deposited in the cavity (thin layer of gas in the detector) to that of the surrounding medium, assuming electronic disequilibrium.

C. The basic cavity theory relates measured dose in a cavity such as an ion chamber to the dose at the same point in the medium in the absence of the cavity, assuming electronic equilibrium.

D. The basic cavity theory relates tissue air ratios in the cavity (sensitive volume of the detector) to that of the surrounding medium.

E. The basic cavity theory relates ionization of air to that of water.

**Answer Question 110: C**

The basic cavity theory relates measured dose in a cavity such as an ion chamber to the dose at the same point in the medium in the absence of the cavity, assuming electronic equilibrium.

In order to measure dose in a medium, a radiation sensitive device (dosimeter) is used. The radiation sensitive part of the dosimeter is called a cavity (which may be gas, liquid or solid.

Cavity theories are used to study the effect of radiation dose produced in the medium by the cavity. Cavity theories also mathematically explain relations between the dose in the sensitive volume of the dosimeter and the dose in the surrounding medium.

The Bragg-Gray cavity theory works best for small cavities and the detector used is the Bragg-Gray (B-G) cavity detector. This theory assumes that the cavity is small compared to the range of charged particles within it. Another assumption is that there are no interactions of uncharged particles in the cavity so that the absorbed dose deposited in the cavity is from charged particles that cross the cavity.

The Spencer and Attix cavity theory allows cavity effects of different sizes.

The Fano theorem also allows for different cavity sizes. The Fano theorem is valid only for infinite media and in conditions where the stopping-power is independent of density

In Burlin's theory different cavity sizes are also accounted for, but all secondary-electron scattering effects are ignored.

All these cavity theories require charged particle equilibrium.

Reference:

Mobit P, Sandison G. Chapter 5-Cavity Theory. In: Microdosimetric Response of Physical and Biological Systems to Low-and High-LET Radiations. Theory and Applications to Dosimetry. 2006, pages 331-366.

**Absorbed-dose to water calibration factor and ADCL calibration**

**Question 111:**

What does the absorbed dose to water calibration factor depend on?

A. Radiation beam (type such as photons, electrons, protons and energy)

B. Dosimeter type

C. Temperature, atmospheric pressure during calibration

D. Standards laboratory performing the calibration [Primary Standards Dosimetry Laboratory (PSDL) or Secondary Standards Dosimetry Laboratory (SSDL)].

E. All the above

**Answer Question 111: E**

The absorbed dose to water calibration factor (ND,w) relates the reading of ionization chamber to the absorbed dose in water.

Reference:

Shiragai A. A proposal concerning the absorbed dose conversion factor. Phys Med Biol. 1978 Mar;23(2):245-52.

**Question 112:**

What is the function of the Accredited Dosimetry Calibration Laboratory (ADCL)?

A. Provides calibrations to users for instruments and radioactive sources for dosimetry in radiotherapy and diagnostic imaging

B. Provides quality assurance for radiation record and verify systems

C. Provides anthropomorphic and dosimetric phantoms for radiation measurement and calibration

D. Provides calibration for radiation planning systems

**Answer Question 112 A.**

References:

https://www.aapm.org/meetings/09SS/documents/17Seuntjens-TheDosimetryChainBIPMPSDLsADCLs.pdf

https://amos3.aapm.org/abstracts/pdf/155-50580-1531640-156396-1569184507.pdf

**Beam quality specification**

**Question 113:**

Which parameters specify beam quality?

Select the correct answer **A, B, C or D** below.

1. Percentage Depth Dose

2. TPR 20 – Tissue Phantom Ratio at 20 cm depth

3. TPR 10 – Tissue Phantom Ratio at 10 cm depth

4. Depth in water at which the dose falls to 50% of maximum for electrons

5. Field size

**A. 1, 2, 3, 4, 5**
**B. 1, 2, 3, 4**
**C. 2 and 3 only**
**D. none**

**Answer Question 113: B**

All describe beam quality EXCEPT field size

Beam quality specification describes the penetrating ability of the radiation beam.

Field size does not specify beam quality.

Reference:

Andreo P. On the beam quality specification of high-energy photons for radiotherapy dosimetry. Med Phys. 2000 Mar;27(3):434-40.

**Reference conditions**

**Question 114:**

What are standard reference conditions as it relates to calibration of megavoltage radiation photon beams, according to TG-51?

A. 10 cm x 10 cm field at a depth of 10 cm in a water phantom, with a source surface distance (SSD) or source axis distance (SAD) of 100 cm
B. 5 cm x 5 cm field at a depth of 5 cm in a water phantom, SSD or SAD of 50 cm
C. 0 degrees Celsius and 1 atmosphere
D. $R_{50}$, the depth in water, which the absorbed dose falls to 50% of the maximum

**Answer Question 114: A**

$R_{50}$ is the beam quality specification for electron beams, not photons.

Reference:

Almond PR, Biggs PJ, Coursey BM et al. AAPM's TG-51 protocol for clinical reference dosimetry of high-energy photon and electron brams. 1999.

**Reference-class Ionization Chambers**

**Question 115:**

Which of the following are features of reference-class ionization chambers?

Select the correct answer **A, B, C or D** below.

1. Minimal volume averaging effects
2. Accuracy
3. Minimal photon energy dependence
4. Reproducibility

**A. 1, 2, 3, 4**
**B. None**
**C. 1 and 2 only**
**D. 3 and 4 only**

**Answer Question 115: A**

All are features of reference-class ionization chambers.

An example of a reference-class ionization chamber is the Farmer ionization chamber

Reference:

Liu K, Holmes S, Khan AU, Hooten B, DeWerd L, Schüler E, Beddar S. Development of novel ionization chambers for reference dosimetry in electron flash radiotherapy. Med Phys. 2024 Dec;51(12):9275-9289.

**Uncertainties**

**Question 116:**

Which of the following are uncertainty factors in the calibration of megavoltage beams.

Select the correct answer **A, B, C or D** below.

1. ion chamber calibration factor

2. depth uncertainty (exceeding 0.5 mm) in which the ion chamber is placed in a phantom

3. beam quality correction factor

4. uncertainty in the source surface distance if it exceeds 1 mm

A. 1, 2, 3 only
B. 1, 2, 3, 4
C. None
D. 1 and 3 only

**Answer Question 116: B**

All are uncertainty factors in the calibration of megavoltage beams.

An ion chamber calibration factor converts an ionization current reading from an ion chamber into dose.

The beam quality correction factor accounts for how ionization chambers respond to various radiation beam qualities and energy.

Reference:

Castro P, García-Vicente F, Mínguez C, Floriano A, Sevillano D, Pérez L, Torres JJ. Study of the uncertainty in the determination of the absorbed dose to water during external beam radiotherapy calibration. J Appl Clin Med Phys. 2008 Jan 22;9(1):70-86.

# Chapter 8: Photon beam characteristics and dosimetry

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Humza Nusrat, May Tsao

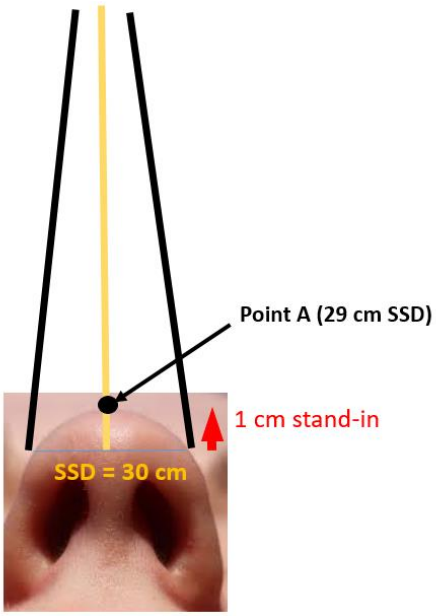
**8.1 Properties of External Radiation Beams**

**Inverse square law**

**Question 117:**

A patient with a basal cell skin cancer was set up for a direct orthovoltage beam of radiation using an open orthovoltage cone. The stand-in dimension measured 1 cm. An SSD of 30 cm was used.

What would be the percentage difference relative to the prescription dose if the patient were mistakenly set up to have the SSD at point A (29 cm) rather than at 30 cm?




A. 107%
B. 93%
C. 105%
D. 95%

**Answer Question 117: A**

Using the ISL:

$(30/29)^2 = 107\%$

Reference:

Gräfe J, Poirier Y, Jacso F, Khan R, Liu HW, Villarreal-Barajas JE. Assessing the deviation from the inverse square law for orthovoltage beams with closed-ended applicators. J Appl Clin Med Phys. 2014 Jul 8;15(4):356–366.

**Beam Profiles and their measurements (scanning water phantoms, flatness and symmetry, penumbra-geometric and physical)**

**Question 118:**

What type of penumbra is due to source size (ie. the source not being a true point source)?

A. Geometric
B. Physical
C. Transmission
D. Topographic

**Answer Question 118: A**

Geometric penumbra arises because the source of radiation is not a point source. As such photons are produced in different points within the source. These photons diverge causing a gradient of dose along the edge of the beam. Source size, SSD and field size contribute to the size of the geometric penumbra.

Transmission penumbra occurs when the radiation beam's edge gets partially attenuated at the edge of the jaws or multileaf collimator.

Physical penumbra arises from scattered photons from the beam's interaction within tissue. Overall physical penumbra takes into account the scattered photons in tissue as well as the geometric and transmission penumbra.

There is no such thing as a topographic penumbra.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 119:**

Which beam has the smallest penumbra?

A. 180 kV orthovoltage with SSD 30 cm to the skin surface
B. 6 MV photons with SSD 100 cm in soft tissue of the leg
C. 6 MeV electrons with SSD 100 cm

D. 6 MV photons with SSD 100 cm in lung of the chest

**Answer Question 119: A**

Orthovoltage has a small penumbra 2 mm (50% to 90% distance) at the skin surface. Orthovoltage x rays interact with tissue primarily through the photoelectric effect resulting in sharper dose falloff and smaller penumbra.

Megavoltage photons interact with tissue primarily through the Compton effect leading to a larger penumbra.

Megavoltage electrons interact with tissue through ionizing collisions and Bremsstrahlung interactions, leading to a larger penumbra.

The shorter SSD with orthovoltage also contributes to a smaller extent to the smaller penumbra.

The penumbra of 6 MV photons or 6 MeV electrons (50% - 90%) at the surface is in the order of 1 cm.

The penumbra of photons in lung is increased due to increased lateral scatter in low density lung.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Percent Depth Dose (PDD)**

**Question 120:** Which of the following is true regarding a photon beam of 6 MV energy?

A. The depth of maximum dose for this beam is 3 cm.
B. The depth of maximum dose for this beam is more superficial compared to that of an 18 MV beam.
C. There is a high risk of neutron contamination at this energy.
D. All of the above are true.

**Answer Question 120: B**

For photon beams, the depth of maximum dose (Dmax) increases as the beam energy increases. In other words, the lower the beam energy, the more superficial the hotspot. A good rule of thumb to calculate the Dmax for photon beams in cm is E/4 up until 10 MV, and for energies >10 MV, E/5. The Dmax for beam energies of 6 MV, 10 MV, 15 MV, and 18 MV are approximately 1.5 cm (6/4), 2.5 cm

(10/4), 3 cm MV (15/5), and 3.5 cm (18/5). Neutron contamination occurs in beams generally >10 MV due to photodisintegration. This would not be a concern at an energy of 6 MV.

Reference:

Mott JHL, West NS. Essentials of depth dose calculations for clinical oncologists. *Clin Oncol (R Coll Radiol)*. 2021;33(1):5-11.

**Question 121:**

Which of the following beam energies would have the highest surface dose at a small field size, <10 x 10 $cm^2$, for the same dose prescription?

A. 6 MV photon beam with flattening filter
B. 10 MV photon beam with flattening filter
C. 6 MV photon beam without flattening filter (flattening filter free, or FFF)
D. 10 MV photon beam without flattening filter (flattening filter free, or FFF)

**Answer Question 121: C**

A 6 MV FFF photon beam would have the highest surface dose of all of the options listed. 6 MV would have higher surface dose than 10 MV, because the depth of maximum dose increases with beam energy. In other words, surface dose increases with decreasing beam energy. A flattening filter free (FFF) beam removes the flattening filter from conventionally flattened beams, thereby increasing the dose rate in the center of the beam, but also decreasing the average energy across the beam. The beam profile becomes more cone-shaped. Due to the lower average energy of the beam, the surface dose of a 6 MV FFF beam would be higher than that of a 6 MV flattened beam.

This phenomenon is responsible for higher surface dose with FFF beams at small field sizes (<10 x 10 $cm^2$ for 6 MV beams, and <25 x 25 $cm^2$ for 10 MV beams). However, as field size increases, the presence of a flattening filter results in formation of scattered electrons which also contribute to raising the surface dose. At larger field sizes, (>10 x 10 $cm^2$ for 6 MV beams, and >25 x 25 $cm^2$ for 10 MV beams**)** the effect of electron contamination predominates and surface dose is actually higher for flattened beams.

Reference:

Cashmore J. Surface dose variations in 6 and 10 MV flattened and flattening filter-free (FFF) photon beams. J Appl Clin Med Phys. 2016 Sep 8;17(5):293-307. doi: 10.1120/jacmp.v17i5.6284. PMID: 27685127; PMCID: PMC5874110.

**Question 122:**

Which of the following is true of a flattened photon beam?

A. Dose increases with increasing field size
B. At any given depth, the horizontal profile of a flattened beam is cone-shaped, with greater dose at the mid axis than at the periphery
C. There is decreased electron contamination compared to an unflattened beam
D. All of the above are true

**Answer Question 122: A**

Dose rate increases with increasing field size in the presence of a flattening filter due to increased electron contamination with increasing field size. This eliminates answer choice C. When the flattening filter is removed (FFF), the horizontal profile of a flattened beam is cone-shaped with greater dose at the mid axis than at the periphery. However, when the flattening filter is in place, the horizontal beam profile is flattened, and at shallow depths, the off-axis dose (at the periphery of the profile) is actually greater than the mid-axis dose.

Reference:

Mohammed M, Chakir E, Boukhal H, Mroan S, El Bardouni T. Evaluation of the dosimetric characteristics of 6 MV flattened and unflattened photon beam. *J King Saud Univ Sci*. 2017;29(3):371-379.

**Dependence on Depth, Beam Energy, Field Size, SSD, Mayneord F –Factor**
**Tissue-air ratio TAR**
**Tissue-maximum ratio TMR**
**Tissue-phantom ratio TPR**
**Back-scatter factor BSF**

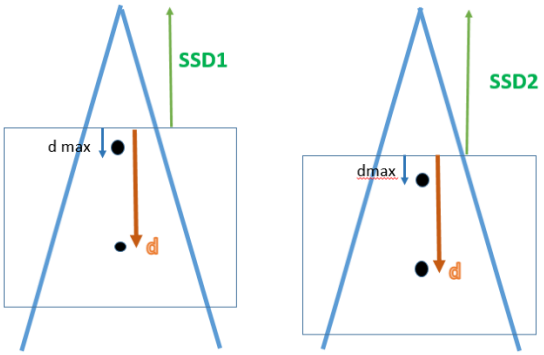
**Question 123:**

The Mayneord F factor:

A. converts percentage depth dose (PDD) values from one source-surface distance (SSD) to another
B. accounts for scatter and attenuation in tissue
C. is independent of the inverse square law
D. accounts for dose in various tissues

**Answer Question 123: A**

The Mayneord Factor converts PDD from one SSD to another by the following formula:




$$F = \left(\frac{SSD_2 + d_{max}}{SSD_1 + d_{max}}\right)^2 \left(\frac{SSD_1 + d}{SSD_2 + d}\right)^2$$

The Mayneord factor is based on the inverse square law.

It does **not** account for scatter nor attenuation.

The Mayneord factor should not be confused with the f-factor (Roentgen to rad conversion factor) which describes the differential deposition of dose in different tissues.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 124:**

Which ratio best describes the scatter contribution to the surface dose within the field, when a direct orthovoltage beam is used to treat a tip of nose basal cell skin cancer?

A. Tissue-air ratio TAR
B. Tissue-maximum ratio TMR
C. Tissue-phantom ratio TPR
D. Back-scatter factor BSF

**Answer Question 124: D**
Back scatter factor (BSF) accounts for the radiation scattered back toward the source by the surrounding medium.

BSF is a type of peak scatter factor (PSF) that is relevant for kilovoltage (orthovoltage) x-ray beams, where the depth of maximum dose is at the surface of the phantom. BSF represents the increase in dose at the surface due to scattered radiation originating from the phantom. BSF is the ratio of dose at the surface of the phantom (at dmax) to the dose in free space at the same point.

PSF accounts for the contribution of scattered radiation to the total dose at a particular depth (eg. dmax). PSF describes how much of the total dose is due to the primary beam and how much is due to scatted photons. PSF is generally used for megavoltage beams (eg. 6MV photons).

Tissue air ratio (TAR) is the ratio of absorbed dose at a specific depth in a tissue phantom to the absorbed dose at the same depth in air. TARs are used to correct for effects of radiation scattering and absorption in tissue.

Tissue maximum ratio (TMR) is the ratio of dose at a specific depth to the dose at the depth of maximum dose, while keeping the source axis distance (SAD) constant.

Tissue phantom ratio (TPR) is the ratio of absorbed dose at a given depth in a phantom to the absorbed dose at a reference depth within the same phantom.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Definitions, Interrelations, Dependence on Depth, SSD, Field size, Converting PDD to TMR**

**Question 125:**

What is the tissue maximum ratio (TMR) for a 6MV beam (10 cm x10 cm field size, SSD =100 cm) at depth 10 cm? The percentage depth dose (PDD) is 67%, with reference depth at 1.5 cm. The phantom scatter factor is 1.

A. 0.78
B. 0.81
C. 0.57
D. 0.52

**Answer Question 125: A**

The TMR equation related to PDD is:

$$\mathrm{TMR} = \mathrm{PDD}\left(\frac{SSD+d}{SSD+d_{ref}}\right)^2\left(\frac{S_p(r_{d_{ref}})}{S_p(r_d)}\right)$$

$$PDD = TMR\left(\frac{SSD+d_{max}}{SSD+d}\right)^2\left(\frac{S_p(r_d)}{S_p(r_{dmax})}\right)$$

TMR = tissue maximum ratio
PDD= percentage depth dose
SSD = source surface distance (100 cm)
d = depth of interest (10 cm)
$d_{ref}$ = reference depth (in this case it is dmax = 1.5 cm for 6 MV)
Sp ($rd_{ref}$) = phantom scatter with field size r, at reference depth $d_{ref}$
Sp ($r_d$) = phantom scatter with field size r, at depth d

TMR = $0.67 \times (110/101.5)^2 \times 1$
=0.67 X 1.17
=0.78

References:
Bedford JL, Hansen VN, Webb S. The derivation of tissue-maximum ratio from percentage depth dose requires peak scatter factor to be considered a function of source-to-surface distance. Br J Radiol. 1998 Aug;71(848):876-81.

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

https://oncologymedicalphysics.com/dose-calculation-hand-methods/

**Off-axis factors**

**Question 126:**

The following applies to off axis ratios (OARs) **EXCEPT:**

A. An off axis ratio is the ratio of dose at a given depth and distance from the central axis compared to the dose at the same depth along the central axis of the radiation beam

B. Off axis ratios should be radially symmetric about the central axis in an open field

C. The depth dependence from off axis ratios are due to beam hardening from the flattening filter

D. Using a wedge will not result in a change to the off axis ratios

**Answer Question 126: D.**

All are correct except D. Wedges change the shape of the beam profile at depth and radically alter the off-axis ratios.

Reference:

Prasad SC, Bassano DA, Ames TE. Calculation of dose in off-central axis planes using off-axis ratios. Med Phys. 1984 Mar-Apr;11(2):204-6.

**Out of field dose**

**Question 127:**

Which of the following are factors, which determine out of field radiation dose?

Choose A, B, C or D below.

1. Lateral scatter in the patient

2. Collimator or linear accelerator treatment head scatter

3. Radiation leakage from the head of the linear accelerator

4. Neutron contamination for photon beams with energy 10 MV or higher

A. 1, 2, 3
B. 2 and 3 only
C. 2, 3 and 4
D. 1, 2, 3, 4

**Answer Question 127: D**

All of these are factors which determine out of field radiation dose.

Reference:

Benzazon N, Colnot J, de Kermenguy F et al. Analytical models for external photon beam radiotherapy out of field dose calculation: a scoping review. Front Oncol 13: 2023. doi: 10.3389/fonc.2023.1197079

## 8.2 System of Monitor Unit Calculations

**Estimating equivalent square:**

**Question 128:**

What is the equivalent square field for a 15 cm x 10 cm field?

A. 12 cm x 12cm

B. 10 cm x 10 cm

C. 15 cm x 15 cm

D. 11cm x 11 cm

**Answer Question 128: A**

$$Equivalent\ Square\ size = 4\frac{Area}{Perimeter}$$

$Equivalent\ Square\ size = 4\frac{A \times B}{2(A+B)}$ Where A and B are the lengths of the two field sizes

$$Equivalent\ Square\ size = 4\frac{15 \times 10}{2(15+10)}$$

$$Equivalent\ Square\ size = 4\frac{150}{50} = 4 \times 3 = 12$$

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**MU calculation (photons)**

**Question 129:**

What is the calculated monitor units for the following photon treatment parameters?
3 Gy prescribed at the 100% isodose line along the central axis; using 6MV direct photon beam
10 cm x 10 cm field with $d_{max}$ of 1.5 cm
Extended SSD =120 cm

Calibration output factor is 1 cGy/MU at $d_{max}$ in a 10 cm x 10 cm field, SSD at 100 cm)
Collimator scatter factor is 1
Phantom scatter factor is 1
No wedges
No beam modifying devices

A. 208 monitor units
B. 430 monitor units
C. 300 monitor units
D. 441 monitor units

**Answer Question 129: B**

Photon Field MU Calculations

$$\mathrm{MU} = \frac{\mathrm{R_x}}{\mathrm{K}\times(\mathrm{factor})\times S_c(r_c)\times S_p(r)\times ISC\times TT\times WF\times TF\times OAF}$$

$$MU = \frac{300\ cGy}{1\frac{cGy}{e}\times 1\times 1\times 1\times (\frac{100+1.5}{120+1.5})^2\times 1\times 1\times 1\times 1}$$

$$MU = \frac{300\ cGy}{0.6979}$$

$$MU = 430$$

MU = monitor units
$R_x$ = prescription dose at depth of interest in cGy
K = calibration output factor in cGy/MU (usually 1 cGy/MU at $d_{max}$ in a 10 cm x 10 cm field, SSD at 100 cm)
Factor = Percentage Depth Dose (PDD) usually used in non-isocentric techniques
or
Tissue Maximum Ratio (TMR) usually used in isocentric techniques
$S_c$ = collimator scatter factor
$S_p$ = phantom scatter factor
ISC = inverse square law correction

For PDD calculation:

$$ISC_{PDD} = (\frac{Source-to-Calibration-Depth}{SSD+d})^2$$

Source to calibration depth = SSD + $d_{max}$ = 100 cm + 1.5 cm
SSD + d = 120 cm extended SSD +1.5 cm (depth of 100% isodose line for 6MV photon)

For TMR calculation:

$$ISC_{TMR} = (\frac{Source-to-Calibration-Depth}{SAD})^2$$

TT =treat to level (=1 if isodose line not specified); if the prescription dose is at the 90% isodose line, TT =0.9

WF = wedge factor (=1 if no wedge used)
TF =transmission factor (=1 if no trays or other beam modifying devices are placed in the beam's path)
OAR = off axis factor (=1 if calculation made alone the central axis)

Reference:
https://oncologymedicalphysics.com/dose-calculation-hand-methods/

**Parallel opposed field plans (calculation maximum dose and exit dose)**

**Question 130:**

For a 6 MV parallel opposed beam treatment arrangement (100 cm SSD), what is the calculated surface dose, maximum dose and midplane dose along the central axis? Each field size is 10 cm x 10 cm. There are no beam modifying devices. 8 Gy is prescribed to d max for each field. The thickness of tissue is 20 cm.

Percentage depth dose (PDD) at surface: 14%

PDD at 1.5 cm depth: 100%

PDD at 10 cm depth: 67%

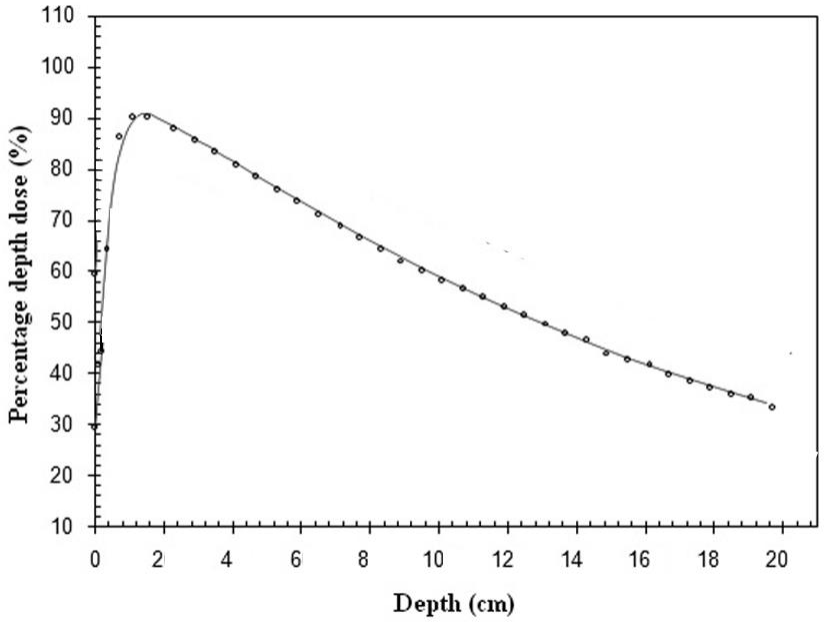
PDD at 18.5 cm depth: 35%

PDD at 20 cm depth: 34%

| | | | |
|---|---|---|---|
| A. | Surface dose: 3.8 Gy | Maximum dose: 10.9 Gy | Midplane dose: 10.7 Gy |
| B. | Surface dose: 1.1 Gy | Maximum dose: 8.0 Gy | Midplane dose: 5.4 Gy |
| C. | Surface dose: 3.8 Gy | Maximum dose: 10.9 Gy | Midplane dose: 5.4 Gy |
| D. | Surface dose: 1.1 Gy | Maximum dose: 10.9 Gy | Midplane dose: 10.7 Gy |

**Answer Question 130: A**

Percent depth dose curve for a single 6MV photon beam (10 cm x 10 cm field) along the central axis is shown below:




**Surface dose:**

14% of incident beam (8 Gy at dmax) = 1.1 Gy

34 % or exit beam (8 Gy at dmax) = 2.7 Gy

Total: 1.1 Gy + 2.7 Gy = **3.8 Gy**

**Maximum dose:**

Occurs at dmax at depth = 1.5 cm

100% of incident beam = 8 Gy

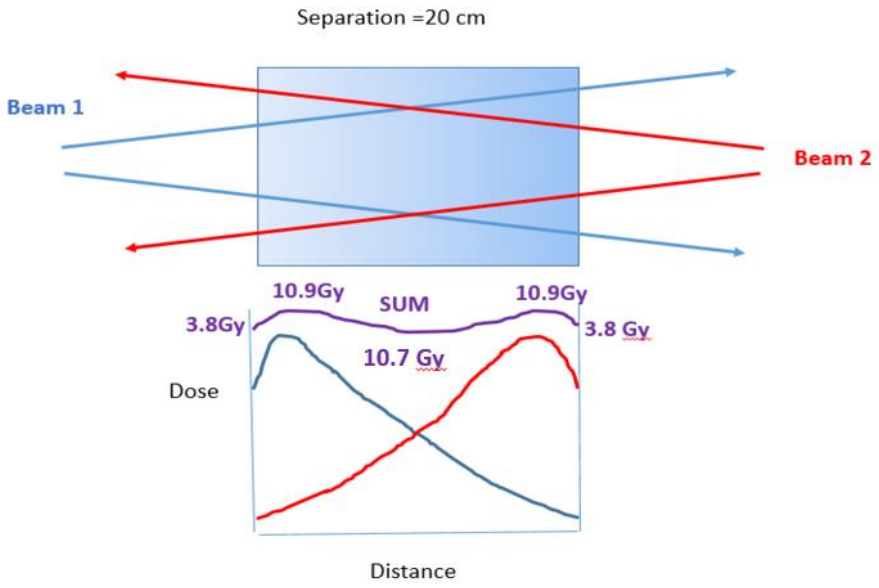
Depth = 20-1.5 cm for exit beam = 18.5 cm depth with corresponding PDD =36% : 8 Gy x 0.36 = 2.9 Gy

Total: 8Gy + 2.9 Gy = **10.9 Gy**

**Midplane dose:**

At depth = 20cm/2 = 10cm

PPD =67%: 8 Gy x 0.67= 5.36 Gy x 2 beams = **10.7 Gy**




Reference:

Chang, D.S., Lasley, F.D., Das, I.J., Mendonca, M.S., Dynlacht, J.R. (2014). Dosimetry of Photon Beams in a Patient. In: Basic Radiotherapy Physics and Biology. Springer, Cham.

**8.3 Treatment planning with photons**

**Algorithms for dose computation**

**Question 131:**

Which of the following are algorithms for dose computation? Choose A, B, C or D below.

1. Monte Carlo
2. Boltzmann transport equation
3. Convolution/Superposition
4. Pencil Beam
5. Anisotrophic analytical algorithm

A. 1 and 2
B. 3, 4 and 5
C. 1, 3, 4, 5
D. 1, 2, 3, 4, 5

**Answer Question 131: D**

All are examples of algorithms for dose computation.

References:

Bosse C, Narayanasamy G, Saenz D, Myers P, Kirby N, Rasmussen K, Mavroidis P, Papanikolaou N, Stathakis S. Dose Calculation Comparisons between Three Modern Treatment Planning Systems. J Med Phys. 2020 Jul-Sep;45(3):143-147.

https://oncologymedicalphysics.com/dose-calculation-algorithms/

**Corrections and limitation for inhomogeneities**

**Radiological depth**

**Question 132:**

What is the radiological depth of 3 cm thick lung with density 0.26 g/cm$^3$?

A. 0.78 g/cm$^2$
B. 3 g/cm$^2$
C. 11. 5 g/cm$^2$
D. 3.26 g/cm$^2$

**Answer Question 132: A**

Radiological depth can be calculated my multiplying physical depth and the density of the medium;

3 cm x 0.26 g/ cm$^3$= 0.78 g/cm$^2$

Reference:

Siddon RL. Calculation of the radiological depth. Med Phys. 1985 Jan-Feb;12(1):84-7. doi: 10.1118/1.595739. PMID: 3974530.

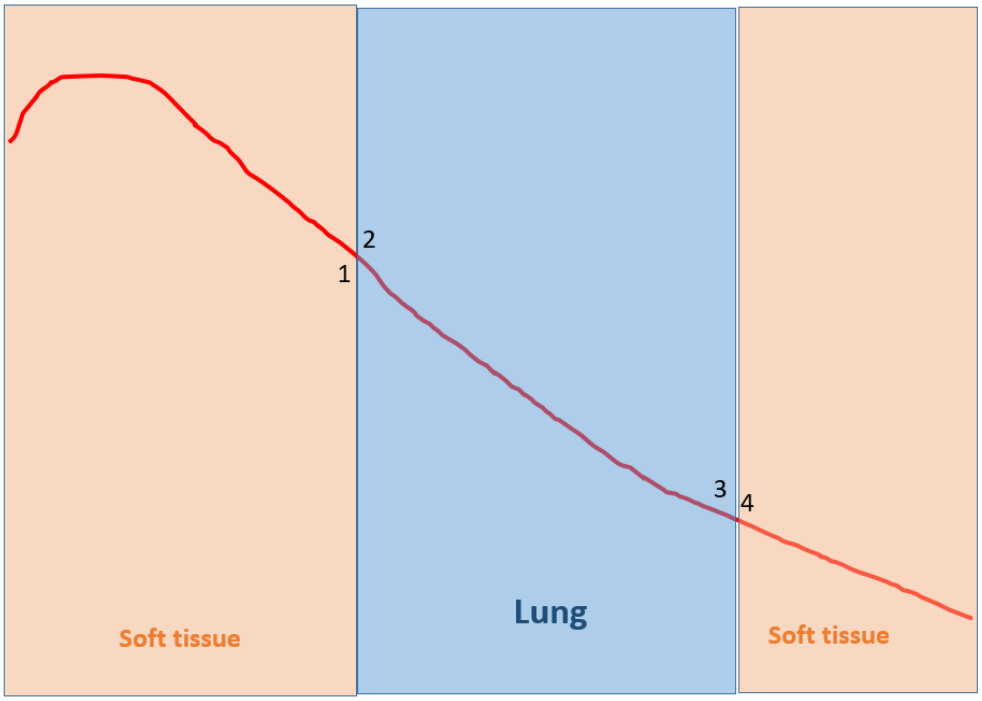
**Dose perturbations at interfaces**

**Question 133:**

In the following diagram, what happens to dose at points 1, 2, 3, and 4 for a 6 MV photon beam traversing the interface between soft tissue, lung and then soft tissue, as compared to a 6 MV photon beam in homogeneous soft tissue? Consider the amount of lung present to be low enough such that the PDD at the second interface is not significantly altered.

Choose the correct answer A, B, C or D below.

The following diagram shows the geometry of the tissue inhomogeneity within the path of a 6 MV photon beam along the central axis. The red line is the percentage depth dose of the 6 MV photon in soft tissue.




The options for the correct answer below are presented in order of the points: 1, 2, 3, 4.

A. Decrease, increase, increase, decrease
B. Increase, decrease, decrease, increase
C. No change, decrease, no change, increase
D. Increase, decrease, increase decrease

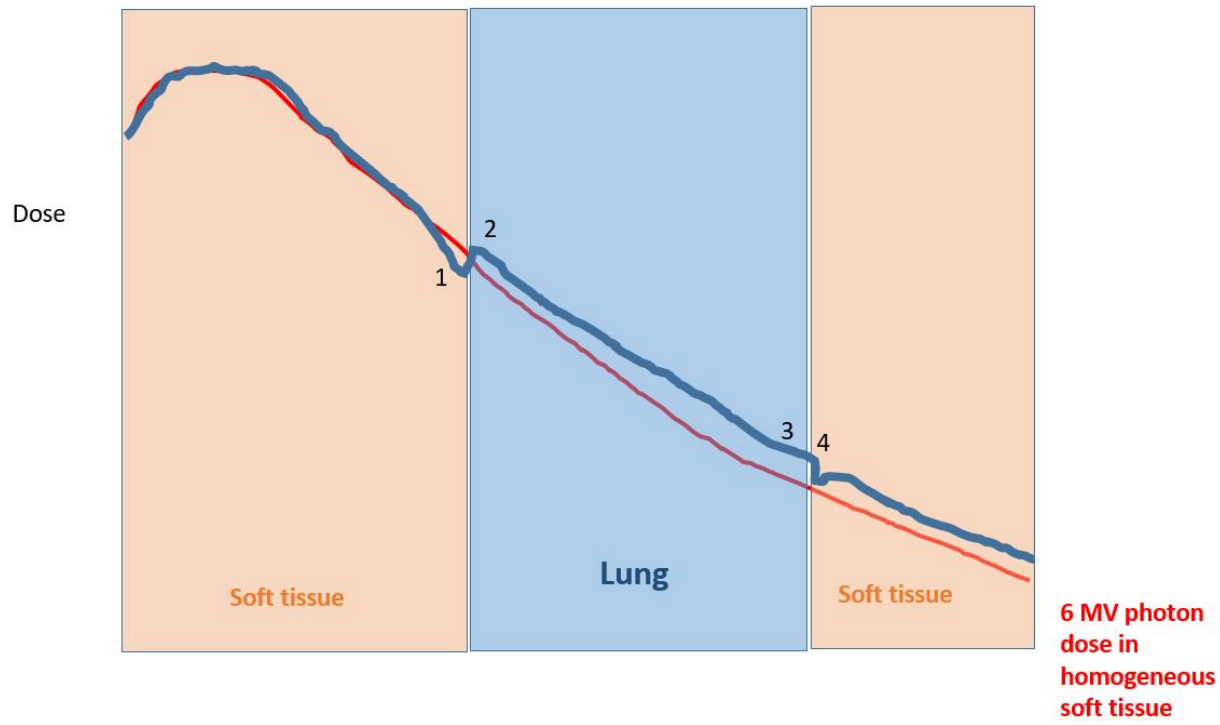
**Answer Question 133: A**

Point 1: Relative decrease due to reduced back scatter from lung compared to soft tissue.

Point 2: Relative increase. Dose in lung is higher than in soft tissue due to comparatively increased buildup from soft tissue. Also, dose in lung is higher than in soft tissue due to decreased primary beam attenuation. The percentage depth dose curve doesn't fall off as quickly, so the lung will have higher dose for a given depth compared to soft tissue.

Point 3: Relative increase due to increased backscatter from soft tissue into lung interface.

Point 4: If the lung tissue is not extensive enough to significantly change the PDD at depth. Relative decrease. Dose beyond lung has a build-up region and so the dose with dip after entering the soft tissue. If the lung tissue IS extensive enough, even though the dose will dip after the interface, it might still be higher as compared to a 6 MV beam in a purely homogeneous soft tissue medium.




Reference:

https://oncologymedicalphysics.com/photon-dose-distributions/

# Chapter 9: Electron beam characteristics and dosimetry

Ria Mulherkar Kraft, Mark Anthony Francis D'Souza, Humza Nusrat, May Tsao

**9.1 Basic characteristics**

**Electron range types**

**Question 134.1:**

What is the continuous slowing down approximation (CSDA) range?

A. The total distance a particle travels until it loses all its energy
B. The average distance a charged particle travels while slowing down to rest (assuming constant energy loss and ignoring fluctuations)
C. The distance where the dose falls to a specific level

**Answer Question 134.1: B**

**Question 134.2:**

What is the practical range ($R_p$) of an electron beam?

A. The total distance a particle travels until it loses all its energy
B. The average distance a charged particle travels while slowing down to rest (assuming constant energy loss and ignoring fluctuations)
C. The distance where the dose falls to a specific level

**Answer Question 134.2: C**

**Question 134.3:**

What is the maximum range (Rmax) of an electron beam?

A. The total distance a particle travels until it loses all its energy
B. The average distance a charged particle travels while slowing down to rest (assuming constant energy loss and ignoring fluctuations)
C. The distance where the dose falls to a specific level

**Answer Question 134.3: A**

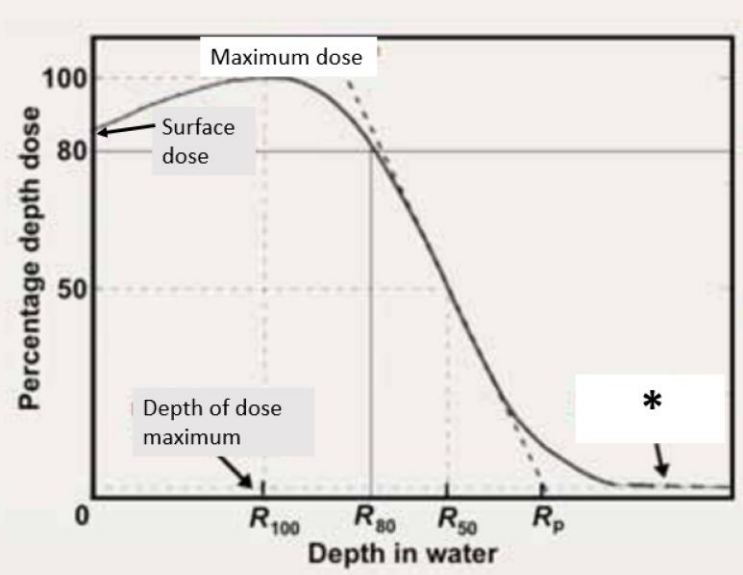
References:
Strydom W, Parker W, Olivares M. Chapter 8 Electron Beams: Physical and Clinical Aspects in: Review of Radiation Oncology Physics: A Handbook for Teachers and Students. Pages 1-24,

https://www.irsn.fr/sites/default/files/documents/professionnels_sante/documentation/syllabus_chapitre_8.pdf

**Question 135:**

What interaction is responsible for the dose deposition at * in the electron depth dose curve?

Electron percent depth dose curve below:




A. Compton
B. Pair production
C. Bremsstrahlung
D. Photoelectric effect

**Answer Question 135: C**

The position of the * and its associated arrow (in the electron percent depth dose curve) points to the Bremsstrahlung electron tail.

References:

Strydom W, Parker W, Olivares M. Chapter 8 Electron Beams: Physical and Clinical Aspects in: Review of Radiation Oncology Physics: A Handbook for Teachers and Students. Pages 1-24,

https://www.irsn.fr/sites/default/files/documents/professionnels_sante/documentation/syllabus_chapitre_8.pdf

**9.2 Dose Rate and Monitor unit calculations:**

**Electron monitor unit calculations**

**Question 136:**

Consider a 9 MeV electron beam with a 1.0 cGy/MU dose rate at the normalization point [source-surface distance (SSD) 100 cm, 10 cm x 10 cm field].

Calculate the monitor units (MU) required to deliver 200 cGy to $d_{max}$ using a 10 cm x 10 cm field with 100 cm SSD. No bolus is used.

A. 220
B. 182
C. 234
D. 200

**Answer Question 136: D**

$$MU = \left( \frac{\left( D_{Rx} / \left( IDL\,\% / 100 \right) \right)}{\left( O_{10,100} \times OF_{FS} \times OF_{SSD} \right)} \right)$$

$$MU = \left( {}^{200}/_{(1.0) \times (1.0) \times \left( 1.0 \right)} \right) = 200$$

MU = monitor units

$D_{Rx}$ = Prescribed dose

$IDL_{\%}$ = Prescription isodose line (in percent)

$O_{10,100}$ = Output factor for 10 cm x 10 cm field at 100 SSD

$OF_{FS}$ = output factor for field size

SSD = source surface distance

$OF_{SSD}$ = output factor for SSD

$OF_{FS}$ =1 when the field size is 10 cm x 10 cm;
$OF_{SSD}$ =1 when SSD = 100 cm

Otherwise:

$$OF_{LxW} = \sqrt{OF_{LxL} \times OF_{WxW}}$$

L = length of field
W = width of field

The output factor is also corrected for distance when SSD is not 100 cm using either the "effective SSD" method or the "air-gap factor" method. These two methods may not give the exact same answer.

The "effective SSD" technique uses the inverse-square law.

The "air –gap factor" uses the inverse-square law but also takes into account, attenuation and scattering in air, in beam collimation and shaping devices, with varying SSD..

The output factor is also corrected when bolus is used.

References:

https://jpneylon.github.io/ABR/PDFs/Electrons_Fundamentals.pdf

Gibbons JP, Antolak JA, Followill DS, Huq MS, Klein EE, Lam KL, Palta JR, Roback DM, Reid M, Khan FM. Monitor unit calculations for external photon and electron beams: Report of the AAPM Therapy Physics Committee Task Group No. 71. Med Phys. 2014 Mar;41(3):031501. doi: 10.1118/1.4864244. PMID: 24593704; PMCID: PMC5148083.

**9.3 Treatment planning with electrons**

**Question 137:**

A 72-year-old woman presents with a basal cell carcinoma of the right cheek. The lesion measures approximately 2 cm wide. If this lesion is treated with 6 MeV electrons and a 0.5 cm bolus, what will be the approximate depth of the 90% isodose line within the patient's cheek?

A. 1 cm
B. 3 cm
C. 4 cm
D. 6 cm

**Answer Question 137: A**

A general rule of thumb for calculating the depth of the 90% isodose line is energy in MeV divided by 4. A useful mnemonic to remember electron rules of thumb is "5-4-3-2" rule.

- E/5 = Dmax
- E/4 = D90%
- E/3 = D80%
- E/2 = practical range

In this case, the 90% isodose line would be expected to be at 6 divided by 4 or 1.5 cm. However, with a 0.5 cm bolus in place, the 90% isodose line would be at approximately 1 cm from the skin surface to within the patient's cheek.

Reference:

Mott JHL, West NS. Essentials of depth dose calculations for clinical oncologists. *Clin Oncol (R Coll Radiol)*. 2021;33(1):5-11.

**Question 138:**

In the question above, assuming a 6 MeV electron beam is used to treat a cancer of the right cheek, and the thickness of the patient's cheek is 1.5 cm, and a 0.5 cm bolus is in place, what should be the thickness of a lead shield inside of the patient's cheek?

A. No shielding is required
B. 1 mm lead shield
C. 2 mm lead shield
D. 3 mm lead shield

**Answer Question 138: C**

Another useful rule of thumb is that mean energy of electrons decreases by approximately 2 MeV per 1 cm that it travels in tissue, and per 1 mm that it travels in lead. In this case, the 6 MeV beam would travel through 2 cm of tissue-equivalent material (0.5 cm bolus + 1.5 cm tissue). The energy of the beam would therefore be reduced by 4 MeV (2 MeV * 2 MeV/cm) down to 2 MeV. By the time it hits the lead, only 1 mm of lead would be required to shield the 2 MeV beam, since lead blocks electrons at approximately 2 MeV/mm of lead. Generally, an additional 1 mm of lead is added an extra precaution. Therefore, the thickness of lead required to shield an electron energy is calculated by the following formula: Thickness = MeV/2 (in mm) + 1 mm.

Reference:

Mott JHL, West NS. Essentials of depth dose calculations for clinical oncologists. *Clin Oncol (R Coll Radiol)*. 2021;33(1):5-11.

**Question 139:**

Which of the following electron energies would be expected to have the highest surface dose at a patient's skin, assuming these are used without any bolus?

A. 6 MeV
B. 9 MeV
C. 12 MeV
D. 15 MeV

**Answer Question 139: D**

For electron beams, the surface entrance dose increases with increasing energy. This is in contrast to photon beams, whose surface dose is higher with decreasing beam energy.

Reference:

Mott JHL, West NS. Essentials of depth dose calculations for clinical oncologists. *Clin Oncol (R Coll Radiol)*. 2021;33(1):5-11.

**Question 140:**

What happens when an electron beam is applied at an angle to the surface?

A. The depth of maximum dose ($d_{max}$) moves closer towards the surface
B. The relative dose at $d_{max}$ decreases
C. The depth of the 90% dose level (R90) or the therapeutic electron range is reduced
D. A and C
E. All of the above

**Answer Question 140: D**

A is correct because, with increased obliquity, shallower points receive more side scatter from electrons that have traveled a longer distance, moving Dmax closer to the surface and **increasing** its value. Additionally, at greater depths, side scatter contributes less to the dose, while the inverse square law and the increased air gap from the angled beam further reduce the dose. These combined effects decrease the effective treatment depth (ie. R90), making Option C correct as well.

In summary, increased angle of obliquity for a single direct electron beam results in:

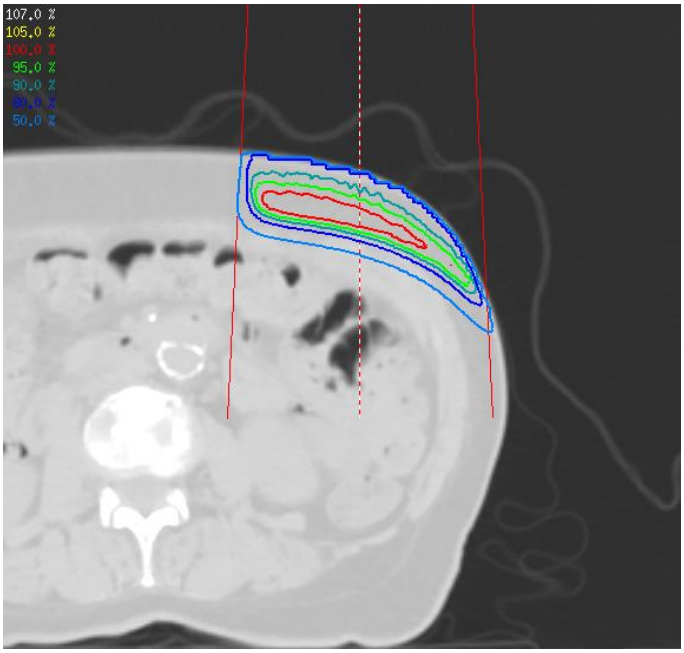
-shift of the depth of maximum dose and R80 toward the surface

-the relative dose at $d_{max}$ increases

-increase in surface dose

-increase in the practical range of electrons due to the contribution of high scatter angle electrons which do not pass through as much tissue

The diagram below shows the dosimetry of an incident electron beam on oblique surface.




References:

Gerbi BJ. Radiation therapy using high energy electron beams. https://oncohemakey.com/radiation-therapy-using-high-energy-electron-beams/

Strydom W, Parker W, Olivares M. Review of Radiation Oncology Physics: A Handbook for Teachers and Students. Chapter 8. Electron Beams: Physical and Clinical Aspects. Page 234.

Khan FM, Deibel FC, Soleimani-Meigooni A. Obliquity correction for electron beams. Med Phys. 1985 Nov-Dec;12(6):749-53. doi: 10.1118/1.595659. PMID: 4079867.

**Question 141:**

A basal cell skin cancer is prescribed a 5500 cGy/20 daily fractions to the 95 % isodose line using a direct 12 MeV electron beam with 1 cm of bolus. On day 1 of treatment, the therapist forgets to place the bolus. On the next treatment, the radiation therapist realizes that bolus was not used on day 1. The therapist makes sure that the bolus is placed for the remainder of the treatment. Considering the skin cancer, the absence of bolus on day 1 will result in:

A. Significant underdose
B. Significant overdose
C. Insignificant underdose
D. Insignificant overdose

**Answer Question 141: C**

The surface dose for an electron beam is approximately 70% + beam energy (E), so for a 12 MeV beam, this results in a surface dose around 82%. Without the bolus, the tumor will receive less dose at the surface, leading to an underdose. The nominal dose per fraction is 275 cGy, so the dose on the first day without bolus would be approximately 275 cGy×0.82=225.5. Over 20 fractions, this results in 225.5 cGy+(275 cGy×19)=5451 cGy, leading to an underdose of about 1%. This slight underdose makes **C** (Insignificant underdose) the correct answer.

Reference:

Mott JHL, West NS. Essentials of depth dose calcualtions for clinical oncologists. Editorial. Clinical Oncology 33: 5-11; 2021.

**Question 142:**

In certain direct electron treatments, lead is used to protect nearby normal tissues. However, there can be an increase of dose at the tissue-shield interface due to backscatter. Which of the following methods can be used to minimize this effect.

A. Increase the thickness of the lead shield
B. Use a higher atomic number material
C. Add a layer of low atomic number material at the interface
D. Increase the electron beam energy

**Answer Question 142: C**

Placing a layer of low atomic number material between the lead shield and the tissue interface absorbs the backscattered electrons, thereby reducing the dose to the tissue.

Increasing the thickness or atomic number of the lead shield does not reduce backscatter; in fact, it may increase it. Increasing the electron beam energy would allow for deeper penetration but would still produce backscatter.

Reference:

Singh S, Semwal MK, Bhatt CP. Estimation of Backscatter from Internal Shielding in Electron Beam Radiotherapy Using Monte Carlo Simulations (EGSnrc) and Gafchromic Film Measurements. J Med Phys. 2019 Oct-Dec;44(4):239-245.

# Chapter 10: Intensity modulated radiation therapy

Erika Kollitz, May Tsao

**10.1 Concepts of IMRT**

**Question 143:**

Why are photon energies 6- 10MV typically used for IMRT?

A. IMRT photon energies greater than 10MV provide worse target coverage for targets which are deep within the patient.

B. More monitor units are needed for IMRT plans using photon energies more than 10 MV compared to monitor units using lower photon energy (< 10MV) for the same IMRT setup.

C. It is not possible to calculate dose modulation for IMRT photon energy > 10 MV as the lateral range of electrons with higher energy widens unpredictably with small IMRT fields.

D. Clinical data indicates that the use of 10MV or lower IMRT energy is associated with less risk of radiation induced malignancy as compared to IMRT with photon energy > 10MV.

E. Higher energy photons (higher than 10MV) results in greater leakage of radiation, treatment head scatter, patient scatter and secondary neutron production.

**Answer Question 143: E**

Higher energy photons (higher than 10MV) results in greater leakage of radiation, treatment head scatter, patient scatter and secondary neutron production. The production of neutrons also has radiation safety implications for therapists who walk in and out of the radiation treatment unit following IMRT treatments. High energy IMRT exposes the radiation treatment machine and hardware to many high energy photons, which can photo-activate elements in the hardware creating primary and daughter products which are radioactive and thereby potentially increase radiation exposure to staff and patients.

IMRT energies more than 10MV tend to provide *better* target coverage for deep targets due to the penetrating percentage depth dose characteristics of high energy photon beams.

Fewer monitor units are needed for high energy photon IMRT compared to low energy (eg. 6 MV or 10 MV) IMRT.

Dose calculation for high energy IMRT plan are possible even though the lateral range of scattered electrons widens the lateral fall-off of IMRT modulated fields.

There is no robust clinical data indicating that the use of higher IMRT energy (greater than 10MV) is associated with a higher rate of radiation induced malignancy. The use of higher energy IMRT resulting in increased treatment head leakage/scatter, internal patient scatter and the production of neutrons theoretically increases the risk of radiation malignancy. However, this risk is theoretical and has not been reported clinically.

References

Madani I, Vanderstraeten B, Bral S et al. Comparison of 6MV and 18 MV photons for IMRT treatment of lung cancer. Radiother Oncol 82(1): 63-69, 2007.

Hussein M, Aldridge S, Guerrero Urbano T, Nisbet A. The effect of 6 and 15 MV on intensity-modulated radiation therapy prostate cancer treatment: plan evaluation, tumour control probability and normal tissue complication probability analysis, and the theoretical risk of secondary induced malignancies. Br J Radiol. 2012 Apr;85(1012):423-32. doi: 10.1259/bjr/24514638. Epub 2011 Oct 18. PMID: 22010028; PMCID: PMC3486667.

Nooshin Banaee, Kiarash Goodarzi, Hassan Ali Nedaie, Neutron contamination in radiotherapy processes: a review study, *Journal of Radiation Research*, Volume 62, Issue 6, November 2021, Pages 947–954

**Question 144:**

Order the following radiation delivery types with respect to the volume of the patient exposed to low dose radiation (V5 Gy) from lowest to highest, given to the same planning target volume dose when treating the prostate. Choose the correct answer **A, B, C or D** below.

1) 6 MV 3D CRT (4 field box)
2) 6 MV VMAT (1 arc)
3) 15 MV IMRT (5 fields with dynamic MLC)

**A. 1, 2, 3**
**B. 1, 3, 2**
**C. 3, 2, 1**
**D. 2, 3, 1**

**Answer Question 144: A**

Lowest whole body dose to highest whole body dose:

6 MV 3D CRT, 6 MV VMAT, 15 MV IMRT

The contributions to whole body dose are:

- Photon imaging dose (CT simulation and image guided radiotherapy using daily kV cone beam CT)
- Scatter therapy photons
- Neutron contamination

6 MV CRT has the lowest body dose as there are typically less monitor units used than IMRT. Collimator scatter, head leakage and neutron dose increases with increased monitor units. There are also less number of beams with CRT compared to IMRT. IMRT and VMAT radiotherapy plans result in larger volumes of low dose outside the treatment volume compared to 3D CRT.

15 MV IMRT has higher whole body dose because of increased neutron scatter with high energy 15 MV as compared to lower energy 6 MV.

Reference:

Hauri P, Schneider U. Whole-body dose equivalent including neutrons is similar for 6 MV and 15 MV IMRT, VMAT, and 3D conformal radiotherapy. J Appl Clin Med Phys. 2019 Mar;20(3):56-70..

**10.2 IMRT Delivery Systems**

**Question 145:**

What is the main purpose of multi leaf collimators in IMRT delivery systems?

A. to filter a polyenergetic beam to a monoenergetic beam

B. to shape and control the intensity of radiation across the field

C. to automate field sizes for 3D CRT

D. to reduce radiation leakage at the head of a linear accelerator

**Answer Question 145: B**

Reference:

Cheung K. Intensity modulated radiotherapy: advantages, limitations and future developments. Biomed Imaging Interv J. 2006 Jan;2(1):e19.

**10.3 Inverse Planning**

**Question 146:**

What is inverse planning?

A. A computerized method of generating beam parameters (number, direction, MLC segments, monitor units) to achieve the desired dose distribution

B. A computerized method to measure dose to tumour and normal tissues with a high degree of accuracy

C. A computerized method to generate beams with shielding for 3D CRT

D. A computerized method to calculate dose in various tissue densities

**Answer question 146: A**

Inverse planning is a computerized method of generating beam parameters to achieve the desired dose distribution.

Forward inverse planning starts with a planner determining the beam configurations (number, energy, angle). The computer then calculates the intensity profile of each beam to achieve the desired dose distribution.

Simulated annealing is an inverse planning computerized optimization technique used to find the best radiation plan. Radiation beam angles and beam weighting are adjusted iteratively to minimize accost function (representing a deviation from the desired dose distribution).

References:

Cheung K. Intensity modulated radiotherapy: advantages, limitations and future developments. Biomed Imaging Interv J. 2006 Jan;2(1):e19.

Morrill SM, Lane RG, Rosen II. Constrained simulated annealing for optimized radiation therapy treatment planning. Comput Methods Programs Biomed. 1990 Nov;33(3):135-44.

**10.4 IMRT Quality Assurance**

**Question 147:**

Which of the following is NOT a quality assurance verification metric for IMRT?

A. Dose difference
B. Distance to agreement
C. Gamma index
D. Off axis ratios

**Answer Question 147: D**

All the above are verification metrics for IMRT EXCEPT D.

Reference:

https://oncologymedicalphysics.com/imrt-quality-assurance/

# Chapter 11: Prescribing, reporting and evaluating radiotherapy plans -including dosimetry and planning

Alexandra DeLeo, May Tsao

### 11.1 Prescribing and Reporting in Radiotherapy

**Question 148:**

The International commission on radiation units and measurements (ICRU)'s principal objectives are:

**A.** To develop internationally accepted recommendations regarding quantities and units of radiation and radioactivity

**B.** To develop international guidelines on recommended dose fractionation schedules for radiation therapy

**C.** To develop international guidelines on contouring target volumes for radiotherapy.

**D.** To develop international recommendations regarding quality assurance of radiation plans.

**E.** To develop procedures suitable for the measurement and application of radiation quantities in diagnostic radiology, radiation oncology/therapy, radiation biology, nuclear medicine, radiation protection, not including industrial and environmental activities

**Answer Question 148: A**

The ICRU principal objectives are to develop internationally accepted recommendations regarding:

1. The quantities and units of radiation and radioactivity
2. Procedures suitable for the measurement and application of these quantities in diagnostic radiology, radiation therapy, radiation biology, nuclear medicine, radiation protection and industrial and environmental activities
3. Physical data needed in the application of these procedures, the use of which assures uniformity in reporting.

Reference:

https://www.icru.org/about-icru/objectives/

**Question 149:**

This patient presents with a biopsy proven basal cell skin cancer of his ear. According to ICRU 50, the blue outline represents:



A. GTV (Gross Tumour Volume)
B. CTV (Clinical Target Volume)
C. PTV (Planning Target Volume)
D. TV (Target volume)

**Answer Question 149: A**

According to ICRU 50 (Prescribing, recording and reporting photon beam therapy), the GTV (gross tumour volume is the grossly visible/palpable tumour. As such, the blue outline represents GTV.

Reference:
https://www.icru.org/report/prescribing-recording-and-reporting-photon-beam-therapy-report-50/

**Question 150:**

A 45 year old woman presents with favourable stage IA Nodular Sclerosis Hodgkin's lymphoma of her mediastinum. She is treated with ABVD (Adriamycin, Bleomycin, Vinblastine and Dacarbazine) chemotherapy for 2 cycles. Her postchemotherapy PET scan shows complete metabolic response (Deauville score 2). The patient is planned for involved site radiation to her mediastinum.

The blue outline in the computed tomography simulation image below represents:




A. GTV (Gross Tumour Volume)
B. CTV (Clinical Target Volume)
C. PTV (Planning Target Volume)
D. TV (Target Volume)

**Answer Question 150: B**

The blue contour represents CTV = clinical target volume or subclinical microscopic spread. The Hodgkin's lymphoma has responded completely to chemotherapy and as such there is no GTV (gross tumour volume) left.

Reference:
https://www.icru.org/report/prescribing-recording-and-reporting-photon-beam-therapy-report-50/

**Question 151:**

This patient with stage I adenocarcinoma of lung has undergone computed tomography (CT) simulation with 4D (4-dimensional) CT. The Clinical Target Volume (CTV) has been contoured on the 0, 50, and maximum projection phase. The summation of these CTV volumes represent:

A. PTV (Planning Target Volume)

B. TV (Target Volume)
C. ITV (Internal Target Volume)
D. SM (Set-up Margin)

**Answer Question 151: C**

The internal target volume is a margin used to account for physiological variations in site, size shapes (for example due to respiration, different bladder/rectal filling, swallowing, heart beat etc.)

Reference:

https://www.icru.org/report/prescribing-recording-and-reporting-photon-beam-therapy-report-62/

**Question 152:**

What is the contouring structure called that is added to the Internal Target Volume (ITV) to encompass geometric day-to-day set-up variation?

A. PTV (Planning Target Volume)
B. TV (Target Volume)
C. IM (Internal Margin)
D. SM (Setup Margin)

**Answer Question 152: A**

The PTV encompasses geometrical variation; margin on CTV to account for day-to-day set-up variation

**Internal margin (IM)**: A margin added to CTV to account for physiological movements (eg. respiratory, heart, digestive) and variation in size, shape and position of the CTV during radiotherapy in relation to the Internal Reference Point ad its corresponding Coordinate System.

**Internal target volume (ITV)**: A margin added to CTV to allow for variation in size, shapes of organs and tissues contained or adjacent to the CTV.

**Setup margin (SM):**

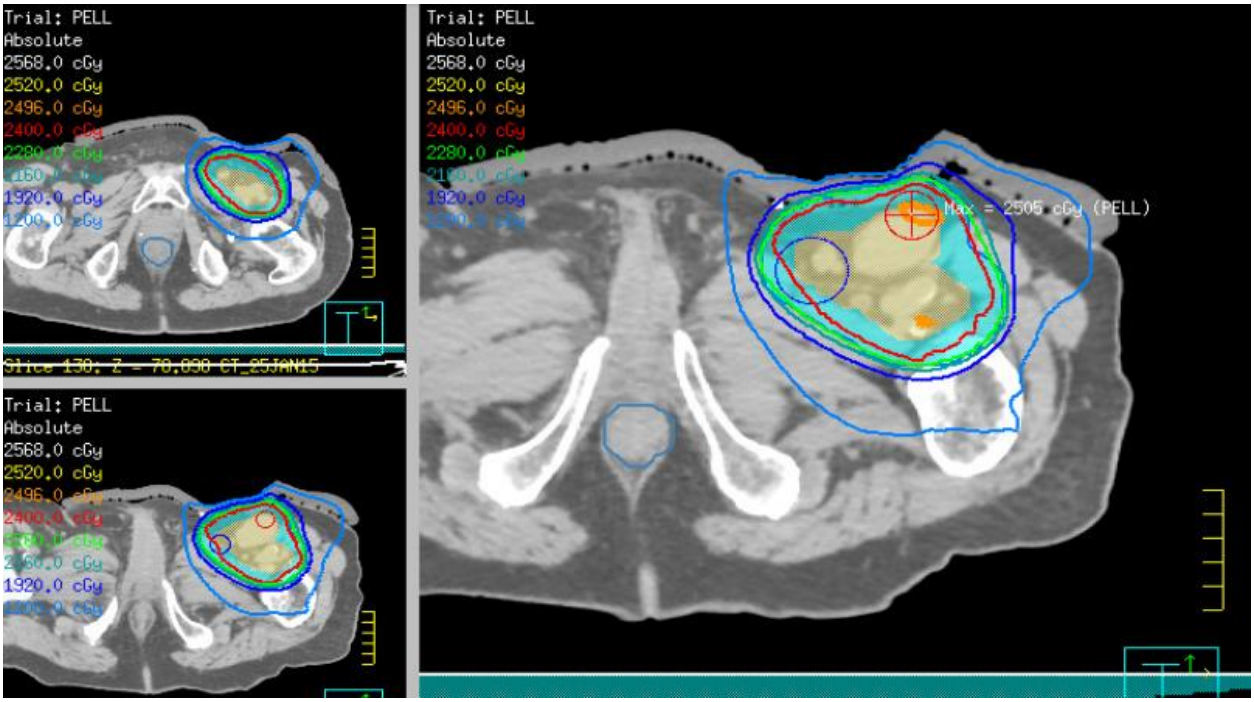
A margin that is added for variations in patient positioning, mechanical uncertainties of the equipment (gantry sagging, collimators, couch).

ITV = CTV + IM
PTV = ITV + SM

Reference:
https://www.icru.org/report/prescribing-recording-and-reporting-photon-beam-therapy-report-62/

**Question 153:**

This patient was planned for radiation therapy to his left groin adenopathy to treat his localized low grade follicular lymphoma. 24 Gy in 12 daily fractions was planned using volumetric modulated arc therapy (VMAT).




The green isodose line (22.8 Gy) represents the:

A. Treated Volume
B. Irradiated Volume
C. Minimum Dose
D. International Commission on Radiation Units and Measurements (ICRU) Reference

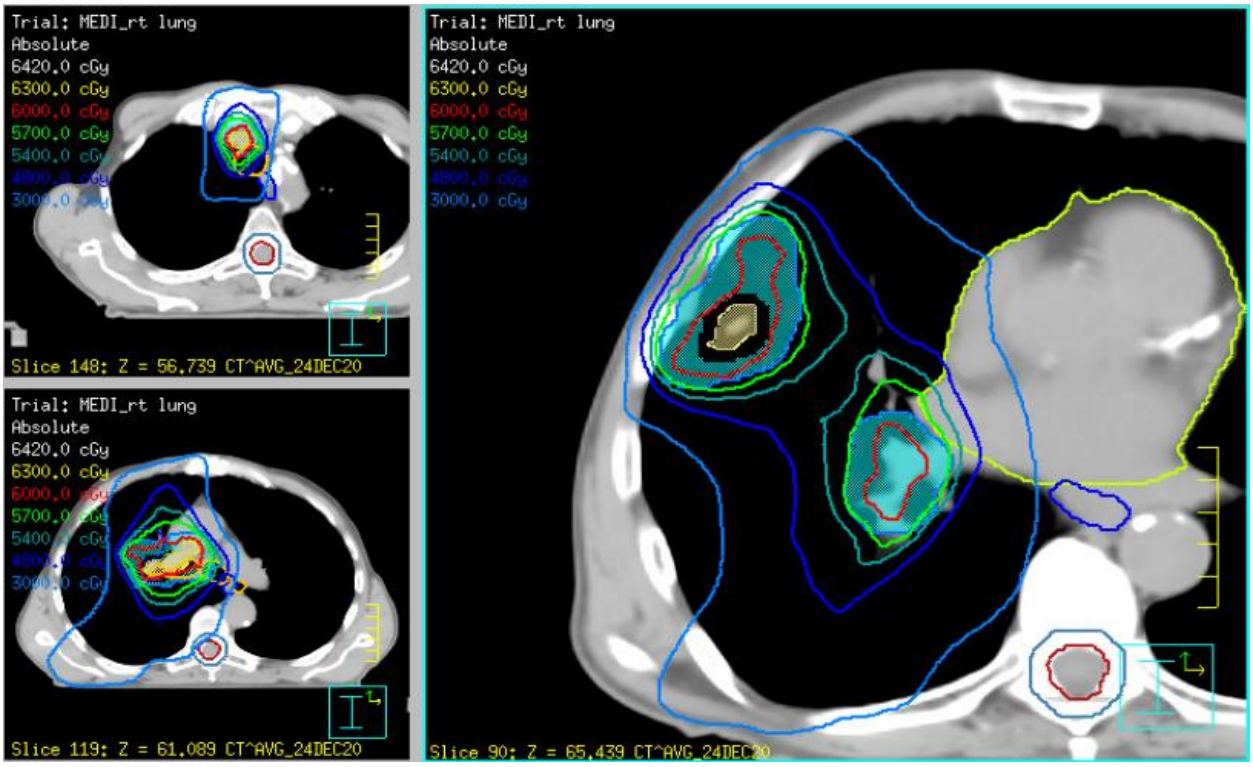
**Answer Question 153: A**
The Treated volume (TV) is the volume enclosed by an isodose line (eg 95%) specified as being appropriate to achieve the purpose of treatment (tumour eradication or palliation).

Reference:
https://www.icru.org/report/prescribing-recording-and-reporting-photon-beam-therapy-report-62/

**Question 154:**

This 70 year old man was planned for radical chemoradiation to treat a stage III non-small lung cancer. The patient was planned with volumetric modulated arc therapy (VMAT) with a prescribed dose of 60 Gy.

The blue isodose line (30 Gy) represents the:




A. Absorbed Dose
B. Irradiated Volume
C. Minimum Dose
D. International Commission on Radiation Units and Measurements (ICRU) Reference

**Answer Question 154: B**

The irradiated volume is the volume of tissue that receives a dose that is considered significant in relation to normal tissue tolerance

Reference:
https://www.icru.org/report/prescribing-recording-and-reporting-photon-beam-therapy-report-62/

**Question 155:**

Which of the following is the correct definition for D $_{near\text{-}max}$ according to the International Commission of Radiation Units and Measurement (ICRU) 83:

A. The maximum dose is the dose in the planning target volume when the minimum dimension is more than 15 mm

B. The maximum dose is the dose in the organ at risk when the minimum dimension is more than 15 mm

C. The dose greater than 100% of the prescribed dose occurring outside the planning target volume when the minimum dimension is more than 15 mm

D. The high dose received by 2% of the planning target volume or D2%

**Answer Question 155: D**

In ICRU 83, the near maximum dose is used instead of ICRU 50 dmax. The D near-max dose in ICRU is defined as the near maximum doses received by 2% of the PTV volume.

In ICRU 50, dmax is defined as the maximum dose in the PTV or OAR when the minimum dimension is more than 15 mm

In ICRU 50, the hot spot represents a dose outside the PTV which receives a dose larger than 100% of the prescribed dose. A hot spot is considered significant only if the minimum diameter exceeds 15 mm. Answer C would be considered the definition of a hot spot.

References:

https://www.icru.org/report/prescribing-recording-and-reporting-intensity-modulated-photon-beam-therapy-imrticru-report-83/

https://www.icru.org/report/prescribing-recording-and-reporting-photon-beam-therapy-report-50/

Das IJ, Andersen A, Chen Z et al. State of dose prescription and compliance to international standard (ICRU-83) in intensity modulated radiation therapy among academic institiutions. Prac Radiat Oncol 7(2), 3145-155, 2017.

**Question 156:**

Which of the following are true statements regarding the International Commission of Radiation Units and Measurement (ICRU) reference point? Choose **A, B, C or D** below.

1. The ICRU reference point is always at the center of the planning target volume (PTV) even if there is a steep dose gradient
2. The dose at a point which is clinically relevant
3. Easy to define in a clear and unambiguous way
4. Selected so that the dose can be accurately determined
5. Should always be reported with the maximum dose to PTV and minimum dose to PTV

**A. All the above are true**
**B. Only 2, 3 and 4 are true**
**C. Only 1 and 5 are true**
**D. Only 2-5 are true**

**Answer Question 156: D**

The ICRU reference point is:

- a dose at a point which is clinically relevant
- should be easy to define in a clear and unambiguous way
- selected so that the dose can be accurately determined
- be a point in a region where there is no steep dose gradient
- always at the centre (or in the central part of the PTV and when possible, at the intersection of the beam axes).

Reference:

https://www.icru.org/report/prescribing-recording-and-reporting-photon-beam-therapy-report-50/

**11.2 Plan Evaluation and Comparison**

**Question 157:**

What is the equation for the conformity index (CI)?

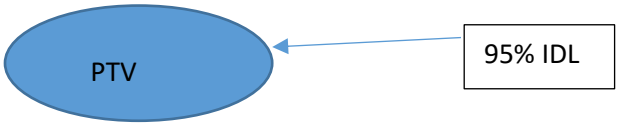
A. Treated volume (TV) divided by the planning target volume (PTV)
B. PTV divided by the TV
C. Irradiated volume (IV) divided by the PTV
D. PTV divided by the IV
E. The 95% isodose line divided by the 50 % isodose line

**Answer Question 157: A**

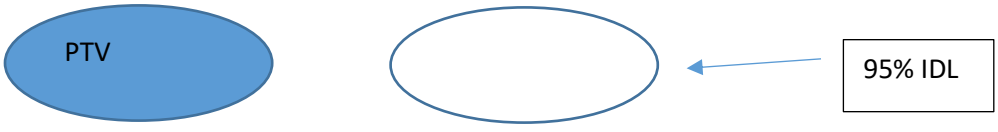
The Conformity Index (CI) =
Treated volume (TV)/ Planning Target Volume (PTV)

The conformity index describes quantitatively how well the treatment isodose (typically 95%) fits the shape of planning target volume, while minimizing dose to adjacent non-target tissues. A conformity index more than 1 means that the treated volume is larger than the planning target volume. A conformity index of less than 1 means that the treated volume is smaller than the planning target volume. The conformity index however, does not describe the degree in which the treated volume and planning target volume overlap.

Conformity index = 1
Treated volume (blue line) completely overlaps planning target volume (blue colorwash)




Conformity index =1
Treated volume is the same in cubic centimeters as the planning target volume but they completely do NOT overlap



Reference:

Feuvret L, Noel G, Mazeron J-J et al. Conformity index: A review. Int J Radait Oncolo Biol Phys 64(2): 333-342, 2006.

**Question 158:**

Consider the following statements regarding organs at risk (OAR). Choose the correct answer **A, B, C or D** below.

1. Liver and bladder are examples of parallel organs at risk
2. The spinal cord is an example of a serial organ at risk
3. The left anterior descending coronary artery is an example of a parallel organs at risk
4. The planning organ at risk volume is obtained by adding a margin to the organ at risk to account for movement, change in size, shape and set-up uncertainties
5. A class 1 radiation injury is mild, transient, reversible and results in no significant morbidity

**A. 1,2, 3 are true; 4 and 5 are false**
**B. 1,2, 4 are true; 3 and 5 are false**
**C. 1,2, 3,4 are true; 5 is false**
**D. All the above are true**

**Answer Question 158: B**

Organs at risk are normal tissues, which influence treatment planning or prescribed dose due the normal tissue's radiation sensitivity.

Class 1: radiation injury that is fatal or result in severe morbidity

Class 2: radiation injury that results in mild or moderate morbidity

Class 3: radiation injury that is mild, transient, reversible that results inn no significant morbidity

Serial: damage at one point will cause damage to the whole organ (eg. spinal cord)

Parallel: if one small part of the organ is damaged, the rest of the organ will continue to function normally (eg. liver, bladder)

Serial-Parallel: kidney (parallel-glomeruli; serial-tubule), heart (parallel-myocardium, serial-coronary arteries)

Planning organ at risk volume:

Margin added to OAR to account for movement, change in size, shape, set-up uncertainties

Reference:

ICRU Report 62. Bethesda, MD: ICRU; 1999. International Commission on Radiation Units and Measurements. Prescribing, recording, and reporting photon beam therapy, Supplement to ICRU Report No. 50

**Question 159:**

The V20 of the lungs is 30%, which means that:

A. 30% of the lungs receives exactly 20 Gy.
B. 30% of the lungs receives at most 20 Gy.
C. 30% of the lungs receives at least 20 Gy.
D. 70% of the lungs receives exactly 20 Gy.

**Answer Question 159: C**

A cumulative dose volume histogram (DVH) describes the amount (usually as a percentage) of a structure receiving doses greater than or equal to a specific dose. In contrast, a differential DVH demonstrates the volume of a structure exactly receiving a particular bin dose. Therefore, on a cumulative DVH plot for a structure, the (x,y) coordinates of (20 Gy, 30%) indicates that 30% of the structure receives 20 Gy or greater, while on a differential DVH, it means that 30% of the structure receives 20 Gy.

Reference:

Wang W, Xu Y, Schipper M, Matuszak MM, Ritter T, Cao Y, Ten Haken RK, Kong FM. Effect of normal lung definition on lung dosimetry and lung toxicity prediction in radiation therapy treatment planning. Int J Radiat Oncol Biol Phys. 2013 Aug 1;86(5):956-63.

**Question 160:**

A radiation oncologist assesses a dose volume histogram (DVH) while evaluating a patient's treatment plan. Which of the following is true?

A. The DVH provides detailed spatial information about dose distribution within a structure.

B. The DVH does not account for anatomical changes or organ motion during treatment.

C. The DVH reflects biological effects on the tumor and normal tissue, allowing for prediction of tumor control probability (TCP) and normal tissue complication probability (NTCP).

D. The DVH demonstrates the conformality of the prescription dose on a given target.

**Answer Question 160: B**

A. A DVH does not provide spatial details about where within a structure that dose is delivered.

B. DVH calculations are typically based on a pre-treatment volumetric imaging, and they do not anticipate potential anatomical changes during treatment.

C. Biological models of tumor control or normal tissue complications are not incorporated into DVHs.

D. Conformality is not accounted for in the DVH, which is why additional equations, such as conformity index, are utilized to evaluate treatment plans.

Reference:
Lawrence TS, Kessler ML, Ten Haken RK. Clinical interpretation of dose-volume histograms: the basis for normal tissue preservation and tumor dose escalation. Front Radiat Ther Oncol. 1996;29:57-66.

**Question 161:** Which of the following is the most commonly evaluated parameter for serial organs at risk?

A. Dmax
B. D98%
C. Dmean
D. V100% (volume receiving at least 100% of the prescription dose)

**Answer Question 161: A**

Serial organs are made up of subunits where each individual subunit is necessary for the function of the entire organ. Therefore, damage to an individual subunit can cause the entire organ to fail. Dmax refers to the maximum dose within a voxel, essentially indicating the maximum dose delivered to a particular structure. Because a high dose to a small volume can disrupt the function of a subunit, and potentially the entire function of a serial organ, Dmax (or doses to small volumes) are often evaluated.

Reference:

Dale E, Olsen DR. Specification of the dose to organs at risk in external beam radiotherapy. Acta Oncol. 1997;36(2):129-35.

# Chapter 12: Imaging fundamentals – Radiography, CT, PET, SPECT, MRI, Ultrasound

Binita R. Chacko, Peter Ferjancic, May Tsao

**12.1 Imaging Fundamentals and X ray Radiography**

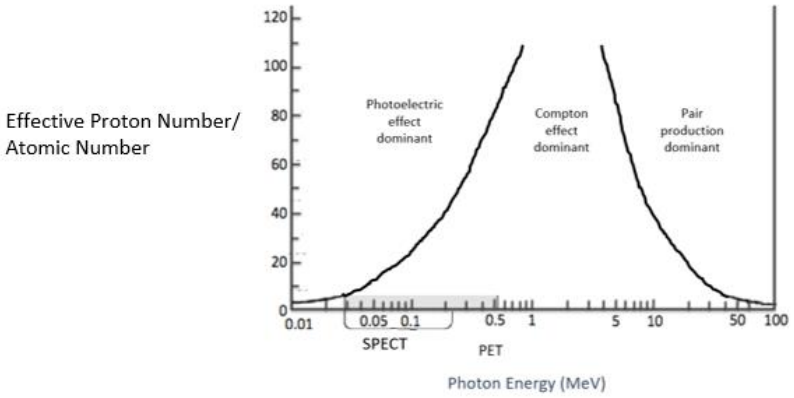
**Question 162:**

What is the primary interaction in diagnostic x-ray radiography?

A. Compton
B. Bremsstrahlung
C. Photoelectric
D. Pair Production

**Answer Question 162: C**

The photoelectric effect is the dominant x ray interaction in diagnostic radiography.




In diagnostic x-ray radiography, the energy range typically falls within the keV (kilo-electron volt) range. In therapeutic radiation, the energy range generally falls within the MeV (mega-electron volt) range. Below 1 MeV, the photoelectric effect is dominant as shown in the figure above.

Reference:

Bichsel H, Schindler H. The interaction of radiation with matter. In: *Particle Physics Reference Library*. Springer International Publishing; 2020:5-44.

**Question 163:**

What image acquisition parameter determines the optical density (darkness) of a diagnostic x-ray radiograph?

A. Light intensity used for viewing the radiograph

B. Roentgen (R)

C. Gray (Gy)

D. milliamperes second (mAs)

**Answer Question 163: D**

Low energy diagnostic x-rays pass through patient tissue to reach an image receptor (diagnostic x-ray film or digital detector). The higher number of x-ray photons which reach the image receptor creates a darker (higher optical density) image. Increasing electric current (mA) and increasing duration of exposure(s), increases the number of photons generated and the number of photons seen by the image receptor.

A milliampere second (mAs) measures the product of electrical current (milliamperes) and exposure time (seconds).

Radiation dose (unit Gy) is not a parameter but rather a consequence of the chosen parameters.

Reference:

Goldman LW. Principles of CT: Radiation Dose and Image Quality. Journal of Nuclear Medicine Technology 35(4): 213-225, 2007.

### 12.2 Computed Tomography (CT)

**Question 164:**

Which of the following statements regarding CT Hounsfield units (HUs) also known as CT number is **false**?

A. HUs are used to calculate therapeutic radiation dose in various tissue densities
B. HUs quantify tissue density in diagnostic CT scans
C. Air is -1000 HU, Water is 0 HU
D. HUs are directly proportional to photon attenuation in tissue

**Answer Question 164: D**

D is false. The other statements are true.

The HU (or CT number) is not directly proportional to photon attenuation in tissue. The calculation of HU is based on the linear transformation of the x-ray attenuation coefficient as follows:

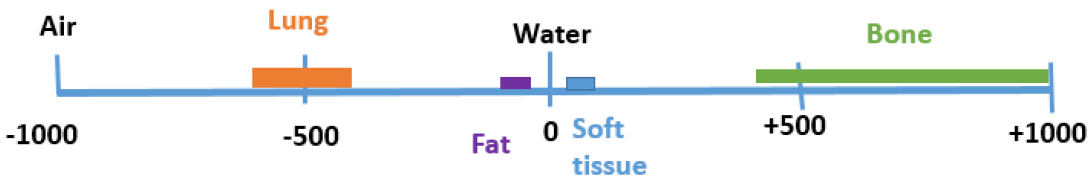
$HU = (\mu tissue - \mu water) / \mu water * 1000$

$\mu tissue$ = the linear coefficient of tissue being measured

$\mu water$ = the linear coefficient of water (=0 HU)

Denser tissues have higher positive HU and less dense tissues have negative HU values.

# Hounsfield Units (HU)




| | |
|---|---|
| Bone | +400 to +1000 |
| Soft tissue | +40 to + 80 |
| Water | 0 |
| Fat | -60 to -100 |
| Lung | -400to -600 |
| Air | -1000 |

References:

Schneider U, Pedroni E, Lomax A. The calibration of CT Hounsfield units for radiotherapy treatment planning. Phys Med Biol. 1996 Jan;41(1):111-24. doi: 10.1088/0031-9155/41/1/009. PMID: 8685250.

Levine ZH, Peskin AP, Holmgren AD, Garboczi EJ. Preliminary X-ray CT investigation to link Hounsfield unit measurements with the International System of Units (SI). PLoS One. 2018 Dec 20;13(12):e0208820.

**Question 165:**

Which of the following does **NOT** determine the image quality of diagnostic CT scans?

A. focal spot size

B. radiation dose

C. metal artifacts

D. small patient's body size

**Answer Question 165: D**

Small body size does not impair CT image quality.

Larger patients generally require higher radiation doses to achieve the same image quality as smaller patients. The radiographer (to help improve image quality in a larger patient) may adjust scan parameters, such as mAs and kVp. However, if the body size exceeds the range adjustable by these parameters, a larger body size can lead to a reduction in image quality.

A larger patient's body size might also influence the ability of the patient to fit into a CT scanner.

The focal spot size affects spatial resolution.

While the higher radiation doses received by the patient in the scanner can improve image quality, there is an increased risk of carcinogenesis. The "Image Gently, Image Wisely" campaign (based on the ALARA principle – As Low As Reasonably Achievable) is focused on reducing unnecessary radiation exposure from medical imaging procedures. For example, low dose CT chest images are used for lung cancer screening in high risk patients. While the image quality is not as good as regular dose CT chest, low dose chest CT makes use of the inherent contrast of lung parenchyma and has high sensitivity and reasonable specificity to detect lung cancer

References:

Zarb F, Rainford L, McEntee MF. Image quality assessment tools for optimization of CT images. Radiography 16(2), 147-153, 2010.

Goldman LW. Principles of CT: Radiation Dose and Image Quality. Journal of Nuclear Medicine Technology 35(4): 213-225, 2007.

**12.3 PET and SPECT**

**Question 166:**

Which one of the following is **NOT** a radiopharmaceutical used in clinical PET/SPECT diagnostic imaging?

A. $^{18}$F-FDG

B. $^{68}$Ga- PSMA-11

C. $^{68}$Ga-Dotatate

D. Tc99m

E. $^{60}$Co

**Answer Question 166: E**

All are radiopharmaceuticals used in PET/SPECT imaging except for cobalt-60.

References:

Zhang, S., Wang, X., Gao, X. *et al.* Radiopharmaceuticals and their applications in medicine. *Sig Transduct Target Ther* **10**, 1 (2025). https://doi.org/10.1038/s41392-024-02041-6

**Question 167:**

What is the difference between PET and SPECT imaging?

A. SPECT scans can assess brain function. PET scans cannot assess brain function.

B. SPECT scans generally have higher resolution (resolution range 2-6 mm) and higher sensitivity to signals than PET scans (resolution range 7-15 mm)

C. PET scans can measure several radiotracers at one time (multi-tracer imaging); whereas SPECT scans can only detect a single radiotracer, at a time

D. PET scans use positron-emitting tracers, SPECT scans use gamma-emitting tracers

**Answer Question 167: D**

PET scans use positron-emitting tracers, SPECT scans use gamma-emitting tracers

Both SPECT and PET are used to assess brain function.

**PET** scans generally have higher resolution (resolution range 2-6 mm) and higher sensitivity to signals than **SPECT** scans (resolution range 7-15 mm).

**SPECT** scans can measure several radiotracers at one time (multi-tracer imaging), whereas **PET** scans can only detect a single radiotracer, at a time.

Reference:

Davis KM, Ryan JL, Aaron VD, Sims JB. PET and SPECT imaging of the brain: History, technical considerations, applications, and radiotracers. Seminars in ultrasound, CT and MRI. 41(6): 521-529, 2020.

**12.4 MRI**

**Question 168:**

What do the MRI sequences T1, T2,TE, TR represent?

A. T1 and T2 are relaxation times. TE and TR are pulse sequence parameters

B. T1 and T2 are pulse times. TE and TR are relaxation parameters

C. T1 and T2 are pulse times. TE and TR are spin sequences.

D. T1 and T2 are excitation times. TE and TR are pulse sequences.

**Answer Question 168: A**

MRI T1, T2 are relaxation times. TE is echo time and TR is repetition time.

T1 represents the longitudinal relaxation time or the time for the magnetic spin of water molecules to return to their normal state after a radiofrequency pulse. Tissues with a shorter T1 relaxation time, such as fat appear bright on T1-weighted images.

T2 represents the transverse relaxation time or the time it takes for magnetic spin of water molecules to lose phase with each other after a radiofrequency pulse. Tissues with longer T2 relaxation times, such as fluid appear bright on T2-weighted images.

TE (Echo) is the time between the radiofrequency pulse and the collection of the peak (echo) signal. The amount of T2 relaxation that occurs before the echo is collected, is determined.

TR (Repetition) is the time between successive radiofrequency pluses in an MRI sequence, which influences the amount of T1 relaxation that occurs before the next pulse.

TR and TE are pulse sequence parameters that can be adjusted to emphasize either T1 or T2 weighting in an image.

Short TE/TR emphasizes T1 weight, improving anatomy and fat images.

Long TE/TR emphasizes T2 weight, improving fluid images.

Reference:

https://radiopaedia.org/articles/mri-sequences-overview?embed_domain=external.radpair.com%252527%25255B0%25255D%252527%25255B0%25255Dfavicon.icofavicon.icofavicon.icofavicon.icofavicon.icofavicon.ico&lang=us

**Question 169:**

Pacemakers or defibrillators (which are not MRI conditional) are contraindicated with diagnostic MRI. Which of the following statement is **false**?

A. The strong magnetic field can lead to cardiac lead heating with may cause myocardial thermal injury

B. Even though pacemakers are made of ferromagnetic materials, pacemakers will not be pulled out of the chest in an MRI unit

C. The strong magnetic field can interfere with the pacemaker's internal circuits and electronic components, causing the pacemaker to malfunction.

D. The radiation dose to pacemakers/defibrillators will exceed a dose of 2 Gy resulting in pacemaker malfunction

**Answer Question 169: D**

All are true EXCEPT D.

MRI units do not use ionizing radiation.

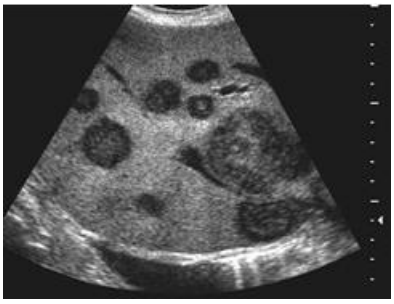
Reference:

Russo RJ, Costa HS, Silva PD et al. Assessing the risks associated with MRI in patients with as pacemaker or defibrillator. NEJM 376(8): 376-755, 2017.

**12.5 Ultrasound**

**Question 170:**

What is used to produce the image below?



A. Radio waves
B. Sound waves
C. Electromagnetic waves
D. Infrared waves

**Answer Question 170: B**

High frequency sound waves are used in ultrasound imaging. A hand held probe (transducer) converts electrical energy to high-frequency sound waves which enter the body.

When the sound wave travels between different tissues, some of the sound waves are reflected back to the transducer as echoes, while the other sound waves continue to travel through the body. The echoes captured in the transducer is processed by a computer to generate a real-time image.

Fluid filled structures (eg. cysts) will produce few echoes and will appear dark (black), as sound is more transmitted. Solid structures (eg. tumours) will produce many echoes and appear lighter, as sound is more reflected.

Reference:
Aldrich JE. Basic physics of ultrasound imaging. Crit Care Med. 2007 May;35(5 Suppl):S131-7.

**Question 171:**

What material is used in a diagnostic ultrasound transducer?

A. TERFENOL-D (an alloy of terbium, dysprosium and iron)

B. silicone carbide

C. piezoelectric crystals

D. indium phosphide

**Answer Question 171: C**

Ultrasound transducers use piezoelectric, often lead zirconate titanate (PZT) crystals which convert electrical energy to high frequency sound (ultrasound) waves.

Reference:
Aldrich JE. Basic physics of ultrasound imaging. Crit Care Med. 2007 May;35(5 Suppl):S131-7.

# Chapter 13: Simulation, motion management and treatment verification

Marian Axente, Victoria (Tori) Doss,

Jordan Holmes, May Tsao

**13.1 Simulation**

**Question 172:**

Which of the following does **NOT** occur during radiotherapy simulation?

A. Positioning the patient in a reproducible manner
B. Obtaining imaging for radiation planning
C. Use of immobilization devices
D. Cone-beam CT imaging

**Answer Question 172: D**

Cone beam CT (CBCT) imaging occurs on the treatment unit to verify treatment. CBCT does not occur during simulation.

Reference:
Berman AT, Plastaras JP, Vapiwala N. Radiation oncology: a primer for medical students. J Cancer Educ. 2013 Sep;28(3):547-53.

**Question 173:**

Which statement below is **true** regarding MRI simulation?

A. synthetic CT images derived from MRI are crucial for dose calculation
B. inferior soft tissue contrast compared to CT simulation
C. higher radiation dose delivered to the patient with MRI simulation compared to CT simulation
D. Inability to use immobilization devices with MRI simulation

**Answer Question 173: A**

All the statements are incorrect, EXCEPT for A.

MRI simulation uses synthetic CT images for tissue inhomogeneity radiation dose calculation. Possible errors in tissue inhomogeneity from synthetic CT images may impact the accuracy of treatment planning. Thus, option A is a correct statement.

References:
Bahloul MA, Jabeen S, Benoumhani S, Alsaleh HA, Belkhatir Z, Al-Wabil A. Advancements in synthetic CT generation from MRI: A review of techniques, and trends in radiation therapy planning. J Appl Clin Med Phys. 2024 Nov;25(11):e14499. doi: 10.1002/acm2.14499. Epub 2024 Sep 26. PMID: 39325781; PMCID: PMC11539972.

Paulson ES, Erickson B, Schultz C, Allen Li X. Comprehensive MRI simulation methodology using a dedicated MRI scanner in radiation oncology for external beam radiation treatment planning. Med Phys. 2015 Jan;42(1):28-39. doi: 10.1118/1.4896096. PMID: 25563245.

### 13.2 Image-based Treatment Verification

**Question 174:**

Which of the following is a **false** statement regarding kV versus MV cone-beam CT (CBCT)?

A. kV CBCT is more susceptible to metal artifacts than MV CBCT
B. kV CBCT has superior soft tissue contrast compared to MV CBCT
C. MV CBCT is associated with higher dose compared to kV CBCT
D. for kV CBCT, the dominant interaction is Compton scattering

**Answer Question 174: D**

All are true except for D.

For kV CBCT, the superior soft tissue contrast as compared to MV CBCT is due to the dominance of the photoelectric effect at kV energies.

For MV energies, the dominant interaction is Compton scattering.

References:

Broderick M, Menezes G, Leech M et al. A comparison of kilovoltage and megavoltage cone beam CT in radiotherapy. Journal of Radiotherapy in Practice 6(3): 173-178, 2007.

Baran G, Dominello MM, Bossenberger T, Paximadis P, Burmeister JW. MVCT versus kV-CBCT for targets subject to respiratory motion: A phantom study. J Appl Clin Med Phys. 2021 Sep;22(9):143-152.

**Question 175:**

Which of the following is **NOT** a possible strategy used to correct for misalignment of CBCT verification with the CT simulation planned image?

A. Re-position the patient
B. Alter the organ at risk anatomy (eg. adjusting bladder/rectal: filling/emptying in prostate radiotherapy)
C. Re-plan (eg. due to tumour enlargement or shift)
D. Use MV CBCT instead of kV CBCT

**Answer Question 175: D**

A, B, C are all strategies used in certain scenarios when the CBCT does not align with the CT simulation planned images.
Changing the CBCT from MV to KV does not correct for this misalignment.

Reference:

Mushonga M, Ung Y, Louie AV, Cheung P, Poon I, Zhang L, Tsao MN. Unanticipated Radiation Replanning for Stage III Non-small Cell Lung Cancer. Adv Radiat Oncol. 2023 May 27;8(6):101275. doi: 10.1016/j.adro.2023.101275. PMID: 38047222; PMCID: PMC10692281.

**13.3 Motion management and Intra-fractional Imaging**

**Question 176:**

What is the primary purpose of motion management in radiotherapy?

A. Increase the size of the radiation hot spot
B. Improve patient comfort during treatment
C. Reduce radiation exposure to non-target tissues
D. Shorten the overall treatment time
E. Enhance imaging resolution

**Answer Question 176: C**

The goal of motion management is to minimize radiation exposure to healthy tissues by controlling or compensating for tumor motion. This helps ensure that the radiation dose is focused on the tumor, minimizing side effects for surrounding organs.

A is incorrect because motion management focuses on targeting accuracy rather than dose increase.

B is incorrect as improving comfort is secondary and not the main goal.

D is incorrect because motion management can sometimes increase treatment time due to necessary adjustments.

E is incorrect as imaging resolution is not directly related to motion management, though accurate targeting helps imaging alignment.

Reference:

Dhont J, Harden SV, Chee LYS, Aitken K, Hanna GG, Bertholet J. Image-guided Radiotherapy to Manage Respiratory Motion: Lung and Liver. Clin Oncol (R Coll Radiol). 2020 Dec;32(12):792-804. doi: 10.1016/j.clon.2020.09.008. Epub 2020 Oct 7. PMID: 33036840.

**Question 177:**

Which of the following best describes abdominal compression as a motion management technique in radiotherapy?

A. It eliminates all tumor motion.
B. It uses breath-hold techniques to stabilize the tumor.
C. It applies pressure to limit diaphragmatic motion, reducing breathing-induced tumor movement.
D. It requires continuous imaging to monitor tumor position.
E. It restricts cardiac motion to stabilize thoracic tumors.

**Answer Question 177: C**.

It applies pressure to limit diaphragmatic motion, reducing breathing-induced tumor movement.

Abdominal compression limits diaphragmatic movement, thereby reducing the range of respiratory motion for tumors in the abdomen and thorax.

A is incorrect because it does not eliminate all motion, only reduces it.

B is incorrect as it is not a breath-hold technique; compression is applied continuously.

D is incorrect because abdominal compression does not require continuous imaging, though imaging may be used to confirm effectiveness.

E is incorrect as it does not target cardiac motion, which requires different approaches.

Reference:

Daly M, McWilliam A, Radhakrishna G, Choudhury A, Eccles CL. Radiotherapy respiratory motion management in hepatobiliary and pancreatic malignancies: a systematic review of patient factors influencing effectiveness of motion reduction with abdominal compression. Acta Oncol. 2022 Jul;61(7):833-841. doi: 10.1080/0284186X.2022.2073186. Epub 2022 May 25. PMID: 35611555.

**Question 178:**

A patient with left-sided breast cancer is undergoing radiotherapy. Which motion management technique would be most appropriate to protect the heart from radiation exposure?

A. Abdominal compression
B. Respiratory gating
C. Fiducial marker tracking
D. Thermoplastic immobilization

E. Deep inspiration breath-hold (DIBH)

**Answer Question 178: E**. Deep inspiration breath-hold (DIBH)

Deep inspiration breath-hold (DIBH) is commonly used for left-sided breast cancer to increase the distance between the heart and the chest wall during radiation delivery, thereby reducing heart exposure to radiation. This technique helps protect cardiac structures while maintaining accurate targeting of the tumor.

A is incorrect because abdominal compression is generally used for abdominal tumors and does not create a significant displacement of the heart.

B is incorrect because respiratory gating synchronizes with the breathing cycle but does not specifically maximize heart-lung separation like DIBH.

C is incorrect as fiducial tracking is typically used for tumors with high motion variability and does not specifically reduce heart dose.

D is incorrect because thermoplastic immobilization restricts patient movement but does not manage respiratory motion or protect the heart during treatment.

Reference:

Salvestrini V, Iorio GC, Borghetti P, De Felice F, Greco C, Nardone V, Fiorentino A, Gregucci F, Desideri I. The impact of modern radiotherapy on long-term cardiac sequelae in breast cancer survivor: a focus on deep inspiration breath-hold (DIBH) technique. J Cancer Res Clin Oncol. 2022 Feb;148(2):409-417. doi: 10.1007/s00432-021-03875-1. Epub 2021 Dec 1. PMID: 34853887.

**Question 179:**

Given a low compliance lung cancer patient with irregular breathing patterns, which motion management method is likely to be more effective?

A. Respiratory gating, as it limits the impact of irregular breathing
B. Fiducial-based tracking, as it allows real-time tracking of the tumor
C. Abdominal compression, which maintains tumor stability
D. Breath-hold technique, as it holds the tumor in a stable position
E. Gated CBCT imaging, which reduces motion blurring

**Answer Question 179: B.** Fiducial-based tracking, as it allows real-time tracking of the tumor

Fiducial-based tracking provides continuous monitoring of tumor position in real time, making it effective for patients with irregular breathing patterns, as it can adapt to changing tumor positions.

A is incorrect because respiratory gating works best with consistent, regular breathing patterns.

C is incorrect because abdominal compression only reduces motion but does not continuously adapt to irregularities.

D is incorrect because breath-hold techniques require patient cooperation and regular breathing control.

E is incorrect because gated CBCT is not continuous and is primarily used for imaging, not real-time motion tracking.

Reference:

Tanabe Y, Kiritani M, Deguchi T, Hira N, Tomimoto S. Patient-specific respiratory motion management using lung tumors vs fiducial markers for real-time tumor-tracking stereotactic body radiotherapy. Phys Imaging Radiat Oncol. 2022 Dec 24;25:100405. doi: 10.1016/j.phro.2022.12.002. PMID: 36655212; PMCID: PMC9841282.

**Question 180:**

Which of the following would be considered the most significant disadvantage of using voluntary breath-hold techniques in radiotherapy?

A. Increased dose to surrounding tissues
B. Reduced tumor localization accuracy
C. Dependency on patient compliance and ability to hold breath
D. Prolonged treatment duration due to pauses
E. Increased imaging requirements

**Answer Question 180: C.** Dependency on patient compliance and ability to hold breath

Voluntary breath-hold techniques require patients to hold their breath at specific times, making the technique dependent on their ability to follow instructions and maintain consistent breath-holds, which can be challenging for some patients.

A is incorrect because breath-hold techniques generally reduce dose to surrounding tissues by stabilizing the tumor.

B is incorrect as breath-hold techniques improve localization accuracy.

D is incorrect as, while treatment may take longer, the primary limitation is patient compliance.

E is incorrect because breath-hold does not inherently require additional imaging.

Reference:

Josipovic M, Persson GF, Dueck J, Bangsgaard JP, Westman G, Specht L, Aznar MC. Geometric uncertainties in voluntary deep inspiration breath hold radiotherapy for locally advanced lung cancer.

Radiother Oncol. 2016 Mar;118(3):510-4. doi: 10.1016/j.radonc.2015.11.004. Epub 2015 Nov 26. Erratum in: Radiother Oncol. 2019 Apr;133:230. doi: 10.1016/j.radonc.2019.01.025. PMID: 26631647.

**Question 181:**

A patient with abdominal cancer experiences significant anxiety with respiratory gating techniques during radiotherapy. Which motion management method might reduce their anxiety while effectively managing tumor motion?

A. Shallow breathing technique
B. Visual biofeedback during gating
C. Abdominal compression device
D. Deep inspiration breath-hold (DIBH)
E. Real-time MRI guidance

**Answer Question 181: C**. Abdominal compression device

Abdominal compression can be effective for managing tumor motion by limiting diaphragmatic movement, which reduces tumor motion without requiring complex respiratory control, making it suitable for patients with anxiety related to active breathing management techniques.

A is incorrect because shallow breathing may reduce motion but lacks consistent tumor stabilization.

B is incorrect as biofeedback during gating could increase anxiety by adding visual stimuli, and still requires respiratory control.

D is incorrect because DIBH generally requires significant breath control, which may exacerbate anxiety.

E is incorrect as real-time MRI guidance can be complex and increase patient anxiety with longer imaging times and confined space.

Reference:

Wu VWC, Ng APL, Cheung EKW. Intrafractional motion management in external beam radiotherapy. J Xray Sci Technol. 2019;27(6):1071-1086. doi: 10.3233/XST-180472. PMID: 31476194.

**Question 182:**

What is the primary purpose of a 4D CT scan in motion management for radiotherapy?

A. To enhance spatial resolution of treatment planning images
B. To capture tumor motion over a complete respiratory cycle
C. To reduce radiation dose to non-target areas
D. To improve organ density measurement for dose calculations
E. To create static images of the tumor

**Answer Question 182: B**
To capture tumor motion over a complete respiratory cycle

A 4D CT scan captures the tumor's full range of motion during the respiratory cycle, allowing for better treatment planning by visualizing the tumor at different points in the cycle.

A is incorrect because spatial resolution is not the primary goal of 4D CT in this context.

C is incorrect as 4D CT itself does not directly reduce radiation dose.

D is incorrect as it does not primarily measure organ density.

E is incorrect as 4D CT captures dynamic motion, not static images.

References:

Davey A, van Herk M, Faivre-Finn C, Brown S, McWilliam A. Automated gross tumor volume contour generation for large-scale analysis of early-stage lung cancer patients planned with 4D-CT. Med Phys. 2021 Feb;48(2):724-732. doi: 10.1002/mp.14644. Epub 2020 Dec 30. PMID: 33290579; PMCID: PMC7986204.

Li F, Li J, Xing J, Zhang Y, Fan T, Xu M, Shang D, Liu T, Song J. Analysis of the advantage of individual PTVs defined on axial 3D CT and 4D CT images for liver cancer. J Appl Clin Med Phys. 2012 Nov 8;13(6):4017. doi: 10.1120/jacmp.v13i6.4017. PMID: 23149795; PMCID: PMC5718544.

Castillo SJ, Castillo R, Castillo E, Pan T, Ibbott G, Balter P, Hobbs B, Guerrero T. Evaluation of 4D CT acquisition methods designed to reduce artifacts. J Appl Clin Med Phys. 2015 Mar 8;16(2):4949. doi: 10.1120/jacmp.v16i2.4949. PMID: 26103169; PMCID: PMC4504190.

**Question 183:**

A patient with prostate cancer is undergoing radiotherapy, and intrafractional organ motion due to bladder and rectal filling poses a challenge for accurate dose delivery. Which strategy would be most effective to mitigate the impact of this organ motion on treatment accuracy?

A. Utilizing implanted fiducials to monitor prostate position via real-time tracking
B. Increasing the planning target volume (PTV) margins to account for organ motion
C. Implementing abdominal compression to stabilize pelvic structures
D. Using deep inspiration breath-hold (DIBH) to reduce abdominal motion
E. Applying thermoplastic immobilization to minimize whole-body movement

**Answer Question 183: A**

Utilizing implanted fiducials to monitor prostate position via real-time tracking

Implanted fiducial markers provide real-time tracking of the prostate, allowing adjustments to radiation

delivery based on the prostate's actual position during treatment. This approach effectively compensates for intrafractional motion due to changes in bladder and rectal filling, ensuring precise targeting throughout the session.

B is incorrect as systematic margin increase would decrease the plan quality. Systematically increasing margins would have to account for worst case scenarios which would include unrealistically large margins.

C is incorrect because abdominal compression is not effective for reducing internal pelvic organ motion, which is primarily due to bladder and rectal filling.

D is incorrect as DIBH is typically used for thoracic motion management and is not effective for managing prostate position.

E is incorrect because thermoplastic immobilization only reduces overall patient movement and does not address internal organ motion related to prostate position.

Reference:

Dang A, Kupelian PA, Cao M, Agazaryan N, Kishan AU. Image-guided radiotherapy for prostate cancer. Transl Androl Urol. 2018 Jun;7(3):308-320. doi: 10.21037/tau.2017.12.37. PMID: 30050792; PMCID: PMC6043755.

**Question 184:**

Compare the effectiveness of gating versus fiducial-based tracking for intra-fraction motion management. In which scenario is each method preferred?

A. Gating for irregular breathing; fiducial tracking for stable breathing
B. Gating for patients with predictable respiratory patterns; fiducial tracking for irregular motion
C. Fiducial tracking for all cases; gating for heart motion control
D. Gating for tumors near the diaphragm; tracking for tumors near the heart
E. Fiducial tracking only for lung tumors; gating only for abdominal tumors

**Answer Question 184: B**.

Gating for patients with predictable respiratory patterns; fiducial tracking for irregular motion

Gating is ideal for patients with consistent respiratory patterns as it synchronizes radiation with the breathing cycle. Fiducial tracking, however, continuously monitors tumor position, making it effective for patients with irregular breathing.

A is incorrect because fiducial tracking is better suited for irregular breathing, not gating.

C is incorrect as fiducial tracking is not used in all cases, nor is gating specific to heart motion.

D is incorrect because gating is effective for thoracic tumors generally, not necessarily proximity to the diaphragm.

E is incorrect as fiducial tracking and gating are not limited to specific tumor sites.

Reference:

Guckenberger M, Richter A, Boda-Heggemann J, Lohr F. Motion compensation in radiotherapy. Crit Rev Biomed Eng. 2012;40(3):187-97. doi: 10.1615/critrevbiomedeng.v40.i3.30. PMID: 22694199.

**Question 185:**

If motion management techniques are unavailable, what adaptive radiotherapy methods could compensate for motion during treatment delivery, and what impact would this have on treatment planning?

A. Use of increased margins to account for motion
B. Patient repositioning between each fraction
C. Continuous imaging throughout treatment
D. Dose escalation for shorter treatment time
E. Use of bolus material to reduce motion

**Answer Question 185: A.**

Use of increased margins to account for motion

Increasing treatment margins can compensate for motion by including the entire range of possible tumor positions, though it also increases radiation exposure to surrounding tissues.

B is incorrect because repositioning does not address intra-fractional motion.

C is incorrect because continuous imaging would increase radiation exposure without directly compensating for motion.

D is incorrect as dose escalation does not reduce the effects of motion.

E is incorrect because bolus material does not prevent or account for motion.

Reference:

Korreman SS. Motion in radiotherapy: photon therapy. Phys Med Biol. 2012 Dec 7;57(23):R161-91. doi: 10.1088/0031-9155/57/23/R161. PMID: 23165229.

# Chapter 14: Clinical brachytherapy

Matthew Studenski, Jordan Holmes, May Tsao

**14.1 Radioactive sources**

**Question 186:**

What is the half-life of Ir-192?

A. 60 hours
B. 74 days
C. 90 days
D. 28 days

**Answer Question 186: B**

74 days.

Ir-192 has an average energy of 380 keV. Other common radionuclides are I-125 (59 day half life and an average energy of around 27 keV) and Pd-103 (17 day half life and an average energy of around 23 keV).

Reference:
Podgorsak MB, DeWerd LA, Paliwal BR. The half-life of high dose rate Ir-192 sources. Med Phys. 1993 Jul-Aug;20(4):1257-9. doi: 10.1118/1.596976. PMID: 8413038.

**Question 187:**

What is the activity of a 3000 Ci Co-60 (half-life of 5.26 years) after 10 years?

A. 119 Ci
B. 320 Ci
C. 555 Ci
D. 811 Ci

**Answer Question 187: D**

To calculate this, you first need to know the decay equation: $A(t) = A_0e^{-\lambda t}$. A(t) is the activity at a certain time, t, and $A_0$ is the original activity. $\lambda$ is the decay constant specific to the particular radionuclide and is related to the half-life ($T_{1/2}$): $\lambda = \ln(2)/T_{1/2}$. $\ln(2) = 0.693$.

The key to these decay calculations is to keep your units consistent, eg. Months, days, years. In this problem, everything is in years, so it is easy.

First we need to calculate $\lambda$, which is 0.132 $years^{-1}$. We can then substitute for $A_0$ and t: A(10 years) = 3000 Ci * $e^{-0.132 * 10}$. Solving for A, we get 811 Ci. This is reasonable as it is around 2 half-lives; 3000 Ci -> 1500 Ci -> 750 Ci.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 188:**

The most appropriate unit to specify brachytherapy source strength is:

A. mg-RaEq
B. mCi
C. Air Kerma
D. R

**Answer Question 188: C**

Air Kerma is a National Institute of Standards and Technology (NIST) traceable quantity and is the recommended unit per AAPM TG-56. The unit of mCi (apparent activity) can be found in many clinics, but is not the recommended unit as two sources with the same apparent activity can have different dose distributions. It is possible to convert from Air Kerma to apparent activity using the Air Kerma rate constant ($\Gamma$), but all measurements and reporting should be done in units of Air Kerma.

Reference:

Nath R, Anderson LL, Meli JA, Olch AJ, Stitt JA, Williamson JF. Code of practice for brachytherapy physics: report of the AAPM Radiation Therapy Committee Task Group No. 56. American Association of Physicists in Medicine. Med Phys. 1997 Oct;24(10):1557-98. doi: 10.1118/1.597966. PMID: 9350711.

**Question 189:**

When will 5 mCi of I-131 (half-life of 8.05 days) and 2 mCi of P-32 (half-life of 14.2 days) have the same activity?

A. 17 days
B. 24 days
C. 33 days
D. 70 days

**Answer Question 189: B**

This is another decay equation asking when two sources will be in equilibrium (equal activity). The trick here is to assume that the sources will have equal activity at 't' days, which we will now calculate. The idea is that A(t) I-131 = A(t) P-32.

First we need to calculate $\lambda$ for both sources keeping the units consistent. We can then substitute: $5 * e^{-0.086t} = 2 * e^{-0.048t}$. If we remember our logarithm rule that $\ln(x^y) = y * \ln(x)$, we get: $\ln(5) - 0.086t = \ln(2) - 0.048t$. Solving algebraically for t, you get 24 days.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**14.2 Radiation Biology**

**Question 190:**

What factors determine the biological effective dose (BED) of high dose rate (HDR) and low dose rate (LDR) brachytherapy?

Select the correct answer **A, B, C or D** below.

1. α/β ratio
2. absorbed dose
3. sub-lethal damage repair
4. overall treatment time

**A. 1, 2, 3, 4**
**B. 3 and 4 only**
**C. 1 and 2 only**
**D. none**

**Answer Question 190: A**

All the options listed determine BED for HDR and LDR brachytherapy.

HDR brachytherapy involves inserting temporary channels into a target volume. Radiation sources are inserted into the channels and computerized methods are used to determine dwell times in various positions to optimize dose distribution. High dose radiation is given over a period of a few minutes and may be repeated over several fractions. An example is iridium-192 used in HDR prostate brachytherapy.

LDR brachytherapy involves permanently (and rarely temporarily) placing radioactive seeds to deliver radiation over an extended period of time. An example is I-125 seeds used in LDR prostate brachytherapy.

For external beam radiation or HDR brachytherapy, the BED is given by the following equation:

$$BED = nd\left(1 + \frac{d}{\frac{\alpha}{\beta}}\right)$$

However, if the HDR brachytherapy is associated with extended irradiation time (>1.5 hours), then repair during irradiation is significant and the following equation should be used instead.

The following equation is also the equation to used for LDR brachytherapy:

$$BED = D\left\{1 + \frac{2D}{\left(\frac{\alpha}{\beta}\right)\mu T}\left[1 - \frac{1}{\mu T}(1 - e^{-\mu T})\right]\right\}$$

D = total dose in Gy

d= dose per fraction in Gy

n=number of fractions

T = overall treatment time in hours

μ = sub-lethal damage repair parameter

$\mu = \ln 2 / T^{1/2}$

$T^{1/2}$ = half-life of sub-lethal damage repair in hours

The following equation converts BED to equivalent dose in 2 Gy fractions:

$$EQD2 = \frac{BED}{\left(1 + \frac{2}{\alpha/\beta}\right)}$$

References:

Yang R, Shao N, Liao A et al. Dosimetric and radiobiological comparison of volumetric modulated arc therapy, high-dose rate brachytherapy and low-dose rate permanent seeds implant for localized prostate cancer. Medical Dosimetry 41(3); 236-241, 2016.

Orton CG. High-dose-rate brachytherapy may be radiobiologically superior to low-dose rate due to slow repair of late-responding normal tissue cells. Int J Radiat Oncol Biol Phys. 2001 Jan 1;49(1):183-9. doi: 10.1016/s0360-3016(00)00810-5. PMID: 11163513.

Skowronek J. Low-dose-rate or high-dose-rate brachytherapy in treatment of prostate cancer - between options. J Contemp Brachytherapy. 2013 Mar;5(1):33-41. doi: 10.5114/jcb.2013.34342. Epub 2013 Mar 29. PMID: 23634153; PMCID: PMC3635047.

Gray C, Campbell K. High Dose Rate Brachytherapy versus Low Dose Rate Brachytherapy for the Treatment of Prostate Cancer: A Review of Clinical Effectiveness and Cost-Effectiveness [Internet]. Ottawa (ON): Canadian Agency for Drugs and Technologies in Health; 2019 Feb 14. Available from: https://www.ncbi.nlm.nih.gov/books/NBK544668/

**Question 191:**

You have a choice of using I-125 or Pd-103 for a low dose rate permanent implant. What is the required prescription dose for I-125 compared to Pd-103 for the same biological effect?

A. Lower
B. Equal
C. Higher

**Answer Question 191: C**

Higher.

To have the same biological effect, you need a higher prescription dose for an I-125 implant. The half-life of I-125 is longer than Pd-103 so the dose will be delivered over a longer period of time. To have the equivalent biological effect, a higher dose is required. This is the same as having a lower dose for an SBRT treatment compared to standard fractionation.

References:

Luxton G. Comparison of radiation dosimetry in water and in solid phantom materials for I-125 and Pd-103 brachytherapy sources: EGS4 Monte Carlo study. Med Phys. 1994 May;21(5):631-41. doi: 10.1118/1.597317. PMID: 7935197.

Peschel RE, Chen Z, Roberts K, Nath R. Long-term complications with prostate implants: iodine-125 vs. palladium-103. Radiat Oncol Investig. 1999;7(5):278-88. doi: 10.1002/(SICI)1520-6823(1999)7:5<278::AID-ROI3>3.0.CO;2-3. PMID: 10580897.

**14.3 Dose Calculation**

**Question 192:**

A common method of calculating dose in brachytherapy is a modular formalism called:

A. Monte Carlo
B. TG-186
C. TG-43
D. Sievert Integral

**Answer Question 192: C**
TG-43.

This modular formalism calculates the dose rate using the source strength, the exposure rate constant, a geometry factor, radial dose function, and an anisotropy factor. Monte Carlo dose calculation is a stochastic simulation of the interaction of radiation in matter. This is the gold standard for dose calculation but is very labor intensive and time consuming. TG-186 is a new report addressing model-based dose calculations. Two examples are Acuros (Grid-based Boltzman Solver) and collapsed-cone convolution. These methods account for heterogeneous media, while TG-43 assumes water, only. The Sievert Integral was used in the past to calculate dose.

Reference:

Rivard MJ, Coursey BM, DeWerd LA, Hanson WF, Huq MS, Ibbott GS, Mitch MG, Nath R, Williamson JF. Update of AAPM Task Group No. 43 Report: A revised AAPM protocol for brachytherapy dose calculations. Med Phys. 2004 Mar;31(3):633-74. doi: 10.1118/1.1646040. Erratum in: Med Phys. 2004 Dec;31(12):3532-3. PMID: 15070264.

**Question 193:**

In TG-43, the radial dose function accounts for attenuation and scatter in the media not accounted for by the geometry factor. Is the radial dose function a larger value for I-125 or Ir-192?

A. I-125
B. Ir-192

**Answer Question 193: B.**

Ir-192.

The radial dose function is required because the inverse square law breaks down in water for low energy sources. The geometry factor accounts for the inverse square falloff, but an additional factor is required as energy decreases. For high energy sources, the radial dose function approaches a value of 1, meaning there is a reduced effect of attenuation and scatter. For low energy sources, the effect is greater, meaning the value of the radial dose function is smaller. I-125 has a lower average energy than Ir-192.

Reference:

Rivard MJ, Coursey BM, DeWerd LA, Hanson WF, Huq MS, Ibbott GS, Mitch MG, Nath R, Williamson JF. Update of AAPM Task Group No. 43 Report: A revised AAPM protocol for brachytherapy dose calculations. Med Phys. 2004 Mar;31(3):633-74. doi: 10.1118/1.1646040. Erratum in: Med Phys. 2004 Dec;31(12):3532-3. PMID: 15070264.

**Question 194**:

In the TG-43 line source formalism, the anisotropy factor accounts for the differences in dose as a function of the radial direction around the source. Where is the anisotropy factor the lowest (ie. largest affect)?

A. Middle of the source (perpendicular bisector).
B. End of the source.
C. 45 degrees from the bisector.

**Answer Question 194: B**

End of the source.

Value of the anisotropy factor is 1 on the perpendicular bisector and decreases (has a larger effect) and you move towards the end of the source. The major contributors are increased attenuation and scattering from the source encapsulation and the source, itself.

Reference:

Rivard MJ, Coursey BM, DeWerd LA, Hanson WF, Huq MS, Ibbott GS, Mitch MG, Nath R, Williamson JF. Update of AAPM Task Group No. 43 Report: A revised AAPM protocol for brachytherapy dose calculations. Med Phys. 2004 Mar;31(3):633-74. doi: 10.1118/1.1646040. Erratum in: Med Phys. 2004 Dec;31(12):3532-3. PMID: 15070264.

**14.4 Brachytherapy delivery**

**Question 195:**

How does pulse dose rate (PDR) brachytherapy differ from high dose rate (HDR) and low dose rate (LDR) brachytherapy?

A. PDR is HDR given in 5 or more fractions

B. PDR is LDR given over 2 fractions

C. PDR uses a lower dose rate than LDR and produces a series of short exposures of 10-30 minutes every hour

D. PDR uses a higher dose rate than LDR and produces a series of short exposures of 10-30 minutes every hour

**Answer Question 195: D**

PDR uses a higher dose rate than LDR and produces a series of short exposures of 10-30 minutes every hour for the same overall time as LDR brachytherapy for the same total dose.

PDR combines the physical advantages of HDR with the use of multi-fractionated temporary brachytherapy.

PDR also has the theoretical radiobiological advantages of LDR, with more normal tissue sparing because of the small pulse size (compared to HDR).

Worldwide, PDR is less common than HDR or LDR brachytherapy.

References:

Skowronek J. Pulsed dose rate brachytherapy - is it the right way? J Contemp Brachytherapy. 2010 Sep;2(3):107-113. doi: 10.5114/jcb.2010.16921. Epub 2010 Oct 13. PMID: 27853471; PMCID: PMC5104829.

Balgobind BV, Koedooder K, Ordoñez Zúñiga D, Dávila Fajardo R, Rasch CR, Pieters BR. A review of the clinical experience in pulsed dose rate brachytherapy. Br J Radiol. 2015;88(1055):20150310. doi: 10.1259/bjr.20150310. Epub 2015 Aug 20. PMID: 26290399; PMCID: PMC4743453.

**Question 196:**

What is the main purpose of remote after-loading units?

A. To reduce radiation exposure staff members

B. To reduce radiation exposure to patients

C. To effectively prevent radioactive source loss

D. To ensure accurate brachytherapy source placement

**Answer Question 196: A**

Reference:

Glasgow GP, Bourland JD. Grigsby PW et al. AAPM Report No. 41. Remote afterloading technology. 1993. https://www.aapm.org/pubs/reports/RPT_41.pdf

**14.5 Prescribing, reporting and evaluating brachytherapy treatments**

**Question 197:**

What are the factors that influence the shape of a brachytherapy dose distribution?

Choose the correct answer **A, B, C or D** below.

1. source strength

2. source position

3. source dwell time

4. applicator size, shape and material

**A. 1, 2, 3, 4**
**B. 1, 2 3 only**
**C. 4 only**
**D. none**

**Answer Question 197: A**

All factors influence brachytherapy dose distributions.

References:

Anderson LL. Plan optimization and dose evaluation in brachytherapy. Seminars in Radiation Oncology 3(4): 290-300, 1993.

Jayarathna S, Hoang M, Badkul R, Hoover A. Dosimetric impact of applicator displacement on three-dimensional image-guided high-dose-rate brachytherapy treatments for cervical cancer. J Contemp Brachytherapy. 2023 Oct;15(5):334-343. doi: 10.5114/jcb.2023.132697. Epub 2023 Oct 31. PMID: 38026073; PMCID: PMC10669918.

Wu CH, Shiau AC, Liao YJ, Lin HY, Liu YW, Hsu SM. Influence of metal of the applicator on the dose distribution during brachytherapy. PLoS One. 2014 Aug 18;9(8):e104831. doi: 10.1371/journal.pone.0104831. PMID: 25133789; PMCID: PMC4136780.

**14.6 Disease sites**

**Question 198:**

What study found that eye plaque brachytherapy was comparable to enucleation for survival in the treatment of ocular melanoma?

A. Ocular Brachytherapy Collaborative Group (OcBr) 93-01 trial
B. Collaborative Ocular Melanoma Study (COMS)
C. Brachytherapy Society Trial 005
D. National Cancer Institute's (NCI) Cancer Therapy Evaluation Program (CTEP) OC-3 trial

**Answer Question 198: B.**

The Collaborative Ocular Melanoma Study (COMS) reported comparable survival between patients treated with plaque brachytherapy and enucleation. Other forms of external beam radiation therapy have also been studied which were associated with excellent local control but very high toxicity.

Reference:

Collaborative Ocular Melanoma Study Group. Ten-year follow-up of fellow eyes of patients enrolled in Collaborative Ocular Melanoma Study randomized trials: COMS report no. 22. Ophthalmology. 2004 May;111(5):966-76. doi: 10.1016/j.ophtha.2003.08.029. PMID: 15121376.

**Question 199:**

What brachytherapy applicators are used for HDR accelerated partial breast irradiation?
Choose the correct answer **A, B, C or D** below.

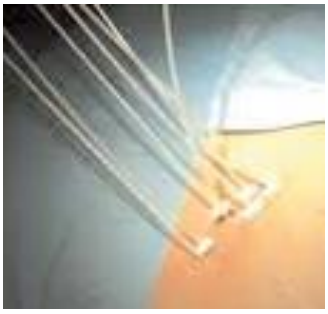
1. Multi-catheters
2. Balloon applicators
3. Strut adjusted volume implant
4. Interstitial catheters
5. Intracavitary applicators

**A. 1, 2, 3, 4, 5**
**B. 1, 4, 5 only**
**C. 1, 2, 4, 5 only**
**D. 4 and 5 only**

**Answer Question 199: A**

All are used for HDR accelerated partial breast irradiation.

Multi-catheters:



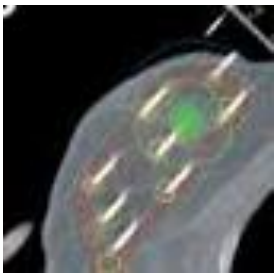
Balloon applicator (intracavitary):



Strut adjusted volume implant



Interstitial catheters:



References:

Cox, J., Swanson, T. Current modalities of accelerated partial breast irradiation. *Nat Rev Clin Oncol* **10**, 344–356 (2013). https://doi.org/10.1038/nrclinonc.2013.65

Weed DW, Edmundson GK, Vicini FA, Chen PY, Martinez AA. Accelerated partial breast irradiation: a dosimetric comparison of three different techniques. Brachytherapy. 2005;4(2):121-9. doi: 10.1016/j.brachy.2004.12.005. PMID: 15893265.

**Question 200.**

What HDR applicator is shown in the following figure?



A. Ring and tandem

B. Cylinder
C. Balloon
D. Interstitial

**Answer Question 200. A**

Ring and tandem.

This is a standard treatment for cervical cancer. The tandem is inserted into the uterus and the ring sits in the vagina, adjacent to the cervical os. A similar setup is done for tandem and ovoids. The patient receives 45 Gy in 1.8 Gy daily fractions of external beam pelvic radiotherapy and then a HDR boost to the cervix delivering about 30 Gy in 4 or 5 fractions.

This is the typical pear shaped dose distribution prescribed to point A. Point A is located 2 cm superior and 2 cm lateral to the cervical os, along the tandem axis. Historically this was the prescription point, but as CT based planning has become more prevalent, volume-based prescriptions are now more common.

Reference:

Otal A, Celada F, Chimeno J, Vijande J, Pellejero S, Perez-Calatayud MJ, Villafranca E, Fuentemilla N, Blazquez F, Rodriguez S, Perez-Calatayud J. Review on Treatment Planning Systems for Cervix Brachytherapy (Interventional Radiotherapy): Some Desirable and Convenient Practical Aspects to Be Implemented from Radiation Oncologist and Medical Physics Perspectives. Cancers (Basel). 2022 Jul 17;14(14):3467. doi: 10.3390/cancers14143467. PMID: 35884528; PMCID: PMC9318845.

**Question 201**:

What imaging modality is typically used for prostate brachytherapy treatment planning?

A. Ultrasound
B. MRI
C. CBCT
D. PET

**Answer Question 201: A**.

Prostate brachytherapy is typically planned using transrectal ultrasound. The ultrasound probe is calibrated to a trans-peritoneal template to assist with needle insertion. Once the needles have been inserted, a final 3D ultrasound image is obtained. Contours can be drawn and dose can be calculated.

Reference:

Crook J, Marbán M, Batchelar D. HDR Prostate Brachytherapy. Semin Radiat Oncol. 2020 Jan;30(1):49-60. doi: 10.1016/j.semradonc.2019.08.003. PMID: 31727300.

**Question 202:**

Cancers in which organ are treated with Yttrium-90 (Y-90) brachytherapy microspheres?

A. Pancreas
B. Brain
C. Liver
D. Kidney

**Answer Question 202: C.**

Y-90 microspheres are used to treat liver cancer, typically HCC or liver metastasis. The procedure takes advantage of the unique circulation in the liver where tumors are typically fed by arterial blood whereas the healthy liver is fed by the portal vein. By traversing the femoral artery into the liver, the microspheres can be selectively injected into the tumor, both embolizing the vessels and delivering a high dose of radiation.

Reference:

Sharma NK, Kappadath SC, Chuong M, Folkert M, Gibbs P, Jabbour SK, Jeyarajah DR, Kennedy A, Liu D, Meyer JE, Mikell J, Patel RS, Yang G, Mourtada F. The American Brachytherapy Society consensus statement for permanent implant brachytherapy using Yttrium-90 microsphere radioembolization for liver tumors. Brachytherapy. 2022 Sep-Oct;21(5):569-591. doi: 10.1016/j.brachy.2022.04.004. Epub 2022 May 20. PMID: 35599080; PMCID: PMC10868645.

# Chapter 15: Brachytherapy QA

Matthew Studenski, Jordan Holmes

**Question 203**:

The most appropriate device to measure brachytherapy source strength is:

A. Well-type ionization chamber.
B. Farmer chamber.
C. Parallel plate chamber.
D. Diode

**Answer Question 203: A**

For measuring source strength, the well-chamber provides the highest level of sensitivity. The chamber is required to be calibrated at a National Institute of Standards and Technology (NIST) traceable lab every 2 years. The calibration converts measured current to Air Kerma strength and defines the location in the chamber where the measurement is done (sweet spot). It is possible to use a Farmer chamber or Parallel plate chamber, but you would not be able to obtain a NIST traceable calibration as your setup would most likely be specific to your institution.

Reference:

Smith BR, DeWerd LA, Culberson WS. On the stability of well-type ionization chamber source strength calibration coefficients. Med Phys. 2020 Sep;47(9):4491-4501. doi: 10.1002/mp.14247. Epub 2020 Jun 17. PMID: 32434273.

**Question 204:**

Following an HDR source exchange, what is the tolerance for the measured source strength versus the calibration certificate from the vendor?

A. ±1%
B. ±3%
C. ±5%
D. ±7%

**Answer Question 204: B**

**±3%.**

If your measured source strength is outside of this tolerance, the first step is always to repeat the measurement to ensure the setup was correct. Double check the pressure and temperature correction and any factors used in your calculation (eg. decay). If possible, check using a different electrometer and well chamber. If it really is out of tolerance, it is important to contact the vendor and determine if there is something systematically wrong with their calibration or if the error is due to fault equipment on the user side.

Reference:

Nair MT, Cheng MC. HDR source calibration methods and discrepancies. Int J Radiat Oncol Biol Phys. 1997 Apr 1;38(1):207-11. doi: 10.1016/s0360-3016(97)00255-1. PMID: 9212025.

**Question 205:**

The process of exposing a film using the HDR source for applicator QA is called:

A. Film QA
B. Exposure
C. Double check
D. Autoradiograph

**Answer Question 205: D**

Autoradiograph.

The process of accepting and commissioning a new applicator is important because it ensures that the appropriate applicator length is entered into the planning system. The applicator is taped to a piece of film and the marker cable is inserted into the applicator. The film and applicator are then exposed to x-rays, creating an image of the applicator and marker cable on the film. The marker cable is then removed and the applicator attached to the afterloader (while still taped to the film). The source is then run out to the most distal dwell position and several more proximal positions.

The source will create a dark mark on the film that should correspond to the location of the marker cable. This is crucial to confirm the distance of the most distal dwell position and that it corresponds to the marker cable as this will link the coordinate system of the afterloader to the CT image of the patient used for treatment planning.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 206:**

The tolerances for source position accuracy and dwell time accuracy are _____ mm and _____ seconds.

A. 1 mm, 5 seconds
B. 5 mm, 5 seconds
C. 1 mm, 1 second
D. 2 mm, 2 seconds

**Answer Question 206: C**

1 mm, 1 second.

It is important to test both the source position and dwell time accuracy during daily QA, post source exchange, and during commissioning. There are multiple methods of doing this.

For source position accuracy, you can do an autoradiograph with film or simply use a positioning ruler and the camera in the room to visually verify the position.

For dwell time accuracy, a stopwatch is usually sufficient for daily QA, but a more strict test can be done by testing the linearity by running the source into the well chamber and measuring the charge collected over different times.

Reference:

Poder J, Rivard MJ, Howie A, Carlsson Tedgren Å, Haworth A. Risk and Quality in Brachytherapy From a Technical Perspective. Clin Oncol (R Coll Radiol). 2023 Aug;35(8):541-547. doi: 10.1016/j.clon.2023.01.001. Epub 2023 Jan 14. PMID: 36682968.

**Question 207**:

In addition to dosimetric accuracy, source position accuracy and dwell position accuracy, what is the final category of brachytherapy QA?

A. Speed.
B. Commissioning.
C. CT number accuracy.
D. Safety.

**Answer Question 207: D**

Safety is always paramount in brachytherapy. Some of the typical tests include, but are not limited to, radiation monitor functionality, emergency stop, catheter interlock, door interlock, audio/visual functionality, radiation shielding survey, transfer tube length, and proper emergency equipment (scissors, lead pig, etc.).

Reference:

Poder J, Rivard MJ, Howie A, Carlsson Tedgren Å, Haworth A. Risk and Quality in Brachytherapy From a Technical Perspective. Clin Oncol (R Coll Radiol). 2023 Aug;35(8):541-547. doi: 10.1016/j.clon.2023.01.001. Epub 2023 Jan 14. PMID: 36682968.

# Chapter 16: Advanced treatment planning and special procedures – including total body irradiation, total skin electron therapy, intraoperative radiation therapy, adaptive radiotherapy

Jill Moreau, May Tsao

### 16.1 Photon Total Body Irradiation

**Question 208:**

Which of the following parameters does **NOT** influence total body radiation (TBI) dosimetry?

A. Field size
B. SSD
C. Tissue inhomogeneity
D. Patient separation
E. Computed tomography (CT) Hounsfield Unit (HU) calibration curve

**Answer Question 208: E**

All are factors which influence total body dosimetry except for CT HU calibration curve.

The unique aspects of extended SSD , large field size, tissue inhomogeneity (e.g. lungs), patient separation make total body radiation (TBI) a unique external beam radiotherapy technique.

Patient positioning options include standing or lying (eg. on one side), An older technique used a sitting position.

Beam options include ant/post, multi-field or IMRT.

Usually, 6-10 MV photons are used.

Doses to lungs (sometimes kidneys, liver and previously irradiated sites) are shielded or planned with attenuated doses.

Total body irradiation is used to treat cancers of the bone marrow such as leukemia, lymphoma, myeloma. It is also used to destroy hematopoietic cells in the bone marrow, which are replaced by cancer free stem cells with a stem cell transplant.

Dose fractionation regimens vary. However, the biological dose is low as hematopoietic cells and hematopoeitic cancers are radiosensitive.

References:

Galvin JM. Total Body Irradiation: Dosimetry and Practical Considerations. 2011. https://www.aapm.org/meetings/2001AM/pdf/7233-71758.pdf

Wong JYC, Filippi AR, Dabaja BS et al. Total Body Irradiation: Guidelines from the International Lymphoma Radiation Oncology Group (ILROG). Int J Radiat Oncol Biol Pys 101(3): 521-529, 2018.

Niroomand-Rad A. Physical aspects of total body irradiation of bone marrow transplant patients using 18 MV x rays. Int J Radiat Oncol Biol Phys 20(3): 605-611, 1991.

**Question 209:**

A patient's dose distribution report shows significant under dosing of the lower extremities during total body irradiation (TBI) using a standing technique with a conventional linac. What is the most likely cause, and what adjustment is most appropriate?

A. Patient height; add beam spoilers to the upper body

B. Beam divergence; adjust patient distance or use tissue compensators

C. Respiratory motion; use a gating system

D. Scatter effects; increase gantry rotation speed

**Answer Question 209: B**

With the patient positioned standing and using a conventional linac, underdosing the lower extremities in total body irradiation occurs because of beam divergence. The inverse square law states that the radiation intensity is inversely proportional to the square of the distance from the source. By adjusting the lower extremities closer to the source, the under dosage may be corrected. Customized compensators can improve dose uniformity in total body irradiation by selectively reducing the dose in specific parts of the body.

The answer A is incorrect. Beam spoilers are used to reduce skin sparing by increasing the lower energy components of the radiation beam and thus the dose to the patient's skin. A beam spoiler is used in total body irradiation to increase the dose to a minimum surface dose, which is at least 90% of the prescribed dose in many total body irradiation protocols. In certain hematological conditions, such as leukemia, leukemic cells may circulate in the skin's blood supply, so that a therapeutic dose is needed to the skin surface.

Respiratory motion and scatter effects do not cause under dosing of the lower extremities in total body irradiation.

References:

Van Dyk, J., et al. Radiation Therapy Committee Task Group 29. The physical aspects of total and half body photon irradiation. AAPM Report No. 17, Task Group 29. College Park (MD): American Institute of Physics, published for the American Association of Physicists in Medicine; 1986.
ISBN: 978-0-883185-13-1. doi: 10.37206/16.

Pierce G, Balogh A, Frederick R, Gordon D, Yarschenko A, Hudson A. Extended SSD VMAT treatment for total body irradiation. J Appl Clin Med Phys. 2019 Jan;20(1):200-211. doi: 10.1002/acm2.12519. Epub 2018 Dec 27. PMID: 30592152; PMCID: PMC6333187.

**Question 210:**

A patient undergoing total body irradiation (TBI) as part of conditioning for hematopoietic stem cell transplantation receives a dose distribution report showing uniform dose coverage throughout the torso but a 30% overdose in regions of the lung. What is the most likely cause of this finding, and what adjustment is most appropriate?

A. The treatment plan lacked adequate lung shielding, and custom blocks or tissue-equivalent compensators should be added.

B. The patient's breathing motion causes inconsistent coverage, and a breath-hold technique should be implemented.

C. The overdose is a result of beam attenuation through the arms, and repositioning the arms closer to the torso is required.

D. Lung tissue is radioresistant, so higher doses are intentional, and no adjustment needed.

**Answer Question 210: A**

In total body irradiation, lung shields are used to reduce the dose to lungs. All other answers are incorrect.

Reference:

Van Dyk, J., et al. Radiation Therapy Committee Task Group 29. The physical aspects of total and half body photon irradiation. AAPM Report No. 17, Task Group 29. College Park (MD): American Institute of Physics, published for the American Association of Physicists in Medicine; 1986.
ISBN: 978-0-883185-13-1. doi: 10.37206/16.

**Question 211:**

A transplant team is concerned about pulmonary toxicity in a patient with pre-existing lung disease. Which conditioning radiation approach would best reduce the risk?

A. Abdominal Irradiation

B. Craniospinal Irradiation

C. Total Body Irradiation (TBI)

D. Total Marrow Irradiation (TMI)

**Answer Question 211: D**

Total marrow irradiation is indicated for patients undergoing stem cell transplant. By irradiating the marrow, hematologic malignancy is destroyed in the marrow and the immune system is suppressed to prevent transplant rejection.

Total marrow irradiation uses intensity modulated radiation therapy to deliver higher doses to the bone marrow while sparing normal organs such as lung.

References:

Köksal M, Kersting L, Schoroth F, Garbe S, Koch D, Scafa D, Sarria GR, Leitzen C, Heine A, Holderried T, Brossart P, Zoga E, Attenberger U, Schmeel LC. Total marrow irradiation versus total body irradiation using intensity-modulated helical tomotherapy. J Cancer Res Clin Oncol. 2023 Aug;149(9):5965-5973. doi: 10.1007/s00432-022-04565-2. Epub 2023 Jan 6. PMID: 36607428; PMCID: PMC10356893.

Mancosu P, Cozzi L, Muren LP. Total marrow irradiation for hematopoietic malignancies using volumetric modulated arc therapy: A review of treatment planning studies. Phys Imaging Radiat Oncol. 2019 Aug 29;11:47-53. doi: 10.1016/j.phro.2019.08.001. PMID: 33458277; PMCID: PMC7807866.

**16.2 Total Skin Electron Therapy**

**Question 212:**

Which one of the following is **NOT** a total skin electron therapy delivery technique?

A. Large electron fields technique

B. Patient shifting technique

C. Rotational techniques

D. Electron transport technique

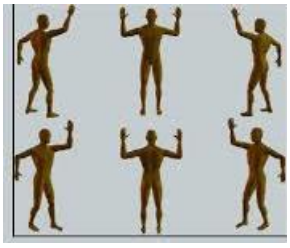
**Answer Question 212: D**

Electron transport is not a total skin electron technique. Electron transport is the transfer of electrons at the molecular level involved in oxidation-reduction reactions.

The total skin electron therapy techniques are:

Large electron fields:

Large electron fields (typically 3-7 MeV) are used with the patient standing and changing their position six times by 60 degrees over the course of a treatment session. The patient stands several meters from the accelerator allowing for large uniform electron fields. This is also known as the Stanford technique.



Patient shifting:

This involves the patient being moved to expose different skin areas to the radiation beam. This is used in the Stanford technique as described above.

Rotational:

This involves the patient being rotated while being irradiated with a single or multiple electron beams to ensure even skin coverage. This is also known as the McGill technique. The patient is rotated at constant speed around a vertical axis while a single or dual electron field delivers the radiation.

Total skin electron therapy is used for hematologic cancers of the skin such as mycosis fungoides.

References:

Piotrowski T, Milecki P, Skórska M, Fundowicz D. Total skin electron irradiation techniques: a review. Postepy Dermatol Alergol. 2013 Feb;30(1):50-5. doi: 10.5114/pdia.2013.33379. Epub 2013 Feb 20. PMID: 24278046; PMCID: PMC3834692.

Karzmark, C.J., et al. American Association of Physicists in Medicine Task Group 30. Total Skin Electron Therapy: Technique and Dosimetry. AAPM Report No. 23. Alexandria, VA: American Association of Physicists in Medicine; 1987. doi:10.37206/22.

Total skin electron radiation in the management of mycosis fungoides: Consensus of the European Organization for Research and Treatment of Cancer (EORTC) Cutaneous Lymphoma Project Group. J Am Acad Dermatol. 2002 Sep;47(3):364-70.

**Question 213:**

A patient undergoing total skin electron therapy (TSET) develops under coverage in dose on the soles of their feet. Which physics strategy is the most appropriate to correct for the under dosage to the soles of feet?

A. Increase the beam energy

B. Use a photon beam instead of electrons

C. Reduce the source surface distance

D. Apply boost fields to the soles

**Answer Question 213: D**

The majority of total skin electron therapy methods assume vertical positioning of the patient, which may result in under dosage of the soles of feet. Increasing the beam energy, using photons instead of electrons and reducing the source-surface distance with the total skin electron therapy technique used will not improve dose to the soles of feet. Typically, under dosage of the soles of feet are supplemented with additional radiation also known as boost fields. D is the correct answer.

References:

Karzmark, C.J., et al. American Association of Physicists in Medicine Task Group 30. Total Skin Electron Therapy: Technique and Dosimetry. AAPM Report No. 23. Alexandria, VA: American Association of Physicists in Medicine; 1987. doi:10.37206/22.

Piotrowski T, Milecki P, Skórska M, Fundowicz D. Total skin electron irradiation techniques: a review. Postepy Dermatol Alergol. 2013 Feb;30(1):50-5. doi: 10.5114/pdia.2013.33379. Epub 2013 Feb 20. PMID: 24278046; PMCID: PMC3834692.

**Question 214:**

A patient undergoing Total Skin Electron Therapy (TSET) presents with unexpected sparing in certain skin folds despite appropriate field setup and dosing. Which of the following best explains this occurrence and influences the next clinical action?

A. The patient has developed radiation resistance in specific regions. Treatment should be escalated in dose.

B. The oblique beam arrangement was insufficient. A switch to conventional photon therapy is required.

C. Skin folds prevent the full dose from the electron beam from reaching the skin surface underneath the fold. Targeted boost fields or supplementary techniques should be considered.

D. Skin folds naturally receive less dose due to increased vascularization. No additional treatment is needed.

**Answer Question 214: C**

Skin folds and other uneven body surfaces may be underdosed with electron beams. Common areas for under dosage include the perineum, axilla and skin folds. To address this, boost radiation to specific areas may be necessary.

Inframammary folds are usually underdosed in patients with pendulous breasts. While supplemental radiation could be given as a targeted boost, a breast sling may also be used. The sling suspends the breasts and allows direct exposure and improved dose deposition in the inframammary folds to the electron beams used in total skin electron therapy.

References:

Karzmark, C.J., et al. American Association of Physicists in Medicine Task Group 30. Total Skin Electron Therapy: Technique and Dosimetry. AAPM Report No. 23. Alexandria, VA: American Association of Physicists in Medicine; 1987. doi:10.37206/22.

King BJ, Deufel CL, Locher SE et a. A method to improve dose uniformity during total skin electron beam therapy in patients with pendulous breasts. Adv Radiat Oncol: 2(3):288-290, 2017.

**Question 215:**

Why is a large source-to-surface distance (SSD) commonly used in total skin electron therapy (TSET)?

A. To reduce treatment time

B. To decrease scatter radiation

C. To achieve dose uniformity across the body

D. To enhance penetration into deeper tissues

**Answer Question 215: C**

A large source-to surface distance is used in total skin electron therapy to achieve a larger, more uniform radiation field that can encompass the skin of the patient's body, reducing the need for multiple smaller electron fields. This approach helps minimize issues with smaller field electron field matching and provides a more homogeneous dose distribution across the skin surface.

References:

Piotrowski T, Milecki P, Skórska M, Fundowicz D. Total skin electron irradiation techniques: a review. Postepy Dermatol Alergol. 2013 Feb;30(1):50-5. doi: 10.5114/pdia.2013.33379. Epub 2013 Feb 20. PMID: 24278046; PMCID: PMC3834692.

Deufel CL, Antolak JA. Total skin electron therapy in the lying-on-the-floor position using a customized flattening filter to eliminate field junctions. J Appl Clin Med Phys. 2013 Sep 6;14(5):115-26. doi: 10.1120/jacmp.v14i5.4309. PMID: 24036864; PMCID: PMC5714577.

**Question 216:**

What is the typical energy range used in total skin electron therapy?

A. 1-2 MeV

B. 3-7 MeV

C. 10-12 MeV

D. 18-20 MeV

**Answer Question 216: B**

Total skin electron therapy typically uses electrons with energies between 3-7 MeV. This energy range is chosen because it can deliver the dose to a few millimeters of skin and subcutaneous tissue. Lower energy would not treat deep to cover skin and subcutaneous tissue. While higher energies would unnecessarily treat deeper normal tissues.

Reference:

Karzmark, C.J., et al. American Association of Physicists in Medicine Task Group 30. Total Skin Electron Therapy: Technique and Dosimetry. AAPM Report No. 23. Alexandria, VA: American Association of Physicists in Medicine; 1987. doi:10.37206/22.

**Question 217:**

Which one of the following areas are commonly shielded during total skin electron therapy (TSET)?

A. Lens

B. Teeth

C. Thyroid

D. Lungs

**Answer Question 217: A**

Lens eye shields, typically in the form of internal eye shields (lead placed on the globe which has been given anesthetic eye drops) are used to shield the lens. This will allow for treatment of the eyelids but will decrease the dose to the lens. However, not every patient will need internal eye shields.

Other shields which may or may not be used are fingernail/toenail shields. Some centres may shield radiation sensitive areas which are not involved with tumour (such as ears, mouth, perineum). In certain circumstances, if the scalp is not involved with tumour, the scalp may sometimes be shielded to prevent total hair loss.

Teeth, thyroid and lungs are not shielded with total skin electron therapy.

Reference:

Karzmark, C.J., et al. American Association of Physicists in Medicine Task Group 30. Total Skin Electron Therapy: Technique and Dosimetry. AAPM Report No. 23. Alexandria, VA: American Association of Physicists in Medicine; 1987. doi:10.37206/22.

https://www.hamiltonhealthsciences.ca/wp-content/uploads/2019/08/TotalSkinElectronBeamRadiation-th.pdf#:~:text=For%20a%20few%20days%2C%20small,the%20completion%20of%20the%20TSEB.

**16.3 Intraoperative Radiotherapy**

**Question 218:**

Which of the following radiation modalities are used in intraoperative radiotherapy (IORT)?

Choose the correct answer **A, B, C or D** below.

1. Orthovoltage

2. Electrons

3. Iridium-192 HDR brachytherapy

4. Electronic brachytherapy

**A. 1, 2, 3, 4**
**B. 1, 2, 3 only**
**C. 2, 3, 4 only**
**D. 3 and 4 only**

**Answer Question 218: A**

All are examples of radiation modalities used in IORT.

Electronic brachytherapy uses a miniaturized x-ray source, placed directly inside or close to the cancer target. Whereas traditional brachytherapy uses radioactive isotopes, electronic brachytherapy uses an electronic x-ray generator.

References:

Shah C. Intraoperative radiation and electronic brachytherapy in Brachytherapy: A look back and forward. Brachytherapy. 2023 Jan-Feb;22(1):4-5. doi: 10.1016/j.brachy.2022.08.003. PMID: 36725199.

Tom MC, Hepel JT, Patel R, Kamrava M, Badiyan SN, Cohen GN, Shah C. The American Brachytherapy Society consensus statement for electronic brachytherapy. Brachytherapy. 2019 May-Jun;18(3):292-298. doi: 10.1016/j.brachy.2018.10.006. Epub 2018 Nov 26. PMID: 30497939.

**16.4 Special Patient Considerations**

**Question 219:**

Based on AAPM TG-203, what cumulative dose to cardiac implantable electronic devices (pacemakers, defibrillators) are considered low risk for device failure, in a patient who is pacing independent?

A. <2Gy and use of < 10 MV photons

B. <5Gy and use of < 10 MV photons

C. < 1Gy and use of <10 MV photons (or any proton beams)

D. < 1Gy and use of <10 MV photons (or any neutron beams)

**Answer Question 219: A**

Malfunction to cardiac implantable electronic devices (CIED) is dependent on the following factors:

- CIED type, model, presence or absence of leads
- Presence of neutrons or other high linear energy transfer (LET) radiation, which may cause single event upsets (SEUs). SEUs occur when radiation interacts with electronic components of the CIED leading to malfunction.
- Cumulative radiation dose
- Patient's pacing dependency
- Dose-rat at the location of the CIED

According to AAPM TG-203, if the patient is pacing independent and if the dose to the CIED is < 2 Gy, the risk of malfunction to the pacemaker is low.

If >10 MV photon, proton, or neutron beams are used, the patient should be managed in the High-Risk category.

Reference:

Miften M, Mihailidis D, Kry SF, Reft C, Esquivel C, Farr J, Followill D, Hurkmans C, Liu A, Gayou O, Gossman M, Mahesh M, Popple R, Prisciandaro J, Wilkinson J. Management of radiotherapy patients with implanted cardiac pacemakers and defibrillators: A Report of the AAPM TG-203[†]. Med Phys. 2019 Dec;46(12):e757-e788. doi: 10.1002/mp.13838. Epub 2019 Nov 1. PMID: 31571229.

**Question 220:**

According to AAPM Task Group Report 36, what is the recommended maximum fetal radiation dose limit to minimize the risk of adverse biological effects?

A. 5 cGy

B. 50 cGy

C. 100 cGy

D. 200 cGy

**Answer Question 220: A**

Reference:

Stovall, M, et al. American Association of Physicists in Medicine. Fetal Dose from Radiotherapy with Photon Beams. Report No. 50. Madison, WI: AAPM; 1995. ISBN: 978-1-563964-53-4. doi: 10.37206/49.

**Question 221:**

According to AAPM Task Group Report 36, which of the following is a recommended strategy to reduce fetal dose during radiotherapy planning?

A. Use photon beam energy greater than 10 MV to improve penetration

B. Increase field size to distribute dose evenly

C. Rotate the collimator or use tertiary MLCs for better field definition

D. Include the abdomen in the treatment field for better accuracy

**Answer Question 221: C**

Only C is correct. All other options do not decrease fetal dose, but may do the opposite and increase fetal dose.

Reference:

Stovall, M, et al. American Association of Physicists in Medicine. Fetal Dose from Radiotherapy with Photon Beams. Report No. 50. Madison, WI: AAPM; 1995. ISBN: 978-1-563964-53-4. doi: 10.37206/49.

**16.5 Contemporary Radiopharmaceutical Therapy**

**Question 222.1:**

What is the radiation emitter for **Lutetium-177 (Lu-177) Prostate Specific Membrane Antigen (PSMA)?**

**A.** Gamma emitter
**B.** Positron emitter
**C.** Beta emitter
**D.** Alpha emitter
**E.** Auger emitter

**Answer Question 222.1: C**

Lu-177 PSMA is a beta emitter that has been used to treat metastatic prostate cancer.

**Question 222.2:**

What is the radiation emitter for **Radium-223 (Ra-223)?**

**A.** Gamma emitter
**B.** Positron emitter
**C.** Beta emitter
**D.** Alpha emitter
**E.** Auger emitter

**Answer Question 222.2:** D

Ra-223 is an alpha emitter that has been used to treat bone metastases from prostate cancer.

**Question 222.3:**

What is the radiation emitter for **Fluorine -18 flurodeoxyglucose ($^{18}$F- FDG)?**

**A.** Gamma emitter
**B.** Positron emitter
**C.** Beta emitter
**D.** Alpha emitter
**E.** Auger emitter

**Answer Question 222.3: B**

$^{18}$F-FDG is a positron emitter, it is used in PET imaging and not for cancer therapy.

Auger electron emitting radiopharmaceuticals (eg.$^{77}$Br) remain experimental and have not been used clinically in patients to treat cancer..

I-131 is a gamma and beta emitter that has been used to treat thyroid cancer.

References:

Sgouros, G., Bodei, L., McDevitt, M.R. *et al.* Radiopharmaceutical therapy in cancer: clinical advances and challenges. *Nat Rev Drug Discov* **19**, 589–608 (2020). https://doi.org/10.1038/s41573-020-0073-9

Bolcaen J, Gizawy MA, Terry SYA, Paulo A, Cornelissen B, Korde A, Engle J, Radchenko V, Howell RW. Marshalling the Potential of Auger Electron Radiopharmaceutical Therapy. J Nucl Med. 2023 Sep;64(9):1344-1351. doi: 10.2967/jnumed.122.265039. Epub 2023 Aug 17. PMID: 37591544; PMCID: PMC10478825.

**Question 223:**

Theranostic pairs in radiopharmaceutical therapy involve using one diagnostic agent for pre-therapy imaging and one corresponding therapeutic agent to deliver the dose. Which combination of imaging and therapeutic isotopes is commonly used together as a theranostic pair?

A. Iodine-131 for imaging and Fluorine-18 for therapy

B. Carbon-11 for imaging and Fluorine-18 for therapy

C. Yttrium-90 for imaging and Radium-223 for therapy

D. Gallium-68 for imaging and Lutetium-177 for therapy

**Answer Question 223: D**

Reference:

Sgouros, G., Bodei, L., McDevitt, M.R. et al. Radiopharmaceutical therapy in cancer: clinical advances and challenges. Nat Rev Drug Discov 19, 589–608 (2020). https://doi.org/10.1038/s41573-020-0073-9.

**Question 224:**

When selecting a radiopharmaceutical agent for targeting small clusters of cancer cells while minimizing damage to surrounding healthy tissue, which type of emitter is generally more appropriate?

A. Alpha emitter, due to its high LET and short range

B. Beta emitter, due to its longer range

C. Beta emitter, because it requires less shielding

D. Both are equally suitable

**Answer Question 224: A**

Reference:

Sgouros, G., Bodei, L., McDevitt, M.R. et al. Radiopharmaceutical therapy in cancer: clinical advances and challenges. Nat Rev Drug Discov 19, 589–608 (2020). https://doi.org/10.1038/s41573-020-0073-9.

**Question 225:**

Which imaging modality is commonly used to perform post-treatment delivery dosimetry in radiopharmaceutical therapy?

A. Magnetic Resonance Imaging (MRI)

B. Positron Emission Tomography (PET)

C. Single Photon Emission Computed Tomography (SPECT)

D. Computed Tomography (CT)

**Answer Question 225: C**

Reference:

Sgouros, G., Bodei, L., McDevitt, M.R. et al. Radiopharmaceutical therapy in cancer: clinical advances and challenges. Nat Rev Drug Discov 19, 589–608 (2020). https://doi.org/10.1038/s41573-020-0073-9.

**16.6 Adaptive Radiotherapy**

**Question 226:**

What is the key purpose of adaptive radiotherapy?

A. To adapt dose fall-off to normal tissue structures, so that the risk of toxicity is minimized

B. To increase to the dose to the target volumes as the target shrinks with radiotherapy

C. To adjust the radiation treatment plan during the course of therapy due to changes in patient anatomy or due to changes in tumour size, shape, position

D. To optimize the treatment plan, particularly for target conformity

**Answer Question 226: C**

Adaptive radiotherapy is used to adjust the radiation treatment plan during the course of therapy due to changes in patient anatomy or due to changes in tumour size, shape, position

Reference:

Glide-Hurst CK, Lee P, Yock AD, Olsen JR, Cao M, Siddiqui F, Parker W, Doemer A, Rong Y, Kishan AU, Benedict SH, Li XA, Erickson BA, Sohn JW, Xiao Y, Wuthrick E. Adaptive Radiation Therapy (ART) Strategies and Technical Considerations: A State of the ART Review From NRG Oncology. Int J Radiat Oncol Biol Phys. 2021 Mar 15;109(4):1054-1075. doi: 10.1016/j.ijrobp.2020.10.021. Epub 2020 Oct 24. PMID: 33470210; PMCID: PMC8290862.

**Question 227:**

Which of the following scenarios is least likely to benefit from adaptive radiotherapy (ART)?

A. Daily anatomical variation in the bladder

B. Rigid, immobile target with no nearby organs at risk

C. Spinal cord moving relative to a tumor target

D. Tumor regression in a head and neck case

**Answer Question 227: B**

Adaptive radiotherapy is best suited for clinical situations where there will be changes in the patient anatomy or changes in the tumour size, shape, position during a course of radiotherapy. As such, adaptive radiotherapy will not be beneficial for a rigid, immobile target with no nearby organs at risk.

Reference:

Glide-Hurst CK, Lee P, Yock AD, Olsen JR, Cao M, Siddiqui F, Parker W, Doemer A, Rong Y, Kishan AU, Benedict SH, Li XA, Erickson BA, Sohn JW, Xiao Y, Wuthrick E. Adaptive Radiation Therapy (ART) Strategies and Technical Considerations: A State of the ART Review From NRG Oncology. Int J Radiat Oncol Biol Phys. 2021 Mar 15;109(4):1054-1075. doi: 10.1016/j.ijrobp.2020.10.021. Epub 2020 Oct 24. PMID: 33470210; PMCID: PMC8290862.

**Question 228:**

You are creating an adaptive radiotherapy (ART) protocol for a patient with a central lung tumor showing significant respiratory motion. Which imaging modality should be utilized for optimal plan adaptation?

A. 4D cone-beam CT

B. Static MRI

C. Weekly fan-beam CT

D. Single pre-treatment PET/CT

**Answer Question 228: A**

4D cone-beam CT is the imaging modality used for lung radiotherapy adaptation. All other options are not.

Reference:

Glide-Hurst CK, Lee P, Yock AD, Olsen JR, Cao M, Siddiqui F, Parker W, Doemer A, Rong Y, Kishan AU, Benedict SH, Li XA, Erickson BA, Sohn JW, Xiao Y, Wuthrick E. Adaptive Radiation Therapy (ART) Strategies and Technical Considerations: A State of the ART Review From NRG Oncology. Int J Radiat Oncol Biol Phys. 2021 Mar 15;109(4):1054-1075. doi: 10.1016/j.ijrobp.2020.10.021. Epub 2020 Oct 24. PMID: 33470210; PMCID: PMC8290862.

**Question 229:**

What is a potential benefit of adaptive radiotherapy (ART) for pelvic treatments, like prostate cancer?

A) It increases the number of fractions required.

B) It allows dose escalation without increasing risk to bladder and rectum.

C) It avoids the need for bladder preparation.

D) It eliminates the need for imaging.

**Answer Question 229: B**

The use of pelvic adaptive radiotherapy allows for precise targeting to the tumour, allowing for dose escalation. By adapting dose to the position of the target and nearby organs at risk, the dose to organs at risk can be kept within safe dose contraints.

Reference:

Glide-Hurst CK, Lee P, Yock AD, Olsen JR, Cao M, Siddiqui F, Parker W, Doemer A, Rong Y, Kishan AU, Benedict SH, Li XA, Erickson BA, Sohn JW, Xiao Y, Wuthrick E. Adaptive Radiation Therapy (ART) Strategies and Technical Considerations: A State of the ART Review From NRG Oncology. Int J Radiat Oncol Biol Phys. 2021 Mar 15;109(4):1054-1075. doi: 10.1016/j.ijrobp.2020.10.021. Epub 2020 Oct 24. PMID: 33470210; PMCID: PMC8290862.

### 16.7 Artificial Intelligence and Automation

**Question 230:**

Which artificial intelligence algorithms are used clinically in radiation oncology? Choose answer **A, B, C or D** below.

1. Machine learning
2. Deep learning
3. Neural networks (CNN, U-Net)
4. Natural language processing

**A. 1, 2, 3 ,4**
**B. 3 only**
**C. None**
**D. 1 only**

**Answer Question 230: A**

All artificial intelligence algorithms are used in radiation oncology.

In particular, AI algorithms, particularly deep learning models such as U-Net are used for automated tumour and organ at risk segmentation for contouring.

Natural language processing is used when radiation oncologists dictate clinical notes which are transcribed by AI through voice recognition. In the future, voice commands may play a larger role in therapeutic radiation practice.

Reference:

Kawamura M, Kamomae T, Yanagawa M, et al. Revolutionizing radiation therapy: the role of AI in clinical practice, *Journal of Radiation Research*, Volume 65, Issue 1, January 2024, Pages 1–9,

# Chapter 17: Particle therapy

Yunjie Yang, May Tsao

**17.1 Particles**

**Question 231:**

What physical property do protons, neutrons and carbon ions share that make these ideal for radiation therapy?

A. Higher oxygen enhancement ratio

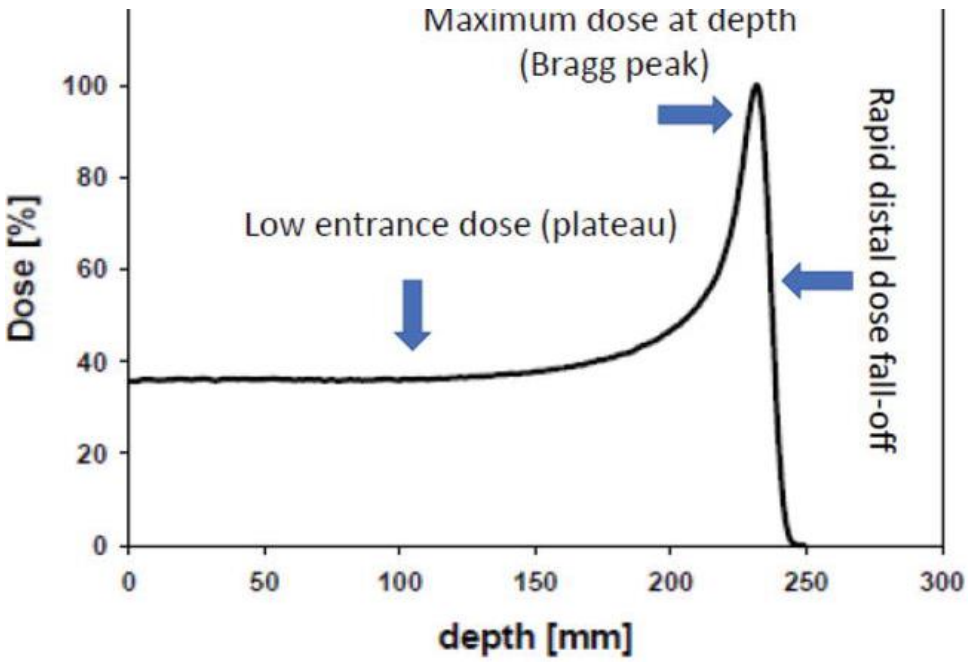
B. Lower linear energy transfer

C. Radiobiological effectiveness >5

D. Bragg peak

**Answer Question 231: D**

Particle therapy share the property of depositing a high dose of radiation at a specific depth (Bragg peak), with minimal dose to tissue superficial and deep to the Bragg peak.




Particles used in radiation therapy have **lower** oxygen enhancement ratio compared to photons.

Particle therapy is also associated with **higher** linear energy transfer compared to photons.

The radiobiological effectiveness (RBE) of particle therapy in clinical use ranges from 1.1-5 (**NOT >5**).

References:

Hazem R. Interaction of proton beam with human tissues in proton therapy [Internet]. Proton Therapy - Scientific Questions and Future Direction. IntechOpen; 2023. Available from: http://dx.doi.org/10.5772/intechopen.1003186

Byun HK, Han MC, Yang K, Kim JS, Yoo GS, Koom WS, Kim YB. Physical and Biological Characteristics of Particle Therapy for Oncologists. Cancer Res Treat. 2021;53(3):611-620. doi: 10.4143/crt.2021.066

**17.2 Radiobiology**

**Question 232:**

What is the Cobalt-Gray-Equivalent (CGE) of 10 Gy proton dose?

A. 20 CGE

B. 5 CGE

C. 11 CGE

D. 30 CGE

**Answer Question 232: C**

To calculate CGE for particle therapy, the following formula is used:

CGE = particle dose in Gy X RBE factor

RBE factor = radiobiological effectiveness factor = 1.1 for protons

CGE = 10 Gy x 1.1 = 11 CGE

| Radiation type | RBE |
|---|---|
| 250 kV x rays | 1 |
| MV x-rays | 1 |
| electrons | 0.9 |
| Protons | 1.1-1.5 |
| Carbon ions | 1.5-5 |
| Fast neutrons | 4-5 |

Reference:

Levin WP, Kooy H, Loeffler JS, DeLaney TF. Proton beam therapy. Br J Cancer. 2005 Oct 17;93(8):849-54. doi: 10.1038/sj.bjc.6602754. PMID: 16189526; PMCID: PMC2361650.

**Question 233:**

As the linear energy transfer increases from protons to heavy ions, what happens to the relative biological effectiveness and the oxygen enhancement ratio respectively?

A. no change, no change

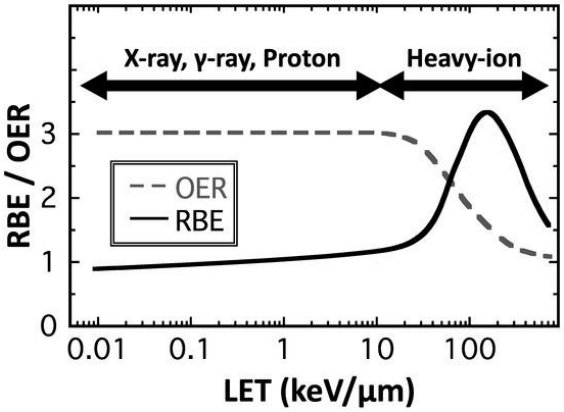
B. decreases, increases

C. increases, decreases

D. increases, no change

**Answer Question 233: C**

As LET increases from protons to heavy ions, RBE increases and the oxygen enhancement ratio (OER) decreases as shown in the figure below:




LET:

Heavy charged particles have high LET as they deposit energy more densely along their path compared to photons.

The Bragg peak at the end of the particle's range allows for precise targeting of tumour.

RBE and OER:

Heavy charged particles have higher RBE than photons (ie. heavy charged particles are more effective at producing biological effects for a given dose).

Heavy charged particles have lower OER, which means that the presence or lack of oxygen has less effect on biological effectiveness. As a result, theoretically heavy charged particles may be more effective at treating hypoxic tumours compared to photons.

Reference:

Mohan R, Held KD, Story MD, Grosshans D, Capala J. Proceedings of the National Cancer Institute Workshop on Charged Particle Radiobiology. Int J Radiat Oncol Biol Phys. 2018 Mar 15;100(4):816-831. doi: 10.1016/j.ijrobp.2017.12.260. Epub 2017 Dec 21. Erratum in: Int J Radiat Oncol Biol Phys. 2018 Jul 15;101(4):1004. doi: 10.1016/j.ijrobp.2018.04.004. PMID: 29485053.

**17.3 Accelerators**

**Question 234:**

What are the different types of particle accelerators used for protons, neutrons and carbon ions?

A. Cyclotrons or Synchotrons

B. Linear accelerator (LINAC)

C. E-linac

D. Magnetron

**Answer Question 234: A**

Cyclical particle accelerators are used to guide charged particles in a circular closed path, accelerating the particles in electric fields.

The two main types of particle accelerators are cyclotrons and synchrotrons.

Cyclotrons:

Path: spiral (centre to outwards)

Magnetic field: constant to bend particles in a circular path

Acceleration: Radiofrequency fields are used to accelerate the particles. Cyclotrons cannot accelerate particles from standstill. Another accelerator is needed to bring the particles up to speed.

Synchrotrons:

Path: circular, oval or polygon

Magnetic field: increases as the particles' speed increases

Acceleration: Radiofrequency cavities placed around the ring accelerate the particles Synchrotrons cannot accelerate particles from standstill. Another accelerator is needed to bring the particles up to speed.

Linear accelerators are for photons and electrons (not protons, neutrons nor carbon ions).

E- linac are electron linear accelerators.

Magnetrons produce microwaves.

References:

Coutrakon GB. Accelerators for heavy-charged-particle radiation therapy. Technol Cancer Res Treat. 2007 Aug;6(4 Suppl):49-54. doi: 10.1177/15330346070060S408. PMID: 17668952.

**17.4 Beam Delivery and Transport**

**Question 235**:

Which of the following is **true** for scattered proton therapy systems?

A. Scattered proton therapy uses intentional scattering over a large area to cover the radiation target.

B. Scattered proton therapy uses a focused beam that is re-positioned to cover the radiation target.

C. Scattered proton therapy uses pencil beam scanning.

D. Scattered proton therapy uses a heavy metal target to scatter protons.

**Answer Question 235: A**

For proton therapy, there are 2 ways to deliver the proton beam to the radiation target.

Scattered proton therapy, the proton beam is initially focused and then intentionally scattered using range compensators to distribute radiation over a larger area to conform to the radiation target shape.

Scanned proton therapy uses pencil beam scanning with magnets to precisely move the narrow proton beam across the target creating a more conformal dose distribution.

References:

DeLaney TF. Proton therapy in the clinic. Front Radiat Ther Oncol. 2011;43:465-485. doi: 10.1159/000322511. Epub 2011 May 20. PMID: 21625169.

Chuong M, Badiyan SN, Yam M, Li Z, Langen K, Regine W, Morris C, Snider J 3rd, Mehta M, Huh S, Rutenberg M, Nichols RC. Pencil beam scanning versus passively scattered proton therapy for unresectable pancreatic cancer. J Gastrointest Oncol. 2018 Aug;9(4):687-693. doi: 10.21037/jgo.2018.03.14. PMID: 30151265; PMCID: PMC6087865.

**Question 236:**

Which device is primarily responsible for achieving lateral beam spreading in a pencil beam scanning proton therapy system?

A. Energy degrader
B. Modulator wheel
C. Scanning magnets
D. Range shifter

**Answer Question 236: C**

The scanning magnets are dipole magnets that can precisely steer the proton pencil beam laterally to point the beam to the desired location. Therefore, the correct answer is C.

Energy degrader, often found in a cyclotron-based proton system, produces the desired beam energy to produce modulation in the depth direction. While the range shifter does scatter the beam, its intended purpose is to further reduce the proton range. As such, the range shifter is not the primary device responsible for lateral beam spreading in a pencil beam scanning system. Modulator wheel is not typically found in a pencil beam scanning system, and its purpose is also to modulate the proton beam range.

Reference:

DeLaney TF. Proton therapy in the clinic. Front Radiat Ther Oncol. 2011;43:465-485. doi: 10.1159/000322511. Epub 2011 May 20. PMID: 21625169.

**17.5 Treatment plannning**

**Question 237:**

What is the OTV in proton therapy planning?

A. Optical target volume

B. Overlap target volume

C. Optimization target volume

D. Over range target volume

**Answer Question 237: C**

In proton therapy, OTV is the optimization of the target volume. The OTV is slightly larger than the Planning Target Volume (PTV) to account for both proton set-up uncertainties but also proton range uncertainties. OTV is used in the optimization process to ensure adequate dose delivery to the target, while minimizing dose to normal structures. For example multi-field optimization intensity-modulated proton therapy (MFO-IMPT) optimizes the dose distribution from multiple proton beams to create a highly conform dose distribution to the OTV.

Reference:

Han D, Hooshangnejad H, Chen CC, Ding K. A Beam-Specific Optimization Target Volume for Stereotactic Proton Pencil Beam Scanning Therapy for Locally Advanced Pancreatic Cancer. Adv Radiat Oncol. 2021 Jul 29;6(6):100757. doi: 10.1016/j.adro.2021.100757. PMID: 34604607; PMCID: PMC8463829.

# Chapter 18: Stereotactic radiosurgery/stereotactic body therapy

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**18.1 Clinical Aspects of SRS/SBRT**

**Question 238:**

Which of the following is **not** a characteristic of stereotactic radiosurgery (SRS) or stereotactic body radiation therapy (SBRT), as compared to conventional IMRT given in 1.8-2 Gy daily fractions?

A. Larger dose per fraction delivered in fewer fractions
B. Smaller planning target volume (PTV) margin
C. Small field dosimetry quality assurance
D. Less steep dose gradients

**Answer Question 238: D**

All these are characteristics of SRS or SBRT that distinguish it from conventionally fractionated IMRT, except for less steep dose gradients.

SRS or SBRT have more steep dose gradients as compared to conventional IMRT

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 239:**

Which one of the following is **NOT** an RTOG 90 05 stereotactic radiosurgery prescription for brain metastases?

A. 24 Gy (<= 2 cm in diameter)
B. 18 Gy (>2cm but <= 3 cm)
C. 15 Gy (> 3 cm but <= 4cm)
D. 10 Gy (>4 cm)

**Answer Question 239: D**

All the above are correct except for D.

For brain metastases larger than 4 cm, stereotactic radiosurgery in 1 fraction may be too toxic (such as radiosurgery induced edema, radionecrosis) and fractionated stereotactic radiotherapy may be preferred.

References:

Shaw E, Scott C, Souhami L, et al. Single dose radiosurgical treatment of recurrent previously irradiated primary brain tumors and brain metastases: final report of RTOG protocol 90-05. Int J Radiat Oncol Biol Phys 2000; 47: 291–98.

Wegner RE, Leeman JE, Kabolizadeh P, Rwigema JC, Mintz AH, Burton SA, Heron DE. Fractionated stereotactic radiosurgery for large brain metastases. Am J Clin Oncol. 2015 Apr;38(2):135-9. doi: 10.1097/COC.0b013e31828aadac. PMID: 23563213.

**Question 240:**

Which one of the following lung stereotactic body radiotherapy (SBRT) prescriptions should **NOT** be used for patients with tumours close to the central airways (within 2 cm of the proximal bronchial tree)?

A. 50 Gy in 5 fractions (1 fraction per day)
B. 60 Gy in 8 fractions (1 fraction per day)
C. 63 Gy in 8 fractions (1 fraction per day)
D. 60 Gy in 3 fractions (1 fraction per day)

**Answer Question 240: D**

The publication by Timmerman et al (cited below), reported a high rate of grade 5 toxicity for centrally located lung cancer SBRT targets when treated with 60 Gy in 3 fractions. Only 54% of patients with central tumours had a 2 year freedom from severe toxicity (grades 3-5) with the use of 60 Gy in 3 fractions of lung SBRT. As such this lung SBRT was **not** recommended for central lung targets.

All other lung SBRT prescriptions listed have been reported to be safe (low risk of toxicity) and effective.

Generally, a biological effective dose (BED) with alpha/beta10 of at least 100 is required for lung SBRT to be considered effective for local control.

References:

Timmerman R, McGarry R, Yiannoutsos C *et al* Excessive toxicity when treating central tumors in a phase II study of stereotactic body radiation therapy for medically inoperable early-stage lung cancer. J Clin Oncol 2006; 24: 4833–9.

Bezjak A, Paulus R, Gaspar LE, Timmerman RD, Straube WL, Ryan WF, Garces YI, Pu AT, Singh AK, Videtic GM, McGarry RC, Iyengar P, Pantarotto JR, Urbanic JJ, Sun AY, Daly ME, Grills IS, Sperduto P, Normolle DP, Bradley JD, Choy H. Safety and Efficacy of a Five-Fraction Stereotactic Body Radiotherapy Schedule for Centrally Located Non-Small-Cell Lung Cancer: NRG Oncology/RTOG 0813 Trial. J Clin Oncol. 2019 May 20;37(15):1316-1325. doi: 10.1200/JCO.18.00622. Epub 2019 Apr 3. PMID: 30943123; PMCID: PMC6524984.

Ahmed N, Hasan S, Schumacher L, Colonias A, Wegner RE. Stereotactic body radiotherapy for central lung tumors: Finding the balance between safety and efficacy in the "no fly" zone. Thorac Cancer. 2018 Oct;9(10):1211-1214. doi: 10.1111/1759-7714.12764. Epub 2018 Aug 10. PMID: 30095228; PMCID: PMC6166089.

Haasbeek CJ, Lagerwaard FJ, Slotman BJ, Senan S. Outcomes of stereotactic ablative radiotherapy for centrally located early-stage lung cancer. J Thorac Oncol 2011; 6: 2036–43.

**Question 241:**

Which of the following does NOT provide organ at risk constraints for stereotactic radiosurgery or stereotactic body radiotherapy?

A. HyTEC
B. AAPM TG-101
C. QUANTEC
D. Emami dose constraints

**Answer Question 241: D**

All listed provide organs at risk constraints for SRS and SBRT, except for Emami. Emami's dose constraints were listed for conventionally fractionated radiation therapy (1.8 Gy – 2 Gy daily fractions) and not for SRS, SBRT.

References:

Emami B, Lyman J, Brown A, Coia L, Goitein M, Munzenrider JE, Shank B, Solin LJ, Wesson M. Tolerance of normal tissue to therapeutic irradiation. Int J Radiat Oncol Biol Phys. 1991 May 15;21(1):109-22. doi: 10.1016/0360-3016(91)90171-y. PMID: 2032882.

Grimm J, Marks LB, Jackson A, Kavanagh BD, Xue J, Yorke E. High Dose per Fraction, Hypofractionated Treatment Effects in the Clinic (HyTEC): An Overview. Int J Radiat Oncol Biol Phys. 2021 May 1;110(1):1-10. doi: 10.1016/j.ijrobp.2020.10.039. PMID: 33864823; PMCID: PMC9447432.

Benedict SH, Yenice KM, Followill D, Galvin JM, Hinson W, Kavanagh B, Keall P, Lovelock M, Meeks S, Papiez L, Purdie T, Sadagopan R, Schell MC, Salter B, Schlesinger DJ, Shiu AS, Solberg T, Song DY, Stieber V, Timmerman R, Tomé WA, Verellen D, Wang L, Yin FF. Stereotactic body radiation therapy: the report of AAPM Task Group 101. Med Phys. 2010 Aug;37(8):4078-101. doi: 10.1118/1.3438081. Erratum in: Med Phys. 2012 Jan;39(1):563. Dosage error in article text. Erratum in: Med Phys. 2023 Jun;50(6):3885. doi: 10.1002/mp.16159. PMID: 20879569.

Marks LB, Yorke ED, Jackson A, Ten Haken RK, Constine LS, Eisbruch A, Bentzen SM, Nam J, Deasy JO. Use of normal tissue complication probability models in the clinic. Int J Radiat Oncol Biol Phys. 2010 Mar 1;76(3 Suppl):S10-9. doi: 10.1016/j.ijrobp.2009.07.1754. PMID: 20171502; PMCID: PMC4041542.

Bisello S, Cilla S, Benini A, Cardano R, Nguyen NP, Deodato F, Macchia G, Buwenge M, Cammelli S, Wondemagegnehu T, Uddin AFMK, Rizzo S, Bazzocchi A, Strigari L, Morganti AG. Dose-Volume Constraints fOr oRganS At risk In Radiotherapy (CORSAIR): An "All-in-One" Multicenter-Multidisciplinary

Practical Summary. Curr Oncol. 2022 Sep 27;29(10):7021-7050. doi: 10.3390/curroncol29100552. PMID: 36290829; PMCID: PMC9600677.

**Question 242:**

Which one of the following regimens allows the highest max dose to spinal cord?

A. Single fraction SRS
B. Three fractions SRS
C. Five fractions SBRT
D. 2Gy/fraction conventional treatment

**Answer Question 242: D**

Rationale: Generally, the dose tolerance for OARs increases with the increase in fraction numbers. According to AAPM TG101's table III, the limit of max dose to spinal cord is 14, 21.9, and 30 Gy for one fraction, three fractions and five fractions SRS/SBRT regimen, respectively. And according to Quantec's dose limit, the max dose for conventional treatment is 50 Gy (0.2% rate of Myelopathy).

Reference:

Benedict, Stanley H., et al. "Stereotactic body radiation therapy: the report of AAPM Task Group 101." Medical physics 37.8 (2010): 4078-4101.

Marks LB, Yorke ED, Jackson A, Ten Haken RK, Constine LS, Eisbruch A, Bentzen SM, Nam J, Deasy JO. Use of normal tissue complication probability models in the clinic. Int J Radiat Oncol Biol Phys. 2010 Mar 1;76(3 Suppl):S10-9. doi: 10.1016/j.ijrobp.2009.07.1754. PMID: 20171502; PMCID: PMC4041542.

**Question 243:**

Which of the following represents the RTOG definition of the conformity index (CI), as used to evaluate brain SRS plans?

A. volume covered by prescription dose / target volume
B. target volume/volume covered by prescription dose
C. volume covered by 50% prescription dose / target volume
D. maximum dose / prescription dose

**Answer Question 243: A**

The RTOG defines conformity index as the [volume covered by prescription dose / target volume]. Ideal conformity index for brain SRS plans are between 1.0 to 2.0. There are alternative definitions of conformity index, such as the Paddick and SALT-Lomax conformity index, but the RTOG definition is

widely accepted. Answer C describes the gradient index, and answer D describes the homogeneity index, which are other measures that can be used to evaluate SRS conformity.

Reference:

Stanley J, Breitman K, Dunscombe P, Spencer DP, Lau H. Evaluation of stereotactic radiosurgery conformity indices for 170 target volumes in patients with brain metastases. J Appl Clin Med Phys. 2011 Feb 1;12(2):3449. doi: 10.1120/jacmp.v12i2.3449. PMID: 21587193; PMCID: PMC5718686.

**18.2 Cranial Stereotactic systems**

**Question 244:**

Which of the following statements is true about Gamma Knife radiosurgery?

A. Treatment is administered by gamma rays, produced by beta plus decay
B. Dose is typically prescribed to the 70% isodose line.
C. Patients can be treated via a frame-based or frameless approach
D. The machine consists of hundreds of radioactive Cobalt-59 sources

**Answer Question 244: C**

A. Cobalt-60 is a radioactive isotope with a half-life of 5.27 years that decays into nickel-60 via beta minus decay. It emits an electron (beta particle) and an antineutrino. Two photons emitted from the nucleus (gamma rays) have energies of 1.17 and 1.33 MeV; these gamma rays are used for therapeutic purposes in Gamma Knife radiosurgery.
B. Dose is typically prescribed to the 50% isodose line.
C. Recent versions of the Gamma Knife delivery device allow for frameless treatment. Traditionally, all treatments were performed with a rigid stereotactic frame, which is fixed with pins to a patient's skull, providing immobilization and target localization.
D. Cobalt-59 is a stable isotope, which is bombarded with neutrons to produce Cobalt-60, which is used for radiotherapy. The gamma knife consists of hundreds of Cobalt-60 sources (not Cobalt-59).

Reference :

Vulpe H, Save AV, Xu Y, et al. Frameless stereotactic radiosurgery on the Gamma Knife Icon: early experience from 100 patients. Neurosurgery. 2020; 86(4): 509-516.

**Question 245:**

Which of the following are characteristics of non-coplanar beams, often used in stereotactic radiosurgery?

A. Improve conformality
B. Allow for steep dose gradients (rapid dose fall-off)

C. Introduce additional setup uncertainty
D. All of the above
E. A and B only

**Answer Question 245: D**

Non-coplanar beams refer to beams that treat the same target (same isocenter) but do not share the same geometric plane relative to the patient. In conventional C-arm LINACs, this is achieved by rotating the couch (i.e., couch kick) so that the beam can enter the patient from different geometric planes.

These beams have been increasingly utilized in planning stereotactic radiotherapy due to their ability to improve conformality and allow for more rapid dose fall-off, generating steeper dose gradients. One disadvantage of noncoplanar beams is that couch rotation of the LINAC must be very carefully checked, as this can introduce an additional degree of uncertainty.

References:

Clark GM, Popple RA, Prendergast BM, Spencer SA, Thomas EM, Stewart JG, Guthrie BL, Markert JM, Fiveash JB. Plan quality and treatment planning technique for single isocenter cranial radiosurgery with volumetric modulated arc therapy. Pract Radiat Oncol. 2012 Oct-Dec;2(4):306-313. doi: 10.1016/j.prro.2011.12.003. Epub 2012 Jan 30. PMID: 24674169.

Smyth G, Evans PM, Bamber JC, Bedford JL. Recent developments in non-coplanar radiotherapy. Br J Radiol. 2019 May;92(1097):20180908. doi: 10.1259/bjr.20180908. Epub 2019 Feb 1. PMID: 30694086; PMCID: PMC6580906.

**Question 246:**

How is CyberKnife therapy performed?

A. A linear accelerator is directly mounted onto a robot to deliver 4MV, coplanar x-rays.
B. Patients are immobilized with a stereotactic frame, while the linear accelerator moves with 6-degrees-of-freedom translation during treatment delivery.
C. The respiratory tracking system enables the beam to be on only when the target is within a prespecified position.
D. Treatment is delivered under image-guidance, using orthogonal x-rays.

**Answer Question 246: D**

A. The robotic manipulator removes the restrictions of beam geometry by allowing for the delivery of independently targeted noncoplanar fields with 6MV x-rays.
B. No stereotactic frame is required during Cyberknife treatment. Patient and/or target movement during treatment is addressed by the ability of the beam to move with 6-degrees-of freedom on the robotic arm, as well as the available tracking systems.

C. Tracking involves localizing and following the target in real time, and the beam is on at all times, allowing for more efficient treatment delivery. Gating involves the beam turning on and off depending on the position of the target, and can result in longer treatment times.
D. The tumor position is determined at the start of treatment via x-ray image guidance, and the patient is aligned using an adjustable treatment table. During treatment, the image guidance system determines additional corrections needed to align the treatment beam to the target, if there is movement. Then, the linear accelerator is repositioned by the robot to align the treatment beam to the patient.

Reference:

Kilby W, Dooley JR, Kuduvalli G, Sayeh S, Maurer CR. The CyberKnife® Robotic Radiosurgery System in 2010. Technology in Cancer Research & Treatment. 2010;9(5):433-452.

**Question 247:**

Which of the following is a potential advantage of cone-based radiosurgery compared to multileaf collimators (MLCs) for small targets?

A. Cone collimators decrease geometric penumbra compared to MLCs.
B. Duration of treatment when using cone collimators is always shorter than with MLCs.
C. Cone collimators have a lower risk of collision than MLCs.
D. Cone collimators improve target coverage (D95%) compared to MLCs.

**Answer Question 247: A**

A. A characteristic feature of cone collimators is the sharper geometric penumbra, resulting in a steep dose fall-off at the tumor margin, which is especially desirable for radiosurgery.
B. Treatment duration is often longer when using cone collimators, especially if multiple isocenters are employed.
C. As cone collimators can be attached to linear accelerators and protrude from the treatment head, the gantry clearance space is smaller, leading to an increased risk of collision.
D. The type of collimation does not influence target coverage (D95%); cone and multileaf collimators can produce similar target coverage.

Reference:

Jang SY, Lalonde R, Ozhasoglu C, Burton S, Heron D, Huq MS. Dosimetric comparison between cone/Iris-based and InCise MLC-based CyberKnife plans for single and multiple brain metastases. J Appl Clin Med Phys. 2016;17:184–99.

**Question 248:**

Which of the following statements is **true** regarding linear accelerator (LINAC) vs. GammaKnife-based SRS for brain metastases?

A. Local control is superior with GammaKnife-based SRS
B. LINAC-based SRS typically results in sharper dose fall-off compared to GammaKnife-based
C. GammaKnife-based SRS typically results in sharper dose fall-off compared to LINAC-based
D. LINAC-based SRS typically takes longer than GammaKnife-based SRS

**Answer Question 248: C**

GammaKnife-based SRS typically results in sharper dose fall-off compared to LINAC-based SRS. In GammaKnife, the prescription dose is typically prescribed to the 50% isodose line, whereas in LINAC-based SRS, the prescription dose is typically prescribed to the 80% isodose line. Hence, dose fall-off is sharper for GammaKnife.

There is no significant difference in local control between the two modalities, based on data from a meta-analysis comparing outcomes across the two modalities. Regarding timing, LINAC-based SRS is typically much faster compared to GammaKnife, especially as the number of targets increases. This is in part because with GammaKnife, each target is treated using a unique isocenter (if not multiple isocenters), whereas LINAC-based SRS allows for multiple targets to be treated with the same isocenter.

References:

Chea, M., Fezzani, K., Jacob, J. *et al.* Dosimetric study between a single isocenter dynamic conformal arc therapy technique and Gamma Knife radiosurgery for multiple brain metastases treatment: impact of target volume geometrical characteristics. *Radiat Oncol* **16**, 45 (2021). https://doi.org/10.1186/s13014-021-01766-w

Dharnipragada, R., Dusenbery, K., Watanabe, Y. *et al.* Comparison of Gamma Knife (GK) and Linear Accelerator (LINAC) radiosurgery of brain metastasis resection cavity: a systematic review and proportional meta-analysis. *Clin Exp Metastasis* **41**, 1–8 (2024). https://doi.org/10.1007/s10585-023-10240-8

**Question 249:**

In Gamma Knife (GK) SRS treatment, magnetic resonance (MR) images are often used in treatment planning. However, MR images are known to have image distortions. Why are MR images still used most of the time in such high accuracy SRS treatment?

A. The modern MR scanner has very high precision and the distortion in images is negligible.
B. MR localizer is mounted to the stereotactic frame in MR scan for GK treatment to guarantee the accuracy.
C. CT images are known to have minimal or no distortion, and they are often registered with MR images for Gamma Knife treatment.
D. The distortion in MR images is ignored in GK SRS treatment.

**Answer Question 249: B**

Rationale:

Anatomic distortion is present in all MRI data due to non-linearity of gradient fields and measures up to several millimeters.

A. Although distortion correction maps might be used in modern MR scanner to reduce the distortion in images, the distortion is still non-negligible especially when the images are further away from the center.
B. MR localizer has fiducial rods whose locations are precisely known with respect to the frame, thus allowing the localization of any point within the MRI image.
C. CT images are not always requested in GK treatment.
D. The distortion in MR images cannot be ignored.

References:

Seibert, Tyler M., et al. "Distortion inherent to magnetic resonance imaging can lead to geometric miss in radiosurgery planning." *Practical radiation oncology* 6.6 (2016): e319-e328.

Huang, Ke Colin, et al. "Phantom-based characterization of distortion on a magnetic resonance imaging simulator for radiation oncology." *Physics in Medicine & Biology* 61.2 (2016): 774.

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**18.3 Extracranial**

**Question 250:**

Which of the following is **NOT** used in lung stereotactic body radiotherapy to account for motion or to manage motion?

A. 4D-CT simulation
B. breath hold
C. abdominal compression
D. fiducial markers
E. helical CT simulation

**Answer Question 250: E**

All methods could be used to account for motion with lung SBRT, except for helical CT simulation. Helical (or spiral) CT simulation creates a detailed 3D images of the patient's anatomy for radiation therapy planning but does not incorporate any motion management strategies.

4D –CT simulation uses CT simulation to capture images of the lung and surrounding tissues during multiple phases of the breathing cycle to take into account tumor motion. An Internal Target Volume (ITV) is generated, to ensure that adequate dose is delivered to a moving target.

Breath hold techniques minimize breathing motion. Different types of breath hold techniques include:

Deep inspiration breath hold (DIBH):

- A patient take a deep breath allowing the lungs to expand. The patient then holds their breath for a specified duration (eg. 20-30 seconds)
- DIBH can move the heart away from the high dose radiation field (eg. for left sided breast cancer or for mediastinal lymphoma)
- DIBH can be voluntary (patient controls breathing) or assisted (eg. Active Breathing Co-ordinator-ABC device)

End expiration breath hold (EEBH):

- A patient holds their breath at the end of exhalation
- EEBH may reduce tumour motion and decrease radiation exposure to healthy tissue. EEBH is not used as frequently as DIBH. EEBH may be useful for some lower esophageal cancers.

Abdominal compression helps reduce breathing motion, and thereby reduce the motion of the lung cancer SBRT target.

Lung SBRT target fiducials have also been used to guide SBRT targeting (eg. with cyberknife).

References:

Caillet V, Booth JT, Keall P. IGRT and motion management during lung SBRT delivery. Phys Med. 2017 Dec;44:113-122. doi: 10.1016/j.ejmp.2017.06.006. Epub 2017 Jun 21. PMID: 28647449.

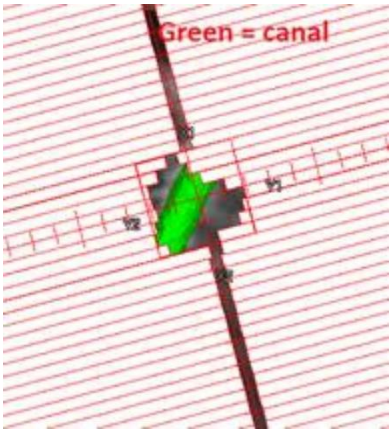
Moran DE, Parikh M, Sheiman RG, Brook OR, Sun MRM, Mahadevan A, Siewert B. Comparison of technical success and safety of transbronchial versus percutaneous CT-guided fiducial placement for SBRT of lung tumors. J Med Imaging Radiat Sci. 2021 Sep;52(3):409-416. doi: 10.1016/j.jmir.2021.06.004. Epub 2021 Jul 3. PMID: 34229986.

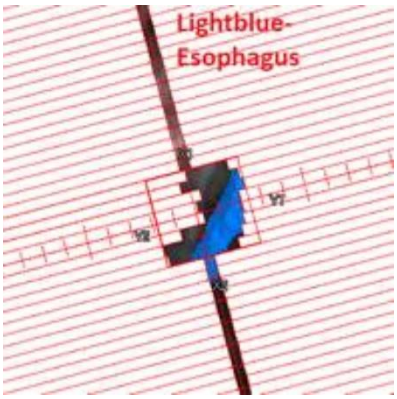
**Question 251:**

This 75 year old man was treated with lung stereotactic body radiation therapy (SBRT) to his stage 1 squamous cell lung cancer of the right upper lobe.

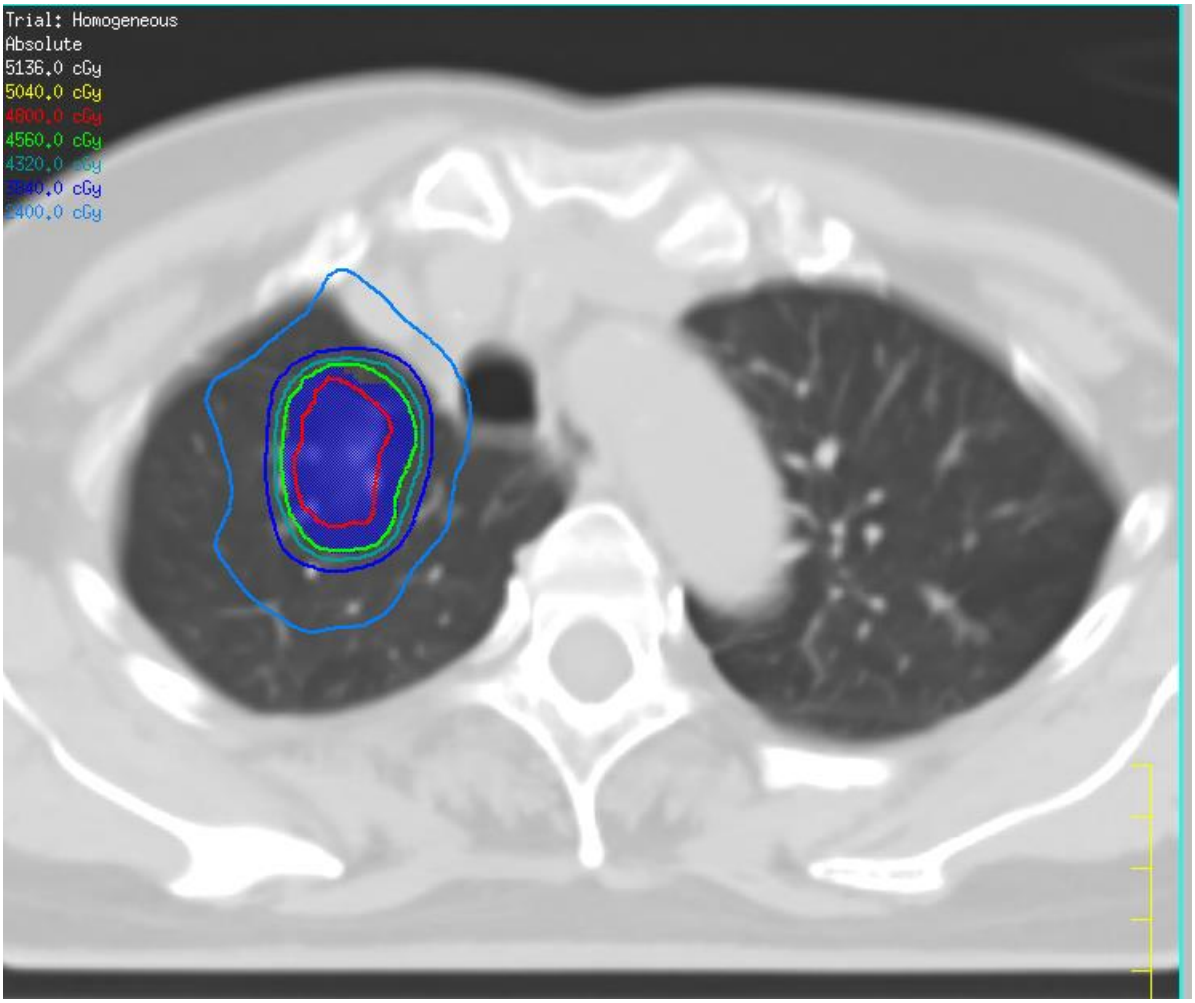
The dosimetry below shows the SBRT plan ***without*** heterogeneity tissue correction. The blue colorwash is the planning target volume (PTV). 48 Gy was prescribed in 4 fractions given every other day.

There are some SBRT segments which pass through the spinal canal and esophagus as shown below. **The heart is completely outside the SBRT segments. In this plan, the dose to the heart was only from scattered radiation as there was no direct entrance or exit dose through the heart.**




**WITHOUT** tissue heterogeneity correction:




**Question 251.1**:

When tissue heterogeneity is turned on, does PTV V95%:

**A.** Increase
**B.** Decrease

**Answer Question 251.1: B**

**PTV V95% decrease-** the volume of PTV that receives 95% of the prescription dose (green isodose line) is decreased due to increase in penumbra in lung (with tissue heterogeneity correction). The only change is turning the tissue correction in the plan "on". The beams and monitor units for the SBRT plan has not changed. This results in under-coverage of the target. In order to account for the lower lung density, the segments in the beams or arcs would need to be larger to account for the increase in penumbra in lung.

**Question 251.2:**

When tissue heterogeneity is turned on, does PTV Dmax:

**A.** Increase
**B.** Decrease

**Answer Question 251.2: A**

**PTV Dmax increase-** The Compton effect is the primary therapeutic radiation-matter interaction which is influenced largely by electron density. With tissue heterogeneity turned on there is less attenuation of the primary photons entering lung tissue (as lung is less dense than soft tissue). The less attenuated beam in lung results in higher Dmax in PTV.

**Question 251.3:**

When tissue heterogeneity is turned on, does ITV D99%:

**A.** Increase
**B.** Decrease

**Answer Question 251.3: B**

**ITV D99% decrease;** The dose that 99% of the ITV gets is reduced (with tissue heterogeneity correction). Because the penumbra is larger in lung, the higher isodose lines (eg 90% IDL and higher) are contracted. The target is under-covered with tissue heterogeneity turned on.

**Question 251.4:**

When tissue heterogeneity is turned on, does PTV D95%:

**A.** Increase
**B.** Decrease

**Answer Question 251.4: B**

**PTV D95% decrease;** The dose that 95% of the PTV receives gets reduced with tissue heterogeneity correction due to the increase in penumbra in lung

**Question 251.5:**

When tissue heterogeneity is turned on does spinal cord, Dmax and esophagus, Dmax:

**A.** Increase
**B.** Decrease

**Answer Question 251.5: A**

**Spinal cord, Dmax and Esophagus Dmax**- **increase**; There are two competing effects playing here.

First, when heterogeneity correction is on, there are more primary photons reaching various organs, due to lower lung density.

Second, when heterogeneity correction is on, there are less scattered photons, due to lower lung density (the less dense the tissue is, the less it is going to scatter)

The final result depends on which effect predominates.
In the case of canal and esophagus, some VMAT beam segments deliver dose directly to those organs. The first effect dominates and turning on inhomogeneity correction increases dose to both organs.

**Question 251.6:**

When tissue heterogeneity is turned on does heart, dmax:

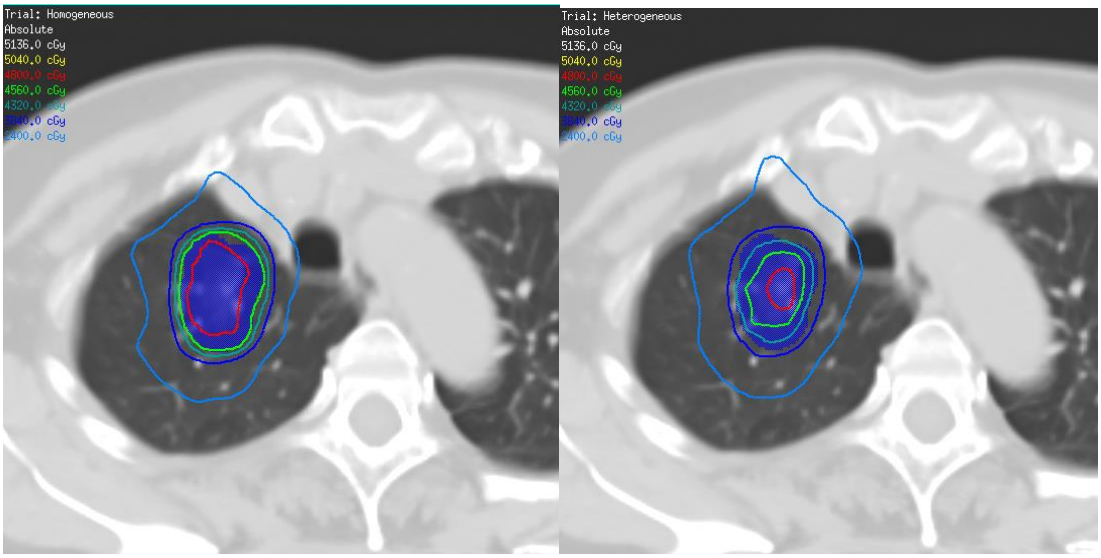
**A.** Increase
**B.** Decrease

**Answer Question 251.6: B**

**Heart Dmax- decrease:** on the other hand, the heart is located way outside of the primary beam. All the dose delivered to the heart is due to scattered photons, which is the predominant effect. In lung tissue, there is less scattered photons due to lower lung density compared to tissue density. There is no primary beam (no segments) going through the heart. So, when heterogeneity correction is turned on, max point dose to the heart decreases.

Dosimetry without tissue correction (LEFT) compared to heterogeneity/tissue correction turned on (RIGHT) below:

**WITHOUT** tissue correction **WITH** tissue correction




| | Homogeneous (no tissue correction) | Heterogeneous (tissue correction turned on) |
|---|---|---|
| Lung-ITV V20 | 3.70% | 3.73% |
| PTV, Dmax | 4952 cGy | 4992 cGy |
| ITV D99% | 4809 cGy | 4375 cGy |
| PTV V95% | 99.80% | 29.84% |
| PTV D95% | 4647 cGy | 4074 cGy |
| Spinal cord, Dmax | 303.8 cGy | 335.4 cGy |
| Heart, Dmax | 63.6 cGy | 59.5 cGy |
| Esophagus, Dmax | 1779 cGy | 2000 cGy |
| Esophagus, V1880cGy | 0 cc | 0.52 cc |

References:

Herman Tde L, Gabrish H, Herman TS, Vlachaki MT, Ahmad S. Impact of tissue heterogeneity corrections in stereotactic body radiation therapy treatment plans for lung cancer. J Med Phys. 2010 Jul;35(3):170-3.

Thibodeau R, Li HK, Tanny S et al. Heterogensous versus homogeneous radiation dose calculations of twice-daily fractionation in small cell lung cancer. Cureus 13(12): 320226, 2021.

Zabihzadeh M, Ghahremanai Z, Hoseini SM et al. Effect of lung inhomogeneity on dose distribution during radiotherapy of patient with lung cancer. Int J Radiat Res 18(3): 579-586, 2020

**18.4 Quality Assurance for SRS/SBRT**

**Question 252:**

What is the most accurate definition of small field for radiosurgery quality assurance, based on the American Association of Physicists in Medicine (AAPM) Task Group (TG) 155 report?

A. ≤ 1 cm x 1 cm
B. ≤ 2 cm x 2 cm
C. ≤ 3 cm x 3 cm
D. ≤ 4 cm x 4 cm

**Answer Question 252: C**

For radiosurgery quality assurance, a small field is 3 cm x 3 cm or less. The small fields used in radiosurgery (SRS) or stereotactic body radiotherapy (SBRT) are associated with the following challenges:

- Accurate dosimetry- there is decreased scattering, absence of electronic equilibrium and steep dose gradients with SRS or SBRT fields compared to conventional external beam fields
- Detector considerations – detectors for radiosurgery need to have high spatial resolution (sub-millimeter resolution). Accuracy is also needed due to the small treatment volumes and steep dose gradients. Detectors need to be of comparable field size to avoid volume averaging. Detectors also need to be made of water equivalent properties (or have accurate correction factors) to minimize the perturbation effect of radiation due to the materials used to fabricate the detectors.
- Output factors: output factors or dose delivered per unit of radiation emitted can be overestimated with treatment planning systems for small fields
- Patient specific quality assurance is used to identify discrepancies between the calculated SRS or SBRT plan by the treatment planning system and the actual dose delivered by the treatment machine.

Reference:

Duggan DM, Coffey CW 2nd. Small photon field dosimetry for stereotactic radiosurgery. Med Dosim. 1998 Fall;23(3):153-9. doi: 10.1016/s0958-3947(98)00013-2. PMID: 9783268.

**Question 253:**

Which of the following is used to verify the accuracy of the linac isocentre for cranial radiosurgery?

A. Water tank measurement
B. Output constancy check
C. Winston-Lutz test
D. IGRT QA

**Answer Question 253: C**

Rationale:

A. Water tank measurement is used for scanning beam PDD and profiles.
B. Output constancy check is used to verify that the machine's output (dose/MU) is within tolerance for accurate treatment
C. The Winston-Lutz (WL) test is a well-described procedure developed for verification of the linac isocenter for cranial SRS.
D. IGRT QA is used for checking the reliability of machine's imaging system (MV and KV).

Reference:

Galvin JM, Bednarz G. Quality assurance procedures for stereotactic body radiation therapy. International Journal of Radiation Oncology* Biology* Physics. 2008 May 1;71(1):S122-5.

# Chapter 19: Quality assurance (QA) in radiation oncology

Yunjie Yang, Jessica Fagerstrom, May Tsao

**19.1 Linac and CT based QA**

**Question 254:**

According to the American Association of Physicists in Medicine (AAPM) Task Group (TG)-142 and the Medical Physics Practice Guidelines (MPPG) report 8b recommendations, which of the following quality assurance (QA) test is recommended to be performed **daily** on a linear accelerator used for patient treatments in radiation oncology?

A. Output constancy check
B. Light to radiation field coincidence check
C. Beam energy check
D. Radiation isocenter check

**Answer Question 254: A**

While many clinics may have a daily QA program that performs more checks compared to AAPM TG-142 and MPPG8b recommendations depending on the availability of tools and systems, only output constancy check is recommended in both guidelines that needs to be checked daily.

References:

Hanley J, Dresser S, Simon W, Flynn R, Klein EE, Letourneau D, Liu C, Yin FF, Arjomandy B, Ma L, Aguirre F, Jones J, Bayouth J, Holmes T. AAPM Task Group 198 Report: An implementation guide for TG 142 quality assurance of medical accelerators. Med Phys. 2021 Oct;48(10):e830-e885. doi: 10.1002/mp.14992. Epub 2021 Aug 11. PMID: 34036590.

Krauss RF, Balik S, Cirino ET, Hadley A, Hariharan N, Holmes SM, Kielar K, Lavvafi H, McCullough K, Palefsky S, Sawyer JP, Smith K, Tracy J, Winter JD, Wingreen NE. AAPM Medical Physics Practice Guideline 8.b: Linear accelerator performance tests. J Appl Clin Med Phys. 2023 Nov;24(11):e14160. doi: 10.1002/acm2.14160. Epub 2023 Oct 4. PMID: 37793084; PMCID: PMC10647991.

**Question 255:**

Which of the following quality assurance (QA) tests is specifically designed to verify the coincidence of the mechanical isocenter with the radiation isocenter of a linear accelerator?

A) Daily laser alignment test
B) Winston-Lutz test
C) Gantry sag measurement
D) EPID (electronic portal imaging device) calibration

**Answer Question 255: B**

The Winston-Lutz test is specifically designed to verify the coincidence of the mechanical isocenter, defined by the movements of the gantry, collimator, and couch, with the radiation isocenter of a linear accelerator. This test involves using a small, high-density ball-bearing phantom placed at the presumed isocenter and acquiring images from multiple gantry, collimator, and couch angles. The test evaluates whether the center of the radiation field, as seen in these images, coincides with the position of the ball-bearing, thus confirming that the mechanical and radiation isocenters are well-aligned.

The daily laser alignment test (Option A) ensures the room lasers, which are used for patient and phantom setup, are aligned with the mechanical isocenter. However, it does not confirm the relationship between the mechanical isocenter and the radiation isocenter. The gantry sag measurement (Option C) checks for deviations in gantry positioning due to mechanical flex, which can affect treatment accuracy, but it is not a direct test of isocenter alignment. EPID calibration (Option D) is primarily focused on the image quality and calibration of the EPID system and does not assess isocenter alignment.

Reference:

Thwaites DI, Mijnheer BJ, Mills JA Chapter 12: Quality assurance of external beam radiotherapy. Review of Radiation Oncology Physics: A Handbook for Teachers and Students.

https://www.irsn.fr/sites/default/files/documents/professionnels_sante/documentation/syllabus_chapitre_12.pdf

**19.2 Treatment related QA**

**Question 256:**

According to ICRU Report 24, what is the overall tolerance for dose accuracy in the treatment delivered to the patient (which takes into account all the uncertainties before treatment delivery)?

A. 1%
B. 2%
C. 3%
D. 4%
E. 5%

**Answer Question 256: E**

The ICRU Report 24, reviewed tumour control probability data and concluded that an uncertainty of 5% in dose accuracy was required in the delivery of absorbed dose to the target volume. Although this has been widely quoted as a standard, it was not explicitly stated the confidence interval that this represented.

Although not specified in ICRU Report 24, in critical situations (eg. steep dose fall-off, or for radiosurgery), 2% may be required.

References:

Thwaites DI, Mijnheer BJ, Mills JA Chapter 12: Quality assurance of external beam radiotherapy. Review of Radiation Oncology Physics: A Handbook for Teachers and Students.

https://www.irsn.fr/sites/default/files/documents/professionnels_sante/documentation/syllabus_chapitre_12.pdf

Wambersie A. The role of the ICRU in quality assurance in radiation therapy. Int J Radiat Oncol Biol Phys. 1984 Jun;10 Suppl 1:81-6. doi: 10.1016/0360-3016(84)90454-1. PMID: 6429103.

INTERNATIONAL ATOMIC ENERGY AGENCY (IAEA), "Quality assurance in radiotherapy", IAEA TECDOC-1040, IAEA, Vienna, Austria (1997).

**19.3 Patient-Specific QA**

**Question 257:**

Which of the following is **not** a component of patient specific quality assurance?

A. Plan and chart check
B. In vivo verification
C. Peer review quality assurance radiotherapy rounds
D. Tumour control probability

**Answer Question 257: D**

All are components of patient specific quality assurance except D.

Tumour control probability are models based on dose-response relationship between radiation dose and tumour control on a number of patients (not individual).

References:

Thwaites DI, Mijnheer BJ, Mills JA Chapter 12: Quality assurance of external beam radiotherapy. Review of Radiation Oncology Physics: A Handbook for Teachers and Students.

https://www.irsn.fr/sites/default/files/documents/professionnels_sante/documentation/syllabus_chapitre_12.pdf

# Chapter 20: Radiation protection and shielding

Steven Babic, May Tsao

## 20.1 Principles of Radiation Protection

**Question 258:**

What is the main purpose of ALARA?

A. To keep organs at risk doses to a minimum
B. To allow adequate doses to tumour for tumour control
C. To minimize radiation exposure
D. To optimize radiation dose distributions

**Answer Question 258: C**

ALARA (As Low as Reasonably Achievable, social and economic factors being taken into account) is a radiation protection principle used in radiation safety to minimize exposure to radiation, while still ensuring the beneficial uses of radiation therapy and considering the practicalities of implementation. All reasonable measures should be taken to reduce exposure, even if the exposure is within legal limit.

Reference:

Frane N, Bitterman A. Radiation Safety and Protection. 2023 May 22. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan–. PMID: 32491431.

**Question 259:**

Which one of the following is not one of the 3 core International Commission on Radiological Protection (ICRP) principles?

A. Justification
B. Optimization
C. Exposure limit
D. Radiation protection quality assurance

**Answer Question 259: D**

The ICRP system of radiation protection is based on 3 core principles: justification, optimization and exposure or dose limit.

Justification: Any activity involving radiation exposure must be justified. This ensures that radiation exposure is only used when the potential benefits outweigh the risks.

Optimization: Radiation doses should be kept as low as reasonably achievable (ALARA) taking into account economic and social factors.

Exposure/ Dose limit: the total dose to any individual from regulated sources in planned exposure situations other than medical exposure of patients should not exceed appropriate limits.

Reference:

https://www.icrp.org/publication.asp?id=ICRP%20Publication%20103#:~:text=They%20maintain%20the%20Commission's%20three,radiological%20protection%20of%20the%20environment.

**Question 260:**

For linac radiation protection purposes, which one of the following is NOT considered a main source of radiation?

A. Primary radiation
B. Scatter radiation
C. Cone beam CT
D. Leakage

**Answer Question 260: C**

The three radiation types used for radiation protection purposes are:

Primary radiation, which comes directly from the source, used to treat patients

Scatter radiation, which is the radiation scattered from the beam by the patient, treatment couch etc.

Leakage, which is the radiation that leaks out of the head of the linac. Leakage is regulated to be no more than 0.1% of the useful beam dose rate at 1 m

The walls of a linac bunker must be shielded for these sources of radiation.

Reference:

Silari M. Special radiation protection aspects of medical accelerators. Radiat Prot Dosimetry. 2001;96(4):381-92. doi: 10.1093/oxfordjournals.rpd.a006626. PMID: 11843087.

**20.2 Concepts and Units**

**Question 261:**

What are quality factors in radiation protection?

A. A number related to biological effectiveness of different types of radiation
B. A list of factors describing radiation protection procedures
C. A description of how well a radiation bunker is shielded as per occupational limits
D. A description of time, distance and shielding factors in a radiation department's radiation protection program

**Answer Question 261: A**

In radiation protection, the quality factor Q is used to quantitatively describe the relative biological effectiveness of different types of radiation. The idea is that not all radiation types cause the same amount of biological damage for the same amount of energy absorbed by tissue. Different radiation types are assigned a quality factor Q (to standardize the biological effect) which is then used to calculate a dose equivalent (measured in Sieverts or Sv) from absorbed dose (measured in Gy). A higher Q factor represents greater biological damage.

| Radiation type and energy range | Quality factor, Q |
|---|---|
| Photons, all energies | 1 |
| Electrons, all energies | 1 |
| Neutrons, thermal | 5 |
| Neutrons, fast | 20 |
| Alpha particles, all energies | 20 |

Reference:

Kerr GD. Quality factors. Health Phys. 1988 Aug;55(2):241-9. doi: 10.1097/00004032-198808000-00016. PMID: 3410691.

**Question 262:**

If a patient receives 1 Gy of alpha particles, what is the dose equivalent?

A. 1 Sv
B. 2 Sv
C. 5 Sv
D. 10 Sv
E. 20 Sv

**Answer Question 262: E**

The Q factor of alpha particles is 20.

Equivalent dose = absorbed dose x Q factor (in Sieverts)
= 1 Gy x 20
= 20 Sv (Sieverts)

Reference:

Kerr GD. Quality factors. Health Phys. 1988 Aug;55(2):241-9. doi: 10.1097/00004032-198808000-00016. PMID: 3410691.

**Question 263:**

Stochastics effects from radiation are tissue effects, in which:

A. there is no effect below a threshold dose
B. the probability of the effect increases with dose
C. the probability of the effect is dependent on lethal damage to the functional subunit
D. the probability of the effect is independent on the lethal damage to the functional subunit

**Answer Question 263: B**

Stochastic effects have no threshold dose. Even low doses can cause these effects. Examples include DNA mutations in somatic cells leading to cancer or mutations in germ cells leading to heritable effects.

Deterministic effects or non-stochastic effects have a threshold dose below which the effect does not occur. Examples include radiation induced sterility, radiation induced cataract formation.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**20.3 Radiation Monitoring, Surveys, Leak Testing**

**Question 264:**

According to the U.S. Nuclear Regulatory Commission (NRC), what is the occupational exposure limit (to the whole body) for a non-pregnant radiation worker in one year?

A. 25 mSv
B. 50 mSv
C. 75 mSv
D. 100 mSv

**Answer Question 264: B**

Occupational radiation exposure limits protect radiation workers from the harmful effects of radiation. The limit to the whole for a non-pregnant worker is 50 mSv in a single year and 100 mSv over 5 years, with a higher limit for certain tissues, such as **500 mSv** for the lens of the eye and **500 mSv** for the skin or extremities.

For pregnant workers, the dose to the embryo/fetus is limited to 5 mSv from the time pregnancy is declared to the end of term.

For minors (under the age of 18), the limits are 10% of the occupational dose limits for adults, often capped at 5 mSv per year.

1 rem = 10 mSv

NRC **occupational** dose limits

| | |
|---|---|
| Whole body | 50 mSv in a year |
| Any organ | 500 mSv in a year |
| Skin | 500 mSv in a year |
| Extremity | 500 mSv in a year |
| Lens of the eye | 150 mSv in ayear |
| Fetus of declared pregnancy worker | 5 mSv in a year |

ICRP **public** dose limits

| | |
|---|---|
| Whole body | 1 mSv in a year |
| | |
| Skin | 50 mSv in a year |
| | |
| Lens of the eye | 15 mSv in a year |
| | |

References:

https://www.nrc.gov/reading-rm/basic-ref/glossary/occupational-dose.html

https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/part020-1301.html

**Question 265:**

A Cs- 137 tube with an activity of 6.2 mCi was mistakenly left in the drawer of a wooden desk for 24 hours before being discovered. A radiation oncologist occupied the desk for 8 hours at an average distance of 1 m from the source. What is the average radiation exposure to the oncologist?

Cs-137decay: 0.512 MeV $\beta^-$ (94.6%) and 0.661 MeV ϒ (85.1%)

Cs-137 half-life is 30 years

Cs-137 gamma ray constant is 3.26 $Rcm^2$ / h mCi

A. 0.02 R
B. 1.2 R
C. 5 R
D. 10 R

**Answer Question 265: A**

Source encapsulation typically shields the electron. The activity decay over the period of exposure is negligible as the half-life for Cs-137 is 30 years and the exposure is 8 hours.

The equation to determine the exposure from a radionuclide can be determined via the gamma-ray constant via the equation:

$$X = \frac{\Gamma A t}{d^2}$$

$$X = \frac{3.26 \frac{R\ cm^2}{h\ mCi}}{10\ 000\ cm^2} \times 6.2\ mCi\ \times 8\ h$$

= 0.0162 R
**≈ 0.02 R**

Where:

*X* = exposure

$$\Gamma\ (gamma\ ray\ constant) = 3.26\ \frac{R\ cm^2}{h\ mCi}$$

A (activity)= 6.2 mCi

t (time of exposure)=8h

d (distance) = 1 m

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**Question 266:**
A T1 N0 M0 squamous cell carcinoma of the floor of mouth is being planned for interstitial brachytherapy. Which of the following is **not** a consideration for radiation protection to minimize exposure?

A. Time
B. Distance
C. Shielding
D. Use low activity sources

**Answer Question 266: D**

All are radiation protection principles to minimize radiation exposure, except for D.

Minimize the time spent near a radiation source to only what it takes to get the job done.

Maximize the distance from the radiation source as much as possible.

Put something between you and the radiation source.

The use of low activity sources may require an increase in treatment time, but in itself, low activity sources are not considered a basic principle for radiation protection when handling sealed radioisotopes.

Reference:

Gibbons, J. P., & Khan, F. M. (2020). *Khan's the physics of radiation therapy* (Sixth edition). Lippincott Williams & Wilkins.

**20.4 Shipping and Receiving Radioactive Sources**

**Question 267:**

Which U.S. agency does **NOT** issue regulations that govern the transport of class 7 radioactive materials?

A. US Department of Transportation
B. Nuclear Regulatory Commission
C. International Atomic Energy Agency

**Answer Question 267: C**

In the U.S., the Department of Transportation and the Nuclear Regulatory Commission are the primary organizations that issue regulation on the transport or radioactive materials based on the standards developed by the International Atomic Energy Agency (IAEA). The **I**AEA plays a central role in establishing international guidelines for the safe transport of radioactive materials. However, the IAEA does not issue regulations that govern the transport of class 7 radioactive materials.

Reference:

https://www.nrc.gov/reading-rm/basic-ref/students/for-educators/11.pdf

**Question 268:**

When receiving a package containing radioactive material, what is the first action required?

A. Store the package in a controlled area
B. Open the package immediately to verify contents
C. Survey the package for contamination and radiation levels
D. Sign and file the shipping paperwork

**Answer Question 268: C**

According to the U.S. Nuclear Regulatory Commission (NRC), when receiving a package containing radioactive material, licensees must follow specific procedures to ensure safety and regulatory compliance.

1) Perform a a visual inspection and examine the package for signs of damage.
2) Survey the external surface for radiation levels using a calibrated survey meter.

3) Perform a wipe test of the package surface to check for removable radioactive contamination
4) Immediately report to the final delivery carrier and to the NRC if any of the following are found: radiation level > 2 mSv/h (200 mrem/h) at the package surface; contamination > limits; evidence of leakage or damaged contents

References:

https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/part020-1906.html
https://www.nrc.gov/reading-rm/doc-collections/cfr/part071/index.html
https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1384_web.pdf

**Question 269:**

What does the Transport Index (TI) represent on a radioactive package label?

A. The number of isotopes in the package
B. The distance in km the package may be transported
C. The dose rate in µSv/h at 1 meter from the package
D. The weight of the package in kg

**Answer Question 269: C**

The Transport Index is a dimensionless number assigned to a package to help control its handling, storage, and transportation.

For example, if the radiation dose rate is 0.015 mSv/h at 1 meter:

TI = 0.015 × 100 = 1.5

Reference:

https://www.nrc.gov/reading-rm/doc-collections/cfr/part071/index.html
https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1384_web.pdf

**20.5 Types of Radiation Exposure- NCRP160**

**Question 270:**

What was the primary purpose of the National Council on Radiation Protection and Measurements (NCRP) 160 report?

A. To detail radiation exposure in the general US population
B. To detail radiation exposure to radiation workers
C. To define radiation safety limits for the general public
D. To detail radiation protection procedures for radiation oncology departments

**Answer Question 270: A**

NCRP 160 is a report titled "Ionizing Radiation Exposure of the Population of the United States". This report detailed radiation exposure to the US population from both natural background radiation (radon, terrestrial and cosmic radiation) as well as radiation from diagnostic imaging (particularly CT scans and nuclear medicine scans). The report found that the average exposure increased from 3.6 mSv in the 1980s to 6.2 mSv in 2006, largely due to increased CT use.

References:

National Council on Radiation Protection and Measurements (NCRP). (2009). *Ionizing radiation exposure of the population of the United States* (NCRP Report No. 160). Bethesda, MD: NCRP.

Bolus NE. NCRP report 160 and what it means for medical imaging and nuclear medicine. J Nucl Med Technol. 2013 Dec;41(4):255-60. doi: 10.2967/jnmt.113.128728. Epub 2013 Oct 31. Erratum in: J Nucl Med Technol. 2014 Jun;42(2):160. PMID: 24179182.

**Question 271:**

According to NCRP Report 160, what is the largest source of natural background radiation exposure to the average person living in the United States?

A. Nuclear power plants
B. Cosmic radiation
C. Medical imaging
D. Radon gas and its decay products

**Answer Question 271: D**

Radon is a radioactive gas that seeps into buildings from soil and rock, especially in basements and ground floors. Radon contributes an average of ~2.3 mSv/year to an individual's effective dose and

accounts for approximately 37% of the total 6.2 mSv annual radiation exposure to a person living in the U.S.

Total Annual Radiation Exposure (NCRP 160):

| Source Category | Component | Average Dose (mSv/year) | % of Total Exposure |
|---|---|---|---|
| **Natural Background** | Radon and Thoron | ~2.3 | ~37% |
| | Cosmic Radiation | ~0.33 | ~5% |
| | Terrestrial Radiation | ~0.21 | ~3% |
| | Internal Radiation (e.g., potassium-40) | ~0.29 | ~5% |
| | **Subtotal Natural** | **~3.1** | **~50%** |
| **Medical Exposure** | Computed Tomography (CT) | ~1.5 | ~24% |
| | Nuclear Medicine | ~0.7 | ~11% |
| | Interventional Fluoroscopy | ~0.4 | ~6% |
| | Conventional Radiography (X-rays) | ~0.33 | ~5% |
| | **Subtotal Medical** | **~3.0** | **~48%** |
| **Other Sources** | Consumer Products (e.g., tobacco, building materials) | ~0.1–0.13 | ~2% |
| | Occupational, Industrial, Environmental | ~<0.01 each | ~<1% total |

Reference:

National Council on Radiation Protection and Measurements (NCRP). (2009). *Ionizing radiation exposure of the population of the United States* (NCRP Report No. 160). Bethesda, MD: NCRP.

**Question 272:**

Which of the following is not a source of ionizing radiation discussed in NCRP 160?

A. CT scan
B. Radon
C. Ultrasound
D. Gamma rays

**Answer Question 272: C**

Ionizing radiation is radiation with enough energy to eject electrons from atoms, creating ions (charged particles).

Ultrasound uses high-frequency sound waves that do not carry enough energy to ionize atoms or molecules.

Reference:

National Council on Radiation Protection and Measurements (NCRP). (2009). *Ionizing radiation exposure of the population of the United States* (NCRP Report No. 160). Bethesda, MD: NCRP.

**20.6 Protection Regulations**

**Question 273:**

Which organization does **not** provide regulation for linear accelerator radiation protection?

A. Nuclear Regulatory Commission (NRC)
B. International Atomic Energy Agency (IAEA)
C. Food and Drug Administration (FDA)

**Answer Question 273: B**

The NRC and FDA regulates linear accelerator radiation protection.

The IAEA and ICRP provide international guidance and standards for radiation protection, which are adopted by regulatory bodies. However, the IAEA and ICRP (International Commission on Radiological protection) do not regulate linear accelerator radiation protection.

**Question 274:**

A "Caution, Radiation Area" sign must be posted when radiation levels could result in an individual receiving:

A. >0.05 mSv in one hour
B. >1 mSv in a year
C. >0.1 mSv in a year
D. >0.5 mSv in one year

**Answer Question 274: A**

The sign must include the trefoil radiation symbol (magenta/purple on a yellow background and it must bear the words: “Caution: Radiation Area”.

Reference:

https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/index.html

**20.7 Radiation Shielding**

**Question 275:**

What thickness of concrete is needed to shield the primary beam from an adjacent waiting room (used by the general public), which is located next to a linear accelerator with the following parameters?

- Distance from the waiting room to the linac isocentre is 6 m
- The linac workload is 57 600 Gy at the isocentre per year
- SAD = 1m
- The tenth value layer of concrete is 30 cm
- The fraction of time the general public (family member) spends in the waiting room is associated with an occupancy factor of 0.25

A. 0.25 m
B. 0.5 m
C. 1 m
D. 1.5 m

**Answer Question 275: D**

The limit value for the general public is 1 mSv per year.

$$P = \frac{WUT}{A} \times \left(\frac{dref}{d}\right)^2$$

P = dose limit in sV
A = minimum attenuation required for the shield/barrier
W= workload; a measure of radiation output in Gy
U = use factor or the fraction of time the primary beam is in a certain direction

U = 1 when the gantry is pointing down
U=0.5 when the gantry is pointing up
U= 0.25 when the gantry is in the lateral direction

T= occupancy factor or the time a particular location is occupied
Dref = the distance from the source to the reference point (eg. isocentre)
D = distance from the point of interest to the isocentre

$$A = \frac{WUT\left(\frac{dref}{d}\right)^2}{P}$$

$$= \frac{57\,600 \times 0.25 \times 0.25}{0.001} \times \left(\frac{1}{6}\right)^2$$

= 100 000

The tenth value layer of concrete is 30 cm=0.03 m

One tenth value layer attenuates dose by 1/10.

Two tenth value layers attenuate the dose by 1/10 x 1/10 = 1/100

Three tenth value layers attenuate the dose by 1/10 x 1/10 x 1/10 = 1/1 000

Four tenth value layers attenuate the dose by 1/10 x 1/10 x 1/10 x 1/10 = 1/ 10 000

Five tenth value layers attenuate the dose by 1/10 x 1/10 x 1/10 x 1/10 x 1/10 = 1/ 100 000

Therefore, 5 tenth value layers are needed or:

5 x 0.3 m = 1.5 m

Reference:

Rodgers JE. Radiation therapy vault shielding calculational methods when IMRT and TBI procedures contribute. J Appl Clin Med Phys. 2001 Summer;2(3):157-64. doi: 10.1120/jacmp.v2i3.2609. PMID: 11602012; PMCID: PMC5726044.

**Question 276:**

For linear accelerators operating above 10 MV, the door shielding must account for:

A. Only photon leakage
B. Scatter from patient only
C. Neutron radiation and associated capture gamma rays
D. Ultrasound interference

**Answer Question 276: C**

At photon energies above approximately 10 MeV, high-energy X-rays can induce photonuclear (γ, n) reactions in the high-Z components of a linear accelerator, such as the target, flattening filter, and collimators. These reactions produce fast neutrons, which are highly penetrating and pose a significant biological hazard. Neutrons can travel down the maze or reach the treatment room door, making appropriate shielding essential.

A typical linac bunker door is constructed with three shielding layers, each serving a specific purpose:

1. The inner layer is made of a high atomic number (high-Z) material, such as lead, which attenuates primary photons from the linac and also helps slow down fast neutrons, enhancing the effectiveness of the next layer.
2. The middle layer consists of borated polyethylene (BPE), a hydrogen-rich material that effectively moderates and captures thermal neutrons. However, during neutron capture, this layer emits capture gamma rays.
3. The outer layer, also composed of a high-Z material like lead, serves to attenuate the secondary capture gamma rays produced in the BPE layer, thereby minimizing radiation leakage at the vault entrance and ensuring the safety of staff and public.

Reference:

National Council on Radiation Protection and Measurements (NCRP). (2005). *Structural Shielding Design and Evaluation for Megavoltage X- and Gamma-Ray Radiotherapy Facilities* (NCRP Report No. 151). Bethesda, MD: NCRP.

**Question 277:**

What does a secondary barrier protect against in a radiation therapy or diagnostic setting?

A. Only from UV rays
B. Neutron capture gamma rays
C. Scattered and leakage radiation
D. Infrared radiation

**Answer Question 277: C**

In radiation therapy or diagnostic imaging, a secondary barrier is designed to protect against scattered radiation—which results from the primary beam interacting with the patient, walls, or other surfaces—and leakage radiation that escapes from the X-ray tube housing or linac head.

Reference:

National Council on Radiation Protection and Measurements (NCRP). (2005). *Structural Shielding Design and Evaluation for Megavoltage X- and Gamma-Ray Radiotherapy Facilities* (NCRP Report No. 151). Bethesda, MD: NCRP.

National Council on Radiation Protection and Measurements (NCRP). (2004). *Structural Shielding Design for Medical X-Ray Imaging Facilities* (NCRP Report No. 147). Bethesda, MD: NCRP.

**Question 278:**

Which material is most effective for shielding high-energy photons (e.g., from linacs)?

A. Aluminum
B. Lead
C. Water
D. Polyethylene

**Answer Question 278: B**

Lead is the most effective material for shielding high-energy photons due to its high atomic number, which enhances the likelihood of radiation interactions that absorb or scatter photons. Its high density allows a greater concentration of atoms in a compact volume, enabling efficient attenuation with less material thickness compared to alternatives. Lead is easily shaped and widely accessible, making it a practical choice for radiation shielding applications.

Reference:

National Council on Radiation Protection and Measurements (NCRP). (2005). *Structural Shielding Design and Evaluation for Megavoltage X- and Gamma-Ray Radiotherapy Facilities* (NCRP Report No. 151). Bethesda, MD: NCRP.

# Chapter 21: Safety and incidents

Steven Babic, May Tsao

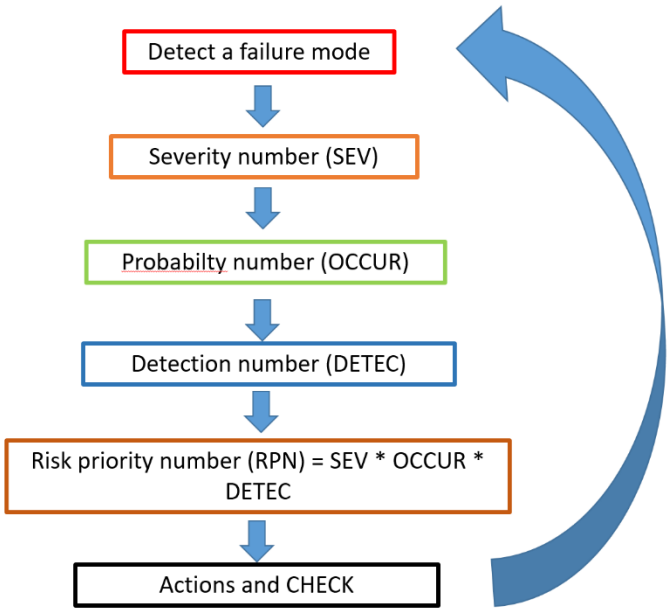
**Question 279:**

A failure mode and effects analysis (FMEA):

A. identifies personnel and human error contributing to mistakes
B. delves into the root causes to find the actual source of the problem,rather than simply addressing the immediate issue
C. is a structured, sequential process of identifying potential failures, their causes and effects
D. begins with the undesired event (e.g. unintended radiation overdose) and then analyzes the conditions that could lead to it

**Answer Question 279: C**

A failure mode and effects analysis (FEMA) is a structured, sequential process of identifying potential failures, their causes and effects.




References:

Chowdhury IH, Garg R, Huber KE, Stambaugh NP, Stambaugh C. Implementation of a patient safety training program in radiation oncology residency: A pilot study. J Appl Clin Med Phys. 2024 Feb 15;25(3):e14286. doi: 10.1002/acm2.14286. Epub ahead of print. PMID: 38358132; PMCID: PMC10929992.

https://en.wikipedia.org/wiki/Failure_mode_and_effects_analysis

**Question 280:**

How is the risk priority number (RPN) utilized during a failure mode and effects analysis?

A. Higher RPN values indicate higher risk and direct organizations to focus on mitigating these first
B. Lower RPN values indicate higher risk and direct organizations to focus on mitigating these first
C. A failure mode and effects analysis does not utilize RPN

**Answer Question 280: A**

The Risk Priority Number (RPN) is a tool for prioritizing risks in Failure Mode and Effects Analysis (FMEA) and other risk management processes. It is determined by multiplying the ratings for severity (S), occurrence (O), and detection (D) of a potential failure. Each factor is typically rated on a scale between 1-10, with 1 being lowest and 10 being highest. Hence, the value of RPN can range between 1 and 1000. Failure modes or causes with the highest RPN values require the most urgent attention and should be dealt with first.

Reference:

AAPM TG-100. The report of Task Group 100 of the AAPM: Application of risk analysis methods to radiation therapy quality management, 2016.

https://aapm.onlinelibrary.wiley.com/doi/10.1118/1.4947547

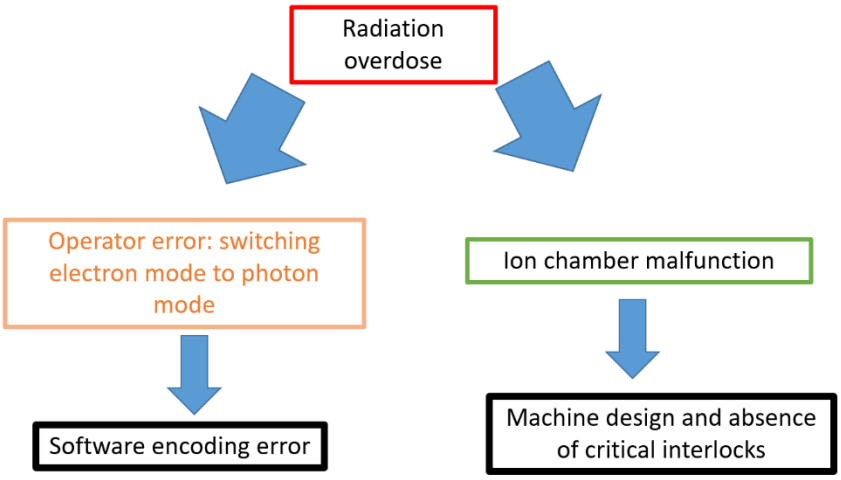
**Question 281:**
What is a Fault Tree Analysis (FTA)?

A. identifies personnel and human error contributing to mistakes
B. delves into the root causes to find the actual source of the problem, rather than simply addressing the immediate issue
C. is a structured, sequential process of identifying potential failures, their causes and effects
D. begins with the undesired event (eg. unintended radiation overdose) and then analyzes the conditions that could lead to it (a systematic top-down method)

**Answer Question 281: D**

A Fault Tree Analysis begins with the undesired event (eg. unintended radiation overdose) and then analyzes the conditions that could lead to it.

An example of a fault tree analysis:




Reference:

Israelski EW, Muto WH. Human factors risk management as a way to improve medical device safety: a case study of the therac 25 radiation therapy system. Jt Comm J Qual Saf. 2004 Dec;30(12):689-95. doi: 10.1016/s1549-3741(04)30082-1. PMID: 15646102.

**Question 282:**

Which of the following is **NOT** an example of radiotherapy Incident Learning Systems?

A. ROSIS
B. CARS
C. SAFRON
D. ROILS
E. FTA

**Answer Question 282: E**

All the above are examples radiation oncology incident learning systems, except for E (FTA- fault tree analysis). A fault tree analysis is a type of failure analysis but, in itself is not a radiotherapy incident learning system.

An Incident Learning System is a structured process for identifying, reporting, analyzing and learning from incidents to improve patient safety and prevent future harm. Incident learning systems are categorized by their scope (local, national, international) and the type of incident addressed (eg. medication, radiation therapy, general patient safety).

ROSIS- Radiation Oncology Safety Information System is an international system for radiation oncology incidents

CARS – Radiotherapy Incident Reporting and Learning System from the Center for Assessment of Radiological Sciences (CARS)

SAFRON- Safety in Radiation Oncology has been established by the International Atomic Energy Agency (IAEA).

ROILS – Radiation Oncology Incident Learning System sponsored by the American Society of Radiation Oncology (ASTRO) and the American Association of Medical Physicists (AAPM)

Reference:

Pawlicki T, Coffey M, Milosevic M. Incident Learning Systems for Radiation Oncology: Development and Value at the Local, National and International Level. Clin Oncol (R Coll Radiol). 2017 Sep;29(9):562-567. doi: 10.1016/j.clon.2017.07.009. Epub 2017 Jul 21. PMID: 28739028.

**Question 283:**

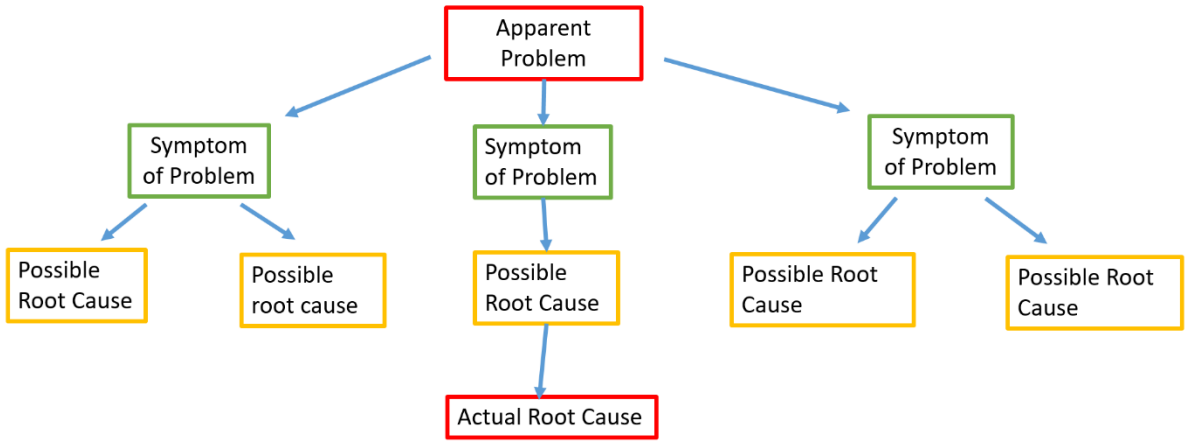
What is a Root Cause Analysis?

A. identifies personnel and human error contributing to mistakes
B. delves into the actual source of the problem, rather than simply addressing the immediate issue
C. is a structured, sequential process of identifying potential failures, their causes and effects
D. begins with the undesired event (eg. unintended radiation overdose) and then analyzes the conditions that could lead to it

**Answer Question 283: B**

A root cause analysis delves into the root causes to find the actual source of the problem, rather than simply addressing the immediate issue

Root cause analysis tree diagram:




Reference:
Chowdhury IH, Garg R, Huber KE, Stambaugh NP, Stambaugh C. Implementation of a patient safety training program in radiation oncology residency: A pilot study. J Appl Clin Med Phys. 2024 Feb 15;25(3):e14286. doi: 10.1002/acm2.14286. Epub ahead of print. PMID: 38358132; PMCID: PMC10929992.

**Question 284:**

What was the key design flaw with Therac -25 that lead to radiation overdose leading to death and disability?

A. Monitor units were incorrectly calculated by 100 fold
B. High dose electron beams were generated incorrectly
C. Therac-25 untis produced excessive amounts of neutron contamination
D. The radiation beam did not “turn off” when the calculated monitor units were reached

**Answer Question 284: B**

The Therac-25 was a computer –controlled radiation machine produced by Atomic Energy of Canada Limited (AECL) in 1982.

There were 6 published accidents reported with the Therac-25 which arose when the high-current electron beam generated in X ray mode was delivered to patients. Two software critical flaws were discovered. When the operator incorrectly selected x-ray mode but then quickly corrected the setting to electron mode, the electron beam remained set for x-ray mode. A second critical fault allowed the electron beam to activate during a field-light mode. In the light field mode, the patient and collimator were positioned by illuminating the treatment area with visible light. There was no ion chamber to detect dose in the light mode, since electrons or photons were not expected in light mode. Although

previous models had hardware interlocks in place to prevent these critical errors, these were removed with Therac-25. Instead, software checks were used.

Patients were overdosed by approximately 100 times the intended radiation dose and oer a narrower area. A patient described the feeling as an "intense electric shock" which caused the patient to scream and run out of the treatment unit. Patients experienced radiation burns, systemic symptoms of radiation over dose and some patients died from radiation overdose.

References:

https://en.wikipedia.org/wiki/Therac-25

Israelski EW, Muto WH. Human factors risk management as a way to improve medical device safety: a case study of the therac 25 radiation therapy system. Jt Comm J Qual Saf. 2004 Dec;30(12):689-95. doi: 10.1016/s1549-3741(04)30082-1. PMID: 15646102.

**Question 285:**

In November 1992, a patient in Indiana Regional Cancer Center was treated for anal carcinoma using a 4.3 Ci Ir-192 source inserted using an HDR brachytherapy afterloader. The room radiation monitor showed high radiation levels following the treatment, but since the monitor was malfunctioning, the readings were dismissed. Additionally, a survey meter was not used to check the patient. The control console showed that the source had safely returned to the HDR unit but the source actually had detached from the wire and was lodged in one of the catheters inside the patient. The patient was transferred back to a nursing home where the catheter containing the radioactive pellet fell out 4 days later.

In what order of magnitude was the dose delivered in comparison to the intended dose?

A. Double
B. Three times
C. Ten times
D. One thousand times

**Answer Question 285: D**

It was estimated that the patient received a dose of 16,000 Gy or a thousand times the intended dose. The patient died 5 days after treatment and the coroner listed the cause of death as "acute radiation exposure and consequences thereof". Many nursing home residents, employees and visitors were also unknowingly exposed to radiation. The incident served as the catalyst for the NRC's regulations on HDR brachytherapy treatments and the NRC now mandates that patients be checked with a survey meter following treatment.

References:

https://www.nrc.gov/reading-rm/doc-collections/gen-comm/bulletins/1992/bl92003.html

https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1084_web.pdf

https://en.wikipedia.org/wiki/List_of_civilian_radiation_accidents

**Question 286:**

In 2010, the New York Times' coverage of radiation therapy accidents highlighted a number of treatment errors that resulted in patient injury and or death. One of the articles "Radiation Offers New Cures, and Ways to Do Harm" reported on the death of a patient in his early 40s who was being treated for oropharyngeal cancer using IMRT. After four fractions that were correctly delivered, the patient's radiation oncologist requested a re-plan with a lower dose to the teeth. During the re-plan, MLC leaf control point data was corrupted. The patient received 13 Gy for 3 fractions with open, unmodulated treatment fields that fully exposed the brainstem and resulted in the patient receiving seven times the prescribed dose.

This error could have been prevented by the following **EXCEPT**:

A. Observing MLC leaf movement during treatment.
B. Performing patient-specific QA or plan validation prior to treatment with the new plan.
C. Not re-planning and treating solely with the original plan.
D. By the Vendor introducing fail safe mechanisms in the treatment planning and Linac delivery software.

**Answer Question 286: C**

An unusual sequence of events led to this error. A minor plan modification was made after several treatment fractions. During the re-plan, the computer kept crashing and the medical physicist did not realize that corrupted MLC leaf control data had been saved. Immediate plan validation was overlooked due to a shortage in staffing. Radiation therapists only noticed after additional fractions that the MLC wasn't modulating the field. Subsequently, the manufacturer of the software introduced fail-safe provisions in the software to prevent this incident from happening again.

References:

https://www.nytimes.com/2010/01/24/health/24radiation.html

McDermott, P and Orton, C. "Physical Quality Assurance and Safety." *The Physics & Technology of Radiation Therapy*. 2nd ed. Madison, WI: Medical Physics Publishing, 2018, pp 21-3.

**Question 287:**

*"Safety is No Accident"* is ASTRO's detailed guide to ensuring safety in radiation oncology. Originally published in 2012 and updated in 2019 to reflect evolving standards, it outlines a comprehensive framework for quality assurance, fostering a culture of safety, and implementing best practices. Which of the following is **not** a key recommendation?

A. Safety culture is foundational
B. Standardized processes and QA are essential
C. Continuous learning and improvement are necessary
D. When errors occur, staff need to be blamed and held accountable for their mistakes

**Answer Question 287: D**

A culture of safety is essential and should be embedded in the department's core values. All staff must be empowered to report errors and concerns without fear of being reprimanded. Well-defined protocols, checklists, and rigorous quality assurance procedures help reduce variability and catch errors before they affect patients. Utilizing tools such as incident learning systems (e.g., RO-ILS) and continuous staff training promotes ongoing improvement and helps the team to stay aligned safely with advancing technologies and clinical practices.

References:

Safety is No Accident: A Framework for Quality Radiation Oncology Care. 2019.

https://www.astro.org/ASTRO/media/ASTRO/Patient%20Care%20and%20Research/PDFs/Safety_is_No_Accident.pdf